



**NAME Institute**

In front of Singhdurbar, Putalisadak

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# **MECEE BL**

## **Model Entrance Exam**

### **(Model: XX)**

### **(Set: A)**

**Date: Baishakh 05<sup>th</sup> 2083 (Saturday)**

**Venue: NAME**

**Duration – 3 hrs**

**Time – 11:00 AM**

#### **INSTRUCTIONS**

There are 200 multiple-choice questions, each having four choices of which only one choice is correct.

Fill ☐ the most appropriate one.

**Note: NAME students must fill their registration number on the OMR sheet.**

1. Mammals are characterized by the presence of
  - a) mammary glands
  - b) pinna
  - c) hair
  - d) diaphragm
2. Agnatha includes:
  - a) hag fishes
  - b) fishes
  - c) jelly fishes
  - d) flying fishes
3. The most appropriate term to describe the life cycle of *Obelia* is:
  - a) Neoteny
  - b) Metagenesis
  - c) Metamorphosis
  - d) Paedomorphism
4. In *Taenia*, area of proliferation is:
  - a) Body
  - b) Scolex
  - c) Neck
  - d) Strobila
5. Endostyle found in Protochordata are modified into which one of the following endocrine gland of vertebrates?
  - a) Pituitary gland
  - b) Thyroid glands
  - c) Pineal gland
  - d) Thymus gland
6. Which one of the following is not hermaphrodite animal
  - a) Leeches
  - b) Polychaetes
  - c) Flatworms
  - d) Earthworm
7. Biramous appendages are typically found in
  - a) Insecta
  - b) Crustacea
  - c) Annelida
  - d) Arachnida
8. A wood boring mollusca/Shipworm is
  - a) Chiton
  - b) Teredo
  - c) Limax
  - d) Patella
9. Which is unrelated.
  - a) Sea Cucumber
  - b) Sea Star
  - c) Sea Urchin
  - d) Sea Squid
10.  $3\frac{1}{2}$ -chambered heart is present in reptiles except:
  - a) Crocodile
  - b) Lizard
  - c) Snake
  - d) Turtle
11. The example of industrial melanism has been cited in which one of the following theory of organic evolution?
  - a) Lamarckism
  - b) Darwinism
  - c) Mutation theory
  - d) Modern synthetic theory
12. Which one of the following ancestor of man first time showed bipedal locomotion?
  - a) Australopithecus
  - b) Cro-Magnon
  - c) Java ape man
  - d) Peking man
13. Menotaxis is shown by
  - a) Ants and wasps
  - b) Wasps and honey bee
  - c) Honey bee and ants
  - d) Euglena and honey bee
14. The sweat gland in mammals are primarily concerned with:
  - a) Regulation of water content
  - b) Regulation of body heat
  - c) Killing of skin bacteria
  - d) Removal of excess of salt
15. The fossil of Cro-magnon man was discovered by
  - a) Mac Gregor
  - b) J. C. Fuhlrott
  - c) W. C. Pei
  - d) E. Dubois
16. Conus arteriosus is a part of the heart because
  - a) It has valves
  - b) It has cardiac muscles
  - c) It is a part of atrium
  - d) It is connected to ventricle
17. Which one of the following vertebrate organs receives the oxygenated blood only?

- a) Gill                      b) Lung                      c) Liver                      d) Spleen
18. Hearts of *Pheretima* are situated in the segments  
a) 10, 13, 16 and 17      b) 7, 9, 12 and 13      c) 4, 5, 10 and 13      d) 11, 14, 17 and 18.
19. In Earthworm the dorsal wall of the intestine from the 26th segment to 95th segment forms a median internal fold called as:  
a) Trochophore      b) Typhlosole      c) Clitellum      d) Trachea
20. Which one of the following pair of items correctly belongs to the category of organs mentioned against it?  
a) Thorn of *Bougainvillea* and tendrils of *Cucurbita* – Analogous organs  
b) Nictitating membrane and blind spot in human eye – Vestigial organs  
c) Nephridia of earthworm and Malpighian tubules of cockroach – Excretory organs  
d) Wings of honey bee and wings of crow – Homologous organs
21. An intracellular parasite is  
a) Entamoeba      b) Ascaris      c) Plasmodium      d) Both a & b
22. Maurer's dots occurs in human RBC, infected by  
a) *P. falciparum*      b) *P. ovale*      c) *P. vivax*      d) *P. malariae*
23. Second vertebra or Axis, vertebra of Human contains  
a) Odontoid process      b) Deltoid ridge      c) Olecranon process      d) Acromian process
24. Acetabulum forms..... Joint with head of femur?  
a) ball & socket      b) pivotal      c) saddle      d) none
25. Which of the following disease is now considered nearly eradicated from world?  
a) Smallpox      b) Polio myelitis      c) Plague      d) Kala-azar
26. The cavities of brain are lined by  
a) Cuboidal cells      b) Squamous cells      c) Ependymal cells      d) Pseudostratified cells
27. Which cells have the shape of signet rings?  
a) mast cells      b) osteocytes      c) adipocytes      d) melanocytes
28. Adrenal medulla has a:  
a) Apolar neuron      b) Bipolar neuron      c) Unipolar neuron      d) Multipolar neuron
29. Gastrocnemius muscle is found attached to:  
a) chest      b) thigh      c) shank      d) forearm
30. Hepatic duct is guarded by:  
a) Sphincter of Oddi      b) Sphincter of Boyden  
c) Eustachian valve      d) Thebasian valve
31. Which one of the following statement is true regarding digestion and absorption of food in humans?  
a) Oxyntic cells in our stomach secrete the proenzyme pepsinogen  
b) Fructose is transported actively in intestine  
c) Chylomicrons are small lipoprotein particles that are transported from intestine into blood capillaries  
d) About 60% of starch is hydrolysed by salivary amylase in our mouth
32. Total lungs capacity is approximately:

- a) 1200ml                      b) 2400ml                      c) 4000ml                      d) 5800ml
33. The cardiac cycle time in a normal man is:  
a) 80 sec                      b) 60 sec                      c) 72 sec                      d) 0.8 sec
34. Crura cerebri are found in:  
a) forebrain                      b) midbrain                      c) Hindbrain                      d) None of these
35. Which layer of the wall of an eyeball contains abundant blood vessels?  
a) Lens                      b) Retina                      c) Choroid                      d) Sclerotic
36. Gubernaculum is the ligamentous connective cord which connects:  
a) Testis to kidney                      c) Muscle to muscle  
b) Testis to scrotum                      d) Ovary to abdominal wall
37. Beginning of archenteron formation in frog's development represents the stage:  
a) morula                      b) early blastula                      c) early gastrula                      d) late gastrula
38. AIDS related complex or ARC is  
a) An early or intermediate stage of HIV infection  
b) Full-blown AIDS with severe opportunistic infections  
c) Not related to HIV infection  
d) A completely asymptomatic stage of HIV
39. 'Pathogens' are also known as  
a) Immune bodies                      b) Antigens                      c) Agglutinins                      d) Autotoxins
40. Filtration pressure in human Kidneys is about:  
a) +10 mm Hg                      b) +70 mm Hg                      c) +45 mm Hg                      d) +55 mm Hg
41. Brinjal, Potato, Tomato, Onion, Ginger belong to:  
a) Five genera                      b) Three genera                      c) Same genus                      d) Same family
42. A bacterium survives extreme heat and radiation mainly due to the presence of  
a) capsule                      b) plasmid                      c) endospore                      d) mesosome
43. Which algal group contributes maximally to marine primary productivity?  
a) Chlorophyceae                      b) Phaeophyceae                      c) Rhodophyceae                      d) Diatoms
44. The ecological role of fungi is best described as  
a) primary producers                      b) nitrogen fixers  
c) decomposers recycling nutrients                      d) secondary consumers
45. Lichens are considered indicators of air pollution because they  
a) absorb heavy metals actively                      b) lack roots and absorb directly from air  
c) require high CO<sub>2</sub> concentration                      d) grow only in polluted regions
46. Which bryophyte feature restricts them to moist habitats?  
a) Absence of roots                      b) Dominant gametophyte  
c) Flagellated sperm requiring water                      d) Lack of vascular tissue
47. Which statement is incorrect about the plant family "Liliaceae"?  
a) Trilocular ovary with axile placentation                      b) Have tetramerous flower.  
c) Epiphyllous stamens.                      d) Usually perennial herbs.
48. Which algae group is found at the deepest depths?

- a) Red algae                      b) Green algae                      c) Brown algae                      d) Blue-green algae
49. In a dicot leaf, reticulate venation is functionally related to  
 a) reduced transpiration                      b) mechanical strength  
 c) efficient transport & photosynthesis                      d) presence of petiole
50. Which floral condition prevents self-pollination but promotes cross-pollination?  
 a) Cleistogamy                      b) Homogamy                      c) Dichogamy                      d) Autogamy
51. Angiosperm taxonomy relies mainly on  
 a) habitat                      b) economic importance  
 c) reproductive characters                      d) plant height
52. In an ecosystem, productivity refers to  
 a) total biomass present                      b) energy flow between trophic levels  
 c) rate of biomass formation                      d) nutrient recycling
53. A population showing a pyramidal age structure is expected to  
 a) decline rapidly                      b) remain stable                      c) show negative growth                      d) increase in future
54. Lichen on tree bark represents which interaction?  
 a) Parasitism                      b) Commensalism                      c) Mutualism                      d) Amensalism
55. Ozone depletion directly increases  
 a) infrared radiation                      b) UV-B radiation                      c) acid rain                      d) greenhouse effect
56. Which practice best supports *in-situ* conservation?  
 a) Seed banks                      b) Botanical gardens                      c) National parks                      d) Cryopreservation
57. The organelle involved in intracellular digestion is  
 a) peroxisome                      b) lysosome                      c) ribosome                      d) centrosome
58. If a tetraploid male plant is crossed with a tetraploid female plant, what will be the ploidy level of the endosperm?  
 a) Haploid (n)                      b) Hexaploid (6n)                      c) Octaploid (8n)                      d) Tetraploid (4n)
59. Which biomolecule acts as biological catalyst?  
 a) Carbohydrate                      b) Lipid                      c) Enzyme                      d) Nucleic acid
60. A cross between  $AaBb \times aabb$  produces four phenotypes in equal ratio due to  
 a) dominance                      b) epistasis                      c) independent assortment                      d) linkage
61. A mutation that does not alter amino acid sequence is called  
 a) missense                      b) nonsense                      c) frameshift                      d) silent
62. DNA replication is termed semi-conservative because  
 a) only one strand replicates  
 b) parental DNA is destroyed  
 c) each daughter DNA has one old and one new strand  
 d) replication occurs in phases
63. The possible blood group of offspring if father has blood group A and mother has blood group B:  
 a) A, AB                      b) B, AB                      c) A, B, O                      d) A, B, AB, O
64. Which one is the correct order of genetic information flow in eukaryotic cells (Central Dogma)?

- a) Protein  $\rightarrow$  mRNA  $\rightarrow$  DNA  
c) DNA  $\rightarrow$  hnRNA  $\rightarrow$  mRNA  $\rightarrow$  Protein  
b) DNA  $\rightarrow$  mRNA  $\rightarrow$  hnRNA  $\rightarrow$  Protein  
d) DNA  $\rightarrow$  mRNA  $\rightarrow$  Protein
65. Collenchyma provides flexibility to plant parts due to  
a) lignified walls  
b) uneven cellulose thickening  
c) dead cells  
d) intercellular spaces
66. Secondary growth in dicot stem occurs due to activity of  
a) apical meristem  
b) intercalary meristem  
c) lateral meristem  
d) ground meristem
67. Vascular bundles in monocot stem are  
a) open and radial  
b) closed and scattered  
c) open and collateral  
d) radial and closed
68. In photosynthesis,  $O_2$  is evolved from water during?  
a) Oxidative Photophosphorylation  
b) Photolysis of water  
c) Excitation of electron  
d) Excitation of cell
69. Respiratory quotient (RQ) is highest when substrate is  
a) protein  
b) fat  
c) carbohydrate  
d) organic acid
70. Which hormone promotes cell elongation in stems?  
a) Cytokinin  
b) Auxin  
c) ABA  
d) Ethylene
71. Opening of stomata during day is mainly due to  
a) loss of  $K^+$  from guard cells  
b) decrease in turgor  
c) influx of  $K^+$  into guard cells  
d) plasmolysis
72. Which of the following has narrow lumen?  
a) Fibers  
b) Sclereids  
c) Tracheid  
d) Vessels
73. When pyruvic acid is changed to acetyl CoA, it is:  
a) Oxidized  
b) Reduced  
c) Isomerized  
d) Rearrangement
74. Which one is used in the culture medium?  
a) Laminaria  
b) Gelidium  
c) Alginic acid  
d) Spirulina
75. A bacterium divides every 35 minutes. If a Culture containing  $10^5$  cells per mL is grown for 175 min, what will be the cell concentration per mL after 175 minutes?  
a)  $32 \times 10^5$  cells  
b)  $175 \times 10^5$  cells  
c)  $64 \times 10^5$  cells  
d)  $16 \times 10^5$  cells
76. Hybrids between two genetically distinct and sexually incompatible plants cannot be produced by traditional breeding methods. Such hybrids can be obtained through:  
a) Somatic hybridization  
b) Genetic Engineering  
c) Inbreeding  
d) Outbreeding
77. If all decomposers are removed from an ecosystem, the first effect will be  
a) increase in producers  
b) nutrient accumulation in soil  
c) disruption of nutrient cycling  
d) increase in consumers
78. All of the following are aneuploidy except:  
a) Turner's syndrome  
b) Down's syndrome  
c) Phenylketonuria  
d) Klinefelter's syndrome
79. Animals found at the bottom of the sea are called:  
a) Benthos  
b) Cryptosoma  
c) Phytoplankton  
d) Littoral
80. If cytokinesis does not occur after karyokinesis in plants, the resulting cell will be  
a) anucleate  
b) multinucleate  
c) polyploid  
d) haploid

81. The number of moles of  $\text{H}_2\text{SO}_4$  present in 4.9 g  $\text{H}_2\text{SO}_4$  is:  
a) 0.5 mole                      b) 0.1 mole                      c) 2 mole                      d) 0.05 mole
82.  $10 \text{ dm}^3$  of  $\text{N}_2$  gas and  $10 \text{ dm}^3$  of gas X at the same temperature contain the same number of molecules, the gas X is  
a)  $\text{CO}_2$                       b)  $\text{CO}$                       c)  $\text{H}_2$                       d)  $\text{NO}$
83. According to Bohr's theory of hydrogen atom, which of the following is quantised for an electron?  
a) Angular momentum                      b) Angular acceleration  
c) Acceleration                      d) Velocity
84. For a, f-orbital, the values of m are  
a) -2, -1, 0, +1, +2                      b) -3, -2, -1, 0, +1, +2, 3  
c) -1, 0, +1                      d) 0, +1, +2, +3
85. Going from fluorine to chlorine, bromine and iodine, the electronegativity  
a) increases                      b) decreases  
c) first decrease then increases                      d) changes randomly
86. For which of the following ionic species radius would be maximum?  
a)  $\text{C}^{4-}$                       b)  $\text{N}^{3-}$                       c)  $\text{O}^{2-}$                       d)  $\text{Mg}^{2+}$
87. Polarization power of a cation increases when  
a) charge on the cation increases                      b) size of the cation increases  
c) charge on the cation decreases                      d) has no relation to its size or charge
88. Which contains both polar and non - polar bonds?  
a)  $\text{NH}_4\text{Cl}$                       b)  $\text{HCN}$                       c)  $\text{H}_2\text{O}_2$                       d)  $\text{CH}_4$
89. The temperature at which real gases obey the ideal gas laws over a wide range of pressure is called  
a) Critical temperature                      b) Boyle's Temperature  
c) Inversion temperature                      d) Reduced temperature
90. Heat exchanged in a chemical reaction at constant temperature and constant pressure is called  
a) internal energy                      b) enthalpy                      c) entropy                      d) free energy
91. Given that bond energies of  $\text{H} - \text{H}$  and  $\text{Cl} - \text{Cl}$  are  $430 \text{ kJ mol}^{-1}$  and  $240 \text{ kJ mol}^{-1}$  respectively and  $\Delta H_f$  for  $\text{HCl}$  is  $-90 \text{ kJ mol}^{-1}$ , bond enthalpy of  $\text{HCl}$  is  
a)  $380 \text{ kJ mol}^{-1}$                       b)  $425 \text{ kJ mol}^{-1}$                       c)  $245 \text{ kJ mol}^{-1}$                       d)  $290 \text{ kJ mol}^{-1}$
92. For reaction  $\text{PCl}_3 (\text{g}) + \text{Cl}_2 \rightleftharpoons \text{PCl}_5 (\text{g})$ , the value of  $K_c$  at  $250^\circ\text{C}$  is  $26 \text{ mol}^{-1} \text{ litre}^1$ . The value of  $K_p$  at this temperature will be  
a)  $0.61 \text{ atm}^{-1}$                       b)  $0.57 \text{ atm}^{-1}$                       c)  $0.83 \text{ atm}^{-1}$                       d)  $0.46 \text{ atm}^{-1}$
93. Which will not affect the degree of ionization?  
a) Temperature                      b) Concentration                      c) Type of solvent                      d) Current
94. A base, as defined by Bronsted theory, is a substance which can  
a) lose a pair of electrons                      b) donate protons  
c) gain a pair of electrons                      d) accept protons
95. The solubility product of silver chloride is  $1.8 \times 10^{-10}$  at  $298 \text{ K}$ . The solubility of  $\text{AgCl}$  in  $0.01 \text{ M HCl}$  solution in  $\text{mol/dm}^3$  is:

- a)  $2.4 \times 10^{-9}$       b)  $3.6 \times 10^{-8}$       c)  $0.9 \times 10^{-10}$       d)  $1.8 \times 10^{-8}$
96. Atomic number of an element is 22. The highest O.S. exhibited by it in its compounds is  
a) 1      b) 2      c) 3      d) 4
97. Action of water or dilute mineral acids on active metals can give  
a) monohydrogen      b) tritium      c) dihydrogen      d) trihydrogen
98. The component present in greater proportion in Coal gas is  
a)  $\text{CH}_4$       b)  $\text{CO}_2$       c)  $\text{CO}$       d)  $\text{H}_2$
99. Which of the following has the highest conductivity ?  
a)  $\text{Li}^+$       b)  $\text{Cs}^+$       c)  $\text{Na}^+$       d)  $\text{K}^+$
100. Which of the following oxides of potassium is not known ?  
a)  $\text{K}_2\text{O}$       b)  $\text{K}_2\text{O}_4$       c)  $\text{K}_2\text{O}_3$       d)  $\text{K}_2\text{O}_5$
101. In the occasion of marriages, the fire works used give green flame. Which one of the following radicals may be present?  
a)  $\text{Na}^+$       b)  $\text{K}^+$       c)  $\text{Ba}^{2+}$       d)  $\text{Ca}^{2+}$
102. Plaster of Paris hardens by  
a) giving off  $\text{CO}_2$       b) changing into  $\text{CaCO}_3$   
c) uniting with water      d) giving out water
103. Al is more reactive than Fe but Al is less easily corroded than Fe because  
a) it is a noble metal  
b) oxygen forms a protective oxide layer  
c) iron undergoes reaction easily with water  
d) Fe form mono and divalent ions
104. The oxide which cannot act as reducing agent is  
a)  $\text{SO}_2$       b)  $\text{NO}_2$       c)  $\text{CO}_2$       d)  $\text{ClO}_2$
105. Phosphate pollution is caused by  
a) sewage and agricultural fertilizers      b) weathering of phosphate rocks only  
c) agricultural fertilizers only      d) phosphate rocks and sewage
106. A match box exhibits  
a) cubic geometry      b) monoclinic geometry  
c) tetragonal geometry      d) orthorhombic geometry
107. The reference electrode is made by using  
a)  $\text{ZnCl}_2$       b)  $\text{CuSO}_4$       c)  $\text{HgCl}_2$       d)  $\text{Fe}_2\text{Cl}_2$
108. During the kinetic study of the reaction,  $2\text{A} + \text{B} \rightarrow \text{C} + \text{D}$ , following results were obtained:

| Run | [A](molL <sup>-1</sup> ) | [B](molL <sup>-1</sup> ) | Initial rate of formation of D (molL <sup>-1</sup> min <sup>-1</sup> ) |
|-----|--------------------------|--------------------------|------------------------------------------------------------------------|
| I   | 0.1                      | 0.1                      | $6.0 \times 10^{-3}$                                                   |
| II  | 0.3                      | 0.2                      | $7.2 \times 10^{-2}$                                                   |
| III | 0.3                      | 0.4                      | $2.88 \times 10^{-1}$                                                  |
| IV  | 0.4                      | 0.1                      | $2.40 \times 10^{-2}$                                                  |

Based on the above data which one of the following is correct?

- a) rate =  $k[A]^2[B]$     b) rate =  $k[A][B]$     c) rate =  $k[A]^2[B]^2$     d) rate =  $k[A][B]^2$

109. Calcination is used in metallurgy for removal of?

- a) Water and sulphide    b) Water and  $CO_2$   
c)  $CO_2$  and  $H_2S$     d)  $H_2O$  and  $H_2S$

110. Which one of the following ores is best concentrated by froth-flotation method?

- a) Galena    b) Cassiterite    c) Magnetite    d) Malachite

111. Iron is obtained on large scale from  $Fe_2O_3$  by

- a) Reduction with CO    b) Reduction with Al  
c) Calcination    d) Passing  $H_2$

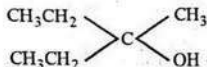
112. After partial roasting the sulphide of copper is reduced by

- a) cyanide process    b) electrolysis  
c) reduction with carbon    d) self reduction

113. One gas bleaches the colour of the flowers by reduction while the other by oxidation. The gases are

- a) CO and  $CO_2$     b)  $H_2S$  and  $Br_2$     c)  $SO_2$  and  $Cl_2$     d)  $NH_3$  and  $SO_3$

114. Name of the following compound is:



- a) 2-ethylbutan-2-ol    b) 1-ethyl-1-methylpropan-1-ol  
c) 3-methyl pentan-3-ol    d) diethylethanol

115. 2-Pentene contains

- a) 15  $\sigma$ - and one  $\pi$ -bond    b) 14  $\sigma$ - and one  $\pi$ -bond  
c) 15  $\sigma$ - and two  $\pi$ -bonds    d) 14  $\sigma$ - and two  $\pi$ -bonds

116. Due to the presence of an unpaired electron, free radicals are:

- a) cations    b) anions  
c) chemically inactive    d) chemically reactive

117. Which of the following acids has the smallest dissociation constant?

- a)  $CH_3CHFCOOH$     b)  $FCH_2CH_2COOH$     c)  $BrCH_2CH_2COOH$     d)  $CH_3CHBrCOOH$

118. The process of separation of a racemic modification into d- and l- enantiomers is called

- a) resolution    b) dehydration    c) revolution    d) dehydrohalogenation

119. Which one of the following heptanols can be dehydrated to hept-3-ene only?

- a) Heptan-3-ol    b) Heptan-4-ol    c) Heptan-2-ol    d) Heptan-1-ol

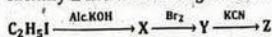
120. Heating a mixture of sodium benzoate or benzoic acid and soda-lime gives

- a) benzene    b) methane    c) sodium benzoate    d) calcium benzoate

121. The addition of HBr is easiest in

- a)  $CH_2=CHCl$     b)  $ClCH=CHCl$     c)  $CH_3-CH=CH_2$     d)  $(CH_3)_2C=CH_2$

122. Identify Z in the following series:



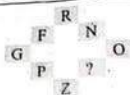
- a)  $\text{CH}_3\text{CH}_2\text{CN}$       b)  $\text{NCCH}_2\text{CH}_2\text{CN}$       c)  $\text{BrCH}_2\text{CH}_2\text{CN}$       d)  $\text{BrCH}=\text{CHCN}$
123. Conversion of cyclohexene to cyclohexanol can be conveniently achieved by  
 a)  $\text{NaOH} - \text{H}_2\text{O}$       b)  $\text{Br}_2 - \text{H}_2\text{O}$   
 c) hydroboration-oxidation      d) hydroboration-hydrolysis
124. n-propyl alcohol and isopropyl alcohol can be chemically distinguished by which reagent  
 a)  $\text{PCl}_5$       b) reduction  
 c) oxidation with potassium dichromate      d) ozonolysis
125. Phenol when heated with concentrated nitric acid gives  
 a) anthranilic acid      b) p-nitrobenzoic acid      c) picric acid      d) salicylic acid
126. The reagent used for the preparation of higher ethers from halogenated ethers is  
 a) Conc.  $\text{H}_2\text{SO}_4$       b) Sodium alkoxide      c) Dry silver oxide      d) Grignard reagent
127. Methyl ketones are usually characterized through  
 a) the Tollen's reagent      b) the iodoform test  
 c) the Schiff's test      d) the Benedict's reagent
128. Which of the following will give Cannizzaro reaction ?  
 a)  $\text{CH}_3\text{CHO}$       b)  $\text{CH}_3\text{COCH}_3$       c)  $(\text{CH}_3)_2\text{CCHO}$       d)  $\text{CH}_3\text{CH}_2\text{CHO}$
129. Which of the following compounds cannot be identified by carbylamine test?  
 a)  $\text{CH}_3\text{CH}_2\text{NH}_2$       b)  $\text{CHCl}_3$       c)  $\text{C}_6\text{H}_5\text{NH}_2$       d)  $\text{C}_6\text{H}_5\text{-NH-C}_6\text{H}_5$
130. Teflon and neoprene are  
 a) copolymers      b) condensation polymers  
 c) homopolymers      d) monomers
131. Angular momentum is  
 a) A scalar      b) A polar vector      c) An axial vector      d) None of these
132. When three forces of 50 N, 30 N and 15 N act on a body, then the body is  
 a) At rest      b) Moving with a uniform velocity  
 c) In equilibrium      d) Moving with an acceleration
133. The greatest height to which a man can throw a stone is h. The greatest distance to which he can throw it, will be  
 a)  $h/2$       b) h      c) 2h      d) 3h
134. A body slides down a frictionless track which ends in a circular loop of diameter D, then the minimum height h of the body in term of D so that it may just complete the loop, is  
 a)  $h = \frac{5D}{2}$       b)  $h = \frac{5D}{4}$       c)  $h = \frac{3D}{4}$       d)  $h = \frac{D}{2}$
135. Gravels are dropped on a conveyor belt at the rate of 0.5 kg/sec. The extra force required in Newton to keep the belt moving at 2 m/sec is  
 a) 1      b) 2      c) 4      d) 0.5
136. A rifle bullet loses  $\frac{1}{20}$ th of its velocity in passing through a plank. The least number of such planks required just to stop the bullet is  
 a) 5      b) 10      c) 11      d) 20
137. A body is projected vertically from the earth's surface with the KE equal to half of escape energy from the earth's surface. The height through which it rises above earth is

- a)  $\frac{R}{2}$                       b) R                      c) 2R                      d)  $\frac{R}{4}$
138. The quality of the material which opposes the change in shape, volume or length is called  
a) Intermolecular repulsion                      b) Intermolecular behaviour  
c) Viscosity                      d) Elasticity
139. Small droplets of a liquid are usually more spherical in shape than larger drops of the same liquid because  
a) Force of surface tension is equal and opposite to the force of gravity  
b) Force of surface tension predominates the force of gravity  
c) Force of gravity predominates the force of surface tension  
d) Force of gravity and force of surface tension act in the same direction and are equal
140. The velocity of a particle performing simple harmonic motion, when it passes through its mean position is  
a) Infinity                      b) Zero                      c) Minimum                      d) Maximum
141. The latent heat of vaporization of a substance is always  
a) Greater than its latent heat of fusion  
b) Greater than its latent heat of sublimation  
c) Equal to its latent heat of sublimation  
d) Less than its latent heat of fusion
142. If on heating liquid through  $80^{\circ}\text{C}$ , the mass expelled is  $(1/100)$  of mass still remaining, the coefficient of apparent expansion of liquid is  
a)  $1.25 \times 10^{-4}/^{\circ}\text{C}$                       b)  $12.5 \times 10^{-4}/^{\circ}\text{C}$                       c)  $1.25 \times 10^{-3}/^{\circ}\text{C}$                       d) none of these
143. 110 J of heat is added to a gaseous system, whose internal energy change is 40 J, then the amount of external work done is  
a) 150 J                      b) 70 J                      c) 110 J                      d) 40 J
144. In adiabatic expansion of a gas  
a) Its pressure increases                      b) Its temperature falls  
c) Its density increases                      d) Its thermal energy increases
145. For cooking the food, which of the following type of utensil is most suitable  
a) High specific heat and low conductivity                      b) High specific heat and high conductivity  
c) Low specific heat and low conductivity                      d) Low specific heat and high conductivity
146. The wavelength of radiation emitted by a body depends upon  
a) The nature of its surface                      b) The area of its surface  
c) The temperature of its surface                      d) All the above factors
147. Image formed by a convex mirror is  
a) Virtual                      b) Real                      c) Enlarged                      d) Inverted
148. A tank is filled with benzene to a height of 120 mm. The apparent depth of a needle lying at a bottom of the tank is measured by a microscope to be 80 mm. The refractive index of benzene is  
a) 1.5                      b) 2.5                      c) 3.5                      d) 4.5
149. If the refractive index of a material of equilateral prism is  $\sqrt{3}$ , then angle of minimum deviation of the prism is

- a)  $30^\circ$                       b)  $45^\circ$                       c)  $60^\circ$                       d)  $75^\circ$
150. Dispersive power depends upon  
a) the shape of prism    b) material of prism    c) angle of prism    d) height of prism
151. On a rainy day, a small oil film on water show brilliant colours. This is due to  
a) Dispersion of light                      b) Interference of light  
c) Absorption of light                      d) Scattering of light
152. The condition for observing Fraunhofer diffraction from a single slit is that the light wavefront incident on the slit should be  
a) spherical                      b) cylindrical                      c) plane                      d) elliptical
153. A big explosion on the moon cannot be heard on the earth because  
a) The explosion produces high frequency sound waves which are inaudible  
b) Sound waves required a material medium for propagation  
c) Sound waves are absorbed in the moon's atmosphere  
d) Sound waves are absorbed in the earth's atmosphere
154. A transverse wave is described by the equation  $y = Y_0 \sin 2\pi \left( ft - \frac{x}{\lambda} \right)$ . The maximum particle velocity is four times the wave velocity if  
a)  $\lambda = \frac{\pi Y_0}{4}$                       b)  $\lambda = \frac{\pi Y_0}{2}$                       c)  $\lambda = \pi Y_0$                       d)  $\lambda = 2\pi Y_0$
155. As an empty vessel is filled with water, its frequency will  
a) Increase                      b) Decrease                      c) Remain unchanged    d) Not predictable
156. The value of electric field due to a point charge at a distance  $r$  is  
a) proportional to  $r^2$                       b) inversely proportional to  $r^2$   
c) proportional to  $r$                       d) independent of  $r$
157. The ratio of charge to potential of a body is  
a) capacitance                      b) conductance                      c) inductance                      d) resistance
158. In a charged capacitor the energy resides  
a) on the positive plate                      b) on both the positive and negative plates  
c) on the field between the plates                      d) around the edge of capacitor plates
159. When there is an electric current passing through a conducting wire along its length, then an electric field must exist  
a) Outside the wire but normal to it                      b) Outside the wire but parallel to it  
c) Inside the wire but parallel to it                      d) Inside the wire but normal to it
160. The equivalent resistance of resistors connected in series is always  
a) Equal to the mean of component resistors  
b) Less than the lowest of component resistors  
c) In between the lowest and the highest of component resistors  
d) Equal to sum of component resistors
161. The resistance of 10 metre long potentiometer wire is  $10\text{ ohm/meter}$ . A cell of e.m.f. 2.2 volts and a high resistance box are connected in series to this wire. The value of resistance taken from resistance box for getting potential gradient of 2.2 millivolt/metre will be

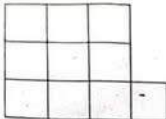
- a)  $790\ \Omega$                       b)  $810\ \Omega$                       c)  $990\ \Omega$                       d)  $1000\ \Omega$
162. Watt-hour meter measures  
a) voltage                      b) power                      c) electric energy                      d) current
163. A particle of charge  $q$  and mass  $m$  moves in a circular orbit of radius  $r$  with angular speed  $\omega$ . The ratio of the magnitude of its magnetic moment to that of its angular momentum depends on  
a)  $\omega$  and  $q$                       b)  $\omega q$  and  $m$                       c)  $q$  and  $m$                       d)  $\omega$  and  $m$
164. Susceptibility is positive and small for a  
a) paramagnetic substance                      b) ferromagnetic substance  
c) diamagnetic substance                      d) non-magnetic substance
165. A diamagnetic material in a magnetic field moves  
a) From weaker to the stronger parts of the field  
b) Perpendicular to the field  
c) From stronger to the weaker parts of the field  
d) In none of the above directions
166. If a copper ring is moved quickly towards south pole of a powerful stationary bar magnet, then  
a) Current flows through the copper ring                      b) Voltage in the magnet increase  
c) Current flows in the magnet                      d) Copper ring will get magnetised
167. The frequency, at which the inductive reactance of  $0.7\text{H}$  inductor is  $220\ \Omega$ , is  
a)  $60\ \text{Hz}$                       B)  $30\ \text{Hz}$                       c)  $50\ \text{Hz}$                       d)  $100\ \text{Hz}$
168. A particle which has zero rest mass and non-zero energy and momentum must travel with a speed  
a) Equal to  $c$ , the speed of light in vacuum                      b) Greater than  $c$   
c) Less than  $c$                       d) Tending to infinity
169. The potential difference applied to an X-ray tube is increased. As a result, in the emitted radiation  
a) The intensity increases                      b) The minimum wavelength increases  
c) The intensity decreases                      d) The minimum wavelength decreases
170. If intensity of incident light is increased in PEE then which of the following is true  
a) Maximum K.E. of ejected electron will increase  
b) Work function will remain unchanged  
c) Stopping potential will decrease  
d) Maximum K.E. of ejected electron will decrease
171. When subjected to a transverse electric field, cathode rays move  
a) Down the potential gradient                      b) Up the potential gradient  
c) Along a hyperbolic path                      d) Along a circular path
172. In Bohr model of the hydrogen atom, the lowest orbit corresponds to  
a) Infinite energy                      b) The maximum energy  
c) The minimum energy                      d) Zero energy
173. What is the half life of lead?

- a) Zero                      b) Infinite                      c) 1590 days                      d) 1590 years
174. In P-type semiconductor, there is  
a) An excess of one electron                      b) Absence of one electron  
c) A missing atom                      d) A donor level
175. Electrical conductivity of a semiconductor  
a) Decreases with the rise in its temperature  
b) Increases with the rise in its temperature  
c) Does not change with the rise in its temperature  
d) First increases and then decreases with the rise in its temperature
176. If the forward voltage in a semiconductor diode is doubled, the width of the depletion layer will  
a) Become half                      b) Become one-fourth                      c) Remain unchanged                      d) Become double
177. The PN junction diode is used as  
a) An amplifier                      b) A rectifier                      c) An oscillator                      d) A modulator
178. Great bear is a  
a) Star                      b) Galaxy                      c) Constellation                      d) Planet
179. Solar energy is due to  
a) fusion reactions                      b) fission reactions  
c) combustion reactions                      d) chemical reactions
180. Quark combination of  $\pi^+$  meson is  
a)  $u\bar{d}$                       b)  $u\bar{u}$                       c)  $\bar{u}d$                       d)  $d\bar{d}$
181. If the digits in the number "298496721" are arranged in descending order, then how many digits remain in same place?  
a) 2                      b) 4                      c) 3                      d) 1
182. Which one is similar as : 47@11, 89@17, 1415@29  
a) 89@5                      b) 145@19                      c) 184@54                      d) 49@25
183. 512 : 8 :: 216 : ?  
a) 9                      b) 8                      c) 5                      d) 7
184. If MQRD is coded as SQRDM, YGTLP is coded as SGTLPY, then QVSRS is coded as:  
a) SVSRSQ                      b) QVSRSS                      c) SSQVRS                      d) QVSRSQ
185. In a row of certain A is ranked 9<sup>th</sup> from left end and B is ranked 10<sup>th</sup> from right end. If C is exactly in between A and B. What could be the minimum number of students in this row?  
a) 9                      b) 10                      c) 19                      d) 13
186. In an examination consisting of 80 questions. Shankar answered 75% of the first 60 questions correctly. What percentage of the remaining 20 question did he need to answer correctly to get 80 % marks in the entire question?  
a) 90%                      b) 98%                      c) 58%                      d) 95%
187. Insert the missing letter.



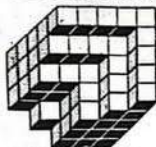
- a) P                      b) W                      c) H                      d) T

188. Count the number of rectangles without squares.



- a) 40                      b) 15                      c) 25                      d) 55

189. Count the number of cubes in the given figure.



- a) 80                      b) 87                      c) 89                      d) 90

190. Identify the figure that completes the pattern.



(X)



(1)



(2)



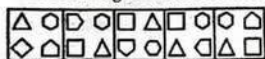
(3)



(4)

- a) 1                      b) 2                      c) 3                      d) 4

191. Choose the figure which is different from the rest.



(1)

(2)

(3)

(4)

(5)

- a) 1                      b) 3                      c) 4                      d) 5

192. Group the given figures into three classes using each figure only once.



1



2



3



4



5



6



7



8



9

- a) 1,3,5 ; 2,4,7 ; 6,8,9    b) 1,5,7 ; 2,3,6 ; 4,8,9    c) 1,3,5 ; 2,6,7 ; 4,8,9    d) 1,6,9 ; 2,4,7 ; 3,5,8

193. 4 men and 6 women can complete a work in 8 days, while 3 men and 7 women can complete it in 10 days. In how many days will 10 women complete it?  
a) 35                      b) 40                      c) 45                      d) 50
194. What will be the day of the week 15th August 2010?  
a) Sunday                      b) Monday                      c) Tuesday                      d) Friday
195. A car owner buys petrol at Rs.7.50, Rs. 8 and Rs. 8.50 per litre for three successive years. What approximately is the average cost per litre of petrol if he spends Rs. 4000 each year?  
a) Rs. 7.98                      b) Rs. 8                      c) Rs. 8.50                      d) Rs. 9
196. A father said to his son, "I was as old as you are at the present at the time of your birth". If the father's age is 38 years now, the son's aged five years back was:  
a) 14 years                      b) 19 years                      c) 33 years                      d) 38 years
197. One morning, Udai and Vishal were talking to each other face to face at a crossing. If Vishal's shadow was exactly to the left of Udai, which direction was Udai facing?  
a) East                      b) West                      c) North                      d) South
198. A, B, C, D and E play a game of cards. A says to B, "If you give me three cards, you will have as many as E has and if I give you three cards, you will have as many as D has." A and B together have 10 cards more than what D and E together have. If B has two cards more than what C has and the total number of cards be 133, how many cards does B have ?  
a) 22                      b) 23                      c) 25                      d) 35
199. Yesterday I saw an ice cube which had already melted due to heat of a nearby furnace.  
a) Always                      b) Never                      c) Often                      d) Sometimes
200. If D is the brother of B, how B is related to C? To answer this question which of the statements is/are necessary?  
The son of D is the grandson of C.  
B is the sister of D.  
a) Only 1                      b) Only 2  
c) Either 1 or 2                      d) 1 and 2 both are required

Result will be published at 8:00 pm

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# MECEE BL

## Model Entrance Exam

**(Model: XXI)**  
**(Set: A)**

Date: Baisakh 12<sup>th</sup> 2083 (Saturday)  
Venue: NAME

Duration – 3 hrs  
Time – 11:00 AM

### INSTRUCTIONS

There are 200 multiple-choice questions, each having four choices of which only one choice is correct.

Fill ☐ the most appropriate one.

26. Which of the following is not correct about life cycle of *Plasmodium*?
- Sporozoite - uninucleated, irregular shaped due to lack of pellicle
  - Rosette stage - arrangement of merozoites like the arrangement of petals in rose flower
  - Trophozoite - feeding stage formed in RBC of human
  - Cryptomerozoite - invade fresh liver cell and starts exo-erythrocytic schizogony
27. Which species of *Plasmodium* has longest incubation period?
- P. vivax*
  - P. ovale*
  - P. malariae*
  - P. falciparum*
28. Anterior abdominal vein in frog is formed by union of
- right and left pelvic veins
  - right and left sciatic veins
  - right and left femoral veins
  - right and left dorso-lumbar veins
29. In frog, pancreas produces
- a) two digestive enzymes, one hormone
  - b) three digestive enzymes, one hormone
  - c) three digestive enzymes, two hormones
  - d) three digestive enzymes, three hormones
30. Which one of the following correctly described the location of some body parts in *Pheretima*?
- Two pairs of accessory glands in 16-18 segments
  - Four pairs of spermathecae in 4-7 segments
  - One pair of ovaries attached at intersegmental septum of 14<sup>th</sup> and 15<sup>th</sup> segments
  - Two pairs of copulatory papillae in 17<sup>th</sup> and 19<sup>th</sup> segments
31. Which of the following release sperm as well as prostatic fluid during copulation in earthworm?
- Male genital pores
  - Spermathecal pores
  - Dorsal pores
  - Male genital papillae
32. On the basis of organization animals are grouped into:
- Metazoa and Eumetazoa
  - Protozoa and Metazoa
  - Protozoa and parazoa
  - Parazoa and Metazoa
33. Which of the following is matching pair of a body feature and the animal possessing it?
- Post anal tail-Octopus
  - Ventral Nervous system - Leech
  - Pharyngeal gill slits absent in embryo of -Chameleon
  - Antenna absent - prawn
34. Dropping of gravid proglottids by Cestodes is called:
- Apolysis
  - Autotomy
  - Paedogenesis
  - Autophagy
35. Which of the following group is Deuterostome?
- Annelida, Arthropoda and Mollusca
  - Echinodermata, Hemichordata and Chordata
  - Annelida, Mollusca and Chordata
  - Arthropoda, Mollusca and Echinodermata
36. The stages, between larval moults in an insect:
- Instar
  - Morula
  - Pupa
  - Larva
37. The main difference between *Gymnophiona* (Apoda) and *Urodela* is that, *Urodela*:
- Have two auricles and one ventricle
  - Have smooth moist skin
  - Have a cloaca
  - Respire by lungs in the adult stage
38. Which of the following feature is universally present in all sponges?
- Marine habitat
  - Presence of spicules

136. Due to capillary action, a liquid will rise in a tube, if the angle of contact is  
 a) Acute b) Obtuse c)  $90^\circ$  d) Zero
137. The work done in stretching an elastic wire per unit volume is  
 a) stress  $\times$  strain b)  $\frac{1}{2} \times$  stress  $\times$  strain c)  $2 \times$  strain  $\times$  stress d) stress/strain
138. The moment of inertia of a solid sphere of radius R about its diameter is same as that of disc of radius 2R about its diameter. The ratio of their masses is  
 a) 5 : 2 b) 5 : 8 c) 4 : 1 d) 2 : 1
139. The coefficient of viscosity for hot air is  
 a) Greater than the coefficient of viscosity for cold air  
 b) Smaller than the coefficient of viscosity for cold air  
 c) Same as the coefficient of viscosity for cold air  
 d) Increases or decreases depending on the external pressure
140. A particle starts SHM from the mean position. Its amplitude is A and time period is T. At the time when its speed is half of the maximum speed, its displacement y is  
 a)  $\frac{A}{2}$  b)  $\frac{A}{\sqrt{2}}$  c)  $\frac{A\sqrt{3}}{2}$  d)  $\frac{2A}{\sqrt{3}}$
141. Mercury boils at  $367^\circ\text{C}$ . However, mercury thermometers are made such that they can measure temperature up to  $500^\circ\text{C}$ . This is done by  
 a) Maintaining vacuum above mercury column in the stem of the thermometer  
 b) Filling nitrogen gas at high pressure above the mercury column  
 c) Filling nitrogen gas at low pressure above the mercury level  
 d) Filling oxygen gas at high pressure above the mercury column
142. The factor not needed to calculate heat lost or gained when there is no change of state is  
 a) Weight b) Specific heat  
 c) Relative density d) Temperature change
143. The root mean square speed of oxygen molecule at certain temperature is 0.5 km/s. The rms speed for hydrogen molecule at the same temperature will be  
 a) 2 km/s b) 4 km/s c) 9 km/s d) 16 km/s
144. Temperature is a measurement of coldness or hotness of an object. This definition is based on  
 a) Zeroth law of thermodynamics b) First law of thermodynamics  
 c) Second law of thermodynamics d) Newton's law of cooling.
145. A Carnot engine takes  $3 \times 10^6$  cal. of heat from a reservoir at  $627^\circ\text{C}$  and gives it to a sink at  $27^\circ\text{C}$ . The work done by the engine is  
 a)  $4.2 \times 10^6$  J b)  $8.4 \times 10^6$  J c)  $16.8 \times 10^6$  J d) zero
146. If the initial temperatures of metallic sphere and disc, of the same mass, radius and nature are equal, then the ratio of their rate of cooling in same environment will be  
 a) 1 : 4 b) 4 : 1 c) 1 : 2 d) 2 : 1
147. In a right angled glass prism, light is normally incident through smaller face. Then deviation is  
 a)  $90^\circ$  b)  $45^\circ$  c)  $30^\circ$  d)  $60^\circ$
148. The power of a convex lens ( $\mu = 1.6$ ) in air is + 6D, the power of the lens in water ( $\mu = \frac{4}{3}$ ) will be

|                        |                                             |
|------------------------|---------------------------------------------|
| c) cartilaginous joint | between frontal and parietal                |
| d) pivot joint         | between third and fourth cervical vertebrae |

14. Which part of the human ear plays no role in hearing as such but is otherwise very much required?  
 a) Eustachian tube      b) organ of corti      c) vestibular apparatus      d) ear ossicles
15. Identify the origin of sympathetic nerve fibres and the location of their ganglia.  
 a) They arise from thoraco lumbar region of spinal cord and form ganglia just beside the vertebral column  
 b) They arise from thoraco cervical region of spinal cord and form ganglia just beside the vertebral column  
 c) They arise from cranio sacral region of spinal cord and form ganglia very close to effector organ  
 d) They arise from thoraco lumbar region of spinal cord and form ganglia very close to effector organ
16. Which of the following teeth are lophodont?  
 a) Incisor and canine      b) Premolar and molar  
 c) Canine and premolar      d) Premolar and incisor
17. Pseudostratified epithelium is always  
 a) single layered      b) double layered      c) multi layered      d) uncertain
18. The value of resting membrane potential is  
 a) - 90 mV      b) - 100mV      c) + 100mV      d) + 90mV
19. Intervertebral disc is made up of  
 a) elastic cartilage      b) fibrous cartilage      c) calcified cartilage      d) hyaline cartilage
20. Choose the correctly matched pair.  
 a) Areolar tissue - loose connective tissue      b) Cartilage - loose connective tissue  
 c) Tendon - specialized connective tissue      d) Adipose tissue - dense connective tissue
21. Under certain conditions scientists have obtained cell like structures. These are known as  
 a) Prebiotic soup      b) Coacervates      c) Protists      d) Microbes
22. Which is incorrect?  
 a) Wings of insects and birds are analogous  
 b) Wings of insects and bats are analogous  
 c) Wings of insects and birds are homologous  
 d) Wings of bats and birds are homologous
23. The cranial capacity of Java ape man was about  
 a) 560 cc      b) 900 cc      c) 1300 cc      d) 1000 cc
24. Species diversity generally increases as one proceeds from  
 a) low altitude to high altitude and from low latitudes to high latitudes  
 b) high altitude to low altitude and from low latitudes to high latitudes  
 c) high altitude to low altitude and from high latitude to low latitude  
 d) low altitude to high altitude and from high latitudes to low latitude
25. Biological organisation starts with  
 a) submicroscopic molecular level      b) cellular level  
 c) organismic level      d) atomic level

55. **Acid rain primarily affects plants by**
  - a) increasing photosynthesis
  - b) damaging leaf cuticle and chlorophyll
  - c) increasing soil fertility
  - d) enhancing transpiration
56. **Sacred groves are important because they**
  - a) increase timber production
  - b) allow hunting
  - c) preserve biodiversity naturally
  - d) promote monoculture
57. **The site of ribosomal RNA synthesis in a eukaryotic cell is**
  - a) nucleoplasm
  - b) nucleolus
  - c) nuclear membrane
  - d) centrosome
58. **Chromosomes are best studied during metaphase because**
  - a) DNA content is doubled
  - b) chromosomes are most condensed
  - c) spindle fibers disappear
  - d) nuclear membrane reforms
59. **When a sample is labeled with radio-labeled C14 Uracil, the molecular component that can be discovered is:**
  - a) DNA
  - b) RNA
  - c) Histone
  - d) Nucleotide
60. **The part of the chromosome after the secondary constriction is known as:**
  - a) Telomere
  - b) Chromomere
  - c) Nuclear organizer region
  - d) Satellite
61. **Which biomolecule shows maximum structural diversity?**
  - a) Lipids
  - b) Proteins
  - c) Carbohydrates
  - d) Nucleic acids
62. **A heterozygous tall plant produces 75% tall offspring. This indicates**
  - a) incomplete dominance
  - b) complete dominance
  - c) codominance
  - d) linkage
63. **How many types of gametes are possible from AaBbDdEE?**
  - a) 2
  - b) 16
  - c) 8
  - d) 4
64. **The enzyme that synthesizes RNA using DNA template is**
  - a) DNA polymerase
  - b) RNA polymerase
  - c) ligase
  - d) helicase
65. **Degeneracy of genetic code means**
  - a) one codon codes multiple amino acids
  - b) multiple codons code same amino acid
  - c) codons overlap
  - d) codons lack punctuation
66. **Which organelle eats old organelles?**
  - a) Lysosome
  - b) Golgi body
  - c) Mitochondria
  - d) Chloroplast
67. **Sclerenchyma cells are dead at maturity due to**
  - a) absence of nucleus
  - b) lignified secondary walls
  - c) presence of vacuole
  - d) cellulose deposition
68. **Vascular cambium is absent in**
  - a) dicot stem
  - b) dicot root
  - c) monocot stem
  - d) gymnosperm stem
69. **In dicot leaf, palisade parenchyma is mainly responsible for**
  - a) transpiration
  - b) gaseous exchange
  - c) photosynthesis
  - d) conduction
70. **Bolting in plants is caused by:**
  - a) Auxin
  - b) Ethylene
  - c) Gibberellins
  - d) Abscissic acid
71. **During anaerobic respiration in plants, pyruvate is converted into**
  - a) lactic acid
  - b) ethanol and CO<sub>2</sub>
  - c) acetyl CoA
  - d) malate

- a) remove all impurities from ores      b) reduce metal oxide  
c) remove silica      d) remove silica and undesirable metal oxide
89. Which of the following is characterized as a heavy metal?  
a) Mg      b) Li      c) Na      d) Hg
90. Which of the following is the most abundant non-metal in the Earth's crust?  
a) Oxygen      b) Silicon      c) Nitrogen      d) Hydrogen
91. The acidic character of oxides of non-metals generally...  
a) Increases down the group  
b) Decreases across a period  
c) Increases across a period from left to right  
d) Is not dependent on the period
92. By passing  $\text{H}_2\text{S}$  gas in acidified  $\text{KMnO}_4$  solution, we get  
a)  $\text{K}_2\text{S}$       b) S      c)  $\text{K}_2\text{SO}_3$       d)  $\text{MnO}_2$
93. What is the maximum covalent capacity of Nitrogen, and why?  
a) 3; due to smaller size      b) 4; due to absence of d-orbitals  
c) 5; due to p-orbital filling      d) 6; due to larger size
94. Which of the following elements exhibits the most basic properties?  
a) F      b) Cl      c) Br      d) I
95. Which of the following is used as an antiseptic?  
a) I      b) Br      c) Cl      d) F
96. In the presence of cobalt chloride ( $\text{CoCl}_2$ ), bleaching powder decomposes to form  
a)  $\text{CaCO}_3$  and  $\text{O}_3$       b)  $\text{ClO}_2$  and  $\text{CaO}$       c)  $\text{Cl}_2\text{O}$  and  $\text{CaO}$       d)  $\text{CaCl}_2$  and  $\text{O}_2$
97. Which one of the following elements is most reactive?  
a) He      b) Ne      c) Ar      d) Xe
98. The valence shell of transition elements consists of  
a) nd orbitals      b) (n-1) d orbitals      c) ns, np, nd orbitals      d) (n-1)d, ns orbitals
99. Which of the following alkali metals burns in air to form only monoxide?  
a) Na      b) Li      c) K      d) Cs
100. Which component of cement sets at the slowest rate  
a) Dicalcium silicate      b) Tricalcium silicate  
c) Tricalcium aluminate      d) Tetracalcium aluminoferrite
101. Which category of salts of alkaline earth metals is not found in solid state, but found in solution state?  
a) Carbonates      b) Bicarbonates      c) Hydroxides      d) Sulphates
102. The IUPAC name of tert-butylchloride is  
a) 4-chlorobutane      b) 2-chlorobutane  
c) 1-chloro-3-methylpropane      d) 2-chloro-2-methylpropane
103. A compound with molecular formula,  $\text{C}_7\text{H}_{16}$  shows optical isomerism, the compound will be  
a) 2, 3-dimethylpentane      b) 2, 2-dimethylpentane  
c) 2-methylhexane      d) 3,4-dimethylpentane
104. Conversion of  $\text{CH}_4$  to  $\text{CH}_3\text{Cl}$  is an example of ..... reaction  
a) electrophilic substitution      b) nucleophilic substitution

N402

 $V_B$ , and the width of charge depleted region  $x$ 

- a)  $V_B$  increases,  $x$  decreases  
c)  $V_B$  increases,  $x$  increases

- b)  $V_B$  decreases,  $x$  increases  
d)  $V_B$  decreases,  $x$  decreases

177. When an input signal 1 is applied to a NOT gate, its output is  
a) 1 b) 0 c) either 0 or 1 d) any positive value
178. Sun radiates continuously and maintains its brightness because  
a) Helium is converted into iron in its core  
b) Of fusion of hydrogen nuclei into helium  
c) Fusion of helium in hydrogen  
d) Burning of carbon, in its core
179. 'Albedo' is  
a) Reflecting power of a heavenly body  
b) Transmittive power of a heavenly body  
c) Absorptive power of a heavenly body  
d) Refracting power of a heavenly body
180. The main constituents of the universe are  
a) ordinary matter b) dark matter  
c) dark energy d) all of them
181. Which is the odd one out?



- a) B                      b) C                      c) E                      d) G

182. Four persons K, L, M and N have a total of Rs 100 among themselves. K and L have as much money as M and N have, but K has more money than L, and M has only half the money that N has. K has in fact Rs 5 more than N has.

**Who has the maximum amount of money.**

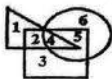
- a) K                      b) L                      c) M                      d) N

183. A fisherman has 5 fishes (namely A, B, C, D, E) each having a different weight.
- |                                           |                                                 |
|-------------------------------------------|-------------------------------------------------|
| (i) A weigh twice as much as B.           | (ii) B weighs four and half times as much as C. |
| (iii) C weighs half as much as D.         | (iv) D weighs half as much as E.                |
| (v) E weighs less than A but more than C. |                                                 |

**Which of the following is the lightest?**

- a) A                      b) B                      c) C                      d) D

- 184. Consider the following figure.**



In the above figure, the triangle represents girls, the square represents sports person, and the circle represents coaches. Which portion in the figure represents girls, who are sports persons, but not coaches?

- a) 4                      b) 2                      c) 1                      d) 3

185. Statement I: The police authority has recently caught a group of house breakers.  
Statement II: The citizens group is the initiator of the police action.

- a) Statement I is the cause and Statement II its effect.
- b) Statement II is the cause, and Statement I is its effect.
- c) Both statements I and II are independent cause.
- d) Both the statements I and II are effects of some common cause.

**Note: NAME students must fill their registration number on the OMR sheet.**

1. Which of the following is the relatively most accurate method for dating of fossils?
  - a) Radio-carbon method
  - b) Potassium-argon method
  - c) Electron-spin resonance method
  - d) Uranium-lead method
2. Which one of the following gas was absent on the primitive Earth?
  - a)  $\text{CH}_4$
  - b)  $\text{NH}_3$
  - c)  $\text{O}_2$
  - d)  $\text{CO}_2$
3. Use of anti-histamine, adrenaline and steroids give a quick relief from
  - a) allergy
  - b) nausea
  - c) cough
  - d) headache
4. Which one of the following pairs is mismatched?
  - a) Hyperthyroidism – exothalamic goitre
  - b) Hyperconitrolism – cretinism
  - c) Hypothyroidism – myxoedema
  - d) Hypercortisolism – cushing's syndrome
5. Which of the following is a pair of viral disease?
  - a) Ring worm, AIDS
  - b) Common cold, AIDS
  - c) Dysentery, common cold
  - d) Typhoid, tuberculosis
6. Widal test is used for susceptibility of
  - a) Malaria
  - b) Cholera
  - c) Yellow fever
  - d) Typhoid
7. Microlecithal eggs are found in
  - a) reptilia + aves
  - b) amphibia + aves + reptilia
  - c) reptilia + aves + chiroptera
  - d) eutheria
8. If for some reason, the vasa efferentia in the human reproductive system get blocked, the gametes will not be transported from
  - a) vagina to uterus
  - b) testes to epididymis
  - c) epididymis to vas deferens
  - d) ovary to uterus
9. Wharton's duct found in
  - a) sublingual
  - b) sub-mandibular
  - c) lacrimal
  - d) parotid
10. The oxygen haemoglobin dissociation curve will show a right shift in case of
  - a) high  $\text{pCO}_2$
  - b) high  $\text{pO}_2$
  - c) low  $\text{pCO}_2$
  - d) less  $\text{H}^+$  concentration
11. Pace maker of the heart is situated
  - a) in wall of right atrium close to eustachian valve
  - b) on intra – auricular septum
  - c) on inter – ventricular septum
  - d) in wall of left atrium close to the opening of pulmonary veins
12. The main function of Henle's loop is
  - a) conservation of water
  - b) filtration of blood
  - c) passage of urine
  - d) formation of urine
13. Select the correct matching of the type of the joint with the example in human skeletal system.

| Types of joint   | Example                             |
|------------------|-------------------------------------|
| a) Hinge joint   | between humerus and pectoral girdle |
| b) Gliding joint | between carpals                     |

- c) Presence of spongocoel  
**39. Zoochlorellae algae in *Hydra* are:**  
 a) Symbionts in nutritive muscular cells  
 c) Symbionts in cnidoblasts  
 b) Symbionts in gut  
 d) Organism that provide hypnotoxin
- 40. Which of the following is not correctly matched pair?**  
 a) *Ascaris* - syncytial epidermis  
 c) *Sycon* - totipotent cells are interstitial cells  
 b) *Planaria* - syncytial & ciliated epidermis  
 d) *Hydra* - totipotent cells are interstitial cells
- 41. Needle-shaped crystals of *Colocasia* are called:**  
 a) sphaeroraphides  
 b) raphides  
 c) cystolith  
 d) silicate
- 42. Which is a false statement about *Spirogyra*?**  
 a) Haploid body  
 c) Star-shaped chloroplast  
 b) Unbranched and filamentous structure  
 d) Pyrenoids in chloroplast
- 43. What is the botanical name of sunflowers?**  
 a) *Tagetes patula*  
 c) *Calendula officinalis*  
 b) *Helianthus annuus*  
 d) *Lycopersicum esculentum*
- 44. Fungi store food material mainly in the form of**  
 a) starch  
 b) glycogen  
 c) cellulose  
 d) glucose
- 45. In lichen, the algal partner mainly contributes**  
 a) protection  
 b) nitrogen fixation  
 c) photosynthesis  
 d) water absorption
- 46. A bryophyte sporophyte remains attached to the gametophyte because**  
 a) it lacks roots  
 c) it is nutritionally dependent  
 b) it is photosynthetically inactive  
 d) fertilization occurs on gametophyte
- 47. Which pteridophyte structure first shows differentiation into root, stem, and leaf?**  
 a) Prothallus  
 b) Zygote  
 c) Sporophyte  
 d) Sporangium
- 48. Wind pollination in gymnosperms is facilitated mainly by**  
 a) sticky pollen  
 c) winged pollen grains  
 b) brightly coloured cones  
 d) nectar secretion
- 49. A modification of root for additional support seen in banyan is called**  
 a) prop root  
 b) stilt root  
 c) pneumatophore  
 d) tuberous root
- 50. Epigynous flower differs from hypogynous flower in having**  
 a) inferior ovary  
 b) superior ovary  
 c) more stamens  
 d) bisexual nature
- 51. Artificial systems of classification are mainly based on**  
 a) phylogeny  
 c) evolutionary relationships  
 b) morphology of few characters  
 d) molecular data
- 52. Standing crop in an ecosystem refers to**  
 a) rate of biomass production  
 c) total biomass at a given time  
 b) energy trapped per unit time  
 d) total number of species
- 53. The presence of more than two allelic forms for a single gene is known as:**  
 a) Epistasis  
 b) Pleiotropism  
 c) Blending Inheritance  
 d) Multiple Allelism
- 54. The transfer of energy between trophic levels follows which principle?**  
 a) Lindeman's 10% law  
 c) Allen's rule  
 b) Gause's principle  
 d) Bergmann's rule

72. The rate of transpiration increases due to:
  - a) Light intensity
  - b) Low temperature
  - c) High humidity
  - d) Decrease in wind velocity
73. Movement of water across semipermeable membrane due to solute gradient is
  - a) diffusion
  - b) active transport
  - c) osmosis
  - d) mass flow
74. The transfer of pollen from another to stigma of same flower is
  - a) geitonogamy
  - b) xenogamy
  - c) autogamy
  - d) cleistogamy
75. Endosperm in coconut is an example of
  - a) nuclear endosperm
  - b) cellular endosperm
  - c) helobial endosperm
  - d) perisperm
76. Endosperm is formed by the fusion of male gamete (AA) and female gamete (aa) in an angiospermic plant. Its ploidy is:
  - a) Aaa
  - b) AAa
  - c) AAA
  - d) aaa
77. Xerophytes lack:
  - a) Well-developed root and sunken stomata
  - b) Well-developed conducting system
  - c) Aerenchyma
  - d) Spiny leaf
78. pH of acid rain:
  - a) 5.5-7
  - b) 3-4.5
  - c) 7
  - d) 8
79. Pyramid of energy is:
  - a) Spindle shape
  - b) always erect
  - c) Inverted
  - d) Sometimes erect
80. If meiosis is replaced by mitosis during gamete formation, the result will be
  - a) sterile offspring
  - b) haploid gametes
  - c) polyploid individuals
  - d) reduced chromosome number
81. The pH of solutions A, B, C, D are respectively 9.5, 2.5, 3.5, 5.5. The most acidic solution is:
  - a) A
  - b) B
  - c) C
  - d) D
82. The value of ionic product of water at 25°C is
  - a) less than  $1 \times 10^{-14}$
  - b) greater than  $1 \times 10^{-14}$
  - c) equal to  $1 \times 10^{-14}$
  - d) equal to  $1 \times 10^{-7}$
83. The compound that can work both as an oxidising as well as a reducing agent is
  - a)  $\text{KMnO}_4$
  - b)  $\text{H}_2\text{SO}_4$
  - c)  $\text{BaO}_2$
  - d)  $\text{H}_2\text{O}_2$
84. If 30 mL of acid is neutralised by 15 mL of 0.2N alkali, the concentration of acid is:
  - a) 0.1 N
  - b) 0.2 N
  - c) 0.3 N
  - d) 0.4 N
85. In voltaic cells, the electrons are lost during the oxidation flow from:
  - a) cathode to anode
  - b) anode to cathode
  - c) remains as it is in the electrolytic solution
  - d) electrons flow from the electrolytic solution to the electrode
86. What is the indicator used in the titration of iodine against hypo ( $\text{Na}_2\text{S}_2\text{O}_3$ )?
  - a) Phenolphthalein
  - b) Starch
  - c) Methyl orange
  - d) Potassium ferricyanide
87. The formation of colloid from suspension is
  - a) Peptization
  - b) Condensation
  - c) Sedimentation
  - d) Fragmentation
88. Flux is used to

- c) free radical substitution  
 105. The addition of HBr is easiest in  
 a)  $\text{CH}_2=\text{CHCl}$  b)  $\text{ClCH}=\text{CHCl}$  c)  $\text{CH}_3-\text{CH}=\text{CH}_2$  d)  $(\text{CH}_3)_2\text{C}=\text{CH}_2$
106. Toluene when refluxed with  $\text{Br}_2$  in the presence of light mainly gives  
 a) o-bromotoluene b) p-bromotoluene  
 c) mixture of o- and p-bromotoluene d) benzyl bromide
107. When chlorine is passed through propene at  $400^\circ\text{C}$ , which of the following is formed?  
 a) PVC b) Allyl chloride c) Vinyl chloride d) 1,2-Dichloroethane
108. Which one of the following does not show resonance?  
 a) benzene b) aniline c) ethane d) nitrobenzene
109. The compound that does not exhibit keto-enol tautomerism is:  
 a) acetone b) acetaldehyde c) benzaldehyde d) acetophenone
110. The alcohol gives the maximum rate of dehydration with concentrated  $\text{H}_2\text{SO}_4$  is:  
 a) ethyl alcohol b) isopropyl alcohol c) tert-butyl alcohol d) Neopentyl alcohol
111. When Anisole is treated with HI, the main products are:  
 a) Iodobenzene and Methanol b) Phenol and Methyl iodide  
 c) Benzene and Methyl iodide d) Phenol and Benzene
112. Formaldehyde when treated with KOH gives methanol and potassium formate. The reaction is known as  
 a) Perkin reaction b) Claisen reaction  
 c) Cannizzaro reaction d) Knoevengel condensation
113. The end product 'C' in the following sequence of chemical reactions is  
 $\text{CH}_3\text{COOH} \xrightarrow{\text{CaCO}_3} \text{A} \xrightarrow{\text{heat}} \text{B} \xrightarrow{\text{NH}_2\text{OH}} \text{C}$   
 a) Acetaldehyde oxime b) Formaldehyde oxime  
 c) Methyl nitrate d) Acetoxime
114. Decarboxylation of carboxylic acids removes which group?  
 a) -OH b)  $-\text{CO}_2$  c)  $\text{H}_2\text{O}$  d)  $\text{O}_2$
115. Rancidity of butter is due to the formation of  
 a) butanoic acid b) butene -1 c) lactic acid d) lactose
116. Which of the following is an azo dye?  
 a) methyl orange b) phenolphthalein c) malachite green d) methylene blue
117. Primary amines on heating with  $\text{CS}_2$  followed by excess of mercuric chloride yields isothiocyanates. The reaction is called  
 a) Hofmann mustard oil reaction b) Perkin reaction  
 c) Fries reaction d) Diels-Alder reaction
118. Acetamide is treated with the following reagents separately. Which one of these would yield methylamine?  
 a)  $\text{NaOH} - \text{Br}_2$  b) Sodalime c) Hot conc.  $\text{H}_2\text{SO}_4$  d)  $\text{PCl}_5$
119. Of the following which one is classified as polyester polymer?  
 a) Terylene b) Bakelite c) Melamine d) Nylon-66
120. The number of molecules in 8.96 litre of a gas at  $0^\circ\text{C}$  and 1 atm pressure is approximately  
 a)  $6.023 \times 10^{23}$  b)  $12.04 \times 10^{23}$  c)  $18.06 \times 10^{23}$  d)  $24.08 \times 10^{23}$

121. If  $3.01 \times 10^{20}$  molecules are removed from 98 mg of  $\text{H}_2\text{SO}_4$ , then the number of moles of  $\text{H}_2\text{SO}_4$  left are:  
 a)  $0.1 \times 10^{-3}$       b)  $0.5 \times 10^{-3}$       c)  $1.66 \times 10^{-3}$       d)  $9.95 \times 10^{-3}$
122. The number of electrons and neutrons of an element is 18 and 20 respectively. Its mass number is  
 a) 2      b) 17      c) 37      d) 38
123. The total number of electrons possible for principal quantum number  $n$  is  
 a)  $n$       b)  $n^2$       c)  $2n$       d)  $2n^2$
124. The correct order of ionization energies is  
 a)  $\text{Zn} < \text{Cd} < \text{Hg}$       b)  $\text{Hg} < \text{Cd} < \text{Zn}$       c)  $\text{Ar} > \text{Ne} > \text{He}$       d)  $\text{Cs} < \text{Rb} < \text{Na}$
125. In which of the following species the interatomic bond angle is  $109^\circ 28'$ ?  
 a)  $\text{NH}_3$ ,  $(\text{BF}_4)^-$       b)  $(\text{NH}_4)^+$ ,  $\text{BF}_3$       c)  $\text{NH}_3$ ,  $\text{BF}_4$       d)  $(\text{NH}_2)^-$ ,  $\text{BF}_3$
126. The pair of species having identical shapes for molecules of both species is  
 a)  $\text{XeF}_2$ ,  $\text{CO}_2$       b)  $\text{BF}_3$ ,  $\text{PCl}_3$       c)  $\text{PF}_5$ ,  $\text{IF}_5$       d)  $\text{CF}_4$ ,  $\text{SF}_4$
127. Which of the following exhibits the weakest inter-molecular forces?  
 a)  $\text{HCl}$       b)  $\text{HF}$       c)  $\text{Xe}$       d)  $\text{H}_2\text{O}$
128. The van der Waal's equation reduces itself to the ideal gas equation at  
 a) high pressure and low temperature      b) low pressure and low temperature  
 c) low pressure and high temperature      d) high pressure alone
129. Enthalpy of neutralisation of  $\text{HCl}$  with  $\text{NaOH}$  is  $x$ . The heat evolved when 500 ml of 2 N  $\text{HCl}$  are mixed with 250 ml of 4N  $\text{NaOH}$  will be.  
 a)  $500x$       b)  $100x$       c)  $x$       d)  $10x$
130. In what manner will increase of pressure affect the following equation?  
 $\text{C (s)} + \text{H}_2\text{O (g)} \rightleftharpoons \text{CO (g)} + \text{H}_2 \text{ (g)}$   
 a) Shift in the forward direction      b) Shift in the reverse direction  
 c) Increase in the yield of hydrogen      d) No effect
131. The magnitude of vector  $\vec{A}$ ,  $\vec{B}$  and  $\vec{C}$  are respectively 12, 5 and 13 units and  $\vec{A} + \vec{B} = \vec{C}$  then the angle between  $\vec{A}$  and  $\vec{B}$  is  
 a) 0      b)  $\pi$       c)  $\pi/2$       d)  $\pi/4$
132. If  $L$  and  $R$  are respectively the inductance and resistance, then the dimensions of  $\frac{L}{R}$  will be  
 a)  $\text{M}^0\text{L}^0\text{T}^{-1}$       b)  $\text{M}^0\text{L}^0\text{T}^0$       c)  $\text{M}^0\text{L}^0\text{T}$   
 d) cannot be represented in terms of  $M$ ,  $L$  and  $T$
133. A car sometimes overturns while taking a turn. When it overturns, it is  
 a) The inner wheel which leaves the ground first  
 b) The outer wheel which leaves the ground first  
 c) Both the wheels leave the ground simultaneously  
 d) Either wheel leaves the ground first
134. In the motion of a projectile freely under gravity, its  
 a) Total energy is conserved      b) Momentum is conserved  
 c) Energy and momentum both are conserved      d) None is conserved
135. A box is placed on an inclined plane and has to be pushed down. The angle of inclination is  
 a) Equal to angle of friction      b) More than angle of friction

- a)  $-6D$                       b)  $+4D$                       c)  $+2D$                       d)  $-2D$
149. The equivalent focal length of a pair of thin lenses of focal length  $f_1$  and  $f_2$  when in contact is
- a)  $f_1 + f_2$                       b)  $f_1 - f_2$                       c)  $\frac{f_1 f_2}{f_1 + f_2}$                       d)  $\frac{f_1 + f_2}{f_1 f_2}$
150. Total internal reflection of light is possible when light enters from
- a) Air to glass                      b) Vacuum to air                      c) Air to water                      d) Water to air
151. The ratio of intensities of two waves is  $9 : 1$ . They are producing interference. The ratio of maximum and minimum intensities will be
- a)  $10 : 8$                       b)  $9 : 1$                       c)  $4 : 1$                       d)  $2 : 1$
152. The condition for observing Fraunhofer diffraction from a single slit is that the light wavefront incident on the slit should be
- a) Spherical                      b) Cylindrical                      c) Plane                      d) Elliptical
153. A wave of frequency  $500\text{Hz}$  has velocity  $360\text{m/sec}$ . The distance between two nearest points  $60^\circ$  out of phase
- a)  $0.6\text{cm}$                       b)  $12\text{cm}$                       c)  $60\text{cm}$                       d)  $120\text{cm}$
154. A resonance air column of length  $20\text{cm}$  resonates with a tuning fork of frequency  $250\text{Hz}$ . The speed of sound in air is
- a)  $300\text{ m/s}$                       b)  $200\text{ m/s}$                       c)  $150\text{ m/s}$                       d)  $75\text{ m/s}$
155. When we hear a sound, we can identify its source from
- a) amplitude of sound                      b) intensity of sound  
c) wavelength of sound                      d) overtones present in the sound
156. The ratio of the forces between two small spheres given same charge in air and in medium of dielectric constant  $K$  is
- a)  $1 : K$                       b)  $K : 1$                       c)  $1 : K^2$                       d)  $K^2 : 1$
157. Number of electrons in one Coulomb of charge will be
- a)  $3.46 \times 10^{29}$                       b)  $6.25 \times 10^{18}$                       c)  $1.6 \times 10^{19}$                       d)  $9 \times 10^{11}$
158. The energy of a charged capacitor is given by the expression ( $q$  = charge on the conductor and  $C$  = its capacity).
- a)  $\frac{q^2}{2C}$                       b)  $\frac{q^2}{C}$                       c)  $2qC$                       d)  $\frac{q}{2C^2}$
159. A potentiometer is an ideal device of measuring potential difference because
- a) It uses a sensitive galvanometer  
b) It does not disturb the potential difference it measures  
c) It is an elaborate arrangement  
d) It has a long wire hence heat developed is quickly radiated
160. The resistance conductivity of a superconductor is
- a) Infinite                      b) Very large                      c) Very small                      d) Zero
161. If the temperature of cold junction of thermocouple is lowered, then the neutral temperature
- a) Increases                      b) Approaches inversion temperature  
c) Decreases                      d) Remains the same
162. A heater coil is cut into two equal parts and only one part is now used in the heater. The heat generated will now be
- a) One fourth                      b) Halved                      c) Doubled                      d) Four times
163. One Tesla is equal to

164. If  $\chi$  is magnetic susceptibility of a material then its relative permeability ( $\mu_r$ ) is equal to
- a)  $\chi + 1$                       b)  $\chi - 1$                       c)  $\frac{1}{\chi}$                       d)  $1 + \chi^2$
165. An ideal transformer has 500 and 5000 turn in primary and secondary windings respectively. If the primary voltage is connected to a 6V battery then the secondary voltage is
- a) 0                      b) 60 V                      c) 0.6 V                      d) 6.0 V
166. Two coils are placed close to each other. The mutual inductance of the pair of coils depends upon
- a) The currents in the two coils  
b) The rates at which currents are changing in the two coils  
c) Relative position and orientation of the two coils  
d) The materials of the wires of the coils
167. Current in the circuit is wattless, if
- a) Inductance in the circuit is zero                      b) Resistance in the circuit is zero  
c) Current is alternating                      d) Resistance and inductance both are zero
168. Which of the following law is used in the Millikan's method for the determination of charge?
- a) Ampere's law                      b) Stoke's law  
c) Fleming's left hand rule                      d) Fleming's right hand rule
169. The de-Broglie wavelength of a particle accelerated with 150 volt potential is  $10^{-10}$  m. If it is accelerated by 600 volts p.d. its wavelength will be
- a)  $0.25 \text{ \AA}$                       b)  $0.5 \text{ \AA}$                       c)  $1.5 \text{ \AA}$                       d)  $2 \text{ \AA}$
170. The minimum energy required to remove an electron is called
- a) Stopping potential                      b) Kinetic energy                      c) Work function                      d) None of these
171. The patient is asked to drink  $\text{BaSO}_4$  for examining the stomach by X-rays because X-rays are
- a) Reflected by heavy atoms                      b) Refracted by heavy atoms  
c) Less absorbed by heavy atoms                      d) More absorbed by heavy atoms
172. The kinetic energy of the electron in an orbit of radius  $r$  in hydrogen atom is ( $e$  = electronic charge)
- a)  $\frac{e^2}{r^2}$                       b)  $\frac{e^2}{2r}$                       c)  $\frac{e^2}{r}$                       d)  $\frac{e^2}{2r^2}$
173. One gram of a radioactive samples, out of 16grms was found left after 2 hours. Half life of the element is
- a) 2 hr                      b) 1 hr                      c) 0.5 hr                      d) 0.25 hr
174. A piece of semiconductor is connected in series in an electric circuit. On increasing the temperature, the current in the circuit will
- a) Decrease                      b) Remain unchanged                      c) Increase                      d) Stop flowing
175. When a PN junction diode is reverse biased
- a) Electrons and holes are attracted towards each other and move towards the depletion region  
b) Electrons and holes move away from the junction depletion region  
c) Height of the potential barrier decreases  
d) No change in the current takes place
176. When forward bias is applied to a P-N junction, then what happens to the potential barrier

Ram and Shyam start walking in opposite directions. Ram covers 4 km and Shyam covers 3 km. Ram turns right and walk 3 km and Shyam turns left and walk 4 km. How far are they from the starting point?

- a) 5 km      b) 6 km      c) 7 km      d) 8 km

187. What number should replace the question mark?



- a) 435      b) 354      c) 453      d) 456

188. If 'A' denotes '+', 'B' denotes '-', 'C' denotes 'X' and 'D' denotes '÷' then which of the following statement is correct?

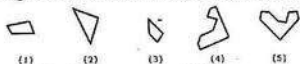
- a)  $10 C 2 A 12 D 2 B 2 = 14$       b)  $10 C 2 A 12 D 2 B 2 = 22$   
c)  $10 C 2 A 12 D 2 B 2 = 24$       d)  $10 C 2 A 12 D 2 B 2 = 32$

189. What number should replace the question mark?



- a) 156      b) 180      c) 188      d) 189

190. It is required to select the alternative which represents three out of the five alternative figures which when fitted into each other would form a complete square.



- a) 123      b) 345      c) 245      d) 234

191. Veena who is the sister-in-law of Ashok, is the daughter-in-law of Kalyani. Dheeraj is the father of Sudeep who is the only brother of Ashok. How is Kalyani related to Ashok?

- a) Mother-in-law      b) Aunt      c) Wife      d) None of these

192. In the two numerical sequences below, one number that appears in the top sequence should appear in the bottom sequence and vice versa. Which two numbers should be changed round?

100, 89, 76, 63, 44, 25,

105, 93, 79, 61, 45, 25

- a) 45, 44      b) 25, 25      c) 61, 63      d) 79, 76

193. Count the number of triangles in the given diagram.



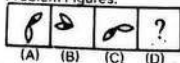
- a) 25      b) 30      c) 35      d) 40

194. Which five letters replace the question mark and give the meaning full word vertically?

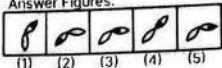
|   |   |   |   |   |
|---|---|---|---|---|
| N | A | E | W | O |
| E | T | R | E | W |
| ? | ? | ? | ? | ? |

- a) TEACHR      b) TICKE      c) WEASE      d) WEATL

- Problem Figures:**



**Answer Figures:**



- (X) (A) (B) (C) (D)

199. Complete the series. 2, 3, 3, 5, 10, 13, ?, 43, 172, 177  
a) 23                      b) 38                      c) 39                      d) 40
200. Arrange the following words according to the dictionary arrangement.  
1. Epitaxy                      2. Episode                      3. Epigene                      4. Epitome                      5. Epilogue  
a) 1,2,3,4,5                      b) 3,2,5,4,1                      c) 3,5,2,1,4                      d) 5,4,2,1,3

NAME Institute

In front of Singhdwar, Patalisadak

Tel: -01-8971544

**MECEE BL**  
**Model Entrance Exam**  
**(Model: XXII)**  
**(Set: A)**

Date: Baishakh 19<sup>th</sup> 2083 (Saturday)

Venue: NAME

Duration – 3 hrs

Time – 11:00 AM

**INSTRUCTIONS**

There are 200 multiple-choice questions, each having four choices of which only one choice is correct.

Fill ☐ the most appropriate one.

1. According to the most widely accepted modern theory, the origin of life on primitive Earth is best explained by:
  - a) Theory of special creation
  - b) Biogenesis theory
  - c) Chemical evolution theory proposed by Oparin and Haldane
  - d) Lamarck's theory of use and disuse
2. The forelimbs of whales, bats, horses and humans show similar bone arrangement but perform different functions. These organs are examples of:
  - a) Analogous organs
  - b) Homologous organs
  - c) Vestigial organs
  - d) Atavistic structures
3. The presence of fossils in different sedimentary rock layers provides the strongest evidence in support of:
  - a) Spontaneous generation
  - b) Darwin's theory of natural selection and descent with modification
  - c) Lamarckism
  - d) Special creation theory
4. An animal possesses jointed appendages, chitinous exoskeleton, open circulatory system and undergoes metamorphosis. This animal belongs to the phylum:
  - a) Mollusca
  - b) Arthropoda
  - c) Annelida
  - d) Echinodermata
5. In earthworm, the coiled tubular structures present in almost every segment that help in excretion and osmoregulation are known as:
  - a) Flame cells
  - b) Nephridia
  - c) Malpighian tubules
  - d) Green glands
6. Peripatus is considered an important connecting link between two phyla because it possesses characters of both. Identify the two phyla:
  - a) Annelida and Mollusca
  - b) Annelida and Arthropoda
  - c) Arthropoda and Echinodermata
  - d) Mollusca and Chordata
7. In cockroach, respiration occurs through a network of branched, chitin-lined air tubes that open externally through small openings called spiracles. These tubes are called:
  - a) Book lungs
  - b) Tracheae
  - c) Gills
  - d) Pulmonary sacs
8. In which of the following chordates does the notochord persist throughout the entire life and never get replaced by a vertebral column?
  - a) Frog
  - b) Amphioxus
  - c) Pigeon
  - d) Human
9. The tissue that has the properties of excitability, conductivity and contractility and is responsible for movement and locomotion in animals is:
  - a) Epithelial tissue
  - b) Muscular tissue
  - c) Connective tissue
  - d) Nervous tissue
10. The thin, delicate epithelium that allows rapid diffusion of gases and is found lining the alveoli of lungs and Bowman's capsule is:
  - a) Stratified squamous epithelium
  - b) Simple squamous epithelium
  - c) Cuboidal epithelium
  - d) Columnar epithelium
11. The longest bone in the human body that bears the maximum body weight is:

- a) Humerus      b) Tibia      c) Femur      d) Fibula
12. The structural and functional unit of the kidney responsible for ultrafiltration, selective reabsorption and tubular secretion is:
- a) Neuron      b) Nephron      c) Alveolus      d) Osteon
13. The natural pacemaker of the human heart that initiates the cardiac impulse is located in the wall of the right atrium and is called:
- a) Atrioventricular node      b) Sinoatrial node  
c) Bundle of His      d) Purkinje fibres
14. In humans, more than 95% of oxygen is transported from the lungs to the body tissues in combination with:
- a) Plasma proteins      b) Haemoglobin in erythrocytes  
c) Dissolved form in plasma      d) Carbonic acid
15. A person suffering from bleeding gums, loosening of teeth and poor wound healing is most likely deficient in:
- a) Vitamin A      b) Vitamin B12      c) Vitamin C      d) Vitamin D
16. The hormone secreted by beta cells of the islets of Langerhans that lowers blood glucose level by promoting uptake of glucose by cells is:
- a) Glucagon      b) Insulin      c) Adrenaline      d) Cortisol
17. People living in hilly areas often suffer from swelling in the neck region due to deficiency of iodine in their diet. This condition is known as:
- a) Diabetes mellitus      b) Goitre      c) Night blindness      d) Rickets
18. Malaria is a protozoan disease caused by Plasmodium species and is transmitted to humans by the bite of:
- a) Male Culex mosquito      b) Infected female Anopheles mosquito  
c) Tse-tse fly      d) Housefly
19. Typhoid fever is a bacterial disease caused by Salmonella typhi. The most common mode of transmission of this disease is:
- a) Inhaling contaminated air      b) Contaminated food and water  
c) Mosquito bite      d) Direct physical contact
20. The Human Immunodeficiency Virus (HIV) primarily attacks and destroys which type of cells of the immune system, leading to severe immunodeficiency?
- a) B-lymphocytes      b) Helper T-lymphocytes (CD4+ cells)  
c) Erythrocytes      d) Thrombocytes
21. The blood group that can be safely donated to a person of any ABO blood group is known as:
- a) A      b) B      c) AB      d) O
22. The process of formation of all types of blood cells from hematopoietic stem cells in red bone marrow is termed as:
- a) Haemolysis      b) Haematopoiesis      c) Haemostasis      d) Agglutination
23. The hormone secreted by parathyroid glands that increases blood calcium level by promoting bone resorption is:
- a) Calcitonin      b) Parathyroid hormone  
c) Thyroxine      d) Insulin

24. The largest gland in the human body that performs detoxification, bile secretion and protein metabolism is:  
 a) Pancreas                      b) Thyroid gland                      c) Liver                      d) Pituitary gland
25. The basic structural and functional unit of the nervous system specialized for reception, conduction and transmission of nerve impulses is:  
 a) Nephron                      b) Neuron                      c) Synapse                      d) Neuroglial cell
26. The sexually transmitted bacterial disease caused by *Treponema pallidum* that can damage multiple organs if left untreated is known as:  
 a) Gonorrhoea                      b) Syphilis                      c) AIDS                      d) Chancroid
27. ELISA test is a highly sensitive immunological test primarily used for the diagnosis of:  
 a) Tuberculosis                      b) HIV infection                      c) Malaria                      d) Typhoid fever
28. The live attenuated vaccine given to infants for the prevention of tuberculosis is:  
 a) DPT vaccine                      b) BCG vaccine                      c) Oral polio vaccine                      d) MMR vaccine
29. The branch of biology that deals with the interactions between living organisms and their physical and biological environment is known as:  
 a) Taxonomy                      b) Ecology                      c) Physiology                      d) Embryology
30. In Nepal, the one-horned rhinoceros is strictly protected because its population has drastically declined due to poaching and habitat destruction. It is currently categorized as:  
 a) Critically endangered                      b) Endangered  
 c) Vulnerable                      d) Extinct in the wild
31. The major cause of depletion of the ozone layer in the stratosphere is the release of:  
 a) Carbon dioxide and methane                      b) Chlorofluorocarbons (CFCs)  
 c) Sulphur dioxide                      d) Nitrogen oxides from vehicles
32. Biofertilizers are considered environment-friendly because they mainly consist of:  
 a) Chemical salts                      b) Living beneficial microorganisms  
 c) Animal dung only                      d) Synthetic plant hormones
33. Which of the following is considered an example of ex-situ conservation of biodiversity?  
 a) National Park                      b) Wildlife Sanctuary                      c) Zoological garden                      d) Biosphere Reserve
34. The normal pH value of human blood that is maintained by buffer systems is:  
 a) 6.8 – 7.0                      b) 7.35 – 7.45                      c) 8.0 – 8.5                      d) 5.5 – 6.0
35. The hormone released from the posterior pituitary that causes strong contraction of the uterus during childbirth is:  
 a) Prolactin                      b) Oxytocin                      c) Vasopressin                      d) Thyroxine
36. The smallest bone in the human body is found in:  
 a) Thigh region                      b) Middle ear                      c) Lower leg                      d) Wrist
37. Which blood vessel carries oxygenated blood from the lungs to the left atrium of the heart?  
 a) Pulmonary artery                      b) Pulmonary vein  
 c) Aorta                      d) Superior vena cava
38. Amoebic dysentery (amoebiasis) is caused by the protozoan parasite:  
 a) *Plasmodium vivax*                      b) *Entamoeba histolytica*  
 c) *Trypanosoma gambiense*                      d) *Giardia intestinalis*

39. The organ known as the "graveyard of RBCs" because old and worn-out red blood cells are destroyed here is:
  - a) Liver
  - b) Spleen
  - c) Kidney
  - d) Bone marrow
40. The total number of chromosomes present in a normal human somatic cell is:
  - a) 23
  - b) 46
  - c) 48
  - d) 44
41. Which of the following is not a nitrogen base present in DNA?
  - a) Adenine
  - b) Guanine
  - c) Uracil
  - d) Thymine
42. Housekeeping genes are mainly responsible for:
  - a) Cell division
  - b) Basic metabolic functions and maintenance
  - c) Defence against pathogens
  - d) Reproduction
43. The most abundant biomolecule in the biosphere is:
  - a) Protein
  - b) Lipid
  - c) Carbohydrate
  - d) Nucleic acid
44. Which pigment is responsible for the red colour in Rhodophyta (red algae)?
  - a) Phycocyanin
  - b) Phycoerythrin
  - c) Fucoxanthin
  - d) Chlorophyll b
45. Lichens represent which type of relationship?
  - a) Parasitism
  - b) Commensalism
  - c) Mutualism
  - d) Amensalism
46. In ecological succession, the first organisms to colonize a barren area are called:
  - a) Climax community
  - b) Pioneer species
  - c) Keystone species
  - d) Dominant species
47. The pyramid of energy in any ecosystem is always:
  - a) Inverted
  - b) Upright
  - c) Spindle shaped
  - d) Variable
48. Biodiversity is maximum in:
  - a) Polar region
  - b) Tropical rain forest
  - c) Desert
  - d) Temperate forest
49. The main difference between prokaryotic and eukaryotic cell is the presence of:
  - a) Cell wall
  - b) Well-defined nucleus
  - c) Ribosome
  - d) Cytoplasm
50. During mitosis, chromosomes are arranged at the equatorial plate in:
  - a) Prophase
  - b) Metaphase
  - c) Anaphase
  - d) Telophase
51. Crossing over occurs during:
  - a) Prophase of mitosis
  - b) Prophase I of meiosis
  - c) Metaphase II of meiosis
  - d) Anaphase I of meiosis
52. The enzyme that unwinds DNA double helix during replication is:
  - a) DNA polymerase
  - b) Helicase
  - c) DNA ligase
  - d) Primase
53. A frameshift mutation is caused due to:
  - a) Substitution of one base
  - b) Insertion or deletion of bases not in multiple of three
  - c) Change in a single codon
  - d) Inversion of chromosome segment
54. In incomplete dominance, the phenotypic ratio in F2 generation is:
  - a) 3:1
  - b) 1:2:1
  - c) 9:3:3:1
  - d) 9:7

55. Which tissue provides mechanical support to young growing parts of dicot stem?  
 a) Parenchyma      b) Collenchyma      c) Sclerenchyma      d) Xylem
56. Vascular bundles in dicot stem are:  
 a) Radial      b) Conjoint, collateral and open  
 c) Conjoint, collateral and closed      d) Scattered
57. The opening and closing of stomata is regulated by:  
 a) Guard cells      b) Subsidiary cells      c) Epidermal cells      d) Companion cells
58. Double fertilization is characteristic of:  
 a) Gymnosperms      b) Angiosperms      c) Pteridophytes      d) Bryophytes
59. The fruit develops from:  
 a) Ovule      b) Ovary      c) Style      d) Stigma
60. Which plant hormone is known as stress hormone?  
 a) Auxin      b) Gibberellin      c) Absciscic acid      d) Cytokinin
61. Krebs cycle occurs in:  
 a) Cytoplasm      b) Mitochondrial matrix  
 c) Inner mitochondrial membrane      d) Chloroplast stroma
62. Photorespiration is more prominent in:  
 a) C4 plants      b) C3 plants      c) CAM plants      d) All plants equally
63. Kranz anatomy is found in:  
 a) C3 plants      b) C4 plants      c) CAM plants      d) Bryophytes
64. Which of the following is a parthenocarpic fruit?  
 a) Mango      b) Banana      c) Pea      d) Wheat
65. Ti plasmid is widely used as a vector in:  
 a) Animal biotechnology      b) Plant genetic engineering  
 c) Bacterial cloning only      d) Virus production
66. Bt cotton is genetically modified to provide resistance against:  
 a) Virus      b) Fungus      c) Bollworm      d) Drought
67. Large scale production of genetically identical plants is done by:  
 a) Grafting      b) Micropropagation      c) Hybridization      d) Selection
68. Alkaloids are examples of:  
 a) Primary metabolites      b) Secondary metabolites  
 c) Storage proteins      d) Carbohydrates
69. Quinine is commercially extracted from:  
 a) Cinchona      b) Papaver somniferum  
 c) Hevea brasiliensis      d) Atropa belladonna
70. In prokaryotes, the genetic material is:  
 a) Linear DNA with histone      b) Circular naked DNA  
 c) RNA only      d) Linear RNA
71. The first stable product of Calvin cycle in C3 plants is:  
 a) Oxaloacetic acid      b) 3-Phosphoglyceric acid (PGA)

- c) Phosphoenol pyruvate  
72. **Collenchyma tissue is characterized by:**  
a) Dead cells with lignified wall  
c) Living cells with thin wall
- d) Malic acid  
b) Living cells with uneven pectin thickening  
d) Dead cells with thin wall
73. **Imbibition is important for:**  
a) Ascent of sap      b) Seed germination  
c) Transpiration      d) Guttation
74. **Pleiotropy means:**  
a) Many genes control one trait  
c) Gene interaction
- b) One gene controls many traits  
d) Incomplete dominance
75. **The respiratory quotient (RQ) of carbohydrate is:**  
a) 0.7      b) 1.0      c) 1.3      d) More than 1
76. **Which of the following is a stem tuber?**  
a) Carrot      b) Potato      c) Turnip      d) Radish
77. **Electron transport chain is located in:**  
a) Cytoplasm  
c) Chloroplast stroma
- b) Mitochondrial inner membrane  
d) Peroxisome
78. **Runner is an example of:**  
a) Underground stem  
c) Root modification
- b) Sub-aerial stem modification  
d) Leaf modification
79. **High CO<sub>2</sub> concentration inside leaf causes:**  
a) Stomatal opening  
c) Increased rate of photosynthesis
- b) Stomatal closing  
d) No effect on stomata
80. **Restriction endonucleases are used in:**  
a) Gel electrophoresis  
c) PCR technique
- b) Recombinant DNA technology  
d) Tissue culture
81. **Among the following pairs of compounds, the one that illustrates the law of multiple proportions is**  
a) NH<sub>3</sub> and NCl<sub>3</sub>      b) H<sub>2</sub>S and SO<sub>2</sub>      c) CS<sub>2</sub> and FeSO<sub>4</sub>      d) CuO and Cu<sub>2</sub>O
82. **7.5 grams of a gas occupy 5.6 litres of volume at STP. The gas is,**  
a) N<sub>2</sub>O      b) NO      c) CO      d) CO<sub>2</sub>
83. **Which of the following does not contain number of neutrons equal to that of <sup>40</sup>Ar<sub>18</sub>?**  
a) <sup>41</sup>K<sub>19</sub>      b) <sup>43</sup>Sc<sub>21</sub>      c) <sup>40</sup>Sc<sub>21</sub>      d) <sup>42</sup>Ca<sub>20</sub>
84. **The values of quantum numbers n, l and m for the 5th electron of Boron will be,**  
a) n = 1, l = 0, m = -1      b) n = 2, l = 1, m = -1      c) n = 2, l = 2, m = -1      d) n = 1, l = 2, m = -1
85. **Which of the following is isoelectronic with carbon atom?**  
a) Na<sup>+</sup>      b) Al<sup>3+</sup>      c) O<sup>2-</sup>      d) N<sup>+</sup>
86. **In XeF<sub>2</sub>, XeF<sub>4</sub> and XeF<sub>6</sub>, the number of lone pairs on Xe are respectively**  
a) 2, 3, 1      b) 1, 2, 3      c) 4, 1, 2      d) 3, 2, 1
87. **Based on kinetic theory of gases, which of the following laws can be proved?**  
a) Boyle's law      b) Charles law      c) Avogadro's law      d) All of these

- 7

- a)  $\text{SiO}_2$                       b)  $\text{SO}_2$                       c)  $\text{Na}_2\text{CO}_3$                       d)  $\text{CaO}$
105. N forms only  $\text{NCl}_3$  whereas P can form both  $\text{PCl}_3$  and  $\text{PCl}_5$ . This is because;
- a) N atom is larger than P in size  
b) P has low lying 3d-orbitals, which can be used for bonding but N does not have any d-orbitals in its valence shell  
c) P is more reactive towards Cl than N  
d) None of these
106. The number of hydrogen atom(s) attached to phosphorous atom in hypophosphorous acid is
- a) Zero                      b) Two                      c) One                      d) Three
107. The increase in boiling points of noble gases from He to Xe is due to the:
- a) Increase in atomic volume                      b) Increase in electron affinity  
c) Increase in polarizability                      d) Decrease in ionization energy
108. Which of the following is the strongest oxidizing agent?
- a)  $\text{HClO}_4$                       b)  $\text{HClO}_3$                       c)  $\text{HClO}_2$                       d)  $\text{HOCl}$
109. Arrange the following acids: 1.  $\text{H}_2\text{SO}_3$ , 2.  $\text{H}_3\text{PO}_3$ , 3.  $\text{HClO}_3$  in the increasing order of acid strength
- a)  $2 > 3 > 1$                       b)  $3 > 2 > 1$                       c)  $1 > 3 > 2$                       d)  $1 > 2 > 3$
110. Which one of the following is present as an active ingredient in bleaching Powder for bleaching action?
- a)  $\text{CaOCl}_2$                       b)  $\text{CaCl}_2$                       c)  $\text{CaO}_2\text{Cl}$                       d)  $\text{Ca}(\text{OCl})_2$
111. What is the general outer configuration of the coinage metals?
- a)  $ns^2np^6$                       b)  $(n-1)d^9ns^2$                       c)  $(n-1)d^{10}ns^1$                       d)  $(n-1)d^{10}ns^2$
112. Which of the following is the main ore of Zinc?
- a) Zinc Blende                      b) Calamine                      c) Bornite                      d) Azurite
113. Which of the following is used as additive in denaturated alcohol?
- a) Ethanol                      b) Petrol                      c) Methanol                      d) Water
114. The C-H bond distance is longest in
- a)  $\text{C}_2\text{H}_2$                       b)  $\text{C}_2\text{H}_4$                       c)  $\text{C}_2\text{H}_6$                       d)  $\text{C}_2\text{H}_5\text{Br}$
115. One recently discovered allotrope of carbon ( $\text{C}_{60}$ ) is commonly known as
- a) Fluorine                      b) Fluorene                      c) Freon                      d) Fullerene
116. Phenol is more reactive than benzene towards electrophilic substitution due to:
- a) Strong mesomeric effect                      b) Hyperconjugative effect  
c) Inductive effect only                      d) Hydrogen bonding
117. Which of the following intermediate has the complete octet around the carbon atom?
- a) Free radical                      b) Carbene                      c) Carbanion                      d) Carbonium ion
118. The compound that does not give a blue colour in Lassaigne's test is:
- a) Aniline                      b) Glycine                      c) Hydrazine                      d) Urea
119. On the addition of  $\text{HBr}$  to propene in the absence of peroxides, the first step involves the addition of:
- a)  $\text{H}^+$                       b)  $\text{Br}^-$                       c)  $\text{H}$  radical                      d)  $\text{Br}$  radical
120. On warming with silver powder, chloroform is converted into

- MCC-EE-BT Model Entrance Exam Set A
- a) Hexachloroethane    b) Acetylene    c) Ethylene    d) Carbon
121. A set of compounds in which the reactivity of halogen atom in the ascending order is
- a) Chloroethane, chlorobenzene, vinyl chloride  
b) Chlorobenzene, vinyl chloride, chloroethane  
c) Vinyl chloride, chloroethane, chlorobenzene  
d) Vinyl chloride, chlorobenzene, chloroethane
122. Alcohol manufactured from water gas is:
- a) Ethanol    b) Butanol    c) Isobutanol    d) Methanol
123. Ethyl chloride is converted into diethyl ether in
- a) Wurtz reaction    b) Grignard reaction    c) Perkin reaction    d) Williamson synthesis
124. The reaction of benzoyl chloride with  $H_2/Pd - BaSO_4$  produces
- a) Benzaldehyde    b) Benzyl alcohol    c) Benzoic acid    d) Benzoyl cyanide
125. Ketones can be obtained in one step by
- a) Hydrolysis of esters    b) Oxidation of primary alcohols  
c) Oxidation of secondary alcohols    d) Reaction of acid halides and alcohols
126. The reagent that can be used to distinguished between methanoic acid and ethanoic acid is
- a) Sodium carbonate solution    b) Ammonical silver nitrate solution  
c) Phenolphthalein    d) Neutral ferric chloride solution
127. Hydrolysis of an ester gives a carboxylic acid which on Kolbe's electrolysis yields ethane. The ester is
- a) Methyl methanoate    b) Methyl propanoate  
c) Ethyl methanoate    d) Methyl ethanoate
128. Which amine of the following will not give carbylamine reaction?
- a) Ethylamine    b) Dimethylamine    c) Methylamine    d) Phenylamine
129. Antipyretics are medical compounds which:
- a) Relieve pain    b) Control malaria  
c) Can kill other organisms    d) Lower body temperature
130. Which of the following is not made of polyamides?
- a) Nylon    b) Natural silk    c) Wool    d) Artificial silk
131. A vector is represented by  $3\hat{i} + \hat{j} + 2\hat{k}$ . Its length in xy plane is
- a) 1    b)  $\sqrt{14}$     c)  $\sqrt{10}$     d)  $\sqrt{5}$
132. The position  $x$  of a particle varies with time  $t$  as  $x = at^2 - bt^3$ . The acceleration at a time  $t$  of the particle will be equal to zero
- a)  $2a/3b$     b)  $a/b$     c)  $a/3b$     d) zero
133. Two bodies of masses 1 kg and 5kg are dropped gently from the top of a tower. At a point 20 cm from the ground both bodies will have same
- a) momentum    b) kinetic energy    c) velocity    d) total energy
134. The maximum range of a projectile is 100 m. The maximum height reached by it is
- a) 100 m    b) 25 m    c) 200 m    d) 75 m
135. When a body moves with a constant speed along a circle
- a) no work is done on it    b) no acceleration produced on the body

- c) no force acting on a body d) its velocity remains constant
136. A block of mass 10 kg placed on rough horizontal surface having coefficient of friction  $\mu = 0.5$ . If a horizontal force of 100 N acting on it, then acceleration of the block will be  
 a)  $10 \text{ m/s}^2$  b)  $5 \text{ m/s}^2$  c)  $15 \text{ m/s}^2$  d)  $0.5 \text{ m/s}^2$
137. If the earth stops rotating, the value of  $g$  at the equator will  
 a) increase b) remains same c) decreases d) none of above
138. For steel  $Y = 2 \times 10^{11} \text{ N/m}^2$ . The force required to double the length of a steel wire of area  $1 \text{ cm}^2$  is  
 a)  $2 \times 10^7 \text{ N}$  b)  $2 \times 10^6 \text{ N}$  c)  $2 \times 10^8 \text{ N}$  d)  $2 \times 10^5 \text{ N}$
139. The potential energy of a molecule on the surface of a liquid compared to one inside the liquid is  
 a) zero b) lesser c) equal d) greater
140. A simple pendulum performs SHM about  $x = 0$  with an amplitude  $A$  and time period  $T$ . The speed of the pendulum at  $x = A/2$  will be  
 a)  $\pi A\sqrt{3}/T$  b)  $\pi A/T$  c)  $\pi A\sqrt{3}/2T$  d)  $3\pi^2 A/T$
141. A wire of cross-sectional area  $A$  at temperature  $T$  is held taut with negligible tension between two rigid supports. If the wire is cooled to a temperature  $(T - \Delta T)$ , what tension is developed in the wire? The coefficient of linear expansion is  $\alpha$  and Young's modulus is  $Y$   
 a)  $Y\alpha\Delta T$  b)  $Y\alpha\Delta T/A$  c)  $A\alpha\Delta T/Y$  d)  $YA/\alpha\Delta T$
142. Two blocks of ice when pressed together join to form one block because  
 a) of heat produced during pressing  
 b) of cold produced during pressing  
 c) melting point of ice decreases with increase of pressure  
 d) melting point of ice increases with increase in pressure
143. If  $Q$ ,  $E$  and  $W$  denote respectively the heat added, change in internal energy and the work done in a closed cycle process, then  
 a)  $W = 0$  b)  $Q = W = 0$  c)  $E = 0$  d)  $Q = 0$
144. 420 J of energy supplied to 10g of water will raise its temperature by nearly  
 a)  $1^\circ\text{C}$  b)  $4.2^\circ\text{C}$  c)  $10^\circ\text{C}$  d)  $42^\circ\text{C}$
145. A Carnot's engine working between  $27^\circ\text{C}$  and  $127^\circ\text{C}$  has a work output of 200 J/Cycle. The energy supplied to the engine from the source in each cycle is  
 a) 400 J b) 800 J c) 600 J d) 500 J
146. A bucket full of hot water is kept in a room and it cools from  $75^\circ\text{C}$  to  $70^\circ\text{C}$  in  $T_1$  minutes, from  $70^\circ\text{C}$  to  $65^\circ\text{C}$  in  $T_2$  minutes and from  $65^\circ\text{C}$  to  $60^\circ\text{C}$  in  $T_3$  minutes. Then  
 a)  $T_1 = T_2 = T_3$  b)  $T_1 < T_2 < T_3$  c)  $T_1 > T_2 > T_3$  d)  $T_1 < T_2 > T_3$
147. Which of the following parameters is constant for the molecules of all gases at a given temperature?  
 a) Mass b) Momentum c) Speed d) Kinetic energy
148. The refractive index of water is  $4/3$  and that of glass is  $5/3$ . What will be the critical angle for the ray of light entering water from glass  
 a)  $\sin^{-1}(4/5)$  b)  $\sin^{-1}(5/4)$  c)  $\sin^{-1}(1/2)$  d)  $\sin^{-1}(2/1)$
149. A virtual image larger than the object can be produced by

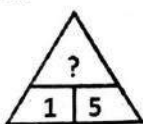
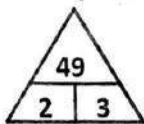
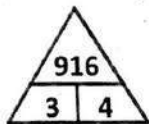
- a) convex mirror      b) concave mirror      c) plane mirror      d) concave lens

150. An achromatic combination of lenses produced  
 a) large in black and white  
 b) colored images  
 c) image unaffected by variation of refractive index with wavelength  
 d) highly enlarged image
151. Two waves of same wavelength and amplitude interference to give a minimum when phase difference is  
 a) 0      b)  $\pi/2$       c)  $\pi$       d)  $3\pi/2$
152. The condition for observing Fraunhofer diffraction from a single slit is that the light wavefront incident on the slit should be  
 a) spherical      b) cylindrical      c) plane      d) elliptical
153. A wave equation which gives the displacement along the Y direction is given by  $Y = 10^{-4} \sin(60t + 2x)$  where x and y are in meters and t in seconds. This represents  
 a) travelling with a velocity of 30 m/s in the negative x direction  
 b) of wavelength  $\pi$  meter  
 c) of frequency  $30/\pi$  Hz  
 d) all
154. The distance between two points differing in phase by  $60^\circ$  on a wave, velocity 360 m/s and frequency 500 Hz is  
 a) 24 cm      b) 12 cm      c) 6 cm      d) 1 cm
155. Standing waves are produced in a 10 m long stretched string. If the string vibrates in 5 segments and the wave velocity is 20 m/s, the frequency is  
 a) 2 Hz      b) 4 Hz      c) 5 Hz      d) 10 Hz
156. The electric field intensity at the surface of a charged conductor is  
 a) zero      b) directed normally to the surface  
 c) directed tangentially to surface      d) directed along  $45^\circ$  to surface
157. A point charge Q is placed at one of the vertices of a cubical block. The electric flux flowing through this cube is  
 a)  $Q/6\epsilon_0$       b)  $Q/4\epsilon_0$       c)  $Q/8\epsilon_0$       d)  $Q/\epsilon_0$
158. In a charged capacitor the energy resides:  
 a) On the positive plate      b) On both positive and negative plates  
 c) In the field between the plates      d) Around the edge of capacitor plates
159. A sphere of radius 10 cm is charged to a potential of 300 volts. The energy of sphere is  
 a)  $5 \times 10^{-7}$  J      b)  $5 \times 10^{-6}$  J      c)  $5 \times 10^{-5}$  J      d)  $5 \times 10^{-3}$  J
160. Two copper wires have their masses in the ratio 2:3 and the length in the ratio 3:4. The ratio of their resistance is  
 a) 4 : 9      b) 27 : 32      c) 16 : 9      d) 27 : 128
161. Kirchhoff's second law is based on law of conservation of  
 a) charge      b) energy  
 c) momentum      d) sum of mass and energy

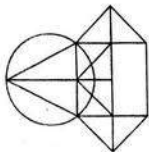
162. If bulbs whose resistance are in ratio 1:2 are connected in series, the power dissipated in them has the ratio of  
 a) 1:1                      b) 1:2                      c) 1:4                      d) 2:1
163. A cell having emf 2 V When connected across resistance  $10\ \Omega$ , p.d. developed across it is 1.5 V. The internal resistance of cell is  
 a)  $1\ \Omega$                       b)  $2.25\ \Omega$                       c)  $2.55\ \Omega$                       d)  $3.33\ \Omega$
164. An electric charge in uniform motion produces  
 a) an electric field only                      b) a magnetic field only  
 c) both electric field and magnetic fields                      d) no such field at all
165. A long wire carries steady current. It is bent into a circle of one turn and the magnetic field at the center of the coil is B. It is then bent into a circular loop of n turns. The magnetic field at the center of the coil will be  
 a) nB                      b)  $n^2B$                       c)  $2nB$                       d)  $2n^2B$
166. The temperature at which ferromagnetic material becomes paramagnetic is called a  
 a) nature temperature                      b) curie temperature  
 c) inversion temperature                      d) critical temperature
167. If the number of turns per unit length of coil of solenoid is doubled, then the self-inductance of solenoid will  
 a) remain same                      b) be halved                      c) be doubled                      d) be four times
168. Current in a circuit is wattless if  
 a) inductance in the circuit is zero                      b) resistance in the circuit is zero  
 c) current is alternating                      d) resistance and inductance both are zero
169. In Millikan's oil drop experiment an oil drop remains stationary between plates when P.d. of V is applied between plates. If radius of oil drop is doubled with same charge, the new p.d. to keep oil drop stationary is  
 a) V                      b) 2V                      c) 4V                      d) 8V
170. A photon in motion has a mass  
 a)  $c/hf$                       b)  $h/f$                       c)  $hf$                       d)  $hf/c^2$
171. The speed of electron in H atom in ground state is  
 a) c                      b)  $c/10$                       c)  $c/2$                       d)  $c/137$
172. The density  $\rho$  of a nuclear matter varies with nucleon number A of nucleus as  
 a)  $\rho \propto A^{2/3}$                       b)  $\rho \propto A^{1/3}$                       c)  $\rho \propto A$                       d)  $\rho \propto A^0$
173. A moderator is used in nuclear reactors in order to  
 a) slow down the speed of neutrons                      b) accelerates the neutrons  
 c) increase the number of neutrons                      d) decrease the number of neutrons
174. In which sequence, the radioactive radiations are emitted in the following reaction  
 $Z^AX^A \rightarrow Z+1Y^A \rightarrow Z-1K^{A-4} \rightarrow Z-1M^{A-4}$   
 a)  $\beta\gamma\alpha$                       b)  $\beta\gamma\alpha$                       c)  $\alpha\beta\alpha$                       d)  $\gamma\alpha\beta$
175. In a semiconductor the forbidden energy gap between the valance and conduction band is of  
 a) 1 eV                      b) 5 eV                      c) 1 KeV                      d) 1 MeV
176. If the forward voltage in the diode is increased, the length of depletion layer  
 a) decreases                      b) increases                      c) no change                      d) none of the above

177. The only function of a NOT gate is to
- stop signals
  - recomplement a signal
  - invert an input signal
  - act as universal gate
178. What is the approximate percentage of the universe's total energy-matter content that is attributed to dark energy?
- 5%
  - 27%
  - 68%
  - 95%
179. According to Hubble's Law, galaxies that are farther away from us tend to have:
- Slower recession velocities
  - Faster recession velocities
  - No recession velocities
  - Variable recession velocities
180. What is the Higgs Boson?
- A fundamental particle that carries the electromagnetic force
  - A particle responsible for the strong nuclear force
  - A particle that gives mass to other fundamental particles
  - A particle responsible for gravity
181. If GOLD is coded as HOME, COME is coded as DONE and CORD is coded as DOSE, how would you code SONS?
- TPOT
  - TOOT
  - TOOS
  - TONT
182. If '-' stands for 'division', '+' for 'multiplication', '-' for 'subtraction' and 'x' for 'addition', then which one of the following equations is correct?
- $4 \times 5 + 9 - 3 + 4 = 15$
  - $4 \times 5 + 9 + 3 + 4 = 17$
  - $4 - 5 + 9 \times 3 - 4 = 11$
  - $4 + 5 + 9 - 3 + 4 = 18$
183. The statements are followed by conclusions. Study the conclusions based on the given statements and select the appropriate answer.
- Statements  
 $V = D < Y < A \geq G \geq E$ ;       $P = A$ ;       $R < V$
- Conclusions  
I.  $E > P$   
II.  $P > E$
- Only conclusion I is true
  - Only conclusion II is true
  - Both conclusion I and II are true
  - No conclusions are true
184. There are 13 couples, 5 single males and 7 single females in party. Every male shakes hand with every female once. But none shakes hand with his wife. How many handshakes in total took place in the party?
- 156
  - 247
  - 347
  - 156
185. Odd one out: Sketch, Poster, Chart, Paper, Diagram
- Sketch
  - Diagram
  - Paper
  - Chart
186. A father told his son, "I was as old as you are at present at the time of your birth". If the father's age is 38 years now, the son's age five years back was:
- 14 years
  - 19 years
  - 33 years
  - 38 years
187. 3 days before yesterday was Sunday. What day will be the day 2 days after tomorrow?
- Sunday
  - Monday
  - Tuesday
  - Saturday
188. A does half as much work as B and C does half as much work as A and B together in the same time. If C alone can do the work in 40 days, all of them together will finish the work in
- 12 days
  - 12.5 days
  - 13.33 days
  - 15 days

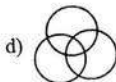
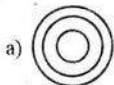
189. If the positions of the first and the sixth letters in the word **DISTRIBUTE** are interchanged; similarly, the positions of the second and the seventh, the third and the eighth and so on, which of the following letters will be the fifth from left after interchanging the positions?
190. Bird : Wing :: Fish : ?
191. What will be in the place of the question mark?



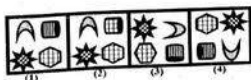
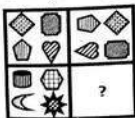
- a) 30                      b) 125                      c) 225                      d) 1125
192. How many triangles are in the figure below?



- a) 13                      b) 15                      c) 12                      d) 14
193. Which of the following diagrams indicates the best relation between Animal, Living thing and Man?



194. Calculate the compound interest on Rs.1000 at a rate of 20% per year for 18 months when interest is compounded semi-annually.
195. A man buys a cycle for Rs. 1400 and sells it at a loss of 15%. What is the selling price of the cycle?
196. Starting from point X, Jack walked 15 m towards the west. He turned left and walked 20 m. He then turned left and walked 15 m. After this he turned to his right and walked 12 m. How far and in which directions is now Jack from X?
197. Identify the figure that completes the pattern.



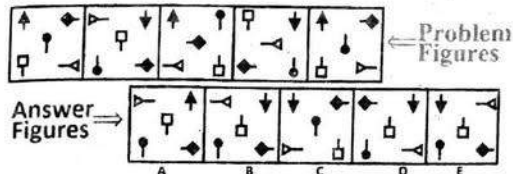
a) 1

b) 2

c) 3

d) 4

198. Select a figure from amongst the Answer Figures which will replace the missing place in the given series as established by the five Problem Figures.



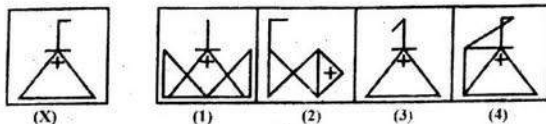
a) A

b) B

c) C

d) D

199. Find out the alternative figure which contains figure X) as its part.



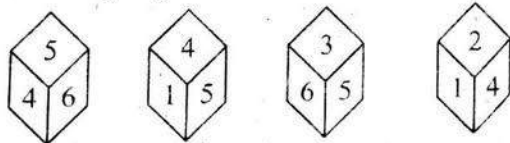
a) 3

b) 2

c) 4

d) 1

200. Which number is on the face opposite 5, if the three different positions of a dice are as shown in the figure given below.



a) 3

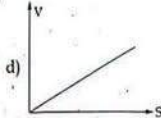
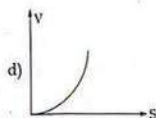
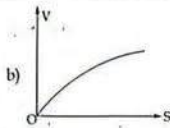
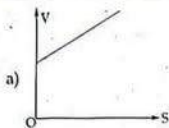
b) 2

c) 1

d) 6

46. Hedge formation can be induced by:  
a) Applying auxin to the shoot apex  
b) Removing the apical bud  
c) Increasing root pressure  
d) Applying abscisic acid
47. Chemically chromosome consists of  
a) DNA only  
b) DNA and histone  
c) DNA, histone and non-histone  
d) histone only
48. Which of the following is called soluble RNA  
a) mRNA  
b) hnRNA  
c) tRNA  
d) rRNA
49. A plant species has a diploid chromosome number of 24. The chromosome number in a monosomic individual will be:  
a) 22  
b) 23  
c) 24  
d) 25
50. The gene causing color blindness is  
a) X-chromosome dominant  
b) X-chromosome recessive  
c) Y-chromosome dominant  
d) Y-chromosome recessive
51. The number of different type of gametes produced by organism with genotype AaBbDdEE is  
a) 4  
b) 8  
c) 16  
d) 2
52. Which is dissimilar in a pair of homologous chromosome?  
a) position of centromere  
b) size of chromosome  
c) No of chromosome  
d) Allele
53. The number of chromatids in a chromosome at metaphase is  
a) one in meiosis and one in mitosis  
b) one in meiosis and two in mitosis  
c) two in meiosis and two in mitosis  
d) two in meiosis and one in mitosis
54. Which of the following stages are opposite to each other?  
a) prophase and anaphase  
b) anaphase and metaphase  
c) prophase and telophase  
d) metaphase and telophase
55. Mitochondria have about .....percentage of total enzyme of cell  
a) 70  
b) 20  
c) 30  
d) 10
56. Which of the following is not found inside nucleus?  
a) nucleosome  
b) nucleolous  
c) nucleoid  
d) nucleoplasm
57. A prokaryotic cell loses its mesosomes due to a mutation. Which function is most likely to be affected?  
a) Photosynthesis  
b) respiration  
c) Protein synthesis  
d) Osmosis
58. Which statement best explains why amino acids are called amphoteric molecules?  
a) They contain both carbon and nitrogen.  
b) They possess both acidic and basic groups.  
c) They form proteins and lipid  
d) They contain both chiral and non-chiral carbon.
59. A carbohydrate on hydrolysis yields only glucose molecules. The carbohydrate is most likely:  
a) Lactose  
b) Starch  
c) Sucrose  
d) Maltose and starch
60. Early blight of potato is caused by

- c) Black Phosphorous d) Violet Phosphorous
106. The process of heating concentrated ore below its melting point in a limited supply of air or in absence of air to convert ore into dry metallic oxide is called .....
- a) Reduction b) Pyrometallurgy c) Calcination d) Rosting
107. Which of the following helps for blood coagulation?
- a)  $\text{Fe}^{+++}$  b)  $\text{Ca}^{++}$  c)  $\text{Na}^+$  d)  $\text{Cu}^{++}$
108. The pH range if methyl red indicator is:
- a) 4.2 to 6.3 b) 8.3 to 10.0 c) 8.0 to 9.6 d) 6.8 to 8.4
109. Aniline is usually purified by:
- a) Chromatography technique b) By addition of oxalic acid  
c) Fractional crystallization d) Steam distillation
110. The Beilstein test for organic compounds is used to detect:
- a) Nitrogen b) Sulphur c) Carbon d) Halogens
111. The compound having only primary hydrogen atoms is
- a) Isobutene b) 2,3-dimethylbutene c) Cyclohexane d) Propyne
112. The most stable carbonium ion is
- a)  $(\text{CH}_3)_2\text{CH}^+$  b)  $(\text{CH}_3)_3\text{C}^+$  c)  $(\text{CH}_3)_3\text{C} - \text{CH}_2^+$  d)  $\text{CH}_3\text{CH}_2^+$
113. Which hydrocarbon has the highest octane number?
- a) Triptane b) Ethane c) Iso-octane d) Methane
114. Identify the germinal dihalide from the following?
- a)  $\text{CH}_2\text{BrCH}_2\text{Br}$  b)  $\text{CH}_3\text{CHBrCH}_2\text{Br}$  c)  $\text{CH}_3\text{CHBr}_2$  d)  $\text{CH}_3\text{CHBrCHBrCH}_3$
115. When benzene diazonium chloride is treated with cuprous chloride / cuprous bromide in presence of  $\text{HCl}/\text{HBr}$ , chlorobenzene is obtained due to replacement of diazonium group by  $-\text{Cl}$ . The reaction is known as,
- a) Sandmeyer reaction b) Williamson reaction  
c) Wurtz reaction d) Friedel -Craft reaction
116. The lowest number of carbon atom in alcohol to show the chain isomer is:
- a) 2 b) 3 c) 4 d) 5
117. In Friedel-Crafts alkylation of phenol, the major product obtained is:
- a) di-Methyl phenol b) ortho-Methyl phenol  
c) meta-Methyl phenol d) para-Methyl phenol
118. Higher homologues of ethers can be prepared by
- a) Diazomethane b) Grignard reagent c) Alkyl halides d) Methane
119. Clemmensen reduction is carried with
- a)  $\text{LiAlH}_4$  in ether b)  $\text{Zn-Hg}$  and conc.  $\text{HCl}$   
c)  $\text{H}_2$  in the presence of  $\text{Pd}$  d)  $\text{NH}_2\text{NH}_2$  / glycol and  $\text{KOH}$
120. Silver mirror test is shown by,
- a)  $\text{HCOOH}$  b)  $\text{CH}_3\text{COOH}$  c)  $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$  d)  $\text{CH}_3\text{CH}_2\text{COOH}$



135. Two vectors of same magnitude have a resultant equal to either one. The angle between the vectors will be  
 a)  $60^\circ$                       b)  $90^\circ$                       c)  $120^\circ$                       d)  $135^\circ$
136. Projectile's time of flight  $T$  and its horizontal range  $R$  are related by equation  $gT^2 = 2R$ . Then angle of projection with horizontal is  
 a)  $30^\circ$                       b)  $45^\circ$                       c)  $60^\circ$                       d)  $90^\circ$
137. A body of mass 1 kg is suspended by a thread. It is (i) lifted up with an acceleration  $5 \text{ m/s}^2$ , (ii) lowered with an acceleration  $5 \text{ m/s}^2$ . The ratio of tensions in the thread is  
 a) 3:1                      b) 1:3                      c) 1:2                      d) 2:1
138. To make the frequency double of a spring oscillator, we have to  
 a) double the mass                      b) half the mass  
 c) quadruple the mass                      d) reduce the mass to one fourth
139. The bulk modulus for an incompressible liquid is  
 a) zero                      b) unity                      c) infinity                      d) between 0 and 1
140. An incompressible fluid flows steadily through a cylindrical pipe which has radius  $2R$  at point A and radius  $R$  at point B. If velocity at point A is  $V$ , then velocity at point B is  
 a)  $V$                       b)  $2V$                       c)  $4V$                       d)  $\frac{V}{2}$
141. The Earth is a spherical conductor of radius  $6.4 \times 10^6 \text{ m}$ . Its capacitance is  
 a)  $7.11 \times 10^4 \text{ F}$                       b)  $7.11 \times 10^{-4} \text{ F}$                       c)  $7.11 \times 10^5 \text{ F}$                       d)  $7.11 \times 10^{-3} \text{ F}$
142. If the distance between the plates of a parallel plate condenser is halved and the dielectric is doubled, then its capacity increases by  
 a) 2 times                      b) 4 times                      c) 8 times                      d) 16 times
143. The electric field intensity at the surface of a solid charged sphere of radius  $R$  and volume charge density  $\rho$  is given by  
 a)  $\frac{\rho R}{\epsilon_0}$                       b)  $\frac{\rho R}{3\epsilon_0}$                       c)  $\frac{5\rho R}{6\epsilon_0}$                       d)  $\frac{\rho}{4\pi\epsilon_0 R}$
144. If an electron is brought towards another electron, then electric potential energy of the system  
 a) increases                      b) decreases                      c) becomes zero                      d) remains the same
145. A potential difference is applied across the ends of a wire. If the potential difference is doubled, then the drift velocity of free electrons will

**Note: NAME students must fill their registration number on the OMR sheet.**

1. Which of the following malarial parasitic stage is found in the stomach wall of the female Anopheles?  
a) Schizont                      b) Trophozoite                      c) Oocyst                      d) Sporozoite
2. All stages in the lifecycle of a malarial parasite are haploid except  
a) Gamont                      b) Ookinete                      c) Schizont                      d) Sporont
3. In the earthworm, the pharyngeal nephridia are situated in the  
a) 3rd, 4th and 5th segment                      b) 4th, 5th and 6th segment  
c) 6th, 7th and 8th segment                      d) 8th, 9th and 10th segment
4. In earthworms, the sperms are stored in the  
a) Testes                      b) Seminal vesicles                      c) Vas deferens                      d) Spermathecae
5. The eyes of which molluscan group resemble Vertebrates eyes?  
a) Bivalvia                      b) Cephalopoda                      c) Gastropoda                      d) Pelecypoda
6. Which is not a typical characteristic of class Nematoda?  
a) Syncytial epidermis                      b) Pseudocoel  
c) Non-living cuticle                      d) Only parasitic animals & plants
7. Hair occurs in all mammals except those belonging to  
a) Rodentia                      b) Chiroptera                      c) Primate                      d) Cetacea
8. A depression or aperture where the digestive, urinary and reproductive ducts open is known as the  
a) Cloaca                      b) Anus                      c) Vestibule                      d) Gonopore
9. Nissl granules are present in  
a) Neutrophils                      b) Axon                      c) Nerve fibre                      d) Cyton
10. Bundle of striated muscle fibres are enclosed in the  
a) Periosteum                      b) Epimysium                      c) Perimysium                      d) Endomysium
11. Histocyte is a connective tissue cell which takes part in  
a) Fibre production                      b) Matrix production                      c) Phagocytosis                      d) Secretion
12. Lamina propria is connected with  
a) Acini                      b) Liver                      c) Graffian follicle                      d) Intestine
13. Which of the following represents class Mastigophora?  
a) Ameoba                      b) Paramecium                      c) Monocystis                      d) trypanosoma
14. Vitamin K is required for the  
a) Change of prothrombin to thrombin                      b) Synthesis of prothrombin  
c) Change of fibrinogen to fibrin                      d) Formation of thromboplastin
15. When the oxygen supply to the tissue is inadequate, the condition is known as  
a) Anoxia                      b) Asphyxia                      c) Hypoxia                      d) Pleurisy
16. If CO<sub>2</sub> concentration is higher, the oxygen dissociation curve of haemoglobin will  
a) Move to the left                      b) Move to the right                      c) Become irregular                      d) Move upward
17. The eustachian valve is a vestige of the semilunar valve

- a) Semilunar valve                      b) Sino-auricular valve  
c) Spiral valve                          d) Sinus venosus
18. Heparin, normally present in blood, is formed by  
a) Granulocytes                      b) Clumped platelets  
c) Mast cell granules                  d) Monocytes
19. The vitamin excreted by urine in higher Vertebrates is  
a) A                      b) D                      c) K                      d) C
20. Filtration fraction is the ratio of  
a) Glomerular filtration rate (GFR) and renal plasma flow (RPF)  
b) Transport maximum (Tm) and clearance factor (Cf)  
c) Hb and HbO<sub>2</sub>  
d) Renal pressure and osmotic pressure
21. Which of the following cranial nerves can regulate heart beat?  
a) X                      b) IX                      c) VIII                      d) VII
22. The parasympathetic nervous system accelerates the activity of the  
a) Heart, adrenal and sweat glands                  b) Lacrymal and sweat glands  
c) Heart, pancreas and lacrymal gland                  d) Gut, iris and urinary bladder
23. The sense of equilibrium is determined by the  
a) Basilar membrane of the cochlea                  b) Tectorial membrane of the ampulla  
c) Sensory crista of the ampulla                  d) Sensory cells of the organ of corti
24. The retina is the most sensitive at the  
a) Optic disc                      b) Periphery                      c) Macula lutea                      d) Fovea centralis
25. Complete failure of the anterior lobe of the pituitary causes  
a) Addison's disease                  b) Acromegaly                      c) Cushing's disease                  d) Simmond's disease
26. Manifestation of masculinity pattern in females due to hormonal effects is known as  
a) Virilism                      b) Masculinity                      c) Castration                      d) Epistaxis
27. The process of release of spermatozoa from Sertoli cells into the cavity of the seminiferous tubule is called  
a) Spermatogenesis                  b) Spermiogenesis                      c) Spermiation                      d) Ejaculation
28. The part of the nephron that is impermeable to water but permeable to electrolyte is  
a) Descending limb of loop of Henle                  b) Ascending limb of loop of Henle  
c) Proximal convoluted tubule (PCT)                  d) Collecting duct
29. Being all mammals, whale, dolphin, bat, monkey and horse have some important common characters, but they also show conspicuous differences. This is due to the phenomenon of  
a) Normalisation                      b) Genetic drift                      c) Convergence                      d) Divergence
30. According to Oparin, the first living organisms were  
a) Autotrophic and aerobic                      b) Heterotrophic and anaerobic  
c) Autotrophic and anaerobic                      d) Heterotrophic and aerobic
31. The fossils of Java man were discovered by

- a) phytophthora infestans                      b) alternaria solani  
c) albugo                                              d) fusarium
61. The number of spores in the sporangium of fern is  
a) 64                                              b) 16                                              c) 32                                              d) 128
62. Compact cone is found in  
a) male cone of pinus                                              b) female cone of pinus  
c) male cone of cycas                                              d) female cone of cycas
63. Which statement best explains why antibiotics generally do not affect viruses?  
a) Viruses are larger than bacteria)                                              b) Viruses lack metabolism.  
c) Viruses contain DNA only                                              d) Viruses possess cell walls.
64. An organism was discovered with the following features:  
a) Circular DNA                                              b) 70S ribosomes  
c) Cell wall absent                                              d) Reproduces by binary fission
65. Which of the following combinations is impossible?  
a) Green alga - starch - chlorophyll b                                              b) Brown alga - laminarin - chlorophyll c  
c) Red alga - floridean starch - chlorophyll d                                              d) Brown alga - floridean starch - chlorophyll d
66. Edible part of banana is  
a) Epicarp                                              b) Epicarp and mesocarp  
c) endocarp and less developed mesocarp                                              d) mesocarp and less developed endocarp
67. Tulip belongs to family  
a) Liliaceae                                              b) Cruciferae                                              c) gramineae                                              d) solanaceae
68. A farmer wants to control mosquito larvae using a fungal biopesticide. Which organism is most suitable?  
a) Rhizobium                                              b) Metarhizium anisopliae  
c) Azotobacter                                              d) Nostoc
69. Pyrethrin is obtained from  
a) Catharanthus                                              b) Chrysanthemum                                              c) Azadirachta                                              d) amaranthus
70. The term humulin refers to  
a) hydrolytic insulin                                              b) powerful insulin                                              c) human insulin                                              d) humerus insulin
71. Vessels differ from tracheid in  
a) having lignified wall                                              b) higher water conducting efficiency  
c) thicker wall than tracheid                                              d) spindle shaped cells
72. Which of the following is not characteristics of meristem  
a) presence of large central vacuole                                              b) lacking reserve food material  
c) continuous state of division                                              d) presence of prominent nucleus
73. Epidermal hair in stem are  
a) unicellular                                              b) multicellular  
c) multinucleated                                              d) helps in absorption of water
74. Function of germ pore is

# NAME

SET  
B

NAME Institute

In front of Singhdurbar, Putalisadak

Tel: 01-5971544

## MECEE BL

### Model Entrance Exam

#### (Model: VII)

Date: 13 Aashadha 2083, Saturday

Venue: NAME

Duration - 3 hrs

Time - 11:00 AM

#### INSTRUCTIONS

There are 200 multiple-choice questions, each having four choices of which only one choice is correct.

Fill ☐ the most appropriate one.



- a) Lies Zies      b) E. Dubious      c) Lamarck      d) W.C. Prie
32. When an organ is transplanted and it is rejected by the body, the lymphocytes produced are  
a) T-cells      b) B-cells      c) Neutrophils      d) Eosinophils
33. Which microorganism is primarily used in bread and alcoholic beverage production?  
a) *Penicillium notatum*      b) *Saccharomyces cerevisiae*  
c) *Lactobacillus acidophilus*      d) *Aspergillus niger*
34. Learning by trial and error is called:  
a) Insight learning      b) Imprinting  
c) Operant conditioning      d) Habituation
35. The main objective of the Ramsar Convention is:  
a) Wildlife trade regulation      b) Wetland conservation and wise use  
c) Forest management      d) Desert reclamation
36. Which vaccine is prepared from a toxoid?  
a) Measles vaccine      b) BCG vaccine      c) Tetanus vaccine      d) OPV
37. Natural Killer cells (NK) are part of  
a) Acquired immunity      b) Humoral immunity  
c) Innate immunity      d) Artificial immunity
38. Which cells are primarily infected by influenza virus?  
a) Hepatocytes      b) Respiratory epithelial cells  
c) Neurons      d) Osteocytes
39. Which of the following is NOT a typical symptom of cholera?  
a) Profuse watery diarrhea      b) Severe dehydration  
c) Vomiting      d) Productive cough
40. Arterial system of frog begins with  
a) Carotid arch      b) Systemic arch  
c) Pulmocutaneous arch      d) Truncus arteriosus
41. Which element helps in the transport of electrons from water in photosynthesis?  
a) Mo & Cl      b) Na & K      c) Mn & Cl      d) Mn & Mo
42. Which enzyme catalyzes the conversion of citrate to isocitrate in the Krebs cycle?  
a) Citrate synthase      b) Aconitase  
c) Succinate dehydrogenase      d) Malate dehydrogenase
43. Dark reaction is called so because it:  
a) Occurs only at night      b) Does not directly require light  
c) Occurs in darkness only      d) Produces no ATP
44. The splitting of rocks by plant roots is mainly due to:  
a) Diffusion pressure      b) Imbibition pressure      c) Root respiration      d) Guttation
45. Transpiration differs from guttation because the water lost during transpiration is:  
a) Rich in minerals      b) Rich in sugars  
c) Nearly pure water vapour      d) Xylem sap

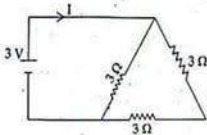
- a) emergence of radicle  
c) initiation of pollen tube
- b) absorption of water for seed germination  
d) release of male gametes
75. The type of ovule present in ranunculaceae family is  
a) hemitropous      b) orthotropous      c) anatropous      d) circinotropous
76. Viviparous seed germination is found in  
a) leguminous plant      b) mangroves plant      c) carnivorous plant      d) parasitic plant
77. Man made ecosystem is  
a) less in diversity  
c) man does not make ecosystem
- b) more in diversity  
d) more stable than natural ecosystem
78. A rat feeding upon potato tuber is  
a) producer      b) carnivore      c) decomposer      d) primary consumer
79. Which of the following is an example of ex-situ conservation?  
a) national park      b) wildlife reserves  
c) zoos and botanical gardens      d) protected area
80. The living organism of all ecosystem collectively constitute  
a) producer      b) decomposer      c) consumer      d) biosphere
81. How many moles of potassium chlorate should be decomposed completely to obtain 67.2 litres of oxygen at STP?  
a) 1      b) 2      c) 3      d) 4
82. Which has maximum number of molecules?  
a) 1.7 g  $\text{NH}_3$       b) 2 g He      c) 4.6 g  $\text{NO}_2$       d) 3.2 g  $\text{SO}_2$
83. The normality of  $\text{H}_2\text{SO}_4$  having 50 milliequivalent in 2 L solution is:  
a) 1.025      b) 1.25      c) 0.050      d) 0.025
84. Rutherford's scattering experiment is related to the size of the:  
a) Atom      b) Nucleus      c) Neutron      d) Electron
85. Which one of the following molecules has the smallest bond angle?  
a)  $\text{H}_2\text{O}$       b)  $\text{NH}_3$       c)  $\text{H}_2\text{Se}$       d)  $\text{H}_2\text{S}$
86. If increase in temperature and volume of an ideal gas is two times, then the initial pressure P changes to:  
a) 4P      b) 2P      c) P      d) 3P
87. The net number of atoms (n) contained within a body centered cubic cell is:  
a) 4      b) 3      c) 2      d) 1
88. When an electric current is passed through acidulated water, 112 mL of hydrogen gas at NTP collects at the cathode in 965 seconds. The current passed, in ampere is:  
a) 0.1      c) 0.5      b) 1.0      d) 2.0
89. For a chemical reaction, which can never be a fractional number?  
a) Order      b) Half-life      c) Molecularity      d) Rate constant
90. In which of the following reactions, equilibrium is independent of pressure:  
a)  $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}); \Delta H = +ve$   
c)  $3\text{H}_2(\text{g}) + \text{N}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}); \Delta H = -ve$   
b)  $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3(\text{g}); \Delta H = -ve$   
d)  $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}); \Delta H = +ve$

91. For the un-balanced chemical reaction given below that occurs in an automobile engine,  $\text{C}_8\text{H}_{18} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$  → The sign of  $\Delta H$ ,  $\Delta S$ , and  $\Delta G$  for the reaction are?  
a) +, -, +                      b) -, +, +                      c) -, -, -                      d) -, +, -
92. The number of  $\text{H}^+$  ions present in  $1\text{cm}^3$  of a solution whose pH is 10 is:  
a)  $10^{-10}$                       b)  $10^{-13}$                       c)  $6.02 \times 10^{10}$                       d)  $6.02 \times 10^{13}$
93. Which of the following is correct?  
a) The pH of one litre solution containing 0.49 g of  $\text{H}_2\text{SO}_4$  is 2.0.  
b) The conjugate base of  $\text{H}_2\text{S}$  is  $\text{S}^{2-}$   
c)  $\text{BF}_3$  is Lewis base.  
d) Phenolphthalein is colourless in basic medium.
94. Which of the following is a redox reaction?  
a)  $\text{NaCl} + \text{KNO}_3 \rightarrow \text{NaNO}_3 + \text{KCl}$                       b)  $\text{CaC}_2\text{O}_4 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{C}_2\text{O}_4$   
c)  $\text{Mg}(\text{OH})_2 + 2\text{NH}_4\text{Cl} \rightarrow \text{MgCl}_2 + \text{NH}_4\text{OH}$                       d)  $\text{Zn} + 2\text{AgCN} \rightarrow 2\text{Ag} + \text{Zn}(\text{CN})_2$
95. 4 g caustic soda is dissolved in 100 cc of solution. The normality of solution is:  
a) 1                      b) 0.8                      c) 0.6                      d) 0.10
96. "Little Boy" was the first nuclear weapon ever used in combat, a fission bomb dropped on Hiroshima in 1945. The radioactive isotope used in the bomb was  
a) U-235                      b) U-239                      c) Pu-238                      d) Pu-239
97. The atomic radius decreases in a period due to  
a) Increase in nuclear attraction                      b) Decrease in nuclear attraction  
c) Increase in number of electrons                      d) Decrease in number of electrons
98. The H - O - H angle in water molecule is about:  
a)  $90^\circ$                       b)  $105^\circ$                       c)  $135^\circ$                       d)  $180^\circ$
99. Identify the Freon from the following:  
a)  $\text{CHCl}_3$                       b)  $\text{C}_2\text{H}_5\text{Cl}$                       c)  $\text{CF}_2\text{Cl}_2$                       d)  $\text{CH}_2\text{Cl}_2$
100. The oxidation number of N in nitrous acid is  
a) +1                      b) +3                      c) +5                      d) +7
101. Halide of Ag most sensitive for the light is:  
a)  $\text{AgBr}$                       b)  $\text{AgCr}$                       c)  $\text{AgCl}$                       d)  $\text{AgI}$
102. Metal which cannot form amalgam is:  
a) Hg                      b) Cu                      c) Fe                      d) Zn
103. Reduction of zinc oxide with economical way can be carried out by which of the following method:  
a) Continuous vertical retort process                      b) blast furnace  
c) Shaft furnace                      d) Muffle furnace
104. Alkaline earth metal present in hydroxyapatite present in living organism is:  
a) K                      b) Ca                      c) Na                      d) Rb
105. Which of the following is used in match box in the match industry?  
a) White or Yellow Phosphorous                      b) Red Phosphorous

121. Dehydration of carboxylic acids gives ———— derivate of the acid.  
a) Amide                      b) Ester                      c) Acid anhydride                      d) Acid chloride
122. When nitrobenzene is reduced at electrolytic condition in presence of strong electrolyte, it gives ————  
a) o-hydroxy aniline                      b) p-amino phenol  
c) phenyl amine                      d) N hydroxyl amine
123. Diethyl oxalate is a reagent used to:  
a) Separation of 1°, 2° and 3° alcohol                      b) Preparation of 1°, 2° and 3° amines  
c) Preparation of 1°, 2° and 3° amines                      d) Separation of 1°, 2° and 3° amines
124. Zeise's salt, is a pioneering organometallic compound. The chemical formula of this salt is:  
a)  $\text{Na}[\text{PtCl}_3(\text{C}_2\text{H}_4)] \cdot \text{H}_2\text{O}$                       b)  $\text{K}[\text{PtCl}_3(\text{C}_2\text{H}_4)] \cdot \text{H}_2\text{O}$   
c)  $\text{K}[\text{PtCl}_3(\text{C}_6\text{H}_6)] \cdot \text{H}_2\text{O}$                       d)  $\text{RMgX}$
125. In esterification, alcohols are converted into esters. In this reaction conc.  $\text{H}_2\text{SO}_4$  acts as,  
a) Positive catalyst                      b) Negative catalyst                      c) Dehydrating agent                      d) Smell enhancing agent
126. Catalytic oxidation of ethylene in presence of silver catalyst gives  
a) Glycol                      b) Acetaldehyde                      c) Acetic acid                      d) Epoxy-ethane
127. Kharasch effect follows free radical mechanism and occurs only in case of .....  
a)  $\text{HI}$                       b)  $\text{HCl}$                       c)  $\text{HBr}$                       d)  $\text{HF}$
128. The time taken to convert a company's investments into cash is called  
a) Entrepreneurship period                      b) Company's investment period  
c) Cash flow cycle                      d) Trail period
129. The fertilizer which consists of all three macronutrients is called .....fertilizer.  
a) NaPK fertilizer                      b) CaPK fertilizer                      c) IPK fertilizer                      d) NPK fertilizer
130. ——— is used to soften water by removing calcium and magnesium ions, which cause hardness.  
a) Phosphine                      b) Caustic soda                      c) Washing soda                      d) Baking soda
131. The escape velocity of 10 g body from the Earth is 11.2 km/s, ignoring air resistance. The escape velocity of 10 kg of the iron ball from the Earth will be  
a) 0.0112 km/s                      b) 0.112 km/s                      c) 11.2 km/s                      d) 0.56 km/s
132. A Couple produces  
a) purely translational motion                      b) purely rotational motion  
c) both translational and rotational motion                      d) no motion
133. A body is initially at rest. It undergoes one dimensional motion with constant acceleration. The power delivered to it at time  $t$  is proportional to  
a)  $t^{\frac{1}{2}}$                       b)  $t$                       c)  $t^{\frac{3}{2}}$                       d)  $t^2$
134. A body starting from rest moves along a straight-line with a constant acceleration. The variation of speed ( $v$ ) with distance ( $s$ ) is represented by the graph

- a) be halved      b) be doubled      c) be quadrupled      d) remain unchanged

146. A 3 volt battery with negligible internal resistance is connected in a circuit as shown in the figure. The current  $I$  in the circuit will be



- a)  $1/3$  A      b) 1 A      c) 1.5 A      d) 2 A

147. A 100W, 200V bulb is connected to a 160 volts supply. The actual power consumption would be

- a) 64 W      b) 80 W      c) 100 W      d) 125 W

148. Kirchhoff's first law deals with the conservation of

- a) charge      b) energy      c) momentum      d) angular momentum

149. Magnetic field intensity in the centre of coil of 40 turns, 0.5 m radius and carrying a current of 2A is

- a)  $0.5 \times 10^{-5}$  T      b)  $1 \times 10^{-4}$  T      c)  $2 \times 10^{-5}$  T      d)  $4 \times 10^{-5}$  T

150. A galvanometer having resistance of  $8\Omega$  is shunted by a wire of resistance  $2\Omega$ . If the total current is 1A, then current passing through the shunt will be

- a) 1.2 A      b) 0.8 A      c) 0.5 A      d) 0.3 A

151. The magnetic susceptibility of a material of a rod is 499. The absolute permeability of vacuum is  $4\pi \times 10^{-7}$  H/m. The absolute permeability of the material of rod is

- a)  $\pi \times 10^{-4}$  H/m      b)  $2\pi \times 10^{-4}$  H/m      c)  $3\pi \times 10^{-4}$  H/m      d)  $4\pi \times 10^{-4}$  H/m

152. The current in a coil of inductance 5H decreases at the rate of 2 A/s. The induced emf is

- a) 2V      b) 5V      c) 10V      d) -10V

153. A capacitor is a perfect insulator for

- a) alternating currents      b) direct current  
c) both AC and DC      d) variable current

154. If a thermometer reads freezing point of water as  $20^\circ\text{C}$  and boiling point as  $150^\circ\text{C}$ , how much thermometer read when the actual temperature is  $60^\circ\text{C}$ ?

- a)  $98^\circ\text{C}$       b)  $110^\circ\text{C}$       c)  $40^\circ\text{C}$       d)  $60^\circ\text{C}$

155. When a bimetallic strip is heated, it

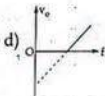
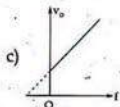
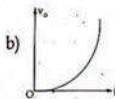
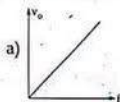
- a) does not bend at all  
b) gets twisted in the form of a helix  
c) bends in the form of an arc with the more expandable metal outside  
d) bends in the form of an arc with the more expandable metal inside

156. The coefficient of superficial expansion of a solid is  $2 \times 10^{-5}/^\circ\text{C}$ . Its coefficient of linear expansion is

- a)  $4 \times 10^{-5}/^\circ\text{C}$       b)  $3 \times 10^{-5}/^\circ\text{C}$       c)  $2 \times 10^{-5}/^\circ\text{C}$       d)  $1 \times 10^{-5}/^\circ\text{C}$

157. At atmospheric pressure, the water boils at  $100^\circ\text{C}$ . If pressure is reduced, it will boil at

- a) higher temperature                      b) lower temperature  
c) at the same temperature                d) at critical temperature
158. A gas is allowed to expand isothermally. The root mean square velocity of the molecules  
a) will increase                                b) will decrease  
c) will remain unchanged                    d) depends on the other factors
159. For a gas  $\frac{R}{C_v} = 0.67$ . This gas is made up of molecules which are  
a) diatomic                                      b) mixture of diatomic and polyatomic molecules  
c) monoatomic                                 d) polyatomic
160. Woolen clothes are used in winter season because woolen clothes  
a) are good sources for producing heat    b) absorb heat from surroundings  
c) are bad conductors of heat                d) provide heat to body continuously
161. If the equation of transverse wave is  $y = 2\sin(kx - 2t)$ , then the maximum particle velocity is  
a) 4 units                                        b) 2 units                                        c) 0                                                d) 6 units
162. The tension in a piano wire is 10N. What should be the tension in the wire to produce a note of double the frequency?  
a) 5N                                              b) 20N                                              c) 40N                                              d) 80N
163. The walls of the halls built for music concerts should  
a) amplify sound                                b) transmit sound                                c) reflect sound                                d) absorb sound
164. The field of view is maximum for  
a) plane mirror                                 b) concave mirror                                c) convex mirror                                d) cylindrical mirror
165. A vessel of depth  $2d$  cm is half filled with a liquid of refractive index  $\mu_1$  and the upper half with a liquid of refractive index  $\mu_2$ . The apparent depth of the vessel seen perpendicularly is  
a)  $d\left(\frac{\mu_1\mu_2}{\mu_1+\mu_2}\right)$                                 b)  $d\left(\frac{1}{\mu_1} + \frac{1}{\mu_2}\right)$                                 c)  $2d\left(\frac{1}{\mu_1} + \frac{1}{\mu_2}\right)$                                 d)  $2d\left(\frac{1}{\mu_1\mu_2}\right)$
166. When white light enters a prism, it gets split into its constituent colours. This is due to  
a) high density of prism material            b) because  $\mu$  is different for different  $\lambda$   
c) diffraction of light                            d) velocity changes for different frequencies
167. For the sustained interference of light, the necessary condition is that the two sources should  
a) have constant phase difference            b) be narrow  
c) be close to each other                        d) of same amplitude
168. A diffraction grating has 100 lines/mm is illuminated normally by a monochromatic light of wavelength 6000 Å. The maximum number of diffraction maxima obtained will be  
a) 4.16                                              b) 4                                                      c) 5                                                      d) 2.96
169. A proton and  $\alpha$ -particle are injected into uniform magnetic field at right angle to the field with equal speeds. Then ratio of radii of their circular paths will be  
a) 1:1                                                b) 2:1                                                c) 1:2                                                d)  $\sqrt{2}$ :1
170. In photo electric cell, the relation between cut-off voltage ( $V_0$ ) and frequency ( $f$ ) of incident light is best represented by



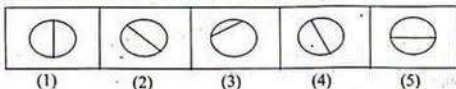
171. Which of the following one is correct?  
 a)  $E^2 = P^2c$       b)  $E^2 = P^2c^2$       c)  $E^2 = Pc^2$       d)  $E^2 = P^2/c^2$
172. On increasing voltage in the reverse bias to large value in a p-n junction diode, the current  
 a) increases slowly    b) remains fixed    c) suddenly increases    d) decreases slowly
173. In a semiconductor p-side is earthed and n-side is applied a potential of  $-2V$ . Then the diode will  
 a) conduct      b) not conduct      c) conduct partially    d) breakdown
174. Which gate is an inverter?  
 a) NOT gate      b) OR gate      c) NAND gate      d) AND gate
175. The ratio of areas within the electron orbits for the first excited state to the ground state for the hydrogen atom is  
 a) 2:1      b) 4:1      c) 8:1      d) 16:1
176. A de-Broglie wave is  
 a) an elastic wave  
 b) an electromagnetic wave  
 c) a matter wave  
 d) both an elastic wave and an electromagnetic wave
177. The wavelength of most energetic x-ray emitted when it is bombarded by 40 kV electron is  
 a) 300 Å      b) 10 Å      c) 4 Å      d) 0.31 Å
178. The half life of a radio isotope is 5 years. What fraction of it will remain active after 20 years?  
 a)  $\frac{3}{4}$       b)  $\frac{1}{4}$       c)  $\frac{1}{8}$       d)  $\frac{1}{16}$
179. In a proton, the number of up quarks is  
 a) 1      b) 2      c) 3      d) 4
180. The mass defect for the nucleus of helium is 0.0303 amu. What is the binding energy per nucleon for helium in MeV?  
 a) 28      b) 7      c) 4      d) 1
181. Arrange the words given below in a meaningful sequence  
 1. Protect      2. Pressure      3. Relief      4. Rain      5. Flood  
 a) 2, 4, 3, 1, 5      b) 2, 4, 5, 1, 3      c) 2, 5, 4, 1, 3      d) 3, 2, 4, 5, 1
182. Statements: Some actors are singers. All the singers are dancers.  
 Conclusions:

1. Some actors are dancers  
2. No singer is actor
- a) only (1) conclusion follows      b) only (2) conclusion follows  
c) either (1) or (2) follows      d) both (1) & (2) follows
183. Flow : River :: Stagnant : ?  
a) Rain      b) Stream      c) Pool      d) Canal
184. Complete the series: 120, 99, 80, 63, 48, ?  
a) 35      b) 38      c) 39      d) 40
185. Introducing a boy, a girl said, "He is son of the daughter of the father of my uncle." How is the boy related to the girl?  
a) Brother      b) Nephew      c) Uncle      d) Son-in-law
186. A train running at the speed of 60km/hr crosses a pole in 9 seconds. What is the length of the train?  
a) 120 m      b) 180 m      c) 324 m      d) 150 m
187. Some articles were bought at 6 articles for Rs. 5 and sold at 5 articles for Rs. 6. Gain percent is  
a) 30%      b)  $33\frac{1}{3}\%$       c) 35%      d) 44%
188. How many times are the hands of a clock at right angle in a day?  
a) 22      b) 24      c) 44      d) 48
189. An error 2% in excess is made while measuring the side of a square. The percentage at error in the calculated area of the square is  
a) 2%      b) 2.02%      c) 4%      d) 4.04%
190. What is the probability of getting a sum of 9 from two throws of a dice?  
a)  $\frac{1}{6}$       b)  $\frac{1}{8}$       c)  $\frac{1}{9}$       d)  $\frac{1}{12}$
191. Which word does not belong with the others?  
a) Parsley      b) Basil      c) Dill      d) Mayonnaise
192. Odometer is to mileage as compass is to  
a) speed      b) hiking      c) needle      d) direction
193. SCD, TEF, UGH \_\_\_\_\_, WKL  
a) CMN      b) UJI      c) VIJ      d) IJT
194. Statements: No women teacher can play, some women teacher are athletes.  
Conclusions:  
(1) male athletes can play  
(2) some athletes can play  
a) only conclusion (1) follows      b) only conclusion (2) follows  
c) either (1) or (2) follows      d) neither (1) nor (2) follows
195. A pineapple costs Rs. 7 each. A watermelon costs Rs. 5 each. X spends Rs. 38 on these fruits.

The number of pineapple purchased is

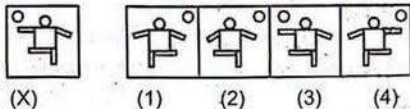
- a) 2                      b) 3                      c) 4                      d) data inadequate

196. Chose the figure which is different from the rest.



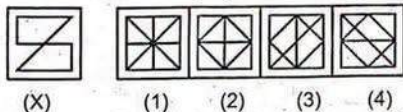
- a) 1                      b) 2                      c) 3                      d) 4

197. Choose the correct mirror image of the given figure (X) from amongst the four alternatives.



- a) 1                      b) 2                      c) 3                      d) 4

198. Find out the alternative figure which contains figure (X) as its part.



- a) 1                      b) 2                      c) 3                      d) 4

199. Complete the given analogy.

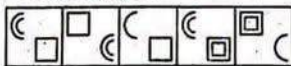
Problem Figures:



- (A) (B) (C) (D)

- a) 1                      b) 2

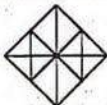
Answer Figures:



- (1) (2) (3) (4) (5)

- c) 4                      d) 5

200. Find the number of triangles in the given figure.



- a) 16                      b) 22                      c) 28                      d) 32

Result will be published on Sunday

Log on to [www.name.edu.np](http://www.name.edu.np)

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Best of Luck

**NAME**

# HINTS AND SOLUTIONS

**SET  
B**

## **MECEE BL** **Model Entrance Exam** **(Model- VII)**

**Date: 13 Aashadha 2083, Saturday**

**NAME Institute**

**In front of Singhdurbar, Putalisadak**

**Tel: - 01-5971544**

Zoology

1. c) Oocyst - After fertilization in the mosquito's gut, the zygote becomes a motile ookinete, which penetrates the stomach wall and develops into a spherical encysted stage called the oocyst on the outer surface of the stomach.
2. b) Ookinete - The ookinete is the motile, elongated zygote formed by the fusion of male and female gametes in the mosquito's gut, making it the only diploid stage ( $2n$ ) in the life cycle; all other major stages are haploid ( $n$ ).
3. b) 4th, 5th and 6th segment - In earthworms (*Pheretima*), pharyngeal nephridia are arranged in three paired tufts located around the anterior digestive tract specifically in segments 4, 5, and 6.
4. d) Spermathecae - While earthworms produce sperms in their testes and mature them in seminal vesicles, during copulation, they receive and temporarily store the partner's sperms inside the spermathecae until cocoon formation.
5. b) Cephalopoda - Cephalopods (like squids and octopuses) possess highly advanced, large, camera-type eyes that show remarkable convergent evolution with vertebrate eyes, featuring a cornea, lens, iris, and retina.
6. d) Only parasitic animals & plants - While class Nematoda contains notorious parasites, it also includes hundreds of thousands of diverse, free-living species that thrive in soil, freshwater, and marine environments.
7. d) Cetacea - Cetaceans (such as whales and dolphins) are marine mammals that have completely lost their adult hair coat as an evolutionary adaptation to reduce hydrodynamic drag and swimming friction.
8. a) Cloaca - The cloaca is a common terminal chamber found in many vertebrates (like amphibians, reptiles, and birds) into which the alimentary canal, urinary ducts, and reproductive tubes empty.
9. d) Cyton - Nissl granules are rich aggregations of rough endoplasmic reticulum and free ribosomes found inside the cyton (cell body) and dendrites of a neuron, but they are completely absent in the axon.
10. c) Perimysium - A skeletal muscle is organized in layers: the entire muscle is wrapped by epimysium, bundles of fibers (fascicles) are enclosed by perimysium, and individual muscle fibers are covered by endomysium.
11. c) Phagocytosis - Histocytes are stationary, specialized tissue macrophages found in connective tissues that act as immune defenders by engulfing cellular debris and foreign pathogens via phagocytosis.
12. d) Intestine - The lamina propria is a thin, supportive layer of loose connective tissue that lies directly beneath the delicate mucosal epithelium of the gastrointestinal tract, most prominently in the intestines.
13. d)
14. b) Synthesis of prothrombin - Vitamin K serves as an essential cofactor in the liver for the post-translational carboxylation and functional synthesis of blood clotting factors, including prothrombin (Factor II).
15. c) Hypoxia - Hypoxia is the physiological state where tissues or cells face an inadequate supply of oxygen, whereas anoxia refers to a total lack of oxygen, and asphyxia implies suffocation from lack of oxygen and  $\text{CO}_2$  buildup.

16. b) Move to the right - An increase in  $\text{CO}_2$  concentration lowers blood pH and decreases hemoglobin's affinity for oxygen, shifting the oxygen dissociation curve to the right to unload oxygen to active tissues (Bohr Effect).
17. b) Sino-auricular valve - The Eustachian valve (valve of the inferior vena cava) in mammalian hearts develops as a functional embryonic remnant of the right valve of the sino-auricular orifice.
18. c) Mast cell granules - Heparin, a powerful natural anticoagulant that keeps blood fluid inside intact vessels, is synthesized and released primarily by the granular structures of circulating basophils and tissue mast cells.
19. d) C - Vitamin C (ascorbic acid) is a highly water-soluble vitamin; because higher vertebrates cannot store excess water-soluble vitamins in fat tissues, any surplus amounts are rapidly excreted out via urine.
20. a) Glomerular filtration rate (GFR) and renal plasma flow (RPF) - The filtration fraction represents the exact ratio of the volume of plasma filtered into Bowman's capsule (GFR) relative to the total plasma passing through the kidneys (RPF), typically averaging around 20%.
21. a) X - The Tenth Cranial Nerve (Vagus nerve) carries parasympathetic motor fibers directly to the heart's sinoatrial (SA) node, releasing acetylcholine to slow down the cardiac rate and regulate heartbeat.
22. d) Gut, iris and urinary bladder - The parasympathetic nervous system governs "rest and digest" responses, which accelerates gastrointestinal gut motility, constricts the pupil iris, and contracts the bladder wall for urination.
23. c) Sensory crista of the ampulla - Dynamic physical equilibrium and rotational balance are determined by the sensory crista ampullaris located inside the semicircular canals, which detect the movement of endolymph fluid.
24. d) Fovea centralis - Located at the center of the macula lutea, the fovea centralis contains an exceptionally high density of closely packed cone cells with no rods, making it the point of maximum visual acuity and sharpest focus.
25. d) Simmond's disease - Panhypopituitarism, which is the complete clinical failure and destruction of the anterior lobe of the pituitary gland, results in tissue wasting and premature aging known as Simmond's disease.
26. a) Virilism - Adrenal virilism is a clinical condition where an overproduction of androgenic steroids from the adrenal cortex triggers the manifestation of secondary male physical characteristics (masculinity) in females.
27. c) Spermiogenesis - Spermiogenesis is the specific terminal process where mature spermatozoa break their physical attachments and are released from the supportive Sertoli cells into the central lumen of the seminiferous tubules.
28. b) Ascending limb of loop of Henle - The ascending limb of Henle's loop features a thick segment that is strictly impermeable to water molecules but actively pumps out electrolytes (sodium and chloride ions) into the medullary interstitium.
29. d) Divergence - Driven by natural selection, adaptive radiation or divergent evolution occurs when closely related mammalian species sharing a common ancestor adapt to entirely different environments (air, sea, land), developing unique structural differences.

30. b) Heterotrophic and anaerobic - Oparin's heterotrophic hypothesis states that the primordial Earth lacked free atmospheric oxygen and was rich in organic molecules, meaning the earliest life forms had to be anaerobic heterotrophs.
31. b) E. Dubious - The famous fossil remains of Java Man (*Pithecanthropus erectus*, now *Homo erectus*) were discovered in Trinil, Java, in 1891 by the Dutch paleoanthropologist Eugène Dubois.
32. a) T-cells - Acute allograft organ transplant rejection is primarily mediated by Cell-Mediated Immunity (CMI), where host T-lymphocytes (specifically cytotoxic T-cells) recognize foreign MHC molecules and attack the donor tissue.
33. b) *Saccharomyces cerevisiae* - Commonly known as brewer's or baker's yeast, *Saccharomyces cerevisiae* is a single-celled fungus widely used for industrial fermentation, breaking down sugars into ethanol and  $\text{CO}_2$ .
34. c) Operant conditioning - First detailed by B.F. Skinner, operant conditioning is a learning mechanism where an organism modifies its voluntary behavior based on a trial-and-error process associated with positive rewards or punishments.
35. b) Wetland conservation and wise use - Signed in 1971, the Ramsar Convention is an international intergovernmental treaty dedicated strictly to providing a framework for the global conservation, protection, and sustainable wise use of wetlands.
36. c) Tetanus vaccine - A toxoid is a chemically modified, non-toxic bacterial toxin that remains antigenic; the Tetanus vaccine (along with Diphtheria) utilizes an inactivated toxin to stimulate protective antibody immunity.
37. c) Innate immunity - Natural Killer (NK) cells are specialized granular lymphocytes that destroy virus-infected and cancerous cells non-specifically without prior exposure, forming a critical component of innate immunity.
38. b) Respiratory epithelial cells - The influenza virus uses its surface hemagglutinin spikes to attach specifically to sialic acid receptors found on the outer surface of ciliated epithelial cells lining the host's respiratory tract.
39. d) Productive cough - Cholera is an acute intestinal infection caused by *Vibrio cholerae*; its classic hallmark symptoms include massive rice-water stool diarrhea, extreme dehydration, and vomiting, whereas a productive cough is a respiratory symptom.
40. d) Truncus arteriosus - The arterial system of a frog originates directly from the truncus arteriosus, a thick muscular vessel arising from the ventricle that splits into two branches, each dividing into carotid, systemic, and pulmocutaneous arches.

### Botany

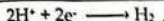
41. c)
42. b) Aconitase catalyzes the reversible isomerization of citrate to isocitrate through cis-aconitate in the Krebs cycle.
43. b) The dark reaction does not directly require light. It utilizes ATP and NADPH generated during the light reaction for carbon fixation.
44. b) Water absorption by root cell walls causes imbibition pressure, which can exert sufficient force to widen cracks and split rocks.
45. c) Transpiration involves loss of water in the form of nearly pure water vapour, whereas guttation releases liquid water containing dissolved minerals.

46. b) Removal of the apical bud eliminates auxin-mediated apical dominance, allowing lateral buds to grow and form a bushy hedge.
47. c) Chromosomes are chemically composed of DNA associated with histone and non-histone proteins.
48. c) Transfer RNA (tRNA) is also known as soluble RNA (sRNA) because it remains dissolved in the cytoplasm and transports amino acids to ribosomes.
49. b) Monosomy is represented as  $2n - 1$ . Therefore,  $24 - 1 = 23$  chromosomes.
50. b) Color blindness is inherited as an X-linked recessive trait and is expressed more commonly in males.
51. b) The number of gamete types equals  $2^n$ , where  $n$  is the number of heterozygous gene pairs. Here  $n = 3$  (Aa, Bb, Dd), so  $2^3 = 8$ .
52. d) Homologous chromosomes have similar size, shape and centromere position but may carry different alleles of the same gene.
53. c) At metaphase of both mitosis and meiosis, each chromosome consists of two sister chromatids joined by a centromere.
54. c) Prophase involves chromosome condensation and disappearance of the nuclear membrane, whereas telophase involves chromosome decondensation and reformation of the nuclear membrane.
55. a) Mitochondria contain approximately 70% of the total enzymes present in the cell because they are the major sites of aerobic respiration.
56. c) A nucleoid is a DNA-containing region found in prokaryotes, whereas nucleolus, nucleoplasm and nucleosomes occur within eukaryotic nuclei.
57. b) Mesosomes are associated with respiration in prokaryotic cells. Mesosome are formed by inward infolding of cell membrane.
58. b) Amino acids contain both an acidic carboxyl group ( $-\text{COOH}$ ) and a basic amino group ( $-\text{NH}_2$ ), making them amphoteric.
59. d) Both starch and maltose yield only glucose molecules upon complete hydrolysis.
60. b) Early blight of potato is caused by *Alternaria solani*. *Phytophthora infestans* causes late blight.
61. a) A typical mature leptosporangiate fern sporangium contains 64 haploid spores.
62. b) The female cone of *Pinus* is compact, woody and persistent, whereas male cones are smaller and less compact.
63. b) Viruses lack their own metabolic and protein-synthesizing machinery. Therefore, antibiotics that target bacterial metabolism, ribosomes or cell walls are ineffective.
64. b) Mycoplasma lacks a cell wall but possesses circular DNA, 70S ribosomes and reproduces by binary fission.
65. d) Floridean starch and chlorophyll d are characteristic of red algae, not brown algae. Therefore this combination is impossible.
66. d) In banana, the edible portion mainly consists of mesocarp and a less-developed endocarp.
67. a) Tulip belongs to the family Liliaceae. Important members of the family Liliaceae include *Lilium* (lily), *Tulipa* (tulip), *Aloe vera*, *Gloriosa*, *Asparagus*, and *Colchicum*.
68. b) *Metarhizium anisopliae* is an entomopathogenic fungus widely used as a fungal biopesticide.

69. b) Pyrethrin is a natural insecticide extracted from *Chrysanthemum* flowers..
70. c) Humulin is genetically engineered human insulin produced through recombinant DNA technology.
71. b) Vessels provide more efficient water conduction because they form continuous tubes with perforated end walls.
72. a) Meristematic cells are actively dividing and therefore possess dense cytoplasm, a prominent nucleus and little or no vacuolation.
73. b) Stem epidermal hairs (trichomes) are generally multicellular and help reduce water loss and herbivory.
74. c) The germ pore serves as the site from which the pollen tube emerges during pollen germination.
75. a) Hemitropus ovules occur in Ranunculaceae. In this ovule body of ovule remains perpendicular to funicle. It is also called half inverted ovule.
76. b) Vivipary is the phenomenon in which seeds germinate while attached to the parent plant. It is characteristics of mangroove plants.
77. a) Artificial ecosystems such as crop fields usually possess lower species diversity and stability than natural ecosystems.
78. d) A rat feeding directly on potato tubers occupies the primary consumer level in the food chain.
79. c) Zoos and botanical gardens conserve species outside their natural habitats and are examples of ex-situ conservation.
80. d) The biosphere represents the sum total of all living organisms and the regions of Earth inhabited by them.

### Chemistry

81. b)  $2\text{KClO}_3 \longrightarrow 2\text{KCl} + 3\text{O}_2$   
2 moles of  $\text{KClO}_3$  gives  $3 \times 22.4$ , that is, 67.2 L of  $\text{O}_2$ .
82. b) 2 g He has maximum molecules as 'n' (mole) is maximum for it.
83. d)  $N = \frac{\text{no. of gm eq}}{v_l} = \frac{50 \times 10^{-3}}{2} = 0.025N$
84. b) The central part consisting whole of the positive charge and most of the mass, called nucleus, is extremely small in size compared to the size of the atom.
85. b) Here-bond angle is as follows:  $\text{NH}_3 > \text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se}$
86. c)  $\frac{PV}{T} = \frac{P'V'}{T'}$   
Here  $V' = 2V$ ,  $T' = 2T$   
 $\frac{PV}{T} = \frac{P' \times 2V}{2T}$  or  $P = P'$   
that is, pressure remains same.
87. c) The body-centered cubic cell consists of 8 atoms at the corners and one atom at the centre.  
So,  $n = (8 \times 1/8) + 1 = 2$
88. c) 22400 mL of hydrogen at STP (or NTP) = 2 g  
So, 112 mL of hydrogen at STP =  $\frac{2 \text{ g} \times 112 \text{ mL}}{22400 \text{ mL}} = 10^{-2} \text{ g}$



$$2\text{F} \longrightarrow 1 \text{ mol}$$

$$= 2 \times 96500 \text{ C} = 2\text{g}$$

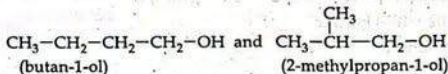
$$2 \text{ g hydrogen will be deposited by } = \frac{2 \times 96500 \times 10^{-2} \text{ g}}{2 \text{ g}} = 965 \text{ C}$$

$$\text{charge (Q)} = \text{IT So, } I = \frac{Q}{T} = \frac{965 \text{ C}}{965 \text{ s}} = 1 \text{ Cs}^{-1} = 1 \text{ A}$$

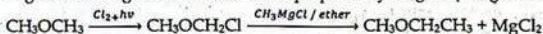
89. c) Molecularity of reaction can never be fraction. It is always equal to number of molecules in whole number only.
90. a) In a), as  $n_p = n_r$  i.e. equal no. of moles between reactant and product, so pressure has no effect on equilibrium.
91. d) Exothermic reaction ( $\Delta H = -ve$ ), greater particle ( $\Delta S = +ve$ , So:  $\Delta G = -ve$ ).
92. c)  $\text{pH} = 10$  means  $[\text{H}^+] = 10^{-10} \text{ mol L}^{-1} = 10^{-13} \text{ g ions/cc}$   
 $= 10^{-13} \times (6.02 \times 10^{23}) \text{ ions/cc}$   
 $= 6.02 \times 10^{10} \text{ ions/cc}$
93. a)
94. d) The oxidation states show a change only in reaction (d).  

$$\overset{0}{\text{Zn}} + \overset{+1}{2\text{A}}\text{gCN} \xrightarrow[-2\text{e}^-]{+2\text{e}^-} \overset{0}{2\text{Ag}} + \overset{0}{\text{Zn}}(\text{CN})_2$$
95. a)  $N = \frac{w \times 1000}{E \times V} = \frac{4 \times 1000}{40 \times 100} = 1\text{N}$
96. a) U-235 was used in little boy and Pu-239 was used in fat man.
97. a)
98. b)
99. c) CFCs are chlorinated and fluorinated methane and ethane derivatives commonly known as Freons.
100. b) Nitrous acid is  $\text{HNO}_2$ . Here, the oxidation number of N is +3.
101. a)  $\text{AgBr}$  is most sensitive as compared to  $\text{AgCl}$ .
102. c) Iron is the only metal which cannot form amalgam. This is because iron neither dissolve in mercury nor react with mercury to form stable alloy.
103. a) Blast furnace is used for Fe whereas shaft furnace is used for Hg.
104. b) Animal bones contain calcium in the form of hydrated compound called hydroxyapatite.  $\text{Ca}_5(\text{PO}_4)_3(\text{OH})$ .
105. b) Red phosphorous is used to make matches in the match industry. White or Yellow Phosphorous is used to make rat poison.
106. c) In calcination, the concentrated ore is heated below its melting point with a limited supply of air or in absence of air to convert ore into dry metallic oxide. This process is carried out in a reverberatory furnace.
107. b)  $\text{Ca}^{++}$  Helps in blood coagulation.
108. a)
109. d) Steam distillation, as aniline is steam volatile.
110. d) It is used for the qualitative test of halogens (Cl, Br, I)
111. d)  $\text{H}_3\text{C}-\text{C}\equiv\text{C}-\text{H}$   
 $\uparrow$   $\alpha$ -hydrogen

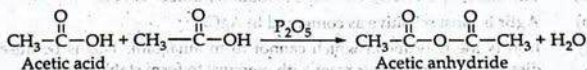
112. b) Being  $3^\circ$  carbocation,  $(\text{CH}_3)_3\text{C}^+$  is the most stable.
113. a) Triptane has an octane number higher than 100, while Iso-octane is assigned an octane number of exactly 100.  
Order of octane number:  
Triptane > Iso-octane > Ethane > Methane
114. c) The dihalides in which two halogen atoms are attached with same carbon atom in a carbon chain, called geminal dihalides or gem-dihalides. For example-  $\text{CH}_3\text{CHBr}_2$ ,  $\text{CH}_2\text{Cl}_2$ ,  $\text{CH}_3\text{CH}_2\text{CCl}_2\text{CH}_3$
115. a) In sandmeyer reaction, benzene diazonium chloride is treated with cuprous chloride / cuprous bromide in presence of  $\text{HCl}$  /  $\text{HBr}$ , chlorobenzene is obtained.
116. c) Chain isomerism is shown by the alcohols having four or more carbon atoms, where the main carbon chain has a different number of carbon atoms.



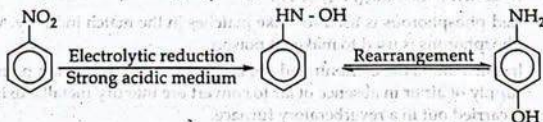
117. d) When phenol is treated with alkyl halide in presence of anhy.  $\text{AlCl}_3$ , ortho- and para-Methyl phenol are formed, where p-methyl phenol is the major product.
118. b) Higher homologues of ethers can be prepared by Grignard reagents.



119. b)
120. a) Other carboxylic acids do not react with Tollen's reagent but formic acid reacts and gives metallic silver which is deposited inner wall of the test tube. So, it is called silver mirror test.
121. c) An acid anhydride is formed by dehydration of carboxylic acids. The byproduct is water.



122. b) When nitrobenzene is reduced at electrolytic condition in presence of strong electrolyte, it gives N phenyl hydroxyl amine (unstable) which undergoes rearrangement and form p-hydroxy aniline a stable product.



Nitrobenzene

N-phenyl hydroxyl amine

p-amino phenol

123. d) Separation of  $1^\circ$ ,  $2^\circ$  and  $3^\circ$  amines by Hoffmann's method uses Diethyl oxalate.
124. b) Zeise's salt is potassium trichloro(ethylene)platinate(II) monohydrate with the formula  $\text{K}[\text{PtCl}_3(\text{C}_2\text{H}_4)] \cdot \text{H}_2\text{O}$

# **NAME**

NAME Institute

In front of Singhdurbar, Putalisadak

Tel: 01- 5971544



## **MECEE BL**

### **Model Entrance Exam**

### **(Model: VIII)**

Date: 20 Aashadha 2083, Saturday

Venue: NAME

Duration - 3 hrs

Time - 11:00 AM

#### **INSTRUCTIONS**

There are 200 multiple-choice questions, each having four choices of which only one choice is correct.

Fill ☐ the most appropriate one.

NAME \_\_\_\_\_

**Note: NAME students must fill their registration number on the OMR sheet.**

1. While studying marine animals during a field visit, a student observes an echinoderm with feather-like arms used for filter feeding. The zoologist identifies it as *Antedon*. What is the common name of this organism?  
 a) Sea star                      b) Brittle star                      c) Sea-lily                      d) Sea-cucumber
2. *Ascaris* is characterized by  
 a) presence of true coelom and metamerism  
 b) presence of neither true coelom nor metamerism  
 c) absence of true coelom but presence of metamerism  
 d) presence of true coelom but absence of metamerism
3. *Sepia* is a cephalopod, in which  
 a) shell is absent                      b) inner shell is found  
 c) a folded shell is found                      d) outer shell is found
4. In a lab experiment, a *Planaria* is cut into three pieces. Each piece regrows into a complete individual. This high regeneration power is primarily due to the presence of  
 a) Flame cells                      b) Neoblasts                      c) Parenchymal cells                      d) Rhabdites
5. In the life cycle of *Plasmodium*, the "Exo-erythrocytic cycle" refers to the phase that occurs in  
 a) Mosquito's gut                      b) Human liver  
 c) Salivary gland of a mosquito                      d) RBCs of man
6. The "spiral valve" in a frog is characteristic feature found in which part of the heart?  
 a) Sinus venosus                      b) Right atrium                      c) Ventricle                      d) Conus arteriosus
7. The body temperature of a frog is  $26^{\circ}\text{C}$  in an environment having a temperature of  $30^{\circ}\text{C}$ . The temperature of the frog in new environment is  
 a)  $25^{\circ}\text{C}$                       b)  $20^{\circ}\text{C}$                       c)  $30^{\circ}\text{C}$                       d) between  $20^{\circ} - 30^{\circ}\text{C}$
8. The archenteron in a frog embryo is formed due to  
 a) Cleavage division                      b) Epiboly                      c) Neurulation                      d) Emboly
9. While observing the external anatomy of an earthworm, a student notices a dark dorsal median line running along the back. This line is formed due to the:  
 a) Supra oesophageal blood vessel                      b) Dorsal blood vessel  
 c) Dorsal nerve cord                      d) Supra intestinal duct
10. The stomach of *Pheretima* is located in the..... segments  
 a) 9<sup>th</sup>-14<sup>th</sup>                      b) 5<sup>th</sup>-7<sup>th</sup>                      c) 8<sup>th</sup>-9<sup>th</sup>                      d) 27<sup>th</sup>-40<sup>th</sup>
11. Which one of the following epithelium is present on inner surface of the hollow organs like the bronchioles and fallopian tubes?  
 a) Columnar epithelium                      b) Ciliated epithelium  
 c) Squamous epithelium                      d) Brush bordered columnar epithelium
12. During muscle contraction, ATP is broken down to provide energy for sliding of the muscle filaments. The ATPase enzyme responsible for this activity is present in:  
 a) troponin                      b) myosin                      c) tropomyosin                      d) actin
13. Blood is liquid connective tissue and is.....in nature  
 a) acidic                      b) alkaline                      c) variable                      d) neutral
14. Muscular tissue is composed predominantly of cells that are specialized for contraction, which result in the movement. The muscles that are immune to fatigue are  
 a) unstriated muscles                      b) cardiac muscles                      c) jaw muscles                      d) skeletal muscles
15. A baby is born with a small tail-like structure, a feature resembling that of distant human ancestors. This phenomenon is known as  
 a) metamorphosis                      b) Regeneration                      c) Atavism                      d) Mutation

16. Which one of the following evidences does not favor the Lamarckian concept of inheritance of acquired characters?  
a) Absent of limbs in snakes      b) Lack of pigment in cave dwelling animals  
c) Webbed toes in aquatic birds      d) Melanisation of peppered moths
17. The famous hominid fossil "Lucy", one of the oldest and best-preserved human ancestor fossils, belongs to the genus  
a) *Oreopithecus*      b) *Dryopithecus*      c) *Pithecanthropus*      d) *Australopithecus*
18. A patient develops a blockage at the site where both bile and pancreatic juice enter the alimentary canal. Which part is most likely blocked?  
a) Jejunum      b) Duodenum      c) Ileum      d) Colon
19. A patient complains of dry mouth due to obstruction of Wharton's duct, reducing the flow of saliva into the oral cavity. Which salivary gland is most likely affected?  
a) Sublingual gland      b) Submandibular gland      c) Parotid gland      d) Lacrimal gland
20. A patient loses the ability to smell after damage to the mucous membrane of the olfactory region. Which of the following structures, found in the nose, normally secretes fluid that keeps the olfactory epithelium moist?  
a) Bowman's gland      b) Bowman's capsule      c) Bowman's membrane      d) Meibomian gland
21. Which of the following statement about the human respiratory system is incorrect?  
a) Each terminal bronchiole gives rise to bronchi.  
b) Alveoli are highly vascularized  
c) Lungs are covered by a double-layered pleura  
d) Pleural fluid reduces friction during breathing.
22. Which layer of the heart wall is primarily responsible for pumping action and is the thickest layer?  
a) Endocardium      b) Epicardium      c) Myocardium      d) Pericardium
23. A patient receives glucose through the bloodstream. The glucose reaches body cells by passing through very thin blood vessels where exchange of nutrients, gases, and wastes occurs. These vessels are:  
a) Arteries      b) Veins      c) Capillaries      d) Aorta
24. During deep sleep, which type of brain waves are predominantly generated?  
a) Alpha waves      b) Beta waves      c) Delta waves      d) Theta waves
25. Which auditory ossicle is attached to the tympanic membrane?  
a) Incus      b) Stapes      c) Malleus      d) Cochlea
26. A dusty particle entering the eye is prevented from reaching the eyeball mainly by:  
a) Retina      b) Cornea      c) Eyelashes      d) Choroid
27. A woman undergoing infertility treatment is found to have blocked tubes that normally transport the ovum to the uterus. Which structures are blocked?  
a) Ureters      b) Fallopian tubes      c) Cervix      d) Vagina
28. The hormone responsible for ovulation is:  
a) FSH      b) LH      c) Estrogen      d) Progesterone
29. Urine finally reaches the urinary bladder through  
a) Urethra      b) Ureter      c) Nephron      d) Renal artery
30. Which of the following is not reabsorbed from the proximal convoluted tubule (PCT)?  
a) Water      b) Sodium ions      c) Amino acids      d) Insulin
31. Sensory nerves carry impulses from  
a) CNS to muscles      b) Receptors to CNS      c) CNS to receptors      d) Brain to brain
32. Which of the following glands atrophies after puberty?  
a) Parathyroid gland      b) Pineal gland      c) Adrenal gland      d) Pituitary gland

NAME \_\_\_\_\_

33. A student develops sudden fever, sore throat, and cough after sitting in a crowded bus next to an infected person. The disease is  
 a) TB b) Influenza c) Cholera d) Typhoid
34. A patient's blood contains proteins that neutralize pathogens. These are:  
 a) Antigens b) Antibodies c) RBCs d) Platelets
35. A patient develops yellowing of skin and eyes (jaundice), fatigue, and signs of liver inflammation after ingesting contaminated food and water. Which infection is most consistent with these clinical features?  
 a) Tuberculosis b) Cholera c) Influenza d) Hepatitis
36. A factory worker exposed to crowded conditions is more likely to get TB because  
 a) Poor digestion b) Airborne transmission c) Food contamination d) Skin contact
37. A 45-year-old patient receives a donor heart from an unrelated human donor. After surgery, the immune system recognizes the transplanted organ as foreign. What type of transplant has been performed?  
 a) Autograft b) Isograft c) Allograft d) Xenograft
38. A fetus is checked for Down syndrome before birth using  
 a) IVF b) Amniocentesis c) Transplantation d) Vaccination
39. When a lizard is threatened by a predator, it detaches its tail and escapes while the detached tail continues to move and distract the predator. This defensive phenomenon is called:  
 a) Mimicry b) Autotomy c) Camouflage d) Apoptosis
40. An eagle possesses lightweight bones, wings, and powerful flight muscles that enable it to spend much of its life in the air. This type of adaptation is known as  
 a) Cursorial b) Fossorial c) Scansorial d) Volant
41. R. Hooke discovered  
 a) Dead cell with no protoplasm b) Living cell with no protoplasm  
 c) Dead cell with protoplasm d) Living cell with protoplasm
42. Hydrolytic enzymes present in lysosomes digest cellular materials and show maximum activity only in a specific pH medium. They are most active in:  
 a) neutral medium b) basic medium  
 c) acidic medium d) Both acidic and basic media
43. In the chromoplast ..... pigments like carotenes and xanthophylls are present  
 a) water and kerosene soluble b) chloroform and fat soluble  
 c) water and fat soluble d) fat soluble
44. Spindle apparatus is completely formed during which stage of mitosis?  
 a) prophase b) metaphase c) anaphase d) telophase
45. The total no of pollen grains in a pollen sac is 32. How many pollen mother cells were there?  
 a) 16 b) 8 c) 64 d) 4
46. Every carbohydrates are  
 a) ribose or deoxyribose b) pentose or hexose  
 c) aldose or ketose d) triose or tetrose
47. Virus can not survive outside host because  
 a) lack of metabolic activity b) separation in two parts  
 c) having thick shield of proteinaceous coat d) due to oxidation
48. The common word for bacteria which are spherical in shape is  
 a) cocci b) bacilli c) spirilla d) pleomorphic
49. Stigma is present in  
 a) Pinus b) Chlamydomonas c) spirogyra d) aspergillus

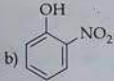
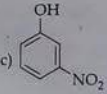
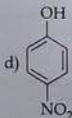
50. The fungus absorbs food by  
a) mycelium      b) root      c) rhizome      d) thallus
51. Which among the following has capacity of absorbing water, used to replace cotton and used as fuel is  
a) Marchantia      b) Riccia      c) sphagnum      d) Funaria
52. The shape of sporangium in fern is  
a) biconvex      b) circular      c) biconcave      d) plano-convex
53. In gymnosperms, the endosperm is formed by the  
a) Germination of a megaspore  
b) Fusion of one male gamete with two polar nuclei  
c) Fusion of one male gamete with one polar nuclei  
d) Fusion of male gamete with egg
54. Rhizome of ginger is modification stem as it bears  
a) adventitious root      b) nodes and internodes  
c) food storing cells      d) axillary bud
55. Which of the following is not true for Cruciferae?  
a) fruit is silique      b) placentation is parietal  
c) flowers are tetramerous      d) inflorescence is capitulum
56. The increase in effect of non-biodegradable pollutants in successive trophic level is called  
a) eutrophication      b) biomagnification      c) ammonification      d) acidification
57. Which one of the following has highest global warming potential?  
a) CFC      b)  $N_2O$       c) methane      d)  $CO_2$
58. Those species which are introduced in a new area and causes harm to native plants is called  
a) endangered species      b) endemic species  
c) allian species      d) rare species
59. Most productive agriculture crop is  
a) maize      b) wheat      c) sugarcane      d) rice
60. A child has blood group O, while both parents have blood group A. Which statement best explains this?  
a) Both parents are homozygous dominant      b) Both parents are heterozygous  
c) One parent has blood group AB      d) Blood group O is dominant
61. In incomplete dominance:  
a) Dominant allele completely masks recessive allele  
b) Both alleles express equally  
c) Heterozygote shows an intermediate phenotype  
d) One allele is lethal
62. Why are males more commonly affected by X-linked recessive disorders?  
a) They inherit two X chromosomes  
b) They possess only one X chromosome  
c) Their Y chromosome carries the defective gene  
d) Their genes mutate more frequently
63. The phenotypic ratio of a monohybrid cross with complete dominance is:  
a) 1:2:1      b) 3:1      c) 9:3:3:1      d) 2:1
64. Which RNA carries amino acids to the ribosome during protein synthesis?  
a) mRNA      b) tRNA      c) rRNA      d) snRNA
65. Which mutation usually has the least effect on protein structure?  
a) Frameshift mutation      b) Nonsense mutation  
c) Silent mutation      d) Deletion mutation

NAME \_\_\_\_\_

66. During heartwood formation, xylem vessels often become blocked by sac-like structures arising from neighboring cells. These structures are called:  
 a) Callose b) Resin ducts c) Tyloses d) Companion cells
67. Secondary growth doesn't occur in  
 a) Yucca b) Agave c) Pea d) Millet
68. The endocarp of coconut is made up of  
 a) Fiber b) cellulose c) sclereids d) parenchyma
69. Pollen grains of a plant having  $2n = 28$  are culture to get callus by tissue culture method. What would be the number of chromosomes in the cells of callus?  
 a) 28 b) 14 c) 7 d) 56
70. DNA fingerprinting has proved useful in forensic science. It involves the use of  
 a) Minisatellite b) ribosomal RNA c) cDNA d) Bacterial DNA
71. A reserve carbohydrate is commonly stored in seeds, tubers, and roots of higher plants. During germination, it is hydrolyzed into simpler sugars to provide energy for the growing embryo. This reserve carbohydrate is:  
 a) Cellulose b) Chitin c) Starch d) Glucose
72. Histologically grafting the union of  
 a) two cambium b) two tissues c) two plant d) two cells
73. In an angiosperm, male plant is diploid and female plant is tetraploid, endosperm will be:  
 a) Haploid b) Triploid c) Tetraploid d) Pentaploid
74. Which of the following experimental material is best suited to study photosynthesis?  
 a) *Chlorella* b) *Chlamydomonas* c) spinach leaf d) *Hydrilla*
75. During the light reaction of photosynthesis,  $O_2$  is evolved by the photolysis of water, which takes place in association with a specific photosystem.  $O_2$  evolution is directly associated with:  
 a) Photosystem-I b) Photosystem-II  
 c) Cytochrome complex d) Both photosystem-I & photosystem-II
76. Krebs cycle is  
 a) aerobic b) anaerobic c) anabolic d) alcoholic
77. High auxin and low cytokinin involves in the formation of  
 a) shoot b) root c) callus d) leaf
78. Ascent of sap will be maximum in  
 a) day time b) night time c) early morning d) noon time
79. A cell in being treated with alcohol and then kept in hypertonic solution, what would happen?  
 a) cell would shrink and die b) cell would swell  
 c) cell would burst immediately d) there would be no change
80. Ovule having funicle lying close to the micropyle and embryo sac horse shoe shape structure is called:  
 a) Anacarpous b) Orthotropous c) Amphitropous d) Circinotropous
81. After appearing CEE, a student ate a piece of chocolate containing sucrose ( $C_{12}H_{22}O_{11}$ ). Analysis show sucrose contain 42.11% Carbon, 6.05% hydrogen and 51.84% oxygen by mass. This data support which law?  
 a) Law of conservation of mass b) Law of Multiple proportion  
 c) Law of reciprocal proportion d) Law of Constant proportion
82. 6g of pure Carbon is completely burnt in excess oxygen to form Carbon dioxide. What is volume of Carbon dioxide gas liberated at normal temperature and pressure (NTP).  
 a) 11.2 L b) 22.4 L c) 14.5 L d) 18.4 L

83. Equal weight of Methane and hydrogen gas are placed in a container at the same temperature and volume. What fraction of the total pressure is exerted by hydrogen gas molecules?  
 a)  $\frac{7}{9}$  b)  $\frac{9}{7}$  c)  $\frac{8}{9}$  d)  $\frac{9}{8}$
84. The pH of 0.10% solution of NaOH is  
 a) 1.6 b) 12.4 c) 8.6 d) 10.4
85. When  ${}_{90}\text{Th}^{228} \rightarrow {}_{83}\text{Bi}^{212}$ , then the number of emitted  $\alpha$  and  $\beta$  particle is respectively  
 a) 4 $\alpha$ , 1 $\beta$  b) 4 $\alpha$ , 4 $\beta$  c) 5 $\alpha$ , 1 $\beta$  d) 5 $\alpha$ , 4 $\beta$
86. The number of ionic, Covalent, and Co-ordinate bonds in  $\text{NH}_4\text{Cl}$  are respectively?  
 a) 1, 3, 2 b) 1, 3, 1 c) 1, 2, 3 d) 2, 3, 2
87. Arrange the following molecules in the decreasing order of bond angle. (i)  $\text{BF}_3$  (ii)  $\text{NH}_3$  (iii)  $\text{CH}_4$  (iv)  $\text{BeCl}_2$   
 a) (iv) > (i) > (ii) > (iii) b) (i) > (iv) > (iii) > (ii)  
 c) (iv) > (ii) > (i) > (iii) d) (iv) > (i) > (iii) > (ii)
88. If  $K_1$  and  $K_2$  are the ionization constants of two different acids. Ratio of degree of dissociation will be:  
 a)  $\frac{K_1}{K_2}$  b)  $\sqrt{\frac{K_1}{K_2}}$  c)  $K_1 \times K_2$  d)  $\frac{K_1^2}{K_2^2}$
89. In Faraday's law, the relation between electrochemical equivalent (ECE) and Chemical equivalent (CE) is:  
 a)  $\text{CE} = F \times \text{ECE}$  b)  $\text{ECE} = F \times \text{CE}$  c)  $\text{ECE} = \text{CE}$  d)  $\text{CF} = \frac{F}{\text{ECE}}$
90. The second order of rate constant usually expressed as  
 a) Mole litre sec b)  $\text{Mole}^{-1} \text{litre}^{-1} \text{sec}^{-1}$  c)  $\text{Mole}^{-1} \text{litre sec}^{-1}$  d)  $\text{Mole litre}^{-2} \text{sec}^{-1}$
91. Heat of neutralization of Strong acid and base is -57.0 KJ. The amount of heat released when 0.5 mol of  $\text{HNO}_3$  solution is added to 0.2 mol of NaOH solution is:  
 a) -28.5 KJ b) 57.0 KJ c) 34.9 KJ d) 11.40 KJ
92. According to the Hess's law, which of the following statement is true:  
 a) The total enthalpy change for both paths will be different because the number of steps varies.  
 b) The total enthalpy change depends only on the initial and final substances, so both paths have the same enthalpy change.  
 c) The enthalpy change along path 1st will always be greater than path 2nd because it has more step.  
 d) Hess's law cannot be applied if there are intermediate substance.
93. The conjugate acid of  $\text{HCO}_3^-$  is  
 a)  $\text{H}_2\text{CO}_3$  b)  $\text{CO}_3^{2-}$  c)  $\text{HCO}_2^-$  d)  $\text{HCO}_3$
94. For a Crystal  $a = b = c$  and  $\alpha = \beta = \gamma = 90^\circ$ , the structure is  
 a) Rhombohedral b) Cubic c) Hexagonal d) Monoclinic
95. The oxidation number of chlorine in sodium chlorate is  
 a) +1 b) +3 c) +5 d) +7
96. For the reaction,  $\text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{PCl}_5(\text{g})$ , the value of  $K_c$  at  $250^\circ\text{C}$  is 26 then, the value of  $K_p$  at this temperature will be  
 a) 0.61 b) 0.57 c) 0.83 d) 0.86
97. Which of the following set of quantum numbers is correct for 4F?  
 a)  $n = 4, l = 3, m = 1, s = +\frac{1}{2}$  b)  $n = 4, l = 4, m = 1, s = -\frac{1}{2}$   
 c)  $n = 4, l = 2, m = +4, s = -\frac{1}{2}$  d)  $n = 3, l = 3, m = -4, s = +\frac{1}{2}$
98. Froth Flootation process is used for Concentration of  
 a) Bauxite b) Chalcopyrite c) Haematite d) Malachite

NAME \_\_\_\_\_

99. M is the molecular mass of  $\text{KMnO}_4$ . The equivalent weight of  $\text{KMnO}_4$ , when it is converted into  $\text{K}_2\text{MnO}_4$ , is
- a) M                      b)  $\frac{M}{3}$                       c)  $\frac{M}{5}$                       d)  $\frac{M}{7}$
100. While moving from left to right in a periodic table, which of the following variable decreases?
- a) Ionization affinity                      b) Ionization potential  
c) Atomic radius                      d) Electronegativity
101. In Solvay process brine solution is saturated with ammonia and excess of  $\text{CO}_2$  to get....?
- a)  $\text{NaHCO}_3$                       b)  $\text{NH}_4\text{HCO}_3$                       c)  $\text{NH}_4\text{Cl}$                       d)  $\text{Na}_2\text{CO}_3$
102. Which one is the suitable method of reduction to obtained magnesium metal from its compound?
- a) Reduction of its oxide by Carbon or Carbon monoxide  
b) Displacement of Magnesium from its aqueous solution  
c) Electrolysis of  $\text{MgCl}_2$  in Aqueous solution  
d) Electrolysis of fused compound
103. The Semi water gas is:
- a)  $\text{CO} + \text{H}_2$                       b)  $\text{CO} + \text{H}_2 + \text{N}_2$                       c)  $\text{CH}_4$                       d)  $\text{H}_2 + \text{CO} + \text{CH}_4$
104. When  $\text{NH}_3$  gas is passed into mercurous nitrate solution
- a) Black Colour of  $\text{Hg}(\text{NH}_2)\text{NO}_3$  and Hg is Formed  
b) Black Colour of  $\text{Hg}(\text{NH}_2)\text{NO}_3$  is Formed  
c) Reddish brown  $\text{HgO} \cdot \text{NH}_2$  is Formed  
d) Black Colour of  $\text{Hg}_2(\text{NH}_2)\text{NO}_3$  is Formed
105. Bleaching property is shown by all except.
- a)  $\text{Cl}_2$                       b)  $\text{O}_2$                       c)  $\text{SO}_2$                       d)  $\text{H}_2\text{O}_2$
106. Which of the following is most volatile?
- a) HF                      b) HBr                      c) HI                      d) HCl
107. Which of the following are d block elements but not regarded as transition elements.
- a) Cu, Ag, Au                      b) Zn, Cd, Hg                      c) Fe, Co, Ni                      d) Ru, Rh, Pd
108. The IUPAC name of  $\text{CH}_3 - \underset{\text{C}_2\text{H}_5}{\text{C}} = \text{CHCH}_3$  is
- a) 2-Ethylbutene                      b) 2-Ethylbut-2-ene  
c) 3-Methylpent-2-ene                      d) 2-Ethylpent-2-ene
109. Which one is the most acidic compound?
- a)                       b) 
- c)                       d) 
110. The number of  $\text{sp}^3$  hybridized Carbon atoms in cyclohexene are
- a) 2                      b) 3                      c) 4                      d) 6

NAME \_\_\_\_\_

111. Which of the following reagents would you prefer used to find out whether the hydrocarbon  $C_3H_4$  contains one triple bond or two double bonds?
- Fehling's solution
  - Ammonical  $AgNO_3$  or  $CuCl$  solution
  - Baeyer's reagent
  - $Br_2/CCl_4$
112. Daily use candles contain:
- Higher saturated hydrocarbon
  - Lower saturated hydrocarbon
  - Higher unsaturated hydrocarbon
  - Lower unsaturated hydrocarbon
113.  $1^\circ, 2^\circ$  and  $3^\circ$  alcohols are distinguished by each other except.
- Hoffmann method
  - Lucas test.
  - Oxidation test
  - Victor Meyer test
114. An aqueous solution of 2,4-dinitrophenyl hydrazine (DNPH) is known as:
- Lucas reagent
  - Brady's reagent
  - Tollen's reagent
  - Fehling's solution
115. Which organic compound is commonly found in nail polish remover and it also effective for removing stubborn stains?
- Acetone
  - Benzene
  - Acetaldehyde
  - Acetic acid
116. Which reagent will bring about the conversion of a Carboxylic acid into an ester?
- $C_2H_5OH$
  - Dry  $HCl + C_2H_5OH$
  - $LiAlH_4$
  - $Al(OC_2H_5)_3$
117. When phenol is heated with Chloroform in the presence of aqueous  $NaOH$  at  $60^\circ-70^\circ C$  followed by acid hydrolysis, what is the major product?
- Sodium Salicylate
  - O-hydroxy benzaldehyde
  - O-hydroxy benzoic acid
  - Salicylic acid
118. Nitro alkanes are reduced to amine by
- $Na/Hg$
  - $Ph_3SnH$
  - $Sn/Hg$
  - $Sn/conc. HCl$
119. Which kind of fission is favoured by sunlight?
- Heterolytic fission
  - Homolytic fission
  - Both heterolytic and homolytic fission
  - Fraction distillation
120. The dehydrobromination of 2-bromobutane gives  $CH_3CH=CHCH_3$ . The product is
- Hoffmann product
  - Saytzeff product
  - Markovnikov's product
  - Hoffmann-Saytzeff product
121.  $CCl_4$  cannot give precipitate with  $AgNO_3$  due to
- Formation of complex with  $AgNO_3$
  - Evolution of  $Cl_2$  gas
  - Chloride ion is not formed
  - $AgNO_3$  does not give silver ion
122. Which of the following acid present in apple?
- Citric acid
  - Malonic acid
  - Malic acid
  - Formic acid
123. Identify Z in the given reaction.  $CH_3CH_2CH_2OH \xrightarrow{Conc. H_2SO_4} X \xrightarrow{Br_2} Y \xrightarrow{alc. KOH} Z$
- $CH_3CH(NH_2)CH_2(NH_2)$
  - $CH_3CHOHCH_3$
  - $CH_3CH(OH)CH(OH)$
  - $CH_3C \equiv CH$
124. Nitration of benzene is

NAME \_\_\_\_\_

- a) Electrophilic addition  
c) Nucleophilic addition
125. In the separation by fractional Crystallization there is difference in their  
a) Density b) Solubility c) Volatility d) Boiling point.
126. Phenyl 2-hydroxybenzoate can be used as  
a) Antiseptic b) Antipyretic c) Analgesic d) Lumino
127. In which of the following, ppt will not be seen when  $H_2S$  gas is passed in:  
a) Acidified  $CuSO_4$  solution b) Neutral  $CuSO_4$  solution  
c) Acidified  $ZnSO_4$  solution d) Ammonical  $ZnSO_4$  solution
128. Caprolactum is the monomer of  
a) Rayon b) Nylon - 6 c) Chloroprene d) Isoprene
129. Which of the following acid radical produces a brisk effervescence of a Colourless, odourless gas when treated with dil.  $H_2SO_4$ ?  
a)  $SO_3^{2-}$  b)  $S^{2-}$  c)  $CO_3^{2-}$  d)  $NO_2^-$
130. Bleaching agent of the pulp is  
a)  $H_2O$  b)  $H_2O_2$  c)  $NaOH$  d)  $TiO_2$
131. Which one is true?  
a) 1 calorie > 1 erg > 1 Joule  
b) 1 erg > 1 calorie > 1 Joule  
c) 1 calorie > 1 Joule > 1 erg  
d) 1 Joule > 1 calorie > 1 erg
132. The vector sum of two forces is perpendicular to their vector differences. In that case, the forces  
a) Are equal to each other in magnitude  
b) Are not equal to each other in magnitude  
c) Are equal to each other in direction  
d) Are equal to each other in both magnitude and direction
133. A wheel of radius 1 meter rolls forward half a revolution on a horizontal ground. The magnitude of the displacement of the point of the wheel initially in contact with the ground is  
a)  $2\pi$  b)  $\sqrt{2}\pi$  c)  $\sqrt{\pi^2 + 4}$  d)  $\pi$
134. On a stationary sail-boat, air is blown at the sails from a fan attached to the boat. The boat will  
a) Remain stationary  
b) Spin around  
c) Move in a direction opposite to that in which air is blown  
d) Move in the direction in which the air is blown
135. A block of mass  $m$  initially at rest is dropped from a height ' $h$ ' onto a spring of force constant  $k$ . The maximum compression in the spring is ' $x$ ', then



a)  $mgh = \frac{1}{2} kx^2$

c)  $mgh = \frac{1}{2} k(x+h)^2$

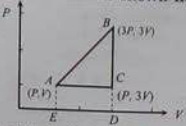
b)  $mg(h+x) = \frac{1}{2} kx^2$

d)  $mg(h+x) = \frac{1}{2} k(x+h)^2$

136. A body of mass 'm' moves in a circular path with uniform angular velocity. The motion of the body has a constant  
a) Acceleration      b) Velocity      c) Momentum      d) Kinetic energy
137. In the rotational motion of a rigid body, all particles move with  
a) Same linear and angular velocity  
b) Same linear and different angular velocities  
c) With different linear velocities and same angular velocities  
d) With different linear velocities and different angular velocities
138. A simple harmonic oscillator has an amplitude 'A' and time period 'T'. The time required by it to travel from  $x = A$  to  $x = A/2$  is  
a)  $T/6$       b)  $T/4$       c)  $T/3$       d)  $T/2$
139. A rod of length  $\ell$  and area of cross-section A is heated from  $0^\circ\text{C}$  to  $100^\circ\text{C}$ . The rod is so placed that it is not allowed to increase in length, then the force developed is proportional to  
a)  $\ell$       b)  $\ell^{-1}$       c) A      d)  $A^{-1}$
140. A ball of mass m and radius r is gently released into a viscous liquid. The mass of liquid displaced by it is  $m'$  such that  $m > m'$ . The terminal velocity ( $v_t$ ) of the ball is directly proportional to  
a)  $\frac{m-m'}{r}$       b)  $\frac{m+m'}{r}$       c)  $\frac{m+m'}{r^2}$       d)  $\frac{m-m'}{r^2}$
141. A capacitor is connected to a battery. If the plate separation is increased while remaining connected to the battery, the capacitance:  
a) Increases      b) Decreases  
c) Remains constant      d) Becomes infinite
142. A hollow conducting sphere of radius R has a charge (+Q) on its surface. What is the electric potential within the sphere at a distance  $r = R/3$  from its centre?  
a) Zero      b)  $\frac{1}{4\pi\epsilon_0} \frac{Q}{r}$       c)  $\frac{1}{4\pi\epsilon_0} \frac{Q}{R}$       d)  $\frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$
143. maximum value of electric field intensity due to a charged sphere is at  
a) Center      b) Surface  
c) Infinity      d) half the radius of the sphere
144. A point charge +q is placed near an uncharged conducting plates. The spacing between adjacent electric lines of force as we move further away from the point charge towards the plate.  
a) Increases continuously  
b) Decreases continuously  
c) Remains perfectly constant  
d) First decreases and then increases
145. Doppler effect is applicable for  
a) moving bodies  
b) one is moving and other are stationary  
c) for relative motion  
d) only when both source and observer are moving
146. The sound carried by air from a sitar to a listener is a wave of the following type:  
a) Longitudinal stationary  
b) Transverse progressive  
c) Transverse stationary  
d) Longitudinal progressive

NAME \_\_\_\_\_

147. A steady current flows in a metallic conductor of non-uniform cross-section. The quantity/quantities constant along the length of the conductor is/are  
 a) Current, electric field and drift speed  
 b) Drift speed only  
 c) Current and drift speed  
 d) Current only
148. 'n' equal resistors are first connected in series and then connected in parallel. What is the ratio of the maximum to the minimum resistance?  
 a) n  
 b)  $\frac{1}{n^2}$   
 c)  $n^2$   
 d)  $\frac{1}{n}$
149. The net resistance of a voltmeter should be large to ensure that:  
 a) It does not get overheated  
 b) It does not draw excessive current  
 c) It can measure a large potential difference  
 d) It does not appreciably change the potential difference to be measured
150. If a high-power heater is connected to the electric mains, then the bulbs in the house become dim, because there is a  
 a) Current drop  
 b) Potential drop  
 c) No current drop  
 d) No potential drop
151. The current is flowing in a south direction along a power line. The direction of the magnetic field above the power line (neglecting the Earth's field) is  
 a) South  
 b) East  
 c) North  
 d) West
152. A superconductor exhibits perfect  
 a) Ferrimagnetism  
 b) Ferromagnetism  
 c) Paramagnetism  
 d) Diamagnetism
153. When the number of turns in a coil is doubled without any change in the length of the coil, its self-inductance becomes  
 a) Four times  
 b) Doubled  
 c) Halved  
 d) Unchanged
154. An L-R circuit has a cell of e.m.f. E, which is switched on at time  $t = 0$ . The current in the circuit after a long time will be  
 a) Zero  
 b)  $\frac{E}{R}$   
 c)  $\frac{E}{L}$   
 d)  $\frac{E}{\sqrt{L^2 + R^2}}$
155. In a purely resistive AC circuit, the current  
 a) Lags behind the e.m.f. in phase  
 b) Is in phase with the e.m.f.  
 c) Leads the e.m.f. in phase  
 d) Leads the e.m.f. in half the cycle and lags behind it in the other half
156. If a thermometer reads freezing point of water as  $20^\circ\text{C}$  and boiling point as  $150^\circ\text{C}$ , how much thermometer read when the actual temperature is  $60^\circ\text{C}$ ?  
 a)  $98^\circ\text{C}$   
 b)  $110^\circ\text{C}$   
 c)  $40^\circ\text{C}$   
 d)  $60^\circ\text{C}$
157. A composite rod made of copper ( $\alpha = 1.8 \times 10^{-5} \text{ K}^{-1}$ ) and steel ( $\alpha = 1.2 \times 10^{-5} \text{ K}^{-1}$ ) is heated. Then  
 a) It bends with steel on concave side  
 b) It bends with copper on concave side  
 c) It does not expand  
 d) Data is insufficient
158. The relation between the gas pressure P and average kinetic energy per unit volume E is  
 a)  $P = \frac{1}{2} E$   
 b)  $P = E$   
 c)  $P = \frac{3}{2} E$   
 d)  $P = \frac{2}{3} E$

159. A perfect gas contained in a cylinder is kept in a vacuum. If the cylinder suddenly bursts, then the temperature of the gas  
 a) Remains constant    b) Becomes zero    c) Increases    d) Decreases
160. An ideal gas is taken around ABCA as shown in the above P-V diagram. The work done during a cycle is  
 a)  $2PV$   
 b)  $PV$   
 c)  $1/2PV$   
 d) Zero
- 
161. If the temperature of a hot body is increased by 50%, then the increase in the quantity of emitted heat radiation will be  
 a) 125%    b) 200%    c) 300%    d) 406%
162. In a given process,  $dW=0$ ,  $dQ<0$ , then for the gas:  
 a) Temperature increases    b) Pressure decreases  
 c) volume decreases    d) Pressure increases
163. A ray of light is incident on a plane mirror at an angle of incidence  $30^\circ$ . The ray after reflection is deviated through.  
 a)  $130^\circ$     b)  $60^\circ$     c)  $90^\circ$     d)  $120^\circ$
164. When a ray of light emerges from a block of glass, the critical angle is  
 a) equal to the angle of reflection  
 b) the angle between the refracted ray and normal  
 c) the angle of incidence for which the refracted ray travels along the glass-air boundary  
 d) the angle of incidence
165. If the refractive angles of two prisms made of crown glass are  $10^\circ$  and  $20^\circ$  respectively, then the ratio of their colour deviation powers will be  
 a) 1 : 1    b) 2 : 1    c) 4 : 1    d) 1 : 2
166. A monochromatic beam of light passes from a denser medium to a rarer medium. As a result  
 a) Its speed decreases    b) Its frequency increases  
 c) Its speed increases    d) Its frequency decreases
167. An unpolarised light wave is travelling along positive x-axis. The electric field vector in the beam vibrates in the direction of  
 a) +ve Y-axis definitely    b) +ve Z-axis definitely  
 c) +ve X-axis    d) Y or Z-axis
168. In the Young double slit experiment, when two light waves form a third minimum, they have  
 a) Phase difference of  $3\pi$     b) Phase difference of  $\frac{5\pi}{2}$   
 c) Path difference of  $3\lambda$     d) Path difference of  $\frac{5\lambda}{2}$
169. In Thomson's experiment, if the value of  $q/m$  is the same for all positive ions striking the photographic plate, then the trace would be  
 a) Straight line    b) Parabolic    c) Circular    d) Elliptical
170. A photon, an electron, and a uranium nucleus all have the same wavelength. The one with the most energy  
 a) Is the photon  
 b) Is the electron  
 c) Is the uranium nucleus

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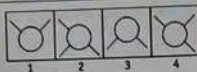
- d) Depends upon the wavelength and the properties of the particle.
171. For intensity of a light of wavelength  $5000\text{\AA}$ , the photoelectron saturation current is  $0.40\text{ }\mu\text{A}$  and the stopping potential is  $1.36\text{ V}$ , the work function of the metal is  
 a)  $2.47\text{ eV}$       b)  $1.36\text{ eV}$       c)  $1.10\text{ eV}$       d)  $0.43\text{ eV}$
172. The characteristic X-ray radiation is emitted when  
 a) The electrons are accelerated to a fixed energy  
 b) The source of electrons emits a monoenergetic beam  
 c) The bombarding electrons knock out electrons from the inner shell of the target atoms and one of the outer electrons falls into this vacancy  
 d) The valence electrons in the target atoms are removed as a result of the collision
173. The ratio of the frequencies of the long wavelength limits of the Lyman and Balmer series of the hydrogen spectrum is:  
 a)  $27:5$       b)  $5:27$       c)  $4:1$       d)  $1:4$
174. The minimum energy required to take out the only one electron from the ground state of  $\text{He}^+$  is  
 a)  $13.6\text{ eV}$       b)  $54.4\text{ eV}$       c)  $27.2\text{ eV}$       d)  $6.8\text{ eV}$
175. If the half-life of a radioactive atom is  $2.3$  days, then its decay constant would be:  
 a)  $0.1$       b)  $0.2$       c)  $0.3$       d)  $2.3$
176. A piece of copper and the other of germanium are cooled from the room temperature to  $80\text{ K}$ , then which of the following would be a correct statement  
 a) Resistance of each increases  
 b) Resistance of each decreases  
 c) Resistance of copper increases while that of germanium decreases  
 d) Resistance of copper decreases while that of germanium increases
177. Zener breakdown takes place if:  
 a) Doped impurity is low      b) Doped impurity is high  
 c) Less impurity in N-type      d) Less impurity in P-type
178. A gate has the following truth table. The gate is

|   |   |   |   |   |
|---|---|---|---|---|
| P | 1 | 1 | 0 | 0 |
| Q | 1 | 0 | 1 | 0 |
| R | 1 | 0 | 0 | 0 |

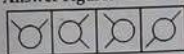
- a) NOR      b) OR      c) NAND      d) AND
179. The "God Particle" is a nickname for which particle?  
 a) Proton      b) Electron      c) Higgs boson      d) Neutron
180. Which of the following particles is classified as a fermion?  
 a) Photon      b) W boson      c) Electron      d) Higgs boson
181. Loiter is related to dandle in the same way as impugn is related to .....  
 a) Challenge      b) Gamble      c) Confiscate      d) Revenge
182. Which one of the following will replace the question mark in the series?  
 $8, 3, 11, 14, 25, ?$   
 a)  $50$       b)  $39$       c)  $29$       d)  $11$
183. Insert the missing character in the place of the question mark(?).

|    |    |     |
|----|----|-----|
| 5  | 9  | 15  |
| 16 | 29 | ?   |
| 49 | 89 | 147 |

184. If in a certain code language, 'Good morning' is written as 11 and 'Good afternoon' is written as 15, then 'Good night' is written as  
a) 15                      b) 7                      c) 12                      d) 9
185. Pointing to a boy in the photograph Reena said, "He is the only son of the only child of my grandfather." How is Reena related to that boy?  
a) Mother                      b) Sister                      c) Aunt                      d) Cannot be determined
186. January 1, 2008 is Tuesday. What day of the week lies on Jan 1, 2009?  
a) Monday                      b) Wednesday                      c) Thursday                      d) Sunday
187. Keshav is taller than Vijay but shorter than Nitin, Nitin is taller than Kishan but shorter than Amar. If Vijay is taller than Kishan, then who is the shortest among them?  
a) Keshav                      b) Vijay                      c) Nitin                      d) Kishan
188. A can do a piece of work 10 days, B can do the same piece of work in 15 days, A does 1<sup>st</sup> day and B does 2<sup>nd</sup> day. How many many days the work will be finish?  
a) 12                      b) 14                      c) 13                      d) 15
189. Decide whether the assertion and the reason are related or not, and choose the correct option.  
Assertion a): The Himalayas once lay under the sea.  
Reason (R): Fossils of marine creatures are traced in the Himalayas.  
a) If both A and R are true and R is the correct explanation of A  
b) If both A and R are true but R is not the correct explanation of A  
c) If A is true but R is false  
d) If A is false but R is true
190. In a tournament, a player has a record of 40% wins, out of the number of games he has played so far, which in turn is  $\frac{2}{5}$  of the total number of games he plays. What is the maximum percentage of the remaining games that the player can lose and still win 50% of all the games played?  
a) 30%                      b) 43.3%                      c) 33.3%                      d) 25%
191. In a mixture of three varieties of tea, the ratio of their weights is 4 : 5 : 8. If 5kg tea of the first variety, 10 kg tea of the second variety and some quantity of tea of the third variety are added to the mixture, the ratio of the weights of three varieties of tea becomes 5 : 7 : 9. Find the quantity of the third variety of tea in the final mixture.  
a) 42 kg                      b) 45 kg                      c) 48 kg                      d) 40 kg
192. In an office, there are two departments. In the first department, there are 10 employees and their average age is 30 years. Similarly, in the second department, there are 5 employees and their average age is 33 years. Then the average age of all employees in that office is:  
a) 30.5 years                      b) 31 years                      c) 31.5 years                      d) 32 years
193. A dog takes 4 leaps for every 5 leaps of a hare, but 3 leaps of a dog are equal to 4 leaps of the hare. Compare their speeds.  
a) 11 : 15                      b) 15 : 11                      c) 16 : 13                      d) 16 : 15
194. A number consists of two digits. If the number formed by interchanging the digits is added to the original number, the resulting number (i.e., the sum) must be divisible by:  
a) 11                      b) 9                      c) 5                      d) 3
195. In how many different ways can the letters of the word 'OPTICAL' be arranged so that the vowels always come together?  
a) 120                      b) 720                      c) 4320                      d) 2160
196. Select the figure from the answer figures that will continue the same series as established by the problem figures.  
Problem Figures

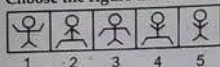


Answer Figures



- a) A                      b) B                      c) C                      d) D

197. Choose the figure that is different from the rest.



- a) 1                      b) 2                      c) 3                      d) 5

198. Count the number of squares in the given figure.

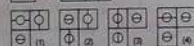


- a) 11                      b) 21                      c) 24                      d) 26

199. Find the water image of: PQ8AF5BZ9

- (a) 608VLE2BZ8                      (b) 608VLE2B56                      (c) 608V79B59                      (d) 608VLE2B58

200. What is the missing segment?



- a) 1                      b) 2                      c) 3                      d) 4

Result will be published on Sunday

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===== Best of Luck =====

# **NAME**



**NAME Institute**

In front of Singhdurbar, Putalisadak

Tel: - 01- 5971544

# **MECEE BL**

## **Model Entrance Exam**

### **(Model: IX)**

**Date: 27 Aashadha 2083, Saturday**

**Venue: NAME**

**Duration - 3 hrs**

**Time - 07:00 AM**

#### **INSTRUCTIONS**

There are 200 multiple-choice questions, each having four choices of which only one choice is correct.

Fill ☐ the most appropriate one.

NAME \_\_\_\_\_

**Note:** NAME students must fill their registration number on the OMR sheet.

1. According to the Oparin-Haldane hypothesis, the organic molecules in the primitive ocean aggregated to form colloidal droplets capable of metabolic-like replication. What are these structures called?
  - a) Microspheres
  - b) Liposomes
  - c) Coacervates
  - d) Protobionts
2. According to Modern Synthetic Theory, the origin of new species is primarily driven by:
  - a) Random genetic drift acting in isolation
  - b) Extensive interspecific hybridization overriding the necessity for selective environmental pressure
  - c) Genetic recombination with natural selection
  - d) Continuous environmental variations & natural selection
3. According to geological time mapping, the separation of birds from reptilian lineages and the sudden dominance of early gymnosperms took place during which geological period?
  - a) Silurian Period
  - b) Carboniferous Period
  - c) Jurassic Period
  - d) Permian Period
4. A true coelom is anatomically distinguished by being completely lined by tissue derived from which embryonic germ layer?
  - a) Ectoderm
  - b) Endoderm
  - c) Mesoderm
  - d) Both Ectoderm and Endoderm
5. Retrogressive metamorphosis is a highly unique developmental phenomenon where an active, free-swimming larva possessing advanced chordate features degenerates into a sedentary, simpler adult. This is characteristically seen in:
  - a) Cephalochordata
  - b) Urochordata
  - c) Cyclostomata
  - d) Amphibia
6. Which of the following features is universally shared by all chordates at some stage of their lifecycle?
  - a) Vertebral column, ventral nerve cord, and open circulatory system
  - b) Notochord, dorsal hollow nerve cord, and pharyngeal gill slits
  - c) Ventral solid nerve cord, heart, and scales
  - d) Exoskeleton, closed circulatory system, and lungs
7. Which of the following characters is common to both Annelida and Arthropoda?
  - a) Segmentation
  - b) Jointed appendages
  - c) Non-chitinous cuticle
  - d) Closed circulatory system
8. Chloragogen cells found surrounding the intestinal wall of an earthworm perform physiological processing comparable to which vertebrate organ?
  - a) Kidney
  - b) Liver
  - c) Spleen
  - d) Pancreas
9. Exo-erythrocytic cycle is absent in:
  - a) Plasmodium vivax
  - b) Plasmodium falciparum
  - c) Plasmodium malariae
  - d) Plasmodium ovale
10. Blood glands are localized structural configurations found in specific anterior segments of Pheretima. In which of the following segment combinations are they situated, and what is their primary function?
  - a) Segments 4, 5, and 6; to produce hemoglobin and blood corpuscles
  - b) Segments 7 and 9; to pump blood into the ventral vessel
  - c) Segments 10 and 11; to secrete calcareous substances to neutralize humic acid

- d) Segments 14, 15, and 16; to produce defensive coelomic fluid
11. Fertilization of Plasmodium occurs in:
    - a) Human liver
    - b) Human RBCs
    - c) Gut of mosquito
    - d) Salivary gland of mosquito
  12. The phenomenon in which a frog changes the color of its skin with respect to its surroundings is known as:
    - a) Nictiation
    - b) Metachrosis
    - c) Ecdysis
    - d) Moulting
  13. Which part of the small intestine is present in humans but absent in frogs?
    - a) Duodenum
    - b) Jejunum
    - c) Ileum
    - d) Colon
  14. During skeletal muscle contraction, which specific structural zone or band within a sarcomere remains entirely constant in length?
    - a) The I-band
    - b) The H-zone
    - c) The A-band
    - d) The distance between two consecutive Z-lines
  15. Osteocytes reside in small spaces called lacunae. Through which microscopic, radiating cytoplasmic channels do these cells communicate and exchange nutrients with each other?
    - a) Haversian canals
    - b) Volkmann's canals
    - c) Canaliculi
    - d) Sarcolemma channels
  16. In which of the following regions would you most expect to find simple ciliated columnar epithelium?
    - a) Inner lining of the stomach
    - b) Fallopian tubes and bronchioles
    - c) Tubular parts of nephrons in kidneys
    - d) Outer surface of the skin
  17. In connective tissue sheaths, what is the correct architectural sequence stretching from the outermost to the innermost layer?
    - a) Epineurium, endoneurium, perineurium
    - b) Perineurium, epineurium, endoneurium
    - c) Perineurium, endoneurium, epineurium
    - d) Epineurium, perineurium, endoneurium
  18. Which of the following components constitutes the structural framework of the first line of defense within the human innate immune system?
    - a) T-lymphocytes and B-lymphocytes
    - b) Anti-bacterial skin secretions, sebum, and mucous membrane linings
    - c) Circulating memory cells and plasma cells
    - d) Interferons and inflammatory interleukins
  19. The pentavalent vaccine DPT, HepB and Hib is routinely administered to infants to provide active artificial immunity against which combination of diseases?
    - a) Diarrhea, Polio, Tetanus, Hepatitis, Haemophilus influenza type B
    - b) Diphtheria, Pertussis, Tetanus, Hepatitis B, Haemophilus influenza type B
    - c) Diphtheria, Pneumonia, Typhoid, Haemophilus influenza type B, Hepatitis B
    - d) Dengue, Polio, Tuberculosis, Hepatitis B, Haemophilus influenza type B
  20. A newborn infant receives passive natural immunity from the mother via the transport of maternal antibodies. Which antibodies cross the placenta during pregnancy, and which are supplied through breast milk (colostrum)?
    - a) IgG crosses placenta; IgD through colostrum
    - b) IgG crosses placenta; IgA through colostrum
    - c) IgE crosses placenta; IgG through colostrum
    - d) IgA crosses placenta; IgM through colostrum
  21. Which of the following is the most commonly used diagnostic test for AIDS?

NAME \_\_\_\_\_

- a) ELISA                      b) Western blot                      c) PCR for HIV RNA                      d) CD<sub>4</sub> count
22. In the context of medical technology, what is a primary potential application of cryopreservation?
- a) Preserving human embryos for future use in IVF  
b) Freezing human tissue for research purposes  
c) Preserving transgenic animals  
d) Store human organs for transplantation
23. How can tissue engineering be utilized dynamically in modern medicine?
- a) Culturing artificial skin  
b) Fabricating prosthetic limbs  
c) Generating biomaterials for transplanting whole organs  
d) Synthesizing chemical-based compounds for targeted chemotherapy
24. Which of the following is a feature unique to secondary aquatic adaptation and is not usually seen in primary?
- a) Presence of gills throughout life                      b) Limbs modified into flippers  
c) Lateral line system present                      d) Body covered with scales
25. Which term describes the movement of an organism in response to a stimulus by comparing the intensity of the stimulus over time with a single sensory organ?
- a) Tropotaxis                      b) Klinotaxis                      c) Kinesis                      d) Chemotaxis
26. Hyperpigmentation of skin, hypotension, low blood glucose, hypokalemia, and hyponatremia are hallmark presentations of which clinical condition caused by hyposecretion of the adrenal cortex?
- a) Grave's disease                      b) Addison's disease                      c) Cushing's disease                      d) Conn's disease
27. Which hormone secretion stimulates the active release of bile into the duodenum?
- a) CCK                      b) Pancreozymin                      c) Secretin                      d) Thyroxine
28. Which of the following visual pigments has the lowest physiological threshold for light activation?
- a) Cyanopsin                      b) Rhodopsin                      c) Iodopsin                      d) Porphyropsin
29. The specialized acoustic spot located within the ampulla of the semi-circular canals is known as the:
- a) Otolith                      b) Organ of Corti                      c) Crista                      d) Macula
30. Cowper's gland secretion is functionally and biochemically analogous to which female structural secretion?
- a) Labia majora                      b) Bartholin's gland                      c) Skene's gland                      d) Clitoris
31. When a male undergoes a vasectomy procedure, which of the following reproductive factors is left unaffected?
- a) Transport of sperms through the distal tract  
b) Fertilizing ability of ejaculated sperm  
c) Testosterone secretion into the bloodstream  
d) Total sperm count within the ejaculate fluid
32. Which part of the ECG corresponds to atrial depolarization?
- a) P wave                      b) QRS complex                      c) T wave                      d) PR interval
33. The cardiac output is given by:
- a) Stroke volume  $\times$  Heart rate                      b) Heart rate  $+$  Stroke volume

- c) Stroke volume  $\times$  Blood pressure      d) Heart rate  $\times$  Blood pressure
34. Which cranial meningeal layer is a highly vascular, delicate connective tissue membrane that closely adheres to the exact contours, gyri, and sulci of the brain surface?  
a) Dura mater      b) Arachnoid mater      c) Pia mater      d) Subdural space
35. Thermoregulation, osmoregulation, circadian rhythms, and the primary control of the anterior pituitary gland via releasing hormones are all vital physiological functions mediated by the:  
a) Thalamus      b) Hypothalamus      c) Basal ganglia      d) Hippocampus
36. If a person is severely dehydrated, which specific hormone is released, and where exactly does it act within the nephron to maximize water conservation?  
a) Aldosterone – Loop of Henle      b) Vasopressin – Collecting duct and DCT  
c) Renin – PCT      d) ANF – Glomerulus
37. The ascending limb of the loop of Henle is highly impermeable to water but actively pumps out NaCl into the interstitium. What is the direct physiological result of this property?  
a) Urine within the ascending limb becomes progressively more concentrated  
b) Tubular fluid becomes hypotonic by the time it reaches the DCT  
c) Water is drawn directly out of the medullary interstitium into the ascending limb  
d) Urea accumulates rapidly in the local tubular fluid
38. Which of the following correctly represents the dental formula for permanent and deciduous (milk) teeth in humans?  
a) Permanent: 2 1 2 3 / 2 1 2 3 (32 teeth); Deciduous: 2 1 2 / 2 1 2 (20 teeth)  
b) Permanent: 2 1 2 / 2 1 2 (20 teeth); Deciduous: 2 1 2 3 / 2 1 2 3 (32 teeth)  
c) Permanent: 2 1 3 3 / 2 1 3 3 (36 teeth); Deciduous: 2 1 2 / 2 1 2 (20 teeth)  
d) Permanent: 2 1 2 3 / 2 1 2 3 (32 teeth); Deciduous: 2 1 3 / 2 1 3 (24 teeth)
39. The vocal cords, which vibrate mechanically to produce sound, are anatomically housed within which cartilaginous framework of the respiratory tract?  
a) Pharynx      b) Larynx      c) Trachea      d) Bronchi
40. According to the Bohr effect, which of the following microenvironmental shifts in a metabolically active skeletal muscle will cause the oxygen-hemoglobin dissociation curve to shift to the right, facilitating rapid oxygen unloading?  
a) Decreased  $p\text{CO}_2$ , increased pH, and decreased temperature  
b) Increased  $p\text{CO}_2$ , decreased pH, and increased temperature  
c) Increased  $p\text{CO}_2$ , increased pH, and decreased temperature  
d) Decreased  $p\text{CO}_2$ , decreased pH, and increased temperature
41. Needle-shaped crystals of Colocasia are called:  
a) sphaeroraphides      b) raphides      c) cystolith      d) silicate
42. A research student collected certain alga and found that its cells contained both chlorophyll a and chlorophyll d as well as phycoerythrin. The alga belongs to  
a) Rhodophyceae      b) Bacillariophyceae      c) Chlorophyceae      d) Phaeophyceae
43. Which one is the wrong pairing for the disease and its causal organism?  
a) Black rust of wheat - *Puccinia graminis*  
b) Loose smut of wheat - *Ustilago nuda*  
c) Root-knot of vegetables - *Meloidogyne species*  
d) Late blight of potato - *Alternaria solani*

NAME \_\_\_\_\_

44. A bryophyte grows in shaded moist areas but dies under prolonged dry conditions. Which adaptation explains this?  
 a) Thick cuticle      b) Vascular tissue      c) Lack of true roots      d) Seeds
45. Specialized cells for fixing atmospheric nitrogen in *Nostoc* are  
 a) Akinetes      b) Heterocysts      c) Hormogonia      d) Nodules
46. *Cycas* and *Adiantum* resemble each other in having:  
 a) Seeds      b) Motile Sperms      c) Cambium      d) Vessels
47. What is the botanical name of sunflowers?  
 a) *Tagetes patula*      b) *Helianthus annuus*  
 c) *Calendula officinalis*      d) *Lycopersicum esculentum*
48. Antherozoid attractant in an archegonial zone of fern is  
 a) Sugar      b) Amino acid      c) Malic acid      d) Mucilage
49. In which aestivation, the posterior largest petal overlaps two lateral free petals, and the lateral ones overlap the fused anterior petals?  
 a) Descending imbricate      b) Ascending imbricate  
 c) Valvate imbricate      d) Quincuncial imbricate
50. Which of the following is the simplest amino acid?  
 a) Alanine      b) Asparagine      c) Glycine      d) Tyrosine
51. The energy - releasing metabolic process in which substrate is oxidized without an external electron acceptor is called:  
 a) glycolysis      b) fermentation      c) aerobic respiration      d) photorespiration
52. A location with luxuriant growth of lichens on the trees indicates that the:  
 a) Trees are very healthy      b) Trees are heavily infested  
 c) Location is highly polluted      d) Location is not polluted
53. In which ecosystem is the biomass of the tertiary consumer larger than the secondary and primary consumers and producer?  
 a) Desert ecosystem      b) Grassland ecosystem  
 c) Forest ecosystem      d) Pond ecosystem
54. Xerophytes lack:  
 a) Well-developed root and sunken stomata      b) Well-developed conducting system  
 c) Aerenchyma      d) Spiny leaf
55. A river with an inflow of domestic sewage rich in organic waste may result in  
 a) drying of the river very soon due to algal bloom  
 b) increased population of aquatic food web organisms.  
 c) an increased production of fish due to biodegradable nutrients.  
 d) death of fish due to lack of oxygen
56. Which Forest is most abundant in the tropical region of Nepal?  
 a) Sal      b) *Shima Castanopsis*      c) *Alnus nepalensis*      d) Pine
57. Which of the following is considered as a polar cell organelle?  
 a) Endoplasmic reticulum      b) Golgi bodies  
 c) Mitochondria      d) Lysosome
58. Which is true for chromosomes in metaphase of mitosis?  
 a) Chromatids unjoined      b) Randomly arranged

- c) Aligned at equatorial plate d) Replication starts
59. A plant exhibits non-disjunction during meiosis. Which condition may arise?  
a) Polyploidy b) Aneuploidy c) Haploidy d) Normal gametes
60. Plant cells are connected by their walls to form a continuous system due to:  
a) Gap junction b) plasmodesmata c) Tight junction d) Middle lamella
61. In sea urchin DNA, which is double stranded, 17% of the bases were shown to be cytosine. The percentages of the other three bases expected to be present in this DNA are :  
a) G 17%, A 16.5%, T 32.5% b) G 17%, A 33%, T 33%  
c) G 8.5%, A 50%, T 24.5% d) G 34%, A 24.5%, T 24.5%
62. The transfer RNA molecule in 3D appears  
a) L-shaped b) E-shaped c) Y-shaped d) S-shaped
63. Mutation causing replacement of one amino acid in a protein is called:  
a) Nonsense mutation b) Missense mutation  
c) Silent mutation d) Frameshift mutation
64. Codon is:  
a) A sequence of three nucleotides which codes for one amino acid  
b) A sequence of three nucleotides which codes for one fatty acid  
c) A sequence of four nucleotides which codes for one amino acid  
d) A sequence of four nucleotides which codes for one fatty acid
65. Which one of the following does not differ in *E.coli* and *Chlamydomonas*?  
a) Ribosomes b) Chromosomal Organization  
c) Cell wall d) Cell membrane
66. If you are provided with root-tips of onion in your class and are asked to count the chromosomes, which of the following stages can you most conveniently look into?  
a) Metaphase b) Telophase c) Anaphase d) Prophase
67. Which is a false statement?  
a) m-RNA has a linear structure with a codon  
b) t-RNA contains an anticodon  
c) r-RNA helps in protein synthesis  
d) t-RNA is the largest among all types of RNA
68. Collagen is  
a) fibrous protein b) globular protein c) lipid d) carbohydrate
69. For genetically distinct plants, the method of plant tissue culture used is:  
a) Shoot culture b) Callus culture c) Embryo culture d) Somatic hybridization
70. Citric acid is extracted from which fungi:  
a) Albigo b) *Acetobacter aceti* c) *Aspergillus niger* d) *Penicillium*
71. Axillary bud and terminal bud are derived from the activity of  
a) lateral meristem b) intercalary meristem c) apical meristem d) parenchyma
72. In a ring girdled plant:  
a) The root dies first b) The shoot and root die together  
c) Neither root nor shoot will die d) The shoot dies first
73. In a dicot root, the endodermis is suberized forming:

NAME \_\_\_\_\_

- a) Casparian strip      b) Lignified walls      c) Phloem sieve      d) Root hairs
74. **Anoxygenic photosynthesis is characteristic of:**  
 a) Rhodospirillum      b) Spirogyra      c) Chlamydomonas      d) Ulva
75. **Photorespiration reduces net photosynthetic efficiency because:**  
 a) Oxygen inhibits Calvin cycle      b) Rubisco fixes  $O_2$  instead of  $CO_2$   
 c) Light is unavailable      d) Water is limiting
76. **How does pruning help in making the hedge dense?**  
 a) It frees axillary buds from apical dominance  
 b) The apical shoot grows faster after pruning  
 c) It releases wound hormones  
 d) It induces the differentiation of new shoots from the rootstock
77. **Movement of ions or molecules in a direction opposite to that of prevailing electrochemical gradient is known as**  
 a) diffusion      b) active transport      c) pinocytosis      d) brownian movement
78. **Total number of ATP formed by aerobic respiration in a higher plant is:**  
 a) 30      b) 36      c) 38      d) 24
79. **Wings of the pollen of Pinus is an outgrowth of:**  
 a) exine      b) intine      c) exo-intine      d) Sporangial wall
80. **Filiform apparatus is characteristic feature of:**  
 a) Nucellar embryo      b) Aleurone cell      c) Synergids      d) Generative cell
81. **An oxide of iodine (I = 127) contains 25.4g iodine for 8g of oxygen. Its formula could be:**  
 a)  $I_2O_3$       b)  $I_2O$       c)  $I_2O_5$       d)  $I_2O_7$
82. **The normality of 4% (wt./vol.) NaOH is:**  
 a) 0.1      b) 1.0      c) 0.05      d) 0.01
83. **If 0.5 mole of  $BaCl_2$  is mixed with 0.2 mole of  $Na_3PO_4$ , the maximum number of moles of  $Ba_3(PO_4)_2$  that can be formed is:**  
 a) 0.7      b) 0.5      c) 0.30      d) 0.1
84. **When  $SO_2$  is passed in a solution of potassium iodate, the oxidation state of iodine changes from:**  
 a) +5 to 0      b) +5 to -1      c) -5 to 0      d) -7 to -1
85. **What fraction of a reactant showing first order remains after 40 minutes if  $t_{1/2}$  is 20 minute?**  
 a)  $\frac{1}{4}$       b)  $\frac{1}{2}$       c)  $\frac{1}{8}$       d)  $\frac{1}{6}$
86. **Active mass of 10 g  $CaCO_3$  is:**  
 a) 1.0      b) 0.5      c) 0.1      d) 0.75
87. **The pH of the solution containing 0.1N NaOH and 0.1N  $NH_4OH$  is:**  
 a) 1      b) 7      c) 2      d) 13
88. **A substance reacts with water and breaks into acid and a base, the process is:**  
 a) Solvation      b) Hydrolysis      c) Dissolution      d) Hydration
89. **Which of the following would not change the pH of  $10\text{ cm}^3$  of dil. HCl, when added to acid?**  
 a)  $5\text{ cm}^3$  of pure water      b)  $20\text{ cm}^3$  of pure water  
 c)  $10\text{ cm}^3$  of conc. HCl      d)  $20\text{ cm}^3$  of same dilute HCl
90. **When a liquid boils, there is an increase in:**

- a) Free energy      b) Kinetic energy      c) Potential energy      d) Heat of vapourisation
91. During electrolysis of  $\text{H}_2\text{O}$ , the molar ratio of  $\text{H}_2$  and  $\text{O}_2$  formed is:  
a) 2:1      b) 1:2      c) 1:3      d) 1:1
92. The value of  $n$  in,  
 $\text{MnO}_4^- + 8\text{H}^+ + ne \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$  is:  
a) 5      b) 4      c) 3      d) 2
93. In a compound C, H, N atoms are present in 9: 1: 3.5 by weight. Molecular weight of compound is 108, its molecular formula is:  
a)  $\text{C}_2\text{H}_6\text{N}_2$       b)  $\text{C}_3\text{H}_4\text{N}$       c)  $\text{C}_6\text{H}_8\text{N}_2$       d)  $\text{C}_9\text{H}_{12}\text{N}_3$
94. 100 mL 0.3N HCl solution were mixed with 200 mL of 0.6N  $\text{H}_2\text{SO}_4$  solution. The final acidic normality is:  
a) 0.9 N      b) 0.6 N      c) 0.5 N      d) 0.4 N
95. Which species has the maximum number of lone pairs of electrons on the central atom?  
a)  $\text{ClO}_3^-$       b)  $\text{XeF}_4$       c)  $\text{SF}_4$       d)  $\text{I}_3^-$
96. Which of the following is isoelectronic with carbon atom?  
a)  $\text{Na}^+$       b)  $\text{Al}^{+3}$       c)  $\text{O}^{2-}$       d)  $\text{N}^+$
97. The weakest chemical force is:  
a) Ionic bond      b) Covalent bond      c) Metallic bond      d) van der Waals force
98. Maximum numbers of electrons in an orbit are given by:  
a)  $n^2$       b)  $2n^2$       c)  $\frac{n^2}{2}$       d)  $3n^3$
99. Kjeldahl method is used to check purity of:  
a) Ammonium phosphate      b) Super phosphate  
c) Muriate of potash      d) Triple superphosphate
100. The correct order of nucleophilicity is:  
a)  $\text{I}^- > \text{Br}^- > \text{Cl}^- > \text{F}^-$       b)  $\text{I}^- > \text{Cl}^- > \text{Br}^- > \text{F}^-$       c)  $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$       d)  $\text{Cl}^- > \text{Br}^- > \text{F}^- > \text{I}^-$
101. Mercury is transported in metal containers made up of:  
a) Fe      b) Pb      c) Zn      d) Sn
102. Which metal cannot be extracted by smelting process?  
a) Pb      b) Fe      c) Zn      d) Al
103. On bubbling  $\text{CO}_2$  through a solution of barium peroxide in water:  
a)  $\text{O}_2$  is formed      b)  $\text{H}_2\text{CO}_3$  is formed      c)  $\text{H}_2\text{O}_2$  is formed      d)  $\text{H}_2$  is formed
104. Sodium carbonate is:  
a) Efflorescent      b) deliquescent      c) Hygroscopic      d) oxidant
105. The coal form containing maximum percentage of carbon is:  
a) lignite      b) anthracite      c) bituminous      d) peat
106. Fractional crystallization is based on:  
a) Boyle's law      b) Boiling points  
c) Dalton's law of partial pressure      d) Solubilities
107. Nitric acid oxidises phosphorous to:  
a)  $\text{H}_3\text{PO}_4$       b)  $\text{P}_2\text{O}_5$       c)  $\text{H}_3\text{PO}_3$       d)  $\text{H}_4\text{P}_2\text{O}_7$
108. The most abundant ore of iron is:  
a) Haematite      b) Limonite      c) Magnetite      d) Siderite

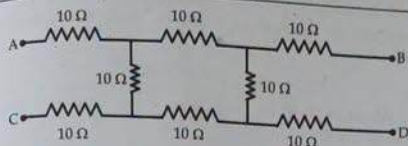
NAME \_\_\_\_\_

109. The solubility of iodine in water is greatly increased by:  
 a) adding an acid  
 b) boiling the solution  
 c) Cooling the solution  
 d) adding potassium iodide
110. Aqueous solution of  $\text{Na}_2\text{S}_2\text{O}_3$  on reaction with  $\text{Cl}_2$  gives:  
 a)  $\text{Na}_2\text{S}_4\text{O}_6$   
 b)  $\text{NaHSO}_4$   
 c)  $\text{NaCl}$   
 d)  $\text{NaOH}$
111. Bleaching powder is:  
 a) Normal salt  
 b) Mixed salt  
 c) Double salt  
 d) Complex salt
112. Which of the following acids has the smallest dissociation constant?  
 a)  $\text{CH}_3\text{CHFCOOH}$   
 b)  $\text{FCH}_2\text{CH}_2\text{COOH}$   
 c)  $\text{BrCH}_2\text{CH}_2\text{COOH}$   
 d)  $\text{CH}_3\text{CHBrCOOH}$
113. A strong base can abstract an  $\alpha$ -hydrogen from:  
 a) Amine  
 b) Ketone  
 c) Alkane  
 d) Alkene
114. What is the primary component of wood that gives structural support and must be removed or separated during pulping?  
 a) Cellulose  
 b) Lignin  
 c) Starch  
 d) Glucose
115. Alkynes can be reduced to alkenes by hydrogenation in presence of:  
 a) Raney Ni  
 b) Anhy.  $\text{AlCl}_3$   
 c) Pd  
 d) Lindlar's catalyst
116. When vinyl chloride is treated with  $\text{NaNH}_2$ :  
 a) It dissolves  
 b) It forms vinyl alcohol  
 c) It forms acetylene  
 d) It has no action
117. The number of methoxy groups in a compound can be determined by treating it with:  
 a) HI and  $\text{AgNO}_3$   
 b) Sodium carbonate  
 c) Sodium hydroxide  
 d) Acetic acid
118. Which of the following on treatment with 50% aq.  $\text{NaOH}$  gives alcohol and acid?  
 a)  $\text{C}_6\text{H}_5\text{CHO}$   
 b)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$   
 c)  $\text{CH}_3\text{COCH}_3$   
 d)  $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$
119. Benzaldehyde reacts with  $\text{NH}_3$  to give:  
 a) Aniline  
 b) Benzamide  
 c) Phenyl cyanide  
 d) Hydrobenzamide
120. To which class of dyes does phenolphthalein belong?  
 a) Azo dyes  
 b) Nitro dyes  
 c) Tetraphenylmethane dyes  
 d) Phthalein dyes
121. The IUPAC name of  $\text{CH}_3 - \text{CH} = \text{C} - \text{CH}_2 - \text{COOH}$  is  

$$\begin{array}{c} | \\ \text{CONH}_2 \end{array}$$
  
 a) 4 - carbamoyl - 3 - ethbut - 3 - enoic acid  
 b) 3 - carbamoylpent - 3 - enoic acid  
 c) 2 - carboxymethylbut - enamide  
 d) 3 - amidopent - 2 - enoic acid
122. Which of the following is an aromatic heterocyclic compound?  
 a) Pyrrole  
 b) Pyrrolidine  
 c) Epoxyethane  
 d) Dioxane
123. Which of the following has maximum acidic strength?  
 a) o-Nitrobenzoic acid  
 b) m-Nitrobenzoic acid  
 c) p-Nitrobenzoic acid  
 d) p-Nitrophenol
124. The homologue of ethyne is:  
 a)  $\text{C}_2\text{H}_4$   
 b)  $\text{C}_2\text{H}_6$   
 c)  $\text{C}_3\text{H}_8$   
 d)  $\text{C}_3\text{H}_4$
125. The conversion of  $\text{ClCH} = \text{CHCl}$  to  $\text{Cl}_2\text{CH} - \text{CHCl}_2$  can be carried out with:

- a)  $\text{Cl}_2$                       b)  $\text{Cl}_2/\text{h}\nu$                       c)  $\text{Cl}_2/\text{AlCl}_3$                       d)  $\text{Cl}_2/\text{aq. NaOH}$
126. Lindane can be obtained by reaction of benzene with:  
 a)  $\text{CHCl}_3/\text{anhy. AlCl}_3$                       b)  $\text{Cl}_2/\text{sunlight}$   
 c)  $\text{C}_2\text{H}_5\text{I}/\text{anhy. AlCl}_3$                       d)  $\text{CH}_3\text{COCl}/\text{AlCl}_3$
127.  $\text{Zn} + \text{very cold dil. HNO}_3$  gives:  
 a)  $\text{Zn}(\text{NO}_3)_2 + \text{N}_2\text{O}$                       b)  $\text{Zn}(\text{NO}_3)_2 + \text{NO}$   
 c)  $\text{Zn}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3$                       d)  $\text{Zn}(\text{NO}_3)_2 + \text{NO}_2$
128. Which of the following organic compounds exhibits positive Fehling test as well as Iodoform test?  
 a) Methanal                      b) Ethanol                      c) Propanone                      d) Ethanal
129. Acetamide and  $\text{NaOBr}/\text{OH}^-$  produce:  
 a) Ethanamine                      b) Methanamine                      c)  $\text{CH}_3\text{CN}$                       d)  $\text{NH}_3$
130. In sodium fusion test of organic compounds, the nitrogen of organic compound is converted into:  
 a) Sodamide                      b) Sodium cyanide                      c) Sodium nitrite                      d) Sodium nitrate
131. If Energy (E), velocity (V) and time (T) are chosen as the fundamental quantities, the dimensional formula of surface tension will be  
 a)  $[\text{E} \text{V}^{-1} \text{T}^{-2}]$                       b)  $[\text{E} \text{V}^{-2} \text{T}^{-2}]$                       c)  $[\text{E}^2 \text{V}^{-1} \text{T}^{-3}]$                       d)  $[\text{E} \text{V}^{-2} \text{T}^{-1}]$
132. A body starts from rest, what is the ratio of the distance travelled by the body during the 4<sup>th</sup> and 3<sup>rd</sup> second?  
 a)  $\frac{7}{5}$                       b)  $\frac{5}{7}$                       c)  $\frac{7}{3}$                       d)  $\frac{3}{7}$
133. If the angle between  $\vec{A}$  and  $\vec{B}$  is  $\theta$ , the value of the product  $(\vec{B} \times \vec{A}) \cdot \vec{A}$  is equal to  
 a)  $BA^2 \sin \theta$                       b)  $BA^2 \cos \theta$                       c)  $BA^2 \sin \theta \cos \theta$                       d) zero
134. The maximum range of a gun on horizontal terrain is 16 km. If  $g = 10 \text{ m/s}^2$ , the muzzle velocity of a shell must be  
 a) 160 m/s                      b)  $200\sqrt{2}$  m/s                      c) 400 m/s                      d) 800 m/s
135. The angular speed of a flywheel making 120 revolutions per minute is  
 a)  $4\pi \text{ rad/s}$                       b)  $4\pi^2 \text{ rad/s}$                       c)  $\pi \text{ rad/s}$                       d)  $2\pi \text{ rad/s}$
136. A man weighs 80 kg. He stands on a weighing scale in a lift which is moving upwards with a uniform acceleration of  $5 \text{ m/s}^2$ . What would be the reading on the scale?  
 a) zero                      b) 400 N                      c) 800 N                      d) 1200 N
137. A point mass  $m$  is moved in a vertical circle of radius  $r$  with the help of a string. The velocity of the mass is  $\sqrt{7rg}$  at the lowest point. The tension in the string at the lowest point is  
 a)  $7mg$                       b)  $8mg$                       c)  $1mg$                       d)  $6mg$
138. Two objects of mass 10 kg and 20 kg respectively are connected to the two ends of a rigid rod of length 10 m with negligible mass. The distance of the centre of mass of the system from the 10 kg mass is  
 a) 5 m                      b)  $\frac{10}{3}$  m                      c)  $\frac{20}{3}$  m                      d) 10 m
139. When a block of mass  $m$  is suspended by a long wire of length  $L$ , the length of the wire becomes  $(L+l)$ . The elastic potential energy stored in the extended wire is  
 a)  $mg/l$                       b)  $mgL$                       c)  $1/2 mg/l$                       d)  $1/2 mgL$
140. A liquid does not wet the solid surface if angle of contact is  
 a) equal to  $60^\circ$                       b) greater than  $90^\circ$                       c) zero                      d) equal to  $45^\circ$

141. Mercury Thermometer can be used to measure temperature up to  
a)  $260^{\circ}\text{C}$                       b)  $100^{\circ}\text{C}$                       c)  $360^{\circ}\text{C}$                       d)  $500^{\circ}\text{C}$
142. The unit of thermal conductivity is  
a)  $\text{JmK}^{-1}$                       b)  $\text{Jm}^{-1}\text{K}^{-1}$                       c)  $\text{WmK}^{-1}$                       d)  $\text{Wm}^{-1}\text{K}^{-1}$
143. Certain quantity of water cools from  $70^{\circ}\text{C}$  to  $60^{\circ}\text{C}$  in the first 5 minutes and to  $54^{\circ}\text{C}$  in the next 5 minutes. The temperature of the surroundings is  
a)  $45^{\circ}\text{C}$                       b)  $20^{\circ}\text{C}$                       c)  $42^{\circ}\text{C}$                       d)  $10^{\circ}\text{C}$
144. In which of the following processes, heat is neither absorbed nor released by a system?  
a) isothermal                      b) adiabatic                      c) isobaric                      d) isochoric
145. First law of thermodynamics is consequence of conservation of  
a) work                      b) Energy                      c) Heat                      d) efficiency
146. The efficiency of a carnot engine depends upon  
a) the temperatures of the source and sink                      b) the volume of the cylinder of the engine  
c) the temperature of the source only                      d) the temperature of the sink only
147. At  $0\text{K}$  which of the following properties of a gas will be zero?  
a) vibrational Energy                      b) Density  
c) kinetic energy                      d) potential Energy
148. In case of a forced vibration, the resonance wave becomes very sharp when the  
a) Damping force is small                      b) restoring force is small  
c) applied periodic force is small                      d) quality factor is small
149. A wave travelling in the +ve x-direction having displacement along y-direction as  $1\text{m}$ , wavelength  $2\pi\text{m}$  and frequency of  $\frac{1}{\pi}\text{Hz}$  is represented by  
a)  $y = \sin(2\pi x + 2\pi t)$                       b)  $y = \sin(x - 2t)$   
c)  $y = \sin(2\pi x - 2\pi t)$                       d)  $y = \sin(10\pi x - 20\pi t)$
150. If the initial tension on a stretched string is doubled, then the ratio of the initial and final speed of a transverse wave along the string is  
a) 1:2                      b) 1:1                      c)  $\sqrt{2}:1$                       d)  $1:\sqrt{2}$
151. A hollow sphere of radius  $1\text{m}$  is given a positive charge of  $10\mu\text{C}$ . The electric field at the centre of hollow sphere will be  
a)  $60 \times 10^3 \text{ V/m}$                       b)  $90 \times 10^3 \text{ V/m}$                       c) zero                      d) Infinite
152. A charge  $Q$  is enclosed by a Gaussian spherical surface of radius  $R$ . If the radius is doubled, then the outward electric flux will  
a) increase four times                      b) be reduced to half                      c) remain the same                      d) be doubled
153. Four point charges  $-Q$ ,  $-q$ ,  $2q$  and  $2Q$  are placed one at each corner of the square. The relation between  $Q$  and  $q$  for which the potential at the centre of the square is zero is  
a)  $Q = -q$                       b)  $Q = -\frac{1}{2}q$                       c)  $Q = q$                       d)  $Q = \frac{1}{q}$
154. Energy per unit volume for a capacitor having area  $A$  and separation  $d$  kept at potential difference  $V$  is given by  
a)  $\frac{1}{2}\epsilon_0 \frac{V^2}{d^2}$                       b)  $\frac{1}{2\epsilon_0} \frac{V^2}{d^2}$                       c)  $\frac{1}{2}CV^2$                       d)  $\frac{Q^2}{2C}$
155. A flow of  $10^7$  electrons per second in a conducting wire constitutes a current of  
a)  $1.6 \times 10^{-12} \text{ A}$                       b)  $1.6 \times 10^{26} \text{ A}$                       c)  $1.6 \times 10^{-26} \text{ A}$                       d)  $1.6 \times 10^{12} \text{ A}$
156. What will be the equivalent resistance between the two points A and D?



- a)  $30\Omega$                       b)  $40\Omega$                       c)  $10\Omega$                       d)  $20\Omega$
157. In a potentiometer circuit a cell of emf 1.5 V gives balance point at 36 cm length of wire. If another cell of emf 2.5 V replaces the first cell, then at what length of the wire, the balance point occurs?  
 a) 21.6 cm                      b) 64 cm                      c) 62 cm                      d) 60 cm
158. The masses of the wires of copper is in the ratio of 1:3:5 and their lengths are in the ratio of 5:3:1. The ratio of their electrical resistance is  
 a) 1:3:5                      b) 5:3:1                      c) 1:25:125                      d) 125:15:1
159. The magnetic field of given length of wire for single turn coil at its centre is B then its value for two turns coil for the same wire is  
 a)  $\frac{B}{4}$                       b)  $\frac{B}{2}$                       c) 4B                      d) 2B
160. Two long parallel wires are at a distance of 1 m. If both of them carry one ampere of current in same direction, then the force of attraction on unit length of the wires will be  
 a)  $2 \times 10^{-7}$  N/m                      b)  $4 \times 10^{-7}$  N/m                      c)  $8 \times 10^{-7}$  N/m                      d)  $10^{-7}$  N/m
161. According to Curie's law, the magnetic susceptibility of a substance at an absolute temperature T is proportional to  
 a)  $\frac{1}{T}$                       b) T                      c)  $T^3$                       d)  $T^2$
162. The magnetic flux linked with a coil (in wb) is given by the equation  $\phi = 5t^2 + 3t + 16$ . The magnitude of induced emf in the coil at the fourth second will be  
 a) 43 V                      b) 108 V                      c) 10 V                      d) 33 V
163. A capacitor is of capacity C and reactance X. If capacitance and frequency become double then reactance will be  
 a) 4X                      b)  $\frac{X}{2}$                       c)  $\frac{X}{4}$                       d) 2X
164. The minimum wavelength of X-rays produced by an electron accelerated through a potential difference of V volts is proportional to  
 a)  $V^2$                       b)  $\sqrt{V}$                       c)  $\frac{1}{V}$                       d)  $\frac{1}{\sqrt{V}}$
165. The half life of a radioactive substance is 20 minutes. In how much time, the activity of substance drops to  $\frac{1}{16}$  th of its initial value?  
 a) 80 minutes                      b) 20 minutes                      c) 40 minutes                      d) 60 minutes
166. The work functions of Caesium (Cs), potassium (K) and sodium (Na) are 2.14 eV, 2.30 eV and 2.75 eV respectively. If incident electromagnetic radiation has an incident energy of 2.20 eV, which of these photosensitive surfaces may emit photoelectrons?  
 a) Na only                      b) Cs only                      c) both Na & K                      d) K only
167. According to Hubble's law, the age of universe is  
 a) approximately 6000 years                      b) less than 6000 years  
 c) roughly 1 billion years                      d) between 10 and 15 billion years

NAME \_\_\_\_\_

168. The de-Broglie wavelength of a neutron in thermal equilibrium with heavy water at a temperature  $T$  (kelvin) and mass  $m$  is
- a)  $\frac{h}{\sqrt{3mkT}}$       b)  $\frac{2h}{\sqrt{3mkT}}$       c)  $\frac{2h}{\sqrt{mkT}}$       d)  $\frac{h}{\sqrt{mkT}}$
169. A 10 eV electron is circulating in a plane at right angle to a uniform field at a magnetic induction  $10^{-4}$  T. The orbital radius of electron is
- a) 18 cm      b) 11 cm      c) 10 cm      d) 12 cm
170. The total energy of an electron in the first excited state of hydrogen is about -3.4 eV. Its kinetic energy in this state is
- a) -3.4 eV      b) -1.7 eV      c) 1.7 eV      d) 3.4 eV
171. Two nuclei have their mass numbers in the ratio of 1:3. The ratio of their nuclear densities would be
- a) 1:1      b) 1:3      c) 3:1      d)  $3^{1/3}$ :1
172. The most penetrating radiation out of the following are
- a)  $\beta$ -rays      b)  $\gamma$ -rays      c) x-rays      d)  $\alpha$ -rays
173. For solids which have the negative temperature coefficient of resistance are
- a) insulators only      b) semiconductors only  
c) insulators and semiconductors      d) metals
174. Following logic gate is
- 
- a) AND      b) NAND      c) OR      d) NOR
175. The increase in the width of the depletion region in a p-n junction diode is due to
- a) reverse bias only      b) both forward bias and reverse bias  
c) increase in forward current      d) forward bias only
176. An object is placed on the principal axis of a concave mirror at a distance of  $1.5 f$  ( $f$  is the focal length). The image will be at
- a)  $1.5 f$       b)  $-1.5 f$       c)  $3f$       d)  $-3f$
177. The frequency of a light wave in a material is  $2 \times 10^{14}$  Hz and wavelength is  $5000 \text{ \AA}$ . The refractive index of material will be
- a) 1.5      b) 3      c) 1.33      d) 1.4
178. A convex lens is dipped in a liquid whose refractive index is equal to the refractive index of the lens. Then its focal length will
- a) become zero      b) become infinite  
c) become small, but non-zero      d) remain unchanged
179. The Brewster's angle  $i_b$  for an interface should be
- a)  $30^\circ < i_b < 45^\circ$       b)  $45^\circ < i_b < 90^\circ$       c)  $i_b = 90^\circ$       d)  $0^\circ < i_b < 30^\circ$
180. Two coherent sources of light interference and produce fringe pattern on a screen. For central maximum, the phase difference between the two waves will be
- a)  $\pi$       b)  $\frac{3\pi}{2}$       c)  $\frac{\pi}{2}$       d) zero
181. If NAME -12 is coded as MZLD-11 then MBBS-07 is coded as:
- a) LAAR-05      b) NCCT-08      c) NGGT-06      d) LAAR-06
182. In a queue of 25 students A is ranked 20<sup>th</sup> position from left end and B is ranked 10<sup>th</sup> position from right end, How many students are there in between A and B?

183. Find odd one out.



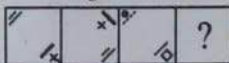
- a) A      b) E      c) C      d) D
184. The ratio of the number of boys and girls in a college is 5 : 7. If the percentage increase in the number of boys and girls be 40% and 50% respectively, what will be the new ratio?
- a) 8 : 9      b) 17 : 18      c) 2 : 3      d) 9 : 10
185. In how many different ways can the letters of the word 'LEADING' be arranged in such a way that the vowels always come together?
- a) 360      b) 480      c) 720      d) 5040
186. Which word does NOT belong with the others?
- a) heading      b) body      c) letter      d) closing
- 187.



Which is the missing tile?



- a) E      b) B      c) D      d) A
188. Look at this series: 31, 29, 24, 22, 17, ... What number should come next?
- a) 15      b) 14      c) 13      d) 12
189. One morning sujata started to walk towards the Sun. After covering some distance she turned to right then again to the right and after covering some distance she again turns to the right. Now in which direction is she facing?
- a) North      b) South      c) North-East      d) South-West
190. complete the given analogy.



(A) (B) (C) (D)

- a) 1      b) 2

Answer Figures:



(1) (2) (3) (4) (5)

- c) 3      d) 4

191. A does 80% of a work in 20 days. He then calls in B and they together finish the remaining work in 3 days. How long B alone would take to do the whole work?
- a) 35      b) 37.5      c) 38      d) 40.5
192. What decimal of an hour is a second ?
- a) 0025      b) 0256      c) 00027      d) 000126
193. On selling 17 balls at Rs. 720, there is a loss equal to the cost price of 5 balls. The cost price of a ball is:

NAME \_\_\_\_\_

a) Rs. 45

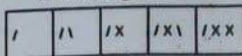
b) Rs. 50

c) Rs. 55

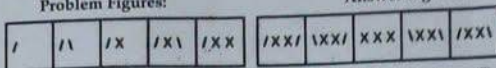
d) Rs. 60

194. Select a figure from amongst the Answer Figures which will continue the same series as established by the five Problem Figures.

Problem Figures:



Answer Figures:



(A)

(B)

(C)

(D)

(E)

(1)

(2)

(3)

(4)

(5)

a) 1

b) 2

c) 3

d) 5

195. Find the number of triangles in the given figure.



a) 16

b) 13

c) 9

d) 7

196. If South-East becomes North, North-East becomes West and so on. What will West become?

a) North-East

b) North-West

c) South-East

d) South-West

197. BEGK : ADFJ :: PSVY : ?

a) ROUX

b) ORUX

c) LQUT

d) LOQT

198. Which of the following diagrams indicates the best relation between Judge, Thieves and Criminals?



a)



b)



c)



d)

199. Which one will replace the question mark?



a) 13

b) 14

c) 12

d) 15

200.

|    |    |    |    |
|----|----|----|----|
| 20 | 22 | 19 | 21 |
| 17 | 19 | 16 | ?  |
| 19 | 21 | ?  | 20 |
| 16 | 18 | 15 | ?  |

Which is the missing section?



A



B



C



D

a) A

b) B

c) C

d) D

Result will be published on Sunday

Log on to [www.name.edu.np](http://www.name.edu.np)

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Best of Luck

**NAME**  
**HINTS AND SOLUTIONS**



**MECEE BL**  
**Model Entrance Exam**  
**(Model- IX)**

**Date: 27 Aashadha 2083, Saturday**

**NAME Institute**  
In front of Singhdurbar, Putalisadak  
Tel: - 01-5971544

## Zoology

1. c) According to the Oparin-Haldane hypothesis, coacervates are the primitive macromolecular aggregates surrounded by a film of water molecules. They exhibited basic metabolic-like properties and are considered precursors to the first living cells.
2. c) Modern Synthetic Theory combines Darwinian natural selection with Mendelian genetics. It states that speciation is primarily driven by genetic variations arising from recombination and mutations, which are then acted upon by natural selection.
3. c) The Jurassic period of the Mesozoic era is characterized by the divergence of birds from theropod reptilian lineages and the rapid proliferation and dominance of early gymnosperms on land.
4. c) A true coelom (eucoelom) is a fluid-filled body cavity that is completely enclosed and lined by the peritoneum, which is tissue derived entirely from the embryonic mesoderm germ layer.
5. b) In Urochordates, the free-swimming larva possesses advanced features like a notochord and dorsal nerve cord. During retrogressive metamorphosis, it loses these traits to become a sessile, simplified adult.
6. b) These structural features form the fundamental diagnostic hallmarks that are universally present in all chordates at some stage of their development.
7. a) True metameric segmentation is a fundamental homologous characteristic shared by both annelids and arthropods. Jointed appendages are unique to arthropods.
8. b) Chloragogen cells in earthworms are involved in glycogen/fat storage, deamination of proteins, and urea synthesis. These metabolic and detoxification roles are directly comparable to the functions of the vertebrate liver.
9. b) The exo-erythrocytic (hypnozoite) stage in the liver is responsible for true relapses and is absent in *Plasmodium falciparum*. It completes its liver phase in one cycle without establishing a dormant liver phase.
10. a) In earthworms (*Pheretima*), blood glands are found in the 4th, 5th, and 6th segments where they manufacture blood corpuscles and hemoglobin.
11. c) While the asexual cycle occurs in humans, the sexual cycle of *Plasmodium* (including fertilization of gametocytes) takes place within the lumen of the female *Anopheles* mosquito's gut.
12. b) Metachrosis is the biological term for the physiological ability of certain animals, like frogs, to change their skin color to blend with their surroundings.
13. b) The small intestine of humans consists of the duodenum, jejunum, and ileum. In frogs, the small intestine is structurally simpler and the jejunum is entirely absent.
14. c) During muscle contraction, actin filaments slide past myosin filaments. Because the A-band corresponds to the total length of the thick myosin filaments, its width remains entirely constant.
15. c) Canaliculi are microscopic, radiating canals crossing the hard bone matrix that house the cytoplasmic extensions of mature osteocytes, allowing them to communicate and exchange nutrients.
16. b) Simple ciliated columnar epithelium features hair-like cilia that move fluids or particles along a tract, lining regions like the fallopian tubes and respiratory bronchioles.
17. d) In peripheral nerve architecture, the correct architectural sequence stretching from the outermost to the innermost layer is the epineurium, perineurium, and endoneurium.

18. b) The first line of defense in innate immunity is composed entirely of physical and chemical barriers that block the initial entry of pathogens into tissues.
19. b)
20. b) IgG is the class of immunoglobulin capable of crossing the placental barrier, while secretory IgA is supplied postnatally through the mother's initial breast milk (colostrum).
21. a) Enzyme-Linked Immunosorbent Assay (ELISA) - screening test
22. a) Diagnostic or confirmatory test - western blot
22. a) The primary potential clinical application of cryopreservation in medical technology is the ultra-low temperature storage of viable embryos for later In-Vitro Fertilization.
23. a) Tissue engineering combines synthetic scaffolds and cellular biology to restore or replace damaged tissues, dynamically applied in modern medicine by culturing functional artificial skin grafts.
24. b) Secondary aquatic animals evolved from terrestrial ancestors. Having pentadactyl limbs modified into flippers is a unique secondary adaptation not seen in primary aquatic animals like fish.
25. b) Klinotaxis involves an organism orienting itself in response to a stimulus by moving its single sensory organ back and forth to compare intensity over time.
26. b) Addison's disease is caused by the hyposecretion of the adrenal cortex, resulting in melanin hyperpigmentation, low blood pressure, low sodium, and high potassium levels.
27. a) Cholecystokinin (CCK) is an enteroendocrine hormone secreted by the duodenal mucosa that triggers contractions in the gallbladder wall to actively release bile into the duodenum.
28. b) Rhodopsin is the photosensitive pigment packed inside rod cells optimized for night vision because it has an incredibly low physiological threshold for light activation.
29. c) The crista ampullaris is a specialized acoustic spot located within the ampulla of each semicircular canal that detects dynamic rotational acceleration.
30. b) Cowper's glands in males secrete an alkaline lubricating fluid, which is functionally and biochemically analogous to the Bartholin's glands located in females.
31. c) A vasectomy involves blocking the vas deferens to stop sperm transport, leaving endocrine Leydig cells completely unaffected as they secrete testosterone directly into the bloodstream.
32. a) On an Electrocardiogram (ECG), the P wave corresponds to the initial electrical impulse spreading across the atrial musculature, signaling atrial depolarization.
33. a) Cardiac output is the total volume of blood pumped out by a ventricle per minute, mathematically calculated by multiplying Stroke Volume by Heart Rate.
34. c) The pia mater is the innermost, highly vascularized meningeal layer that closely adheres to the exact contours, gyri, and sulci of the brain surface.
35. b) The hypothalamus serves as the master control center for vital homeostatic operations, including thermoregulation, osmoregulation, circadian rhythms, and primary control of the anterior pituitary gland.
36. b) During severe dehydration, the hormone Vasopressin (ADH) is released to act directly upon the Distal Convoluted Tubules (DCT) and collecting ducts to maximize water conservation.
37. b) As fluid moves up the water-impermeable ascending limb, NaCl is actively pumped out. Because solute is lost but water cannot follow, the remaining tubular fluid becomes progressively diluted and hypotonic.

38. a) The human permanent dental formula is (32 teeth) while the temporary deciduous milk dentition lacks molars entirely, expressed as (written as or 20 total teeth).
39. b) The larynx (voice box) is the cartilaginous framework of the respiratory tract housing the vocal cords which vibrate mechanically to produce sound.
40. b) According to the Bohr effect, an increase in  $pCO_2$ , high acidity (low pH), and elevated temperature in working skeletal muscle will shift the oxygen-hemoglobin curve to the right to facilitate rapid oxygen unloading.

Botany

41. b) Colocasia (taro or elephant ear) is known for its needle-like calcium oxalate crystals, which are a type of idioblast called raphides. These crystals can cause irritation in the mouth and throat if the plant is not cooked properly.
42. a) In green algae, the photosynthetic pigments are chlorophyll a & d, carotenoids and xanthophylls. Phycoerythrin gives red colour to Rhodophyceae.
43. d) Late blight of potato is caused by *Phytophthora infestans*.
44. c) Bryophytes lack true roots, so they depend on water absorption through surface cells, making them susceptible to desiccation.
45. b)
46. b) Cycas (a gymnosperm) and Adiantum pteridophyte, known as Maiden hair fern resemble each other in having motile sperm. Seeds, cambium are common in gymnosperms and absent in pteridophytes. True vessels are absent in both pteridophytes and gymnosperms.
47. b) Tagetes patula is the French marigold, *Calendula officinalis* is the pot marigold, and *Lycopersicon esculentum* is the former name for the tomato plant (*Solanum lycopersicum*).
48. c) In ferns (Pteridophytes), the archegonium secretes malic acid, which acts as a chemotactic attractant for the antherozoids (male gametes).
49. a) This describes the descending imbricate aestivation, also known as vexillary aestivation. It is a defining feature of the pea family, Leguminosae. The large posterior petal (vexillum) overlaps two smaller lateral petals (alae), which in turn overlap the two smallest anterior fused petals (carinae).
50. c)
51. b) In fermentation, there is no external electron acceptor like oxygen. The cell uses an internal organic molecule (e.g., pyruvate or acetaldehyde) as the electron acceptor.
52. d) Lichens are very good pollution indicators, they do not grow in polluted areas. They are sensitive to sulphur dioxide pollution.
53. d) In Pond ecosystems, the pyramid of biomass is inverted.
54. c)
55. d) Domestic sewage rich in organic waste leads to increase in Biological Oxygen Demand, which leads to decrease D.O. (dissolved oxygen) which leads to death of fish.
56. a) Sal (*Shorea robusta*) forests are the most widespread and dominant forest type in the tropical and sub-tropical regions of Nepal, particularly in the Terai and inner Terai regions.
57. b) Golgi bodies has cis and trans face making it a polar cell organelle.
58. c) During metaphase, sister chromatids align at the equatorial plate, ensuring correct segregation. Replication occurs in interphase, not metaphase.

59. b) Non-disjunction = failure of homologs or chromatids to separate → gametes with abnormal chromosome number (aneuploidy). Polyploidy involves whole sets.
60. b) Plasmodesmata are microscopic channels that pass through the cell walls of adjacent plant cells, connecting their cytoplasm and allowing for the transport of molecules and communication between cells. This forms a continuous cytoplasmic network known as the symplast.
61. b) Chargaff's rule states that  $A = T$  and  $G = C$ . The molar amount of adenine = molar amount of thymine. The molar amount of guanine = molar amount of cytosine. Hence, G is 17%, so, C = 17%, A = 33%, so, T = 33%.
62. a)
63. b) Missense mutation changes a codon to code for a different amino acid, nonsense mutation creates stop codon; silent mutation does not alter amino acid; frameshift mutation alters reading frame.
64. a)
65. d) Cell membrane of *E. coli* (a bacteria) and *Chlamydomonas* (a unicellular green alga) does not differ. As in both cell membrane is composed of lipids and proteins.
66. a) Best stage to count chromosomes is metaphase.
67. d) The statement is false because rRNA is the largest and most abundant type of RNA, making up a significant portion of a ribosome. tRNA is the smallest of the three major types of RNA).
68. a)
69. d) Somatic hybridization is a technique that involves the fusion of protoplasts from two different plant species or varieties. This process creates a hybrid plant with a unique combination of genetic traits from both parents, resulting in a genetically distinct plant.
70. c) *Aspergillus niger*, a common black mold, is used commercially on a large scale for the fermentation process that produces citric acid for the food and pharmaceutical industries.
71. c) Apical meristem is located at the apex of stem, root and branches, and forms axillary and terminal bud. Intercalary meristems take part in linear growth. Lateral meristem occurs parallel to the circumference of the organs. They increase the girth of the plant.
72. a) In a ring girdled plant, roots die first due to damage of phloem causing inhibition of translocation of food.
73. a) Casparian strip regulates water movement into xylem. Lignification occurs in xylem; phloem sieve and root hairs have different functions.
74. a) In Rhodospirillum, electron donor is organic compound during photosynthesis.
75. b) Rubisco oxygenates RuBP, producing 3-phosphoglycerate & 2-phosphoglycolate that is metabolically costly, reducing efficiency.
76. a) When an apical bud is present on a plant, it suppresses the growth of axillary buds, this is called apical dominance. When in pruning apical bud is cut off the axillary buds start growing & hedge become dense.
77. b) Movement of ions against concentration gradient involves the expenditure of higher energy. Diffusion involves the movement of solute particles from region of higher concentration to a region of lower concentration. Pinocytosis is cell drinking. Brownian movement is the random to and fro movement of atoms and molecules.
78. b) In a eukaryotic cell, the complete aerobic respiration of one glucose molecule yields a net of 38 ATP molecules. This includes 2 ATP from glycolysis, 2 from the Krebs

79. a) cycle, and 34 from oxidative phosphorylation. Note: However 2 ATP are lost due to the energy cost of transporting NADH into the mitochondria.  
The wings on the pollen grains of Pinus are a balloon-like outgrowth of the exine, the outer layer of the pollen wall. These wings aid in wind dispersal

80. c)

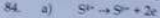
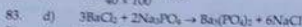
## Chemistry

81. c)  $g \text{ atom of } I = \frac{254}{127} = 2$

$g \text{ atom of oxygen} = \frac{8}{16} = 0.5$

$\therefore \text{Ratio of } g \text{ atoms } I:O = 2:5$

82. b)  $N = \frac{4 \times 1000}{40 \times 100} = 1.0$



85. a) For first order rxn,

Amount left after  $n$  - half lives  $= \left(\frac{1}{2}\right)^n \times \text{initial amount} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$ .

86. a) Since  $CaCO_3$  is solid, its active mass is taken as unity.

87. d) If strong base is present in a solution, then pH is calculated from its concentration,  $\therefore [OH^-] = 10^{-1}$  or  $pOH = 1$

88. b) Salts on hydrolysis forms acids and bases.

89. d) Both have already same pH.

90. c) At boiling point liquid and vapour phase have same kinetic energy  $\left(\frac{3}{2}RT\right)$  but vapour phase has more potential energy.



92. a) Equate charge on both sides.

93. c)  $-9 + 1 + 3.5 = 13.5$

$\therefore 13.5 \text{ g contains } \frac{9}{12} \text{ mole carbon}$

$\therefore 108 \text{ g contains } = \frac{9}{12} \times \frac{108}{13.5} = 6 \text{ mole carbon}$

$\therefore \text{formula is } C_6H_6N_2$

94. c)  $M_{eq} \text{ of } HCl = 100 \times 0.3 = 30$

$M_{eq} \text{ of } H_2SO_4 = 200 \times 0.6 = 120$

$\therefore N_{mixture} = \frac{30 + 120}{300} = \frac{1}{2}$

95. d)  $I_2^-$ ,  $XeF_4$ ,  $Si_4$  and  $ClO_4^-$  have 3, 2, 1, 1 lone pair of electrons respectively.

96. d) Both C and  $N^+$  have six electrons.

97. d) Van der Waals force is weakest force.

98. b) Maximum numbers of electrons in an orbit are given by  $2n^2$ .

99. a) Kjeldahl's method is specifically designed to determine nitrogen content in a substance (Ammonium phosphate contains nitrogen).

100. a) Stronger is acid, more is the nucleophilicity of conjugate base.

101. a) Fe does not form amalgam with mercury.

102. d) Al has more affinity for oxygen.



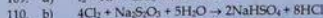
104. a)  $Na_2CO_3$  loses water on standing in air.

105. b) The purest variety of carbon is anthracite.

106. d) Fractional crystallization separates components based on differences in their solubilities.

107. a)  $HNO_3$  is a powerful oxidising agent. It oxidises phosphorous to orthophosphoric acid ( $H_3PO_4$ ).

108. a) Haematite is most abundant ore of iron.

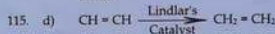


111. b) Bleaching powder ( $CaOCl_2$ ) is a mixed salt because it contains two distinct anions ( $Cl^-$  and  $OCl^-$ ).

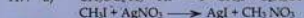
112. c)  $BrCH_2CH_2COOH$  is the weakest acid and have lowest dissociation constant because I.E. of Br is lesser than F and is far away from -COOH group.

113. b) Removal of H from ketone gives resonance stabilized carbanion.

114. b) Lignin is a complex polymer that binds wood fibers together; it must be separated during the pulping and bleaching stages, so the paper doesn't yellow or become brittle.

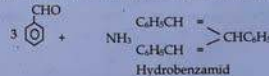


Lindlar's catalyst prevents further reduction of ethene to ethane.



118. a) It is Cannizzaro reaction shown by aldehydes lacking with  $\alpha$ -H atom

119. d)



120. d) It belongs to phthalein dyes.

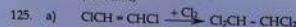
121. b) 3 - Carbamoylpent - 3 - enoic acid.

122. a) Pyrrole is an aromatic heterocyclic compound, since it has 6  $\pi$  electrons.

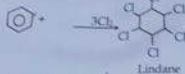


123. a) Due to electron withdrawing nature of -NO<sub>2</sub> group and 'Ortho' effect, o-nitrobenzoic acid has maximum acidic strength.

124. d) Ethyne ( $C_2H_2$ ) and propyne ( $C_3H_4$ ) are members of same homologous series alkynes ( $C_nH_{2n-2}$ ).

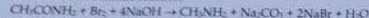


126. b)

127. c) Zinc reacts with very cold and dilute  $\text{HNO}_3$  to produce ammonium nitrate ( $\text{NH}_4\text{NO}_3$ ).

128. d) Ethanal gives both Fehling's test and iodoform test.

129. b) Hoffmann Bromamide reaction.

130. b) Carbon and nitrogen from the organic compound fuse with sodium to form sodium cyanide ( $\text{NaCN}$ ).

Physics

131. b) surface tension  $\left[ \frac{F}{l} \right] = \left[ \frac{N}{m} \right] = \left[ \frac{kg \cdot m \cdot s^{-2}}{m} \right] = \left[ \frac{kg}{s^2} \right] = [E V^{-2} T^{-2}]$ 132. a)  $S_{20} = u + \frac{a}{2}(2t-1) = 0 + \frac{a}{2}(2t-1) = \frac{a}{2}(2t-1)$ 

$$\frac{S_{20}}{S_{10}} = \frac{\frac{a}{2}(2 \times 20 - 1)}{\frac{a}{2}(2 \times 10 - 1)} = \frac{39}{19}$$

133. d)  $(\vec{B} \times \vec{A}) \perp \vec{A}$ 

$$\text{So } (\vec{B} \times \vec{A}) \cdot \vec{A} = 0$$

134. c)  $R_{\text{max}} = \frac{u^2}{g} \Rightarrow 16000 = \frac{u^2}{10} \Rightarrow u = 400 \text{ m/s}$ 135. a)  $\omega = 2\pi f = 2\pi \times \frac{120}{60} \text{ radian/sec} = 4\pi \text{ rad/s}$ 136. d)  $N = m(g+a) = 80(10+5) = 1200 \text{ N}$ 137. b)  $T_{\text{max}} = \frac{mv^2}{r} + mg = \frac{m(\sqrt{7}g)^2}{r} + mg = 8mg$ 138. c)  $m_1 x_1 = m_2 x_2 \Rightarrow 10x_1 = 20(10-x_1) \Rightarrow 30x_1 = 200 \Rightarrow x_1 = 20/3 \text{ cm}$ 139. c)  $U = \frac{1}{2} F \times e = \frac{1}{2} mgl$ 140. b) when angle of contact  $> 90^\circ$  Then liquid doesn't wet solid141. c) Volume of the mercury thermometer changes with rise of temperature and can measure temperature ranging from  $-30^\circ\text{C}$  to  $357^\circ\text{C}$  ( $= 360^\circ\text{C}$ )142. d)  $\frac{dQ}{dt} = \frac{KA\Delta\theta}{d}$ 

$$\Rightarrow K = \frac{\frac{dQ}{dt} d}{A \Delta\theta}$$

$$\Rightarrow \text{unit of } K = \text{Wm}^{-1}\text{K}^{-1}$$

143. a)  $\frac{d\theta}{dt} = K(\theta_{\text{avg}} - \theta_0)$ 

$$\frac{10-40}{5} = K \left( \frac{10+40}{2} - \theta_0 \right) \quad \text{--- (1)}$$

$$\frac{40-54}{5} = K \left( \frac{40+54}{2} - \theta_0 \right) \quad \text{--- (2)}$$

on solving  $\theta_0 = 45^\circ\text{C}$ 

144. b) An adiabatic process occurs without exchange of heat or mass between a thermodynamic system and its surroundings. Therefore, the given process is adiabatic.

145. b)

146. a)  $\eta = 1 - \frac{T_2}{T_1}$  where $T_1$  = temp. of source,  $T_2$  = temp. of sink

147. c) According to kinetic theory of gases, at zero kelvin all molecular motions of a gas stops. Therefore kinetic energy will be zero.

148. a) The phenomena of increase in amplitude when the frequency of driving force is equal to the natural frequency of the oscillator is called resonance. In forced vibration, the resonance wave becomes very sharp when damping force is small (i.e. negligible).

149. b) The standard equation of a wave travelling along +ve x-direction is given by

$$y = A \sin(kx - \omega t)$$

$$y = 1 \sin \left( \frac{2\pi}{\lambda} x - 2\pi f t \right) = \sin \left( \frac{2\pi}{\lambda} x - 2\pi \left( \frac{1}{2} \right) t \right) = \sin(x - 2t)$$

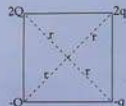
150. d)  $v = \sqrt{\frac{T}{\mu}} \Rightarrow v \propto \sqrt{T} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$ 

151. c) Electric field will be zero at the centre of hollow sphere.

$$\Phi = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

Electric flux does not depend on area of Gaussian surface. Hence  $\Phi$  remains same.

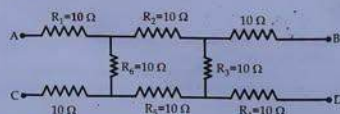
153. a)



$$V_c = 0 \Rightarrow \frac{k(-Q)}{r} + \frac{k(-Q)}{r} + \frac{k(2Q)}{r} + \frac{k(2Q)}{r} = 0 \Rightarrow Q + Q + 0 = 0 \Rightarrow Q = -Q$$

154. a) Energy density  $= \frac{1}{2} \epsilon_0 E^2 = \frac{1}{2} \epsilon_0 \left( \frac{V}{d} \right)^2 = \frac{1}{2} \epsilon_0 \frac{V^2}{d^2}$ 155. a)  $\frac{n}{t} = \frac{10^7}{\text{sec}}, I = \frac{Q}{t} = \frac{nq}{t} = \frac{n}{t} \times e = 10^7 \times 1.6 \times 10^{-19} = 1.6 \times 10^{-12} \text{ A}$ 

156. a)

 $R_2$  &  $R_3$  are in series  $R' = R_2 + R_3 = 10 + 10 = 20\Omega$  $R_5$  &  $R_6$  are in series  $R'' = R_5 + R_6 = 10 + 10 = 20\Omega$  $R'$  &  $R''$  are in parallel  $R''' = \frac{R'R''}{R'+R''} = \frac{20 \times 20}{20+20} = 10\Omega$  $R_{AD} = R_1 + R''' + R_4 = 10 + 10 + 10 = 30\Omega$ 157. d)  $\frac{v_1}{v_2} = \frac{1}{2} \Rightarrow \frac{1.5}{2.5} = \frac{36}{l_2} \Rightarrow l_2 = 36 \times \frac{2.5}{1.5} = 60 \text{ cm}$ 158. d)  $R = \frac{1}{\frac{1}{A} + \frac{\rho l}{A l}} = \frac{\rho l}{A} \Rightarrow R \propto \frac{l}{A}$  $R_1 : R_2 : R_3 = \frac{l_1^2}{A_1} : \frac{l_2^2}{A_2} : \frac{l_3^2}{A_3} = 25 : 3 : \frac{1}{5} = 125 : 15 : 1$ 159. c)  $2 \times 2\pi R = 2\pi R \Rightarrow r = R/2, B_1 = \frac{\mu_0 I}{2R}, B_2 = \frac{\mu_0 (2I)}{2r} = \frac{\mu_0 (2I)}{2(R/2)} = 4 \frac{\mu_0 I}{2R} = 4B_1$

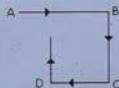
160. a)  $F = \frac{1}{2}mv^2 = \frac{6.6 \times 10^{-34} \times 10^6}{2 \times 9.1} = 2 \times 10^{-22} \text{ N/m}$
161. a) Curie law  $\chi_m \propto \frac{1}{T}$
162. a)  $E = \frac{dq}{dt} = \frac{d(3t^2 + 20 + 10)}{dt} = 10t + 3$   
At  $t = 4 \text{ sec}$ ,  $E = 10 \times 4 + 3 = 43 \text{ Volt}$
163. c)  $X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC}$   
 $X_C' = \frac{1}{2\pi f' C} = \frac{1}{2\pi(20)(20)} = \frac{1}{4(2\pi \times 20)} = \frac{1}{4} \times \frac{1}{2\pi \times 20}$   
 $= \frac{1}{4} X_C = \frac{X_C}{4}$
164. c)  $eV = \text{Energy of electron. For minimum wavelength, maximum loss of energy}$   
 $eV = \frac{hc}{\lambda} \Rightarrow \lambda \propto \frac{1}{V}$
165. a)  $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n \Rightarrow \frac{1}{16} = \left(\frac{1}{2}\right)^n$   
 $\Rightarrow \left(\frac{1}{2}\right)^4 = \left(\frac{1}{2}\right)^n \Rightarrow n = 4, t = nT_{1/2} = 4 \times 20 = 80 \text{ minutes}$
166. b) Incident energy = 2.20 eV. If  $\phi_0 < 2.20 \text{ eV}$  electron will emit.  
If  $\phi_0 > 2.20 \text{ eV}$  No electron emission, only Caesium will emit electron.
167. d)
168. a)  $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meE}} = \frac{h}{\sqrt{2m \times 2T}} = \frac{h}{\sqrt{4mT}}$
169. c)  $\frac{1}{2}mv^2 = 10 \text{ eV} = 10 \times 1.6 \times 10^{-19} \text{ J}$   
 $\Rightarrow v = \sqrt{\frac{2 \times 10 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}} = \sqrt{\frac{20 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}} = 1.87 \times 10^6 \text{ m/s}$   
 $r = \frac{mv}{eB} = \frac{9.1 \times 10^{-31} \times 1.87 \times 10^6}{1.6 \times 10^{-19} \times 10^{-4}} = 0.1 \text{ m} = 10 \text{ cm}$   
 $E_K = -T, E = -(-3.4) = +3.4 \text{ eV}$
170. d)  $\rho = \frac{\text{mass of nucleus}}{\text{volume of nucleus}}$   
 $\rho = \frac{mA}{\frac{4}{3}\pi R^3}$   
 $\rho = \frac{1}{\frac{4}{3}\pi (R_0 A^{1/3})^3}$   
 $= \frac{3m}{4\pi R_0^3}$
- Hence Nuclear density is independent of mass number.
172. b) The most penetrating radiation is  $\gamma$ -rays.
173. c) For insulators and for semiconductors temperature coefficient of resistance is negative but for metals temperature coefficient of resistance is positive.
174. b)
175. a)
176. c)  $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{1.5} = \frac{1}{f} \Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{1.5} = \frac{0.5}{1.5f} = \frac{1}{3f} \Rightarrow v = 3f$
177. b)  $\mu = \frac{c}{v} = \frac{3 \times 10^8}{2 \times 10^8} = 1.5$
178. b)  $\mu_{\text{liquid}} = \mu_{\text{glass}} = \frac{1}{2 \times 10^8 \times 10000 \times 10^{-10}} = 1$   
 $\Rightarrow f_{\text{liquid}} = \infty$   
 $\tan i_p = \frac{22}{21}$   
 $\mu_2 > 1 \Rightarrow \tan i_p > 1$   
 $\therefore 90^\circ > i_p > 45^\circ$
180. d)  $l = l_1 + l_2 + 2\sqrt{l_1 l_2} \cos \phi \Rightarrow l_{\text{max}} = (\sqrt{l_1} + \sqrt{l_2})^2$

$\phi = 0$

Phase difference between the two waves will be zero for central maximum.

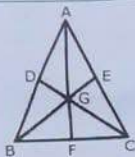
MAT

181. d) Each letter and number are decreased by 1.
182. b)  $(20 + 10 - 25 - 2 = 3)$
183. c) In all the others the black circle is connected to three white circles. In C it is only connected to two white circles
184. c) Originally, let the number of boys and girls in the college be  $5x$  and  $7x$  respectively. Their increased number is  $(140\% \text{ of } 5x)$  and  $(150\% \text{ of } 7x)$   
 $\Rightarrow \frac{140}{100} \times 5x + \frac{150}{100} \times 7x = 2:3$
185. c) The word LEADING has 7 different letters. When the vowels EAI are always together, they can be supposed to form one letter. Then, we have to arrange the letters LNDG (EAI). Now,  $5(4 + 1 = 5)$  letters can be arranged in  $5! = 120$  ways. The vowels (EAI) can be arranged among themselves in  $3! = 6$  ways.  
 $\therefore$  Required number of ways =  $(120 \times 6) = 720$ .
186. c) Heading, body, and closing are all parts of a letter; the letter is the whole, not a part.
187. b) In each line and column, each of the three rings is shaded black once
188. a) This is a simple alternating subtraction series, which subtracts 2, then 5.
189. a)



Hence finally Sujata will face towards North.

190. b) The combination of two symbols placed at the lower-right corner, rotates 90°-CW and moves to the Upper-right corner. Also, the combination of two symbols placed at the upper-left corner, moves to the lower-right corner.
191. b) Whole work is done by A in  $20 \times \frac{1}{4} = 25$  days  
Now,  $1 - \frac{4}{25} = \frac{21}{25}$  work is done by A and B in 3 days  
Whole work will be done by A and B in  $(3 \times 5) = 15$  days.  
A's 1 day's =  $\frac{1}{25}$  (A + B)'s 1 day's work =  $\frac{1}{15}$   
 $\therefore B$ 's 1 day's work =  $\frac{1}{15} - \frac{1}{25} = \frac{4}{75}$   
So, B alone would do the work in  $\frac{75}{4} = 18 \frac{3}{4}$  days.
192. c) Required decimal =  $\frac{1}{60 \times 60} = \frac{1}{3600} = .00027$
193. d) (C.P. of 17 balls) - (S.P. of 17 balls) = (C.P. of 5 balls)  
 $\Rightarrow$  C.P. of 12 balls = S.P. of 17 balls = Rs. 720.  
 $\Rightarrow$  C.P. of 1 ball = Rs.  $\frac{720}{12} = \text{Rs. } 60$
194. d) One extra line is added in each step in a set order
195. a) The figure may be labelled as shown.

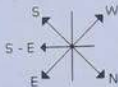
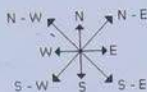


The simplest triangles are AGE, EGC, GFC, BGF, DGB and ADG i.e. 6 in number.  
The triangles composed of two components each are AGC, BGC and ABG i.e. 3 in number.

The triangles composed of three components each are AFC, BEC, BDC, ABF, ABE and DAC i.e. 6 in number.

There is only one triangle i.e. ABC composed of six components.  
Thus, there are  $6 + 3 + 6 + 1 = 16$  triangles in the given figure.

196. c)



It is clear from the diagrams that new name of West will become South-East.

197. b)

As

B  $\xrightarrow{-1}$  A

E  $\xrightarrow{-1}$  D

G  $\xrightarrow{-1}$  F

K  $\xrightarrow{-1}$  J

Similarly

P  $\xrightarrow{-1}$  Q

S  $\xrightarrow{-1}$  R

V  $\xrightarrow{-1}$  U

Y  $\xrightarrow{-1}$  X

198. b) All the thieves are criminals while judge is different from these.

199. d) Sum of numbers in lower half of the circle =  $11 + 9 + 3 + 7 = 30$

Sum of numbers in upper half of the circle =  $? + 4 + 5 + 6 = ? + 15$

Upper half = Lower half

$30 = ? + 15$

$? = 30 - 15$

$? = 15$ .

Therefore, 15 is the answer.

200. b) Line across proceeds +2, -3, +2 and line down proceed -3, +2, -3.

Result will be published on Sunday

Best of Luck

# **NAME**

# **B**

**NAME Institute**

In front of Singhdurbar, Putalisadak

Tel: - 01- 5971544

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# **MECEE BL**

## **Model Entrance Exam**

### **(Model: II)**

**Date: Jestha 09<sup>th</sup> 2083 (Saturday)**

**Venue: NAME**

**Duration - 3 hrs**

**Time - 11:00 AM**

#### **INSTRUCTIONS**

There are 200 multiple-choice questions, each having four choices of which only one choice is correct.

Fill ☐ the most appropriate one.

101. a) Ethylamide      b) Ethylamine      c) Methylamine      d) Acetamide
125. Which is a naturally occurring polymer?  
a) Polythene      b) PVC      c) Teflon      d) Protein
126. Which of the following gases is the primary contributor to the "Greenhouse effect"?  
a)  $\text{CO}_2$       b)  $\text{CH}_4$       c)  $\text{O}_3$       d)  $\text{N}_2\text{O}$
127. A broad spectrum antibiotic is  
a) Paracetamol      b) Penicillin      c) Aspirin      d) Chloramphenicol
128. Minamata disease is caused by water pollution from which heavy metal?  
a) Lead      b) Cadmium      c) Mercury      d) Arsenic
129. Paper chromatography is a classic example of  
a) Column chromatography      b) Adsorption chromatography  
c) Thin-layer chromatography      d) Partition chromatography
130. Silver halide which is readily soluble in  $\text{HH}_4\text{OH}$  is  
a)  $\text{AgF}$       b)  $\text{AgBr}$       c)  $\text{AgCl}$       d)  $\text{AgI}$
131. The unit of surface tension is  
a)  $\text{Nm}^{-1}$       b)  $\text{Nm}^{-2}$       c)  $\text{N}^2\text{m}^{-1}$       d)  $\text{Nm}^{-3}$
132. The displacement-time graphs for two particles A and B are straight lines inclined at angles of  $30^\circ$  and  $60^\circ$  with the time axis. The ratio of velocities of  $v_A : v_B$  is  
a) 1:2      b)  $1:\sqrt{3}$       c)  $\sqrt{3}:1$       d) 1:3
133. A cold soft drink is kept on the balance. When the cap is open, the weight  
a) increases      b) decreases  
c) first increases then decreases      d) remains the same
134. Two equal masses  $m_1$  and  $m_2$  moving along the same straight line with velocities +3 m/s and -5 m/s respectively collide elastically. Their velocities after the collision will be respectively  
a) +4 m/s for both      b) -3 m/s and +5 m/s      c) -4 m/s and +4 m/s      d) -5 m/s and +3 m/s
135. In the case of uniform circular motion, which of the following physical quantities does not remain constant?  
a) Speed      b) Momentum      c) Kinetic energy      d) Mass
136. The radius of gyration is  
a)  $\frac{I}{M}$       b)  $\sqrt{\frac{I}{M}}$       c)  $\sqrt{\frac{M}{I}}$       d)  $\frac{I^2}{M}$
137. There are two bodies of masses 100 kg and 10000 kg separated by a distance of 1m. At what distance from the smaller body will the intensity of the gravitational field be zero  
a)  $\frac{1}{9}$  m      b)  $\frac{1}{10}$  m      c)  $\frac{1}{11}$  m      d)  $\frac{10}{11}$  m
138. The period of a simple pendulum is measured as 'T' in a stationary lift. If the lift moves upwards with an acceleration of 5 g, the period will be  
a) The same      b) Increased by  $\frac{3}{5}$   
c) Decreased by  $\frac{2}{3}$  times      d)  $\frac{T}{\sqrt{6}}$
139. The spherical shape of raindrop is due to  
a) Density of the liquid      b) Surface tension  
c) Atmospheric pressure      d) Gravity

15. During the development of a frog, opposite to the point of entry of the sperm into the egg during fertilization
- a) Establishes the axis of cleavage      b) Forms the centre of rotation of the egg  
c) forms the gray crescent      d) forms the dorsal lip of the blastopore
16. PAP smear examination is used for the detection of:
- a) Breast cancer      b) Cervical cancer      c) lung cancer      d) Oral cavity cancer
17. Which of the following diseases is characterized by fever, delirium, slow pulse, abdominal tenderness and rose-coloured skin rashes?
- a) Measles      b) Tetanus      c) Typhoid      d) Chickenpox.
18. Transmission of Herpes simplex virus to a newborn most commonly occurs
- a) During the birth process      b) From contaminated formula  
c) Through airborne droplets from the mother      d) From infected toys
19. Which of the following drugs is used in the treatment Ascariasis?
- a) Metronidazole      b) Zidovudine      c) Albendazole      d) Isoniazid
20. A pregnant woman who has done amniocentesis test find an extra Barr body in her embryo. The syndrome which is likely to be associated with the embryo as
- a) Down syndrome      b) klinefelter syndrome      c) Patau syndrome      d) edward syndrome
21. Chlorination of water is mainly done at
- a) Increase taste      b) Remove salt      c) kill microorganism      d) Increase oxygen level.
22. Ramsar sites are related to conservation of
- a) Forests      b) Mountain      c) Deserts      d) Wetlands
23. A fish travels a long distance in large rivers for spawning is called
- a) Anadromous migration      b) Catadromous migration  
c) Potamodromous migration      d) Oceanodromous migration.
24. Blood vessels near a wound dilate and become more permeable in response to which substance released from damaged cells?
- a) Pyrogens      b) Antibodies      c) Histamine      d) Interferon.
25. Which of the following are non-excitabile cells found in nervous tissue?
- a) Dendrites      b) Glial cells      c) Nissl bodies      d) Kupffer cells.
26. The protein deposited in the dead superficial cells of the skin that makes the epithelium impervious to water is
- a) Mucus      b) Elastin      c) Collagen      d) keratin
27. If the chyme of a person who has consumed only starch as food is analyzed before it enters the duodenum, which of the following substances will be present?
- a) Maltose and glucose      b) Dextrin and Maltose  
c) Starch, dextrin and maltose      d) Starch, dextrin and glucose
28. The normal volume of air inhaled or exhaled during quiet breathing is called
- a) Tidal volume (TV)      b) Vital capacity (VC)  
c) Expiratory Reserve volume (ERV)      d) Dead Air space.
29. Which of the following gases forms the most stable combination with haemoglobin in RBC?
- a) CO<sub>2</sub>      b) CO      c) O<sub>2</sub>      d) N<sub>2</sub>
30. A healthy person notices that their urine has a pale yellow colour during a routine health check-up. Which pigment is mainly responsible for this yellow colour of urine?

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**Note: NAME students must fill their registration number on the OMR sheet.**

1. Which one of the following is likely to have been absent in free form at the time of origin of life?  
a) Oxygen ( $O_2$ )      b) Methane ( $CH_4$ )      c) Hydrogen ( $H_2$ )      d) Ammonia ( $NH_3$ )
2. The most obvious and consistent differences between Old world monkey and New world monkey lies in the form of  
a) Hands      b) legs      c) Nose      d) Colour
3. Which one of the following doesn't confirm to the theory of "Biogenesis"?  
a) Francesco Redi's experiment      b) Spallanzani's experiment  
c) Louis pasteur's experiments      d) Van Helmont's experiment
4. Uric acid is the main nitrogenous waste in  
a) Fish      b) birds      c) amphibian      d) mammals
5. Which one of the following animals belongs to phylum Echinodermata  
a) Sea Urchin      b) sea horse      c) sea hare      d) Sea mouse
6. In most simple type of canal system of porifera, water flows through which one of the following ways?  
a) spongocoel → Ostia → Osculum → Exterior  
b) Ostia → Spongocoel → Osculum → Exterior  
c) Osculum → Ostia → spongocoel → Exterior  
d) Osculum → Spongocoel → Ostia → Exterior
7. The free swimming larval stage of Cnidarians is  
a) Blastula      b) Medusa      c) Polyp      d) Planula
8. *Fasciola hepatica* occurs in  
a) Bile duct of sheep      b) Liver of Sheep      c) Liver of man      d) Human bile ducts.
9. In many arthropods, the head and thorax are fused to form a cephalothorax. In which of the following classes is the body distinctly divided into head, thorax and abdomen  
a) Insecta      b) Crustacea      c) Myriapoda      d) Arachnida & Crustacea
10. The motile stage of zygote formed by fertilization of macrogamete by a microgamete in plasmodium is called  
a) sporozoite      b) Ookinete      c) Oocyst      d) Gamont
11. Malarial parasites are best detected in the blood of a patient  
a) One hour before the rise of temperature      b) At the time of rise of temperature with rigor  
c) At any time      d) Five hours after the temperature becomes normal
12. The intestine of an earthworm contains an internal fold like structure called the typhlosole. What is the main function of the typhlosole?  
a) Emulsification of food      b) Destruction of bacteria  
c) Increasing the absorptive surface area      d) Secretion of digestive enzyme
13. In an Earthworm, which glands open into the genital papillae?  
a) Prostate gland      b) Accessory gland      c) Cowper's gland      d) Albumin gland
14. The shedding of the outermost layer of the skin, the stratum corneum is called  
a) Regeneration      b) Autolysis      c) Ecdysis      d) Autotomy

NAME \_\_\_\_\_

31. **Columns of Bertini in the kidneys of mammals are formed by the extension of which part?**  
 a) Biliverdin      b) Urochrome      c) Stercobilin      d) Bilirubin
32. **The groove that externally separates atria from ventricles in the human heart is called**  
 a) Cortex into medulla      b) Cortex into pelvis  
 c) Medulla into pelvis      d) pelvis into ureter
33. **A fall in blood pressure due to loss of blood is quickly compensated by**  
 a) Interventricular sulcus      b) Interatrial sulcus  
 c) Coronary sulcus      d) Fossa ovalis.
34. **The "defensive castle" of the brain associated with anger and rage is**  
 a) Thalamus      b) Amygdala      c) Hypothalamus      d) Hippocampus
35. **Which cranial nerve is the longest and most branched?**  
 a) Trigeminal nerve      b) Trochlear nerve      c) Vagus nerve      d) Facial nerve.
36. **Which endocrine gland stores its secretion in the extra-cellular space before discharging it into the blood.**  
 a) Adrenal      b) Thyroid      c) Pancreas      d) Testis.
37. **The eye ball is moved in the orbit by**  
 a) The ciliary muscles  
 b) The ciliary body along with suspensory ligaments  
 c) 4 rectus and 2 oblique muscles  
 d) 3 circular muscles and 4 oblique muscles
38. **Scala vestibuli and Scala tympani are filled with**  
 a) Endolymph      b) Perilymph      c) Lymph      d) Haemolymph
39. **Hormone responsible for the attachment of embryo to the wall of uterus is known as**  
 a) Estrogen      b) Progesterone      c) Relaxin      d) Epinephrine
40. **How many sperms and ova are produced from 25 primary spermatocytes and 25 primary oocytes**  
 a) 100 sperms and 100 ova      b) 100 sperms and 25 ova  
 c) 100 sperms and 50 ova      d) 50 sperms and 25 ova.
41. **The transfer of a pollen grain from an anther to the stigma of a different flower of the same plant is:**  
 a) Ecologically cross-pollination and genetically self-pollination  
 b) Ecologically self-pollination and genetically cross-pollination  
 c) Both ecologically and genetically self-pollination  
 d) Both ecologically and genetically cross-pollination
42. **Which of the following represents the male gametophyte in higher plants?**  
 a) microspore      b) microsporangium with microspore  
 c) pollen grain only      d) pollen grain with tube
43. **The Establishment of a plant in a new area during secondary succession is known as:**  
 a) migration      b) Ecesis      c) Invasion      d) Aggregation
44. **plants that grow on cold soil are:**  
 a) psammophytes      b) oxylophytes      c) chasmophytes      d) psychrophytes.

45. Ecosystem is mainly concerned with energy flow and recycling of matter, in which the flow of energy is:  
 a) unidirectional      b) Bidirectional      c) multidirectional      d) polydirectional
46. The rate at which light energy is converted into chemical energy of organic molecules in the ecosystem is:  
 a) net primary productivity      b) gross primary productivity  
 c) net secondary productivity      d) gross secondary productivity
47. Which one of the following is monosaccharide?  
 a) Maltose      b) Lactose      c) Glucose      d) Starch
48. The utilization of living organism for producing materials useful to human society is known as:  
 a) Applied biology      b) Biotechnology  
 c) Biomedical engineering      d) Genetic engineering
49. Which among the following is preferred to obtain virus-free plants  
 a) Grafting      b) Root tip culture      c) shoot tip culture      d) Micropropagation
50. Emasculation is the process of removal of stamens or anthers without affecting the female reproductive structures before maturity to prevent:  
 a) self-pollination      b) cross pollination      c) Fertilization      d) Germination
51. pyruvic acid, the key product of glycolysis, can have many metabolic fates under aerobic conditions, it forms:  
 a) lactic acid      b)  $\text{CO}_2 + \text{H}_2\text{O}$       c) Acetyl CoA      d) ethanol +  $\text{CO}_2$
52. splitting of water is associated with:  
 a) photosystem I      b) lumen of the thylakoid  
 c) Both photosystem I and II membrane      d) Inner surface of thylakoid.
53. chemoautotrophs obtain energy for photosynthesis by the oxidation of:  
 a)  $\text{H}_2\text{S}$       b) S      c)  $\text{N}_2$       d)  $\text{CO}_2$
54. Which of the following maintains the shape of the cell?  
 a) osmotic pressure      b) wall pressure      c) turgor pressure      d) osmosis
55. Application of phenyl mercuric acetate acts as a:  
 a) weedicide      b) Antitranspirant      c) Hormone      d) Growth regulator.
56. The secondary growth in thickness of plants is brought about initially by the division of the:  
 a) medullary ray      b) xylem      c) pericycle      d) cambium
57. Which of the following plant hormones is a growth inhibitor?  
 a) Ethylene      b) Auxin      c) Gibberellin      d) Cytokinin
58. The major event that occurs during the anaphase of mitosis, which brings about the equal distribution of chromosomes, is:  
 a) splitting of the chromatids      b) splitting of the centromeres  
 c) condensation of the chromatin      d) Replication of the genetic material
59. Terminalization occurs in:  
 a) Diakinesis      b) Cytokinesis      c) leptotene      d) pachytene
60. In meiosis, a bivalent is attached to the spindle fibers at:  
 a) chromatid      b) centromere      c) chromonema      d) chromere
61. synthesis of ATP in mitochondria takes place:

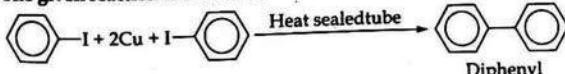
- a) upon cristae      b) In matrix      c) In intracristal space      d) At outer membrane
62. Glyoxysomes are concerned with the metabolism of:  
a) proteins      b) carbohydrates      c) fats      d) All of these
63. The short segment of genetic material having ability to synthesize a polypeptide is called as:  
a) DNA      b) Allele      c) Genome      d) Gene
64. Which of the following individuals is not pure but a hybrid individual for a given character?  
a) Homozygous dominant      b) Homozygous recessive  
c) Heterozygous individual      d) Hemizygous individual
65. In the  $F_2$  generation, if both parental characters are seen along with an intermediate phenotype, then it is known as:  
a) Incomplete dominance      b) co-dominance  
c) sex-linked      d) linkage
66. The enzyme required for removing RNA primer during DNA replication is:  
a) ligase      b) DNA polymerase II  
c) DNA polymerase I      d) RNA primase
67. In a human syndrome, in which there is an XXY chromosome in the body:  
a) Turner's syndrome      b) Klinefelter's syndrome  
c) Down's syndrome      d) super male
68. The strand of DNA on which the synthesis of DNA occurs discontinuously is called:  
a) leading strand      b) discontinuous strand  
c) lagging strand      d) complementary strand
69. When a pyrimidine is replaced by purine, it is called:  
a) Frame shift mutation      b) Transition  
c) Self duplication      d) Transversion
70. Doctors advise athletes to eat Carbohydrate-rich food before competition because  
a) Repair tissue      b) Provide quick energy  
c) Produce enzymes      d) Prevent blood clotting
71. A scientist rings a plant stem by removing bark and phloem. Food transport to roots stops because  
a) Xylem is removed      b) Cambium is destroyed  
c) Phloem conducts food      d) Cortex stores starch
72. Algae used for the construction of sound proof room is:  
a) diatoms      b) volvox      c) Ulva      d) Chara
73. A group of algal cells imprisoned in fungal mycelium serving for vegetative multiplication of a lichen is called:  
a) Isidium      b) soredium      c) cephalodium      d) Helotism
74. In Funaria, the leaves surrounding the archegonial clusters are called:  
a) perichaetial leaves      b) perigonal leaves      c) paraphyses      d) foliage leaves
75. The dominant phase in the lifecycle of the pteridophytes is the sporophyte, which is:  
a) Haploid      b) Diploid  
c) multicellular and Independent      d) photosynthetic and long lived
76. The antherozoids of the cycas are:  
a) uniciliated      b) Biciliated      c) multiciliated      d) Absent

77. wings in pinus seeds develop from:  
 a) Bract scales      b) cone axis      c) seed coat      d) ovuliferous scale
78. vegetative reproduction occurs by bulbil in:  
 a) climbing rose      b) Agave      c) mint      d) citrus
79. The edible part of the litchi fruit is:  
 a) Endosperm      b) cotyledons      c) juicy testa      d) Fleshy aril
80. Bicarpellary, syncarpous gynoecium with bilocular ovary axile swollen placenta and oblique septum occurs in the family:  
 a) cruciferae      b) solanaceae      c) Liliaceae      d) compositae
81. One gram mole of a gas at NTP occupies 22.4 litre. This fact was derived from:  
 a) Law of gaseous volumes.      b) Avogadro's hypothesis.  
 c) Berzelius hypothesis.      d) Dalton's atomic theory.
82. The equivalent weight of  $H_3PO_3$  is  
 a)  $M/2$       b)  $M/3$       c)  $M/6$       d)  $M$
83. Which of the following set of quantum numbers is not possible?  
 a)  $n=2, \ell=1, m=0, s=+1/2$       b)  $n=2, \ell=2, m=+1, s=-1/2$   
 c)  $n=2, \ell=1, m=-1, s=+1/2$       d)  $n=2, \ell=1, m=0, s=-1/2$
84. The bond angle in given compound is



- a)  $109.5^\circ$       b)  $117^\circ$       c)  $115^\circ$       d)  $120^\circ$
85. Which type of bond are present in  $NH_4Cl$  and  $NaCN$  molecules?  
 a) Only ionic      b) Only Co-ordinate  
 c) Only ionic and Covalent      d) Ionic, Covalent and Co-ordinate
86. What is the oxidation state of central element in oleum compound?  
 a) +2      b) +5      c) +6      d) Zero.
87. How many mL of 1 M  $H_2SO_4$  solution is required to neutralize 10 mL of 1 M NaOH Solution?  
 a) 2.5 mL      b) 5.0 mL      c) 10.0 mL      d) 20.0 mL
88. The Crystal for which  $a=b \neq c$  and  $\alpha=\beta=90^\circ$  and  $\gamma=120^\circ$  is  
 a) Cubic System      b) Triclinic System.      c) Hexagonal System      d) Monoclinic System.
89. For a reversible reaction, if concentration of the reactants is doubled, the equilibrium constant will be  
 a) one-fourth      b) Halved      c) Doubled      d) The Same
90. The equilibrium constant,  $K_c$  for the reaction:  $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$  at temperature 'T' is  $4 \times 10^{-4}$ . The value of  $K_c$  of the reaction:  $NO(g) \rightleftharpoons \frac{1}{2} N_2(g) + \frac{1}{2} O_2(g)$  at the same temperature is :  
 a)  $2.5 \times 10^2$       b) 0.02      c)  $4 \times 10^{-4}$       d) 50.
91. Nuclear Reactor is based on  
 a) Nuclear fission      b) Nuclear fusion  
 c) Natural radioactivity      d) Transmutation.
92. Which indicator is used in titration of strong base and weak acid?  
 a) Methyl orange      b) Methyl red      c) Phenolphthalein      d) Starch.

93. If 0.023 gm of Na is dissolved in 100 cm<sup>3</sup> of water then resulting pH of solution is  
a) 12                      b) 14                      c) 8                      d) 10.
94. Which of the following is not used in salt bridge?  
a) Na<sub>2</sub>CO<sub>3</sub>                      b) NH<sub>4</sub>Cl                      c) NH<sub>4</sub>NO<sub>3</sub>                      d) KCl
95. The mixing of gases is generally accompanied by  
a) Change in heat content                      b) Decrease in entropy  
c) Decrease in free energy                      d) Increase in free energy.
96. If  $\Delta H$  and  $\Delta S$  values of the reactants X, Y and Z are given below:  
X:  $\Delta H = 10.5 \text{ kJ}$ ,  $\Delta S = 30 \text{ Jk}^{-1}$   
Y:  $\Delta H = 1.08 \text{ kJ}$ ,  $\Delta S = -100 \text{ Jk}^{-1}$   
Z:  $\Delta H = -11.7 \text{ kJ}$ ,  $\Delta S = -100 \text{ Jk}^{-1}$   
The reaction that will be spontaneous at 300K is  
a) X only                      b) Z only  
c) Y only                      d) All reactions are non-spontaneous
97. Which of the following is strongest acid?  
a) H<sub>2</sub>SO<sub>4</sub>                      b) H<sub>3</sub>PO<sub>4</sub>                      c) HNO<sub>3</sub>                      d) HClO<sub>4</sub>
98. Which of the following is largest ion?  
a) Mg<sup>2+</sup>                      b) O<sup>2-</sup>                      c) Na<sup>+</sup>                      d) Ne
99. Formaldehyde is produced when atomic hydrogen reacts with.  
a) CO                      b) CO<sub>2</sub>                      c) HCN                      d) HCOOH
100. Which of the following is extracted by zone refining process?  
a) Si                      b) V                      c) Ti                      d) C
101. In Solvay process brine solution is saturated with ammonia and excess of CO<sub>2</sub> to get -  
a) NaHCO<sub>3</sub>                      b) NH<sub>4</sub>HCO<sub>3</sub>                      c) NH<sub>4</sub>Cl                      d) Na<sub>2</sub>CO<sub>3</sub>
102. Bachmann's plant is used in the manufacture of  
a) CaOCl<sub>2</sub>                      b) NaOH                      c) Epsom salt                      d) Chlorine
103. The cementing property of cement is due to?  
a) Lime                      b) Silica                      c) Iron oxide                      d) Alumina
104. Which allotrope of Carbon is used in Superconductor?  
a) Diamond                      b) Fullerene                      c) Graphite                      d) Anthracite
105. Ammonia react with Nessler's reagent to give  
a) A violet precipitate                      b) A deep blue precipitate.  
c) A white precipitate.                      d) A brown precipitate.
106. The formula of Marshall's Acid is  
a) H<sub>2</sub>SO<sub>5</sub>                      b) H<sub>2</sub>S<sub>2</sub>O<sub>8</sub>                      c) H<sub>2</sub>S<sub>2</sub>O<sub>7</sub>                      d) CuCO<sub>3</sub>
107. What is the correct sequence of boiling point?  
a) HF > HCl > HBr > HI                      b) HF > HI > HBr > HCl  
c) HI > HCl > HBr > HF                      d) HI > HBr > HCl > HF
108. A blue colour is obtained when a copper wire is immersed in silver nitrate solution. It is due to the formation of:  
a) Ag<sup>+</sup>                      b) Cu<sup>+</sup>                      c) Cu<sup>2+</sup>                      d) A<sup>+</sup> and Cu<sup>+</sup>
109. Which of the following does not give Lassaigne's test?

- a) Azobenzene      b) Glycine      c) Phenylhydrazine      d) Hydrazine
110. In the given structure, number of  $\sigma$  and  $\pi$  bonds are:  $\text{C}(\text{CN})_2 = \text{C}(\text{CN})_2$   
 a) 9 $\sigma$  and 9 $\pi$       b) 9 $\sigma$  and 11 $\pi$       c) 11 $\sigma$  and 9 $\pi$       d) 11 $\sigma$  and 11 $\pi$
111. The IUPAC name of  $(\text{CH}_3)_3\text{C} - \text{CH}(\text{OH}) - \text{CH}_3$  is  
 a) 3,3-dimethyl-2-butanol      b) 1,2-dimethyl-2-butanol  
 c) 1,2-dimethyl-4-propanol      d) 1,1,1-trimethyl-2-propanol
112. Which of the following is optically inactive?  
 a) Meso-compound      b) Enantiomers  
 c) Diastereomers.      d) Compound having one chiral carbon only.
113. Alkynes react with Lindlar's Catalyst to give  
 a) Alkene      b) Cis-alkene      c) Trans-alkene      d) Alkane.
114. Bayer's Reagent is  
 a) Alkaline  $\text{KMnO}_4$       b) Alkaline  $\text{K}_2\text{Cr}_2\text{O}_7$   
 c) Acidified  $\text{K}_2\text{Cr}_2\text{O}_7$       d) Acidified  $\text{KMnO}_4$
115. Which of the following compounds has the highest boiling point?  
 a)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$       b)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$   
 c)  $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{Cl}$       d)  $(\text{CH}_3)_3\text{CCl}$
116. An alkyl halide can be converted into alcohols by  
 a) Addition      b) Elimination      c) Substitution      d) Dehydrogenation.
117. The conversion of glucose or fructose into ethyl alcohol is brought about by  
 a) Invertase      b) Diastase      c) Maltase      d) Zymase
118. Ether is obtained from ethyl alcohol  
 a) In presence of  $\text{H}_2\text{SO}_4$  at 413 K      b) In presence of  $\text{H}_2\text{SO}_4$  at 473 K  
 c) In presence of  $\text{H}_2\text{SO}_4$  at 383 K      d) In presence of  $\text{H}_2\text{SO}_4$  at 273 K.
119. Aldehyde and ketones are distinguished by  
 a) Ammonia      b)  $\text{H}_2\text{SO}_4$  solution.      c) Alkaline  $\text{KMnO}_4$       d) Fehling's solution.
120.  $\text{C}_2\text{H}_5\text{COCl} + \text{H}_2 \xrightarrow{\text{Pd/BaSO}_4} \text{X} + \text{Y}$ . Compound X and Y are  
 a)  $\text{CH}_3\text{CHO}$  and  $\text{HCl}$       b)  $\text{C}_2\text{H}_5\text{CHO}$  and  $\text{HCl}$   
 c)  $\text{CH}_3\text{COCH}_3$  and  $\text{HCl}$       d)  $\text{CH}_3\text{COC}_2\text{H}_5$  and  $\text{HCl}$ .
121. Which one is used as a food preservative?  
 a) Sodium acetate      b) Sodium propionate      c) Sodium benzoate      d) Sodium oxalate
122. The given reaction is known as  
  
 a) Ullmann reaction.      b) Fittig reaction  
 c) Sandmeyer reaction      d) Gattermann reaction.
123. Nitrobenzene can be converted into hydrazobenzene by  
 a) Zn and  $\text{HCl}$       b) Zn and alc.  $\text{NaOH}$   
 c) Zn and aq.  $\text{NaOH}$       d)  $\text{NH}_2\text{NH}_2$  and alc.  $\text{KOH}$
124. Identify the 'C' in the series. Ethylamine  $\xrightarrow{\text{HNO}_2} \text{A} \xrightarrow{\text{PCl}_5} \text{B} \xrightarrow{\text{HNH}_2} \text{C}$

NAME \_\_\_\_\_

140. Two spherical balls of the same material, but with their radii in the ratio 1:2 are falling in a viscous medium with respective terminal velocities. Find the ratio of their momenta.  
 a) 1:2                      b) 1:4                      c) 1:8                      d) 1:32
141. If 'E' is the electric field intensity of an electrostatic field, then the electrostatic energy density is proportional to  
 a) E                      b)  $E^2$                       c)  $1/E^2$                       d)  $E^3$
142. In a parallel-plate capacitor with plate area 'A' and charge 'Q', the force on one plate because of the charge on the other is equal to  
 a)  $\frac{Q^2}{\epsilon_0 A^2}$                       b)  $\frac{Q^2}{2\epsilon_0 A^2}$                       c)  $\frac{Q^2}{\epsilon_0 A}$                       d)  $\frac{Q^2}{2\epsilon_0 A}$
143. A spherical drop of capacitance  $1\mu\text{F}$  is broken into eight drops of equal radius. Then, the capacitance of each small drop is  
 a)  $\frac{1}{8}\mu\text{F}$                       b)  $8\mu\text{F}$                       c)  $\frac{1}{2}\mu\text{F}$                       d)  $\frac{1}{4}\mu\text{F}$
144. The law, governing the force between electric charges is known as  
 a) Ampere's law                      b) Ohm's law                      c) Faraday's law                      d) Coulomb's law
145. Beats are produced by two waves given by  $y_1 = a \sin 2000 \pi t$  and  $y_2 = a \sin 2008 \pi t$ . The number of beats heard per second is  
 a) zero                      b) one                      c) four                      d) eight
146. In a stationary wave the distance between two successive antinodal points is  
 a)  $\frac{\lambda}{4}$                       b)  $\frac{\lambda}{2}$                       c)  $\lambda$                       d) integral multiple of  $\lambda$
147. A potential difference of 'V' is applied at the ends of a copper wire of length 'l' and diameter 'd'. On doubling only 'd', drift velocity  
 a) becomes two-times                      b) becomes half                      c) does not change                      d) becomes one fourth
148. The length of the resistance wire is increased by 10% by stretching. What is the corresponding change in the resistance of the wire?  
 a) 10%                      b) 25%                      c) 21%                      d) 9%
149. Two cells, when connected in series, are balanced on 8 m on a potentiometer. If the cells are connected with polarities of one of the cell is reversed, the balance is 2 m. The ratio of e.m.f. of the two cells is  
 a) 4:1                      b) 5:3                      c) 3:4                      d) 4:3
150. For a given temperature difference, which of the following pairs will generate maximum thermo e.m.f.?  
 a) Antimony-bismuth                      b) Silver-gold                      c) Iron-copper                      d) Lead-nickel
151. If a copper rod carries a direct current, the magnetic field associated with the current will be  
 a) Only inside the rod                      b) Only outside the rod                      c) Both inside and outside the rod                      d) Neither inside nor outside the rod
152. The material used for permanent magnet has  
 a) high retentivity, high coercivity                      b) low retentivity, high coercivity  
 c) low retentivity, low coercivity                      d) high retentivity, low coercivity
153. The magnetic flux linked with a coil is given by an equation  $\phi$  (in weber) =  $8t^2 + 3t + 5$ . The induced e.m.f. in the coil at the fourth second will be  
 a) 16 units                      b) 39 units                      c) 67 units                      d) 145 units

154. The root mean square value of the alternating current is equal to  
a) Twice the peak value  
b) Half the peak value  
c)  $\frac{1}{\sqrt{2}}$  times the peak value  
d) Equal to the peak value
155. Core of transformer is made up of  
a) Soft iron  
b) Steel  
c) Iron  
d) Alnico
156. Mercury boils at  $367^\circ\text{C}$ . However, mercury thermometers are made such that they can measure temperature up to  $500^\circ\text{C}$ . This is done by  
a) Maintaining a vacuum above the mercury column in the stem of the thermometer  
b) Filling nitrogen gas at high pressure above the mercury column  
c) Filling nitrogen gas at low pressure above the mercury level  
d) Filling oxygen gas at high pressure above the mercury column
157. The apparent coefficient of expansion of a liquid when heated in a copper vessel is  $C$ , and when heated in a silver vessel is  $S$ . If  $A$  is the linear coefficient of expansion of copper, then the linear coefficient of expansion of silver is  
a)  $\frac{C+S-3A}{3}$   
b)  $\frac{C+3A-S}{3}$   
c)  $\frac{S+3A-C}{3}$   
d)  $\frac{C+S+3A}{3}$
158. Calorie is defined as the amount of heat required to raise the temperature of 1g of water by  $1^\circ\text{C}$ , and it is defined under which of the following conditions  
a) From  $14.5^\circ\text{C}$  to  $15.5^\circ\text{C}$  at 760 mm of Hg  
b) From  $98.5^\circ\text{C}$  to  $99.5^\circ\text{C}$  at 760 mm of Hg  
c) From  $13.5^\circ\text{C}$  to  $14.5^\circ\text{C}$  at 76 mm of Hg  
d) From  $3.5^\circ\text{C}$  to  $4.5^\circ\text{C}$  at 76 mm of Hg
159. The kinetic theory of gases provides a base for  
a) Charle's law  
b) Boyle's law  
c) Charle's law and Boyle's law  
d) Kirchhoff's law
160. For an adiabatic process, the wrong statement is  
a)  $dQ = 0$   
b)  $dU = -dW$   
c)  $Q = \text{constant}$   
d) Entropy is not constant
161. If the wavelengths of maximum intensity of radiations emitted by the sun and the moon are  $0.5 \times 10^{-6} \text{ m}$  and  $10^{-4} \text{ m}$  respectively, the ratio of their temperature is  
a) 1/100  
b) 1/200  
c) 100  
d) 200
162. The specific heat of a gas in an isothermal process is  
a) Infinite  
b) Zero  
c) Negative  
d) Remains constant
163. When a light wave suffers reflection at the interface from air to glass, the change in phase of the reflected wave is equal to  
a) 0  
b)  $\frac{\pi}{2}$   
c)  $\pi$   
d)  $2\pi$
164. A convex mirror has a focal length ' $f$ '. A real object is placed at a distance ' $f$ ' in front of it from the pole, producing an image at  
a) infinity  
b)  $f$   
c)  $-f/2$   
d)  $2f$
165. The refractive indices of violet and red light are 1.54 and 1.52, respectively. If the angle of the prism is  $10^\circ$ , then the angular dispersion is  
a) 0.02  
b) 0.2  
c) 3.06  
d) 30.6
166. Two coherent monochromatic light beams of intensities  $I$  and  $4I$  are superposed. The maximum and minimum possible intensities in the resulting beam are  
a)  $5I$  and  $I$   
b)  $5I$  and  $3I$   
c)  $9I$  and  $I$   
d)  $9I$  and  $3I$

If the yellow light in the Young's double slit experiment is replaced by red light, the fringe width will

- a) Decrease
- b) Remain unaffected
- c) Increase
- d) First increase and then decrease

168. Diffraction and interference of light suggest

- a) The nature of light is electromagnetic
- b) Wave nature
- c) Nature is quantum
- d) The nature of light is transverse

169. Two nuclei have their mass number in the ratio of 1 : 3. The ratio of their nuclear density would be

- a) 1 : 3
- b) 3 : 1
- c)  $\frac{1}{4} : 1$
- d) 1 : 1

170. If the kinetic energy of a free electron doubles, its de-Broglie wavelength changes by the factor

- a)  $\frac{1}{\sqrt{2}}$
- b)  $\sqrt{2}$
- c)  $\frac{1}{2}$
- d) 2

171. When the light source is kept 20 cm away from a photo cell, a stopping potential of 0.6 V is obtained. When the source is kept 40 cm away, the stopping potential will be

- a) 0.3 V
- b) 0.6 V
- c) 1.2 V
- d) 2.4 V

172. Molybdenum is used as a target element for the production of X-rays because it is

- a) A heavy element that can easily absorb high velocity electrons
- b) A heavy element with a high melting point
- c) An element having low thermal conductivity
- d) Heavy and can easily deflect electrons

173. In a working nuclear reactor, Cadmium rods, (control rods) are used to

- a) Speed up neutrons
- b) Slow down neutrons
- c) Absorb some neutrons
- d) Absorb all neutrons

174. A nucleus of atomic mass A and atomic number Z emits  $\beta$ -particles. The atomic mass and atomic number of the resulting nucleus are

- a) A, Z
- b) A + 1, Z
- c) A, Z + 1
- d) A - 4, Z - 2

175. A sample contains 16 gm of a radioactive material, the half-life of which is two days. After 32 days, the amount of radioactive material left in the sample is

- a) Less than 1 mg
- b)  $\frac{1}{4}$  gm
- c)  $\frac{1}{2}$  gm
- d) 1 gm

176. A piece of copper and the other of germanium are cooled from the room temperature to 80 K, then which of the following would be a correct statement

- a) Resistance of each increases
- b) Resistance of each decreases
- c) Resistance of copper increases while that of germanium decreases
- d) Resistance of copper decreases while that of germanium increases

177. The band gap in Germanium and silicon in eV respectively, is

- a) 0.7, 1.1
- b) 1.1, 0.7
- c) 1.1, 0
- d) 0, 1.1

178. In N-Type semiconductor, the valency of impurity is:

- a) 1
- b) 3
- c) 4
- d) 5

NAME \_\_\_\_\_

179. If A and B are two inputs in an AND gate, then the AND gate has an output of 1 when the values of A and B are  
 a) A = 0, B = 0      b) A = 1, B = 1      c) A = 1, B = 0      d) A = 0, B = 1
180. Which of the following particles is a type of quark?  
 a) Muon      b) Gluon      c) Bottom      d) Tau
181. In a code language, SINGER is written as AIBCED. How can GINGER be written in the same code?  
 a) CBIECD      b) CIBCED      c) CBICED      d) CIBECD
182. Arrange the given words in a logical and meaningful sequence.  
 1. Presentation      2. Recommendation      3. Arrival      4. Discussion  
 5. Introduction  
 a) 5, 3, 4, 1, 2      b) 3, 5, 4, 2, 1      c) 3, 5, 1, 4, 2      d) 5, 3, 1, 2, 4
183. Identify which option correctly replaces the question mark in the given series?  
 100, 50, 52, 26, 28, ?, 16, 8  
 a) 30      b) 36      c) 14      d) 32
184. A man walks 6km towards the North, then turns towards his left and walks for 4 km. He again turns left and walks for 6 km. At this point he turns to his right and walks for 6 km. How many km and in what direction is he from the starting point?  
 a) 10 km and West      b) 6 km and South      c) 4 km and South      d) 8 km and West
185. Eleven students A, B, C, D, E, F, G, H, I, J and K are sitting in the first line facing to the teacher. D who is just to the left of F, is to the right of C at the second place. A is second to the right of E who is at one end. J is the nearest neighbour of A and B and is to the left of G at the third place. H is next to D to the left and is in the third place to the right of I. Who is just in the middle?  
 a) I      b) B      c) J      d) G
186. There are two clocks, both set to show 10 pm on 21<sup>st</sup> January 2010. One clock gains 2 minutes in an hour and the other clock loses 5 minutes in an hour. Then by how many minutes do the two clocks differ at 4 pm on 22<sup>nd</sup> January 2010?  
 a) 126 minutes      b) 136 minutes      c) 96 minutes      d) 106 minutes
187. Pointing to a girl Sandeep said, "She is the daughter of the only sister of my father." How is Sandeep related to the girl?  
 a) Uncle      b) Cousin      c) Father      d) Grandfather
188. Select the option that best replaces the question mark (?).
- |    |    |    |
|----|----|----|
| 16 | 28 | 29 |
| 13 | 12 | 16 |
| 14 | 10 | 15 |
| 15 | 30 | ?  |
- a) 60      b) 30      c) 2      d) 45
189. Seema and Meena divide a sum of Rs. 25000 in the ratio of 3 : 2 respectively. If Rs. 5000 is added to each of their shares, what would be the new ratio formed?  
 a) 2 : 3      b) 3 : 4      c) 5 : 4      d) 4 : 3
190. Two pipes P and Q can fill a tank in 24 and 30 minutes respectively. Both pipes are opened together and the first pipe is closed after 8 minutes. After how many minutes the tank will be filled?  
 a) 20 minutes      b) 18 minutes      c) 22 minutes      d) 24 minutes

191. Kunal bought a suitcase with 15% discount on the labelled price. He sold the suitcase for Rs. 2880 with 20% profit on the labelled price. At what price did he buy the suitcase?  
 a) Rs. 2040                      b) Rs. 2400                      c) Rs. 2604                      d) Rs. 2640
192. The average of 7 consecutive numbers is 20. The largest of these numbers is:  
 a) 20                                  b) 23                                  c) 24                                  d) 26
193. A motorist travels to a place 150 km away at an average speed of 50 km/hr and returns at 30 km/hr. His average speed for the whole journey in km/hr is:  
 a) 35 km/hr                      b) 37 km/hr                      c) 37.5 km/hr                      d) 40 km/hr
194. A statement followed by two conclusions numbered I and II. You must assume the statement to be true, then consider both conclusions and decide which of them logically follows beyond a reasonable doubt based on the given information.

Statements: The serious accident in which a person was run down by a car yesterday had again focused attention on the most unsatisfactory state of roads.

Conclusions:

I. The accident that occurred was fatal.

II. Several accidents have so far taken place because of unsatisfactory state of roads.

- a) If only conclusion I follows                      b) If only conclusion II follows  
 c) If either I or II follows                      d) If both I and II follow

195. A statement followed by two courses of action numbered I and II. You have to assume that everything in the statement is true and on the basis of given information, decide which of them logically follows.

Statement: The car dealer found that there was a tremendous response for the new XYZ's car-booking with long queues of people complaining about the duration of business hours and arrangements.

Courses of Action:

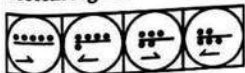
I. People should make their arrangement of lunch and snacks while going for car XYZ's booking and be ready to spend several hours.

II. Arrangement should be made for more booking desks and increased business hours to serve more people in less time.

- a) Only I follows                      b) Only II follows                      c) Either I or II follows                      d) Neither I nor II follows

196. Select the figure from the answer figures that will continue the same series as established by the problem figures.

Problem Figures



1                      2                      3                      4

Answer Figures



A

B

C

D

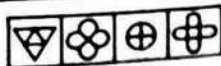
a) A

b) B

c) C

d) D

197. Choose the figure that is different from the rest.



- 1  
2  
3  
4
- a) 1  
b) 2

c) 3

d) 4

198. Find the mirror image.



(N)

a) 1



(1)

(2)

(3)

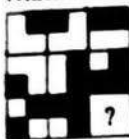
(4)

b) 2

c) 3

d) 4

199. Which one of the following figures will replace the question mark in the figure matrix?



(1) (2) (3) (4)

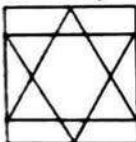
a) 1

b) 2

c) 3

d) 4

200. How many triangles are there in the figure given below?



a) 11

b) 9

c) 12

d) 18

# **NAME**

**NAME Institute**

**In front of Singhdurbar, Putalisadak**

**Tel: - 01- 5971544**

## **HINTS AND SOLUTIONS**



# **MECEE BL**

## **Model Entrance Exam**

**(Model- II)**

**Date: 2083/02/09**

## Zoology

- a) The primitive earth had a reducing atmosphere without free Oxygen ( $O_2$ ). The gases present in primitive atmosphere as  $CH_4$ ,  $NH_3$ ,  $H_2$ ,  $H_2O$  etc.
- c) The major distinguishing feature is the shape of the nose.
- ♦ Old world monkey: Narrow nostril directed downward.
  - ♦ New world monkey: Broad nostril directed sideways.
- d) Theory of biogenesis was proven by three major scientist excluding Van Helmont
- b) Uricotelic animals: Excrete uric acid e.g. most insects, birds, land snail, land reptiles (Lizard & snake).
- ♦ Ureotelic animals: Excrete urea; e.g. Cartilaginous fish, Semi aquatic amphibians (e.g. frog and toad), mammals.
- Note: *Ascaris* and Earthworm are both ammonotelic and ureotelic (More Ureotelic).
- ♦ Ammonotelic animals: Excrete ammonia, eg protozoans, sponges, Cnidarians, Platyhelminthes, annelida, roundworm, bony fish, amphibian tadpoles, crocodiles etc.
  - ♦ Aminotelic: Excrete amino acid, e.g. some molluscs (*Unio*) and some echinoderms (*Asterias*).
  - ♦ Guanotelic: Excrete guanine, e.g. Spiders and scorpions.
- a) Sea urchin- Echinodermata
- ♦ Sea horse (*Hippocampus*) → Pisces.
  - ♦ Sea hare (*Aplysia*) → Mollusca.
  - ♦ Sea mouse (*Aphrodite*) → Annelida.
- b) Ascon type: Most simple and primitive type.
- Flow of water: Water → Ostia → spongocoel → osculum → Exterior. In Ascon type, paragastric cavity is lined by Choanocytes e.g. *Leucosolenia*.
- d) ♦ Ephyra is a larvae of Jellyfish.
- ♦ Planula is free swimming larva, it is common larva of Cnidarians.
- a) *Fasciola hepatica* is a digenetic parasite. The favourite habitat inside the primary host it's liver bile duct and bile passages.
- a) The phylum Arthropoda include animals with segmented bodies and joined appendages. Their body division varies among different classes.
- ♦ Insecta: Body is divided into three distinct parts Head, thorax and Abdomen.
  - ♦ Arachnida (spider, Scorpions) and many Crustacea (crabs, Prawns) have the head and thorax fused together forming a cephalothorax.
  - ♦ Myriapoda (centipedes and millipedes) have a body divided into head and trunk, not into thorax and abdomen.
- b) In life cycle of Malaria parasite (*Plasmodium*), inside the stomach of the Female Anopheles mosquito, the Male gamete (microgamete) fuse with the female gamete

(macrogamete). This fusion forms a zygote. The zygote becomes elongated and motile, and this active stage is called Ookinete.

11. b) In Malaria, infected RBCs rupture during fever with chills (rigor) and release many malarial parasites into the blood. Therefore parasites are mostly easily detected at the time of rise of temperature with rigor. Trick: Rigor = Rupture of RBCs = Maximum parasites in blood.
12. c) In Earthworm, the intestine has an internal fold called the typhlosole. It increases the surface area of the intestine which helps in efficient absorption of digested nutrient.
13. b) In earthworm, the accessory glands open through the genital papillae. Their secretion helps during copulation by aiding attachment between two worms.
  - ♦ Prostate glands open into the male genital pore, not in genital papillae.
  - ♦ Cowper's glands are found in mammals, not in earthworm.
14. c)
  - ♦ Ecdysis → shedding or moulting of the outer covering.
  - ♦ Regeneration → Replacement of lost body parts or tissues.
  - ♦ Autolysis → Self-digestion of cells by their own enzymes.
  - ♦ Autotomy → self-amputation of a body part (e.g. tail loss in lizard).
15. c)
  - ♦ Gray crescent forms opposite to the point of sperm entry due to cytoplasmic rearrangement.
  - ♦ Dorsal lip of blastopore develops later during gastrulation from the gray crescent region.
  - ♦ The sperm entry point doesn't form the centre of rotation.
16. b) A PAP smear test is a screening technique used to detect cancerous or precancerous changes in the cervix of females. It is mainly used for the early detection of cervical cancer.
  - ♦ Breast cancer → detected by mammography.
  - ♦ Lung cancer → detected by X-ray, CT-scan, biopsy.
  - ♦ Oral cavity cancer → detected by oral examination and biopsy.
17. c) Typhoid is an infectious disease caused by bacterium Typhoid fever, mainly due to *Salmonella typhi*. It is commonly spreads through contaminated food and water.
18. a) Herpes simplex virus (HSV), especially HSV-2 is commonly transmitted from an infected mother to the baby during passage through the birth canal at the time of delivery. This is called neonatal herpes. The risk is highest when the mother has active genital herpes lesions during child birth.
19. c) Albendazole is the correct drug:
  - ♦ Metronidazole → used against amoebiasis and anaerobic bacterial infections.
  - ♦ Zidovudine → Antiretroviral drug used in HIV / AIDS.
  - ♦ Isoniazid → Used in tuberculosis treatment.

NAME \_\_\_\_\_

20. b) In Klinefelter syndrome, the chromosome pattern is XXY. Since there are two X chromosomes, one becomes inactive and forms a Barr body.  
 ♦ Normal male  $\rightarrow XY \rightarrow$  No Barr body.  
 ♦ Klinefelter male  $\rightarrow XXY \rightarrow$  one Barr body. Therefore, the presence of an extra Barr body indicates Klinefelter syndrome.
21. c) Chlorination is the process of adding chlorine to water to destroy harmful microorganisms such as bacteria, virus and protozoa. It is an important step in drinking water treatment to prevent water-borne disease like cholera and typhoid.
22. d) Ramsar sites are wetlands of international importance protected under the Ramsar Convention. Examples of Ramsar sites in Nepal include Koshi Tappu Wildlife Reserve and Ghodaghodi Lake.
23. c) ♦ Anadromous migration  $\rightarrow$  Fish migrate from sea to freshwater for spawning (e.g. salmon).  
 ♦ Catadromous migration  $\rightarrow$  Fish migrate from freshwater to sea for spawning (e.g. eel).  
 ♦ Oceanodromous migration  $\rightarrow$  Migration occurs entirely within the ocean.
24. c) When tissues are damaged, cells such as mast cells release histamine. Histamine causes dilation of blood vessel (Vasodilation). Increased permeability of capillaries. This allows white blood cells and plasma proteins to reach the injured area and initiate inflammation and healing.
25. b) Glial cells (neuroglia) are supporting cells of the nervous tissue. They do not conduct nerve impulses, so they are called non-excitabile cells. Their functions include supporting neurons, providing nutrition, forming myelin sheath and protection and repair of nervous tissue.
26. d) ♦ Mucus  $\rightarrow$  A slippery secretion that lubricates surfaces.  
 ♦ Elastin  $\rightarrow$  Provides elasticity to tissues like skin and blood vessels.  
 ♦ Collagen  $\rightarrow$  Structural protein providing strength to connective tissue.
27. c) Starch digestion starts in the mouth by the enzyme salivary amylase (ptyalin). It converts Starch  $\rightarrow$  Dextrin  $\rightarrow$  Maltose. When food reaches the stomach, salivary amylase acts for a short time and then becomes inactive due to acidic medium. Therefore, before the chyme enters the duodenum, digestion of starch is only partial. So, the chyme contain some undigested starch, Dextrin and maltose.
28. a) Tidal volume is the amount air inspired or expired during normal breathing i.e. TV = 500mL.
29. b) Carbon monoxide (CO) binds with haemoglobin about 200 times more strongly than oxygen, forming carboxyhaemoglobin, which is very stable and prevents oxygen transport.  
 ♦  $CO > O_2 > CO_2$  in strength of binding with haemoglobin.
30. b) The normal yellow colour of urine is due to the pigment Urochrome. Urochrome is produced from the breakdown of haemoglobin in old RBCs and excreted through urine.

31. a) The columns of Bertini are the extensions of the renal cortex that project inward between the renal pyramids of the medulla.  
♦ Renal pyramids → part of medulla.  
♦ Column of Bertini → cortical tissue between pyramids.
32. c) The coronary sulcus is an irregular groove present externally between the atria and ventricles. It contains coronary blood vessels supplying the heart.
33. c) When blood is lost, blood pressure decreases. To maintain adequate blood supply to vital organs, the blood activates the sympathetic nervous system which causes increased heart rate.
34. b) The almond-shaped amygdala is associated with emotional responses such as fear, anger and rage.
35. c) The Vagus nerve extensively supplies thoracic and abdominal organ is the longest and mostly branched.  
♦ Largest cranial nerve → Trigeminal nerve.  
♦ smallest cranial nerve → Trochlear nerve.
36. b) The thyroid gland is unique among endocrine glands because it stores its hormones extracellularly in the lumen of thyroid follicles as colloid (thyroglobulin). Later, the hormones  $T_3$  (triiodothyronine) and  $T_4$  (thyroxine) are released into the blood when needed.
37. c) Eye ball is moved in the orbit by 4 rectus and 2 oblique muscles. These muscles are controlled by III, IV and VI cranial nerve.
38. b) ♦ Scala vestibule and scala tympani along with bony labyrinth is filled with perilymph.  
♦ Scala Media, vestibule, semicircular canals are filled with endolymph.
39. b) Progesterone is responsible for implantation of embryo to the wall of uterus.
40. b) 1 Primary spermatocyte yields 4 sperms while 1 primary oocyte yield only 1 ova.

### Botany

41. a) This type of pollination is called geitonogamy, where pollen is transferred between different flowers of the same plant. It requires a pollinating agent, so it is ecologically cross-pollination, but since the pollen and ovule belong to the same plant, it is genetically self-pollination.
42. d) In higher plants (angiosperms), the male gametophyte is the pollen grain after germination, i.e., the pollen grain with a pollen tube, which contains the generative cell that forms two male gametes.
43. b) Ecesis is the successful establishment and growth of a species in a new area after its arrival during succession.

60. b) In meiosis, the bivalent (paired homologous chromosomes) is attached to the spindle fibers at the centromere, which helps in proper separation during cell division.
61. a) ATP synthesis in mitochondria occurs on the cristae (inner mitochondrial membrane) where the ATP synthase enzyme is located, facilitating oxidative phosphorylation.
62. c) Glyoxysomes are specialized peroxisomes found in germinating seeds that are involved in the metabolism of fats, converting stored lipids into carbohydrates via the glyoxylate cycle.
63. d) A gene is a short segment of DNA that carries the information necessary to synthesize a specific polypeptide or functional RNA, making it the basic unit of heredity.
64. c) A heterozygous individual carries two different alleles for a character and is therefore a hybrid, whereas homozygous dominant and recessive individuals are pure lines.
65. a) In incomplete dominance, the F<sub>2</sub> generation shows the parental phenotypes along with an intermediate phenotype (a blend of both). For example, crossing red (RR) and white (rr) flowers gives pink (Rr) in F<sub>1</sub>, and F<sub>2</sub> shows red, pink, and white in a 1:2:1 ratio.
66. c) During DNA replication, the RNA primers laid down on the lagging strand are removed by DNA polymerase I, which also fills the gaps with DNA nucleotides.
67. b) Klinefelter's syndrome occurs in males with an extra X chromosome (XXY). Individuals show tall stature, small testes, reduced fertility, and sometimes gynecomastia. Turner's syndrome is XO (female), Down's syndrome is trisomy 21, and super male is XYY.
68. c) During DNA replication, the lagging strand is synthesized discontinuously in short Okazaki fragments away from the replication fork, whereas the leading strand is synthesized continuously toward the replication fork.
69. d) When a pyrimidine (C or T) is replaced by a purine (A or G), it is called a transversion mutation. In contrast, a transition is a substitution of pyrimidine ↔ pyrimidine or purine ↔ purine.
70. b) Carbohydrate are the primary and fastest energy source.
71. c) Phloem transports prepared food from leaves to other parts.
72. a) Diatoms have silica-rich cell walls (frustules), which are hard, porous, and lightweight. These properties make diatomaceous earth useful as an insulating material in soundproof rooms and for filtration.
73. b) A soredium is a small reproductive structure in lichens consisting of a few algal cells surrounded by fungal hyphae, which helps in vegetative propagation.
74. a) In *Funaria*, the archegonial clusters (female reproductive organs) are surrounded by specialized leaves called perichaetial leaves, which protect and support the archegonia.
75. b) In pteridophytes (ferns and their relatives), the sporophyte is the dominant phase of the lifecycle and is diploid (2n).

76. c) In *Cycas*, the male gametes (antherozoids) are multiciliated and motile, which is a characteristic feature of gymnosperms like *Cycas*.
77. d) In *Pinus*, the wings of seeds develop from the ovuliferous scales of the female cone. These wings help in wind dispersal (anemochory) of the seeds. Bract scales, cone axis, and seed coat do not form the wings.
78. b) *Agave*: In *Agave*, vegetative reproduction occurs through bulbils, which are small buds formed in the axils of leaves or at the tip of the inflorescence.
79. d) In litchi, the edible white, juicy part is the fleshy aril, which is a special outgrowth from the seed and surrounds it.
80. b) In Solanaceae, the gynoecium is bicarpellary and syncarpous with a bilocular ovary showing axile placentation and a swollen placenta. The ovary is divided by an oblique septum, which is a typical diagnostic feature of this family.

### Chemistry

81. b)
82. a)  $\text{H}_3\text{PO}_3$  (phosphorus acid) is dibasic acid. So, its basicity is 2. Equivalent weight =  $\frac{M}{2}$
83. b) For a given  $n$ , the possible values of  $\ell$  range from 0 to  $n-1$  and value of  $m$  range from  $-\ell$  to  $+\ell$ , including zero. For  $n=2$ ,  $\ell \neq 2$  so, option (b) is not possible.
84. b) Ozone has a bent or V-shaped geometry and its bond angle  $116.8^\circ$  ( $\approx 117^\circ$ ).
85. d) In  $\text{NaCl}$  and  $\text{NH}_4\text{Cl}$  all the ionic, covalent and Co-ordinate bond Present.
86. c) Oleum, also known as pyrosulfuric acid, and formula  $\text{H}_2\text{S}_2\text{O}_7$ , Sulphur (S) is central element, let O.S of S be  $x$ ,  $2(+1) + 2(x) + 7(-2) = 0 \therefore x = +6$ .
87. b)  $1 \text{ M H}_2\text{SO}_4 = 2 \text{ N H}_2\text{SO}_4$  [ $\because \text{H}_2\text{SO}_4$  is diprotic acid donating 2 protons ( $\text{H}^+$ )]  
 $1 \text{ M NaOH} = 1 \text{ N NaOH}$   
 $N_1 V_1 = N_2 V_2$   
 $2 \times V_1 = 1 \times 10$   
 $V_1 = 5 \text{ mL}$
88. c) Cubic system  $\Rightarrow a = b = c$  and  $\alpha = \beta = \gamma = 90^\circ$   
 Triclinic system  $\Rightarrow a \neq b \neq c$  and  $\alpha \neq \beta \neq \gamma \neq 90^\circ$   
 Monoclinic system  $\Rightarrow a \neq b \neq c$  and  $\alpha = \gamma = 90^\circ, \beta \neq 90^\circ$
89. d) Equilibrium constant only dependent on temperature
90. d)  $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}); K_c = 4 \times 10^{-4}$   
 $\text{NO}(\text{g}) \rightleftharpoons \frac{1}{2} \text{N}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}); K'_c = \frac{1}{K_c}$  ( $\because$  Reaction is reversed  $K'_c = \frac{1}{K_c}$ )  
 Now,  $K'_c = \frac{1}{\sqrt{K_c}} = \frac{1}{\sqrt{4 \times 10^{-4}}} = 50$ .

91. a) A nuclear reactor is a device used to initiate and control a fission nuclear chain reaction.
92. c) Phenolphthalein is used for strong base and weak acid. Its pH range is 8.2-10.
93. a)  $W = \frac{VEN}{1000}$   
 $N = \frac{W}{VE} \times 1000 = \frac{0.023}{23 \times 100} \times 1000 = 10^{-2} \text{ M}$   
 $[\text{OH}^-] = 10^{-2} \text{ M}$   
 $\text{pOH} = -\log[10^{-2}] = 2, \text{pH} = 14 - 2 = 12$
94. a) Generally salts like KCl, KNO<sub>3</sub> and NH<sub>4</sub>NO<sub>3</sub> etc are used in salt bridge.
95. c) Mixing of gases is a spontaneous process. for spontaneous process → Free energy must be decreased ( $\Delta G < 0$ ) & entropy must be increased ( $\Delta S > 0$ ).
96. d) i.  $\Delta G$  is -ve → process spontaneous  
 ii.  $\Delta G$  is +ve → process spontaneous in reverse direction not in forward direction.  
 iii.  $\Delta G$  is zero → process equilibrium.  $\Delta G = \Delta H - T\Delta S$   
 For X:  $\Delta G = 10.5 \times 10^3 - 300 \times 30 = 1500$  i.e. +ve For Y:  $\Delta G = 1.08 \times 10^3 + 300 \times 100 = 31080$  i.e. +ve  
 For Z:  $\Delta G = -11.7 \times 10^3 + 300 \times 100 = 18300$  i.e. +ve  
 Hence, all values of  $\Delta G$  are +ve all the given are non-spontaneous.
97. d) Strongest acid = HClO<sub>4</sub>, weakest Conjugate base = ClO<sub>4</sub><sup>-</sup>  
 (v strongest acid have weakest conjugate base & vice versa).
98. b) All four of these species are isoelectronic, meaning they all have exact same number of electrons.  
 O<sup>2-</sup> has the fewest protons (8) to pull on the 10 e<sup>-</sup>, resulting in the weakest pull and the largest size.
99. a)  $\text{CO} + 2\text{H} \rightarrow \text{HCHO}$
100. a) Zone refining is used to obtain elements in an extremely high state of purity. It is primarily applied to semiconductor like Si, Ge.
101. a) NaCl (brine solution).  
 $\text{NaCl} + \text{NH}_3 + \text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{NaHCO}_3 + \text{NH}_4\text{Cl}$ .
102. a) Bachmann's plant is used in the manufacture of CaOCl<sub>2</sub> (Bleaching powder).
103. a)
104. b) Fullerene (C<sub>60</sub>) is known for its superconducting properties when doped with alkali metal.
105. d)  $\text{K}_2[\text{HgI}_4] \rightarrow$  Nessler's reagent.  
 $\text{NH}_3 + 2\text{K}_2[\text{HgI}_4] + 3\text{KOH} \rightarrow [\text{HgO Hg}(\text{NH}_2)\text{I}] \text{ (Brown ppt)} + 7\text{KI} + 2\text{H}_2\text{O}$
106. b)
107. b) Due to H-bond. HF has high est boiling point.

- In a tree ecosystem it is observed that several insect feed on a single large tree. These insects are eaten by small birds, which in turn are eaten by larger birds. The inverted pyramid is:
54. a) Pyramid of biomass  
b) Pyramid of energy  
c) Pyramid of number  
d) Pyramid of two conservation
  55. Which pollutant causes eutrophication of water bodies?  
a) Sulphur dioxide  
b) Nitrogen and phosphorus compounds  
c) Carbon monoxide  
d) Heavy metals
  56. Ozone layer in upper atmosphere (stratosphere) is destroyed by  
a) Fluorochloroalkanes  
b) Chlorofluorocarbons  
c) Sulphur dioxide  
d) Hydrochloric acid
  57. Which organelle modifies and packages proteins?  
a) Ribosome  
b) Smooth ER  
c) Golgi apparatus  
d) Lysosome
  58. A chromosome seen during anaphase resembles 'V' shape. What type of chromosome is this?  
a) metacentric  
b) submetacentric  
c) acrocentric  
d) telocentric
  59. Synapsis refers to  
a) separation of chromatids  
b) pairing of homologous chromosomes  
c) crossing over  
d) terminalization
  60. Failure of spindle fiber formation during mitosis will result in  
a) no DNA replication  
b) no chromosome condensation  
c) improper chromosome segregation  
d) cytokinesis without karyokinesis
  61. Which biomolecule shows peptide bond?  
a) Lipid  
b) Protein  
c) Carbohydrate  
d) Nucleic acid
  62. Test cross is used to determine  
a) phenotype  
b) dominance  
c) genotype of dominant individual  
d) linkage
  63. In incomplete dominance, heterozygote shows  
a) same phenotype as dominant  
b) same phenotype as recessive  
c) intermediate phenotype  
d) two phenotypes
  64. DNA-dependent RNA polymerase is active during  
a) replication  
b) transcription  
c) translation  
d) recombination
  65. Start codon for protein synthesis is  
a) UAA  
b) UGA  
c) UAG  
d) AUG
  66. A woman partner was hemophilic and marries a normal man. The resulting progeny will be  
a) All offspring normal  
b) All daughter normal  
c) All sons normal  
d) Some sons normal
  67. Polyphidy is commonly used in plants because it  
a) causes sterility  
b) reduces size  
c) increases vigor and yield  
d) prevents mutation
- 5
68. Independent assortment occurs when genes are located  
a) close on same chromosome  
b) on same chromosome  
c) on different chromosomes  
d) on homologous loci
  69. Which tissue provides mechanical strength without losing flexibility?  
a) Sclerenchyma  
b) Parenchyma  
c) Collenchyma  
d) Xylem
  70. Annual rings in trees are formed due to  
a) secondary phloem  
b) seasonal activity of cambium  
c) primary growth  
d) cork cambium
  71. Radial vascular bundles are characteristic of  
a) dicot stem  
b) monocot stem  
c) dicot root  
d) leaf
  72. Photorespiration occurs because RubisCO acts as  
a) dehydroxylase  
b) carboxylase  
c) diol enzyme  
d) dehydrogenase
  73. How many ATP molecules (in equivalents) are produced from the Krebs cycle alone, during the complete oxidation of one glucose molecule?  
a) 22  
b) 24  
c) 36  
d) 38
  74. Which hormone breaks seed dormancy?  
a) ABA  
b) Ethylene  
c) Cytokinin  
d) Gibberellin
  75. Osmotic pressure is measured by  
a) Potometer  
b) Osmometer  
c) Photometer  
d) Awaometer
  76. The triple fusion in angiosperms results in formation of  
a) zygote  
b) embryo  
c) endosperm  
d) perisperm
  77. Most common type of ovule in angiosperm is:  
a) Orthotropous  
b) Anatropous  
c) Campylotropous  
d) Amphitropous
  78. Seed dispersal by wind is best adapted in seeds having  
a) hooks  
b) wings or hairs  
c) fleshy fruits  
d) heavy endosperm
  79. Biofertilizers are preferred over chemical fertilizers because they  
a) increase pollution  
b) are non-renewable  
c) improve soil fertility naturally  
d) kill pathogens
  80. Golden rice is genetically engineered to be rich in  
a) protein  
b) iron  
c) vitamin A  
d) vitamin C
  81. Total number of electron present in 18 ml of Water is  
a)  $6.02 \times 10^{23}$   
b)  $6.02 \times 10^{24}$   
c)  $6.02 \times 10^{25}$   
d)  $6.02 \times 10^{26}$
  82. Hydrogen and oxygen combine to form  $H_2O_2$  and  $H_2O$  containing 5.98% and 11.2% hydrogen respectively. The data illustrates  
a) law of conservation of mass  
b) law of constant proportion  
c) law of reciprocal proportion  
d) law of multiple proportion
- 6

- a) gas becomes liquid  
b) gas cannot be liquified  
c) molecular motion ceases  
d) random molecular motion occurs
142. In gases of diatomic molecules, the ratio of two specific heats of gases is  
a) 1.2  
b) 1.3  
c) 1.4  
d) 1.6
143. The coefficients of linear expansion of two rods are  $\alpha$  and  $\beta$  respectively. If the length of first rod at  $0^\circ\text{C}$  is  $l_1$  and the difference of their lengths will remain the same at  $T^\circ\text{C}$ , then length of second rod at  $0^\circ\text{C}$  is  
a) 25 cm  
b) 35 cm  
c) 45 cm  
d) 55 cm
144. A gas performs no work, when it expands.  
a) isothermally  
b) isochorically  
c) isobarically  
d) adiabatically
145. The efficiency of a Carnot engine working between the temperature of steam and ice is  
a) 73.8%  
b) 98.9%  
c) 36.2%  
d) 26.8%
146. A body which emits radiations of all possible wavelengths, is known as  
a) good conductor  
b) partial radiator  
c) absorber of photons  
d) perfect black body
147. The energy radiated by a black body is directly proportional to  
a) its temperature  
b) second power of its temperature  
c) third power of its temperature  
d) fourth power of its temperature
148. An open resonating tube has a fundamental frequency of  $f$ . When half of the length is dipped into water, then its fundamental frequency will be  
a)  $f$   
b)  $2f$   
c)  $f/2$   
d)  $3/2 f$
149. If temperature of the atmosphere is increased, then which of following characteristic of sound waves is affected?  
a) velocity  
b) frequency  
c) amplitude  
d) wavelength
150. The voice of women is shriller because  
a) the pitch of their organs is more  
b) they have more energy  
c) the amplitude of vibration of their organs is more  
d) they have control over their tongue
151. If a glass rod is immersed in a liquid of same refractive index, then it will  
a) disappear  
b) look bent  
c) look longer  
d) look shorter
152. If the power of a lens is +5D, then its focal length is  
a) +0.2 cm  
b) -0.2 cm  
c) +20 cm  
d) -20 cm
153. In an equilateral triangular prism, the angle of minimum deviation for a certain wavelength is  $40^\circ$ . The corresponding angle of incidence is  
a)  $30^\circ$   
b)  $40^\circ$   
c)  $45^\circ$   
d)  $30^\circ$
154. Which nature of the wavefront is associated with a parallel beam of light?  
a) plane  
b) spherical  
c) cylindrical  
d) circular
155. At what angle of incidence, the light reflected from glass ( $\mu = 1.5$ ) is completely polarized?

- a)  $36.3^\circ$   
b)  $44.6^\circ$   
c)  $56.3^\circ$   
d)  $90^\circ$
156. What is the value of charge on a body which carries 20 excess electrons?  
a)  $3.2 \times 10^{-19}\text{ C}$   
b)  $3.2 \times 10^{19}\text{ C}$   
c)  $-3.2 \times 10^{-19}\text{ C}$   
d)  $-3.2 \times 10^{19}\text{ C}$
157. Electric field strength due to a point charge of 5  $\mu\text{C}$  at a distance of 80 cm from the charge is  
a)  $3 \times 10^3\text{ N/C}$   
b)  $3 \times 10^6\text{ N/C}$   
c)  $7 \times 10^3\text{ N/C}$   
d)  $9 \times 10^6\text{ N/C}$
158. Electric potential of earth is taken to be zero, because the earth is considered to be a good  
a) dielectric  
b) insulator  
c) conductor  
d) semi conductor
159. Two capacitors of capacitance 3  $\mu\text{F}$  and 6  $\mu\text{F}$  are connected in a series combination. Their equivalent capacitance will be  
a) 2  $\mu\text{F}$   
b) 3  $\mu\text{F}$   
c) 6  $\mu\text{F}$   
d) 9  $\mu\text{F}$
160. A current of 4A flows through 12  $\Omega$  resistor. What is the rate at which heat energy is produced in the resistor?  
a) 48 J  
b) 144 J  
c) 192 J  
d) 312 J
161. A 50V battery is connected across a 10  $\Omega$  resistor. If the current is 4.5 A, then internal resistance of the battery is  
a) zero  
b) 0.5  $\Omega$   
c) 11  $\Omega$   
d) 2.2  $\Omega$
162. Antimony and Bismuth are usually used in a thermocouple, because  
a) higher thermo emf is produced  
b) lower thermo emf is produced  
c) Constant thermo emf is produced  
d) negative thermo emf is produced
163. A 25W, 220V bulb and a 100W, 220V bulbs are joined in parallel and connected to the 220V mains. Which bulb will glow more brightly?  
a) 25W bulb  
b) 100W bulb  
c) both will glow with the same brightness  
d) first 25W bulb then 100W bulb
164. A solenoid has 1000 turns per meter length. If a current of 5A is flowing through it, then magnetic field inside the solenoid is  
a)  $2\pi \times 10^{-3}\text{ T}$   
b)  $2\pi \times 10^{-5}\text{ T}$   
c)  $4\pi \times 10^{-3}\text{ T}$   
d)  $4\pi \times 10^{-5}\text{ T}$
165. To send 10% of the main current through a moving coil galvanometer of resistance 99  $\Omega$ , the shunt required is  
a) 9.5  $\Omega$   
b) 10  $\Omega$   
c) 11  $\Omega$   
d) 9  $\Omega$
166. If material placed in a magnetic field is thrown out of it, then the material is  
a) diamagnetic  
b) paramagnetic  
c) ferromagnetic  
d) non-magnetic
167. What will be the phase difference between rms current and rms voltage when the current in the circuit is wattless?  
a)  $0^\circ$   
b)  $45^\circ$   
c)  $90^\circ$   
d)  $180^\circ$
168. The best material for the core of a transformer is  
a) soft iron  
b) mild steel  
c) hard steel  
d) stainless steel
169. If the velocity of an electron increases, then its de-Broglie wavelength will  
a) increase  
b) decrease

1

- Note: NAME students must fill their registration number on the OMR sheet.
- Flapper of whale and wing of hawk are
    - Hemoglobin
    - Myoglobin
    - Avianlike
    - Vestigial
  - As the proportion of gill rakers progressed, the increased length of their neck was also inherited in their offspring. This theory was given by:
    - Darwinism
    - Lamarckism
    - Mendelism
    - Morgan
  - Bighead lousecomen was derived from:
    - Human earlobe
    - Human nose
    - Nonhuman Man
    - Ameliodiploicus
  - Placental cells are present as excretory organ in:
    - Amniote
    - Protochordates
    - Reptera
    - Mammals
  - Locomotory organ of *Sponges* is:
    - Pseudopodia
    - Cilia
    - They lack locomotory organ
    - Mammals
  - Universal character of Coelenterates is:
    - Tentacles
    - Gelular Cell
    - Cnidoblast
    - Nephridia
  - Which among the following is viviparous:
    - Snake
    - Echidna
    - Seagull
    - Shagull
  - Basal cavity is lined by:
    - Keratinized stratified squamous epithelium
    - Non keratinized stratified squamous epithelium
    - Simple Columnar Epithelium
    - Simple Cuboidal Epithelium with microvilli
  - Haematan canals* are connected by:
    - Lacuna
    - Canaliculi
    - Osteocyte
    - Volkman Canal
  - Brush bordered epithelium contains:
    - Sarcocolla
    - Keracolla
    - Microvilli
    - Ovalith
  - Fibro Cartilage is present in:
    - Vertebral Disc
    - Fingertails
    - Pinnae
    - Eustachian Tube
  - The fertilization of malarial parasite occurs in:
    - Liver of man
    - Blood of man
    - Salivary gland of mosquito
    - Stomach of mosquito
  - Which is not multiple fission in *Plasmodium vivax*?
    - Gamogony
    - Schizogony
    - Oogony
    - Sporogony
  - Congulium* in earthworm is present in which of the following segments?
    - 10, 11 and 12
    - 11, 12 and 13
    - 14, 15 and 16
    - First 4

2

- Earthworm takes in solid food and crush it with which one of the following structure?
  - Nephridia
  - Gizzard
  - Ciliated
  - Seminal Vesicle
- Stomach of frog lacks:
  - Pylorus
  - Body sphincter
  - Cardia
  - Pancreas
- Fertilization in frog takes place in:
  - Water
  - Angula of Fallopian Tube
  - Bilder's Canal
  - Large Intestine
- Which of the following structures absorb the maximum amount of water?
  - Duodenum
  - Ileum
  - Caecum
  - Large Intestine
- Cardiac sphincter is present in:
  - Between Right Atricle and Ventricle
  - Between Left Atricle and Ventricle
  - Caeco-Duodenal Junction
  - Caeco-Duodenal Junction
- A man came running and sat in a bench. He took a deep breath in. This deep breath is:
  - Tidal Volume
  - Inspiratory Capacity
  - Respiratory Reserve Volume
  - Vital Capacity
- Shape of oxyhaemoglobin dissociation curve is:
  - Exponential
  - Linear
  - Sigmoid
  - Normal
- Pericardium of heart is:
  - SA Node
  - AV Node
  - Bundle of His
  - Purkinje Fibers
- The number of pulmonary vein draining to left atricle is:
  - 2
  - 3
  - 4
  - 5
- A female has menstrual cramps. This is due to contraction of which layer:
  - Endometrium
  - Myometrium
  - Perimetrium
  - Dartos Muscle
- A couple are unable to bear children. On sperm analysis, lower than normal count of sperm was found. This condition is known as:
  - Oligozoospermia
  - Polyzoospermia
  - Azoospermia
  - Annulla
- Blood brain barrier is formed by:
  - Endothelial Cells
  - Oligodendrocytes
  - Astrocites
  - Schwann Cells
- A child has fever. A doctor prescribes paracetamol. This drug decrease his temperature to normal by acting on:
  - Venicles
  - Primary Gland
  - Posterior Hypothalamus
  - Anterior Hypothalamus
- You are writing your CEE exam. Only half an hour is remaining and you have not solved Zoology questions yet. Suddenly you feel increased heart rate, your pupils are now dilated. Which hormone has caused this?
  - Epinephrine and Norepinephrine Both
  - Epinephrine Only
  - Norepinephrine Only
  - Thyroxine

## Zool383

1. a) They contain same basic anatomical framework but differ in function.
2. b) Lamarckian states that the characters acquired in one's lifetime can be inherited by offspring. Its major drawback is that the characters that the changes affecting germ cells are only transferred, not those affecting somatic cells.
3. d) First homoeoid to show bipedal locomotion was *Australopithecus Homo erectus* was not the origin of bipedal locomotion but they had advanced bipedal locomotion able to modern humans thus called erectus.
4. b) Amnion, excreted by Nephridia, pores — general body surface, Mammals — kidneys.
5. d) Sarcostia (amoeboid), Mastigophora (flagellates), Ciliophora (ciliates), and Sporozoa (non-motile/parasitic)
6. c) Cnidaria or Coelenterates have Cnidoblast with stinging organelle called nematocyst.
7. a) Although shark is Pisces it is viviparous. Though echidna and duck billed platypus are mammals they are oviparous.
8. b) Cervical Canal, Vagina and Anal Canal are lined by NKSS Epithelium.
9. d) Haversian Canal system is made up of Osteocyte, Lacuna and Canaliculi but connected by Volkmann canal.
10. c) Cilia are present in ciliated epithelium, any epithelium with microvilli is called brush bordered epithelium
11. a) Fibrocartilage are present in vertebral disc and pubic symphyses
12. c) Elastic Cartilage 3E: Eustachian Tube, Epiglottis, Ear Pinnae
13. c) Schizogony, Gamogony and Sporogony all are multiple fission.
14. c) Crangulum also known as chitellum is present in 14,15 and 16 segments and help in formation of cocoon.
15. b) Girazad present on 8th segment is highly muscular structure helping in crushing of food.
16. d) Anterior part of stomach of frog is cardia and posterior is pyloric. It lacks fundus and body.
17. a) Fertilization in frogs is external and occurs in water. This is one of the major reasons for its amphibious nature.
18. b) Ippurum has tallest and largest villi so it absorbs maximum amount of water and nutrients. Large intestine absorbs water only but not the maximum amount
19. c) Gastro-Oesophageal junction, Cardiac Sphincter, Gastro-Duodenal junction, Pyloric Sphincter

20. b) IC, IRV + IV, IRV. Forced inspiration after normal inspiration i.e. TV
21. c) Shape of Oxyhaemoglobin dissociation curve is sigmoidal, sigmoid, sigmoid, 25-30mm of Hg
22. a) SA node is where the impulse for heart beat gets generated. So, it is called Pacemaker of heart. External pacemakers are used when SA node can't generate proper impulse.
23. c) 2 pulmonary veins from each lung, then oxygenated blood into left atricle
24. b) Muscle cramps are due to contraction of myofibrillum. The layer that is sloughing off is Endothelium.
25. a) Polyzoopermia Poly- high number of sperm  
Oligozoopermia Oligo- low or little, little number of sperm
26. c) Acoelomate A. no. no amount of measurable space in between  
Astrocyte form Blood brain barrier that does not allow blood to directly enter the brain parenchyma. Only required nutrients and compounds can diffuse through it. It is a protective mechanism to prevent toxins and infectious agents to prevent from entering brain.
27. c) Anterior Hypothalamus is responsible for temperature regulation in the body. The function of Hypothalamus can be summarized by mnemonic: (SEAT)2  
Adrenal Medulla produces a mixture of 80% epinephrine and 20% norepinephrine. Both of these hormones are collectively called catecholamines and they stimulate sympathetic nervous system producing above mentioned features.
28. a) Hyperparathyroidism means high amount of parathormone in blood. Parathormone leads to reabsorption of bone minerals leading to condition called osteoporosis.
29. b) Pulse is felt due to distension of artery(which are elastic) after ventricles contract. It is most commonly felt at radial artery. Its value corresponds to the heart rate, ranging from 60bpm to 100bpm.
30. b) Presence of D antigen makes the blood group +ve. Its absence is termed as -ve blood group.
31. c) Vertical transmission is transfer of disease from mother to her child through placenta, blood or milk secretions.
32. c) Mycobacterium is called Acid Fast Bacilli.
33. d) Acid Fast: Wall contains Mycolic Acid
34. c) Bacilli: Because they are rod shaped
35. a) Eosinophil is increased in allergic reactions. Asthma, Allergic Rhinitis, Hay fever are allergic reactions.
36. a) Typhoid is called enteric fever. Its causative agent is *Salmonella typhi*

29. On X-ray you see thin bones and bone lesions. The condition that might be causing the bones to be thin might be:
- Hypoparathyroidism
  - Hypoparathyroidism
  - Hyperparathyroidism
  - Hyperparathyroidism
30. Distribution of Artery is:
- Heart Rate
  - Valve
  - Blood Pressure
  - ECG
31. A person's RBC has A antigen but no D antigen. What is the blood group of the person?
- A +ve
  - B +ve
  - A -ve
  - B -ve
32. A man feels a strong urge to urinate and urine starts passing without voluntary control. Which sphincter is responsible for this involuntary control?
- Internal Anal Sphincter
  - External Anal Sphincter
  - Internal Urethral Sphincter
  - External Urethral Sphincter
33. A female is easily fatigued. Has lost 10 kg in last 3 months. She has a history of risky sexual behavior. You did blood tests and found out HIV. Incidentally she is pregnant as well. There is a risk of transfer of disease to the fetus. This mode of transfer is called:
- Horizontal Transmission
  - Vertical Transmission
  - Female Transmission
  - Vertical Transmission
34. Which is known as Acid Fast Bacillus?
- Schistosoma*
  - Giardia*
  - Mycobacterium*
  - Infusoria*
35. A person is asthmatic. He complains of frequent allergies and hives all over the body. You collected blood sample and tested. Which of the following cells are expected to be high?
- Eosinophil
  - B Lymphocyte
  - T Lymphocyte
  - Basophil
36. Which among the following cause enteric fever?
- Salmonella typhi*
  - Shigella*
  - Giardia lamblia*
  - Infusoria*
37. Recently a kidney from a pig was transplanted to human in New York University. Which type of transplants is it?
- Autograft
  - Isograft
  - Allograft
  - Xenograft
38. Bear hootles are present in:
- Male
  - Female
  - Both male and female
  - Absent in mammalian male and female
39. Kallimandua has been affected by air pollution a lot in recent days. Which of the following is not an air pollutant?
- PM
  - CO
  - Ozone in stratosphere
  - Ozone in troposphere
40. Nepal has \_\_\_\_\_ number of national parks.
- 10
  - 11
  - 12
  - 13
41. A virus becomes metabolically active only after entering a host cell because
- It can be crystallized
  - It contains nucleoprotein
  - It is facultative parasite
  - It depends on host machinery for protein synthesis
42. Mycoplasma differs from other bacteria because it
- lacks DNA
  - lacks ribosomes
  - lacks cell wall
  - lacks plasma membrane
43. Organisms found in root of leguminous plant is:
- Azobacter*
  - Clostridium*
  - Rhizobium*
  - Frankia*
44. A fungus reproducing both sexually and asexually but with unknown sexual stage is classified as
- Ascomycete
  - Basidiomycete
  - Zygomycete
  - Deuteromycete
45. *Columnella* is found in the fruiting body of:
- Fungi
  - Muscor
  - Marchantia
  - Lichens
46. Spores and elaters of *Marchantia*:
- Haploid and diploid
  - Both diploid
  - Diploid and haploid
  - Both haploid
47. Which structure in pteridophytes is haploid?
- Sporophyte
  - Prothallus
  - Sporangium
  - Zygote
48. Presence of pollen tube was a major evolutionary advancement because it
- increased seed size
  - prevented embryo abortion
  - eliminated water requirement for fertilization
  - increased genetic variation
49. Which modification helps leaves reduce water loss in xerophytes?
- Broad lamina
  - Thin cuticle
  - Reduced surface area
  - Increased stomatal density
50. In axile placentation, ovules are attached to
- inner wall of ovary
  - central axis
  - base of ovary
  - margins of carpel
51. Phylogenetic classification is considered superior because it is based on
- morphology only
  - habitat preference
  - evolutionary relationships
  - economic utility
52. Net primary productivity is always
- greater than gross productivity
  - equal to respiration
  - less than gross productivity
  - zero in ecosystems
53. In area devastated by forest fire, which grows first?
- Liverworts
  - Moss
  - Fern
  - Algae

110. Which of the following is the weakest base?  
 a)  $\text{ClCH}_2\text{NH}_2$  b)  $\text{CH}_3\text{NH}_2$  c)  $\text{CH}_3\text{CH}_2\text{NH}_2$  d)  $\text{NH}_3$
111.  $\text{X} + \text{H}_2$  produces phenol as one of the products. X can be  
 a) Chlorobenzene b) Benzenechloride c) Benzene d) Benzene dichloride
112. Calcium benzoate and calcium acetate when together gives  
 a) Acetophenone b) Benzophenone c) Acetone d) Formaldehyde
113. Which of the following is decomposed by  $\text{KMnO}_4$  solution?  
 a)  $\text{CCl}_4$  b)  $\text{C}_2\text{Cl}_6$  c)  $\text{C}_2\text{H}_6$  d)  $(\text{CH}_3)_3\text{C}-\text{Cl}$
114. Which of the following oxides reacts with both  $\text{HCl}$  and  $\text{NaOH}$ ?  
 a)  $\text{CaO}$  b)  $\text{ZnO}$  c)  $\text{Na}_2\text{O}$  d)  $\text{CO}_2$
115. Which of the following oxides of nitrogen is anhydride of  $\text{HNO}_3$ ?  
 a)  $\text{NO}$  b)  $\text{NO}_2$  c)  $\text{N}_2\text{O}_4$  d)  $\text{N}_2\text{O}_5$
116. Chlorine acts as bleaching agent only in presence of  
 a) dry air b) Moisture c) Sunlight d) Pure  $\text{O}_2$
117. When  $\text{SO}_2$  gas is passed through acidified solution of  $\text{H}_2\text{S}$  it produces  
 a)  $\text{H}_2\text{SO}_4$  b)  $\text{H}_2\text{SO}_3$  c)  $\text{S}$  d)  $\text{H}_2\text{SO}_5$
118. Which of the following metal cannot be extracted by carbon reduction process?  
 a)  $\text{Cu}$  b)  $\text{Al}$  c)  $\text{Hg}$  d)  $\text{Zn}$
119. Which of the following is used as scavenger in metallurgy?  
 a)  $\text{Fe}$  b)  $\text{Na}$  c)  $\text{Ca}$  d)  $\text{Sr}$
120. Which of the following is not coloured salt?  
 a)  $\text{CuCl}_2$  b)  $\text{ZnCl}_2$  c)  $\text{CuSO}_4$  d)  $\text{MgSO}_4$
121. Which of the following is not an ore of silver?  
 a) Silver glance b) Horn silver c) German silver d) Pyrrhotite
122. Excess of  $\text{NaOH}$  reacts with  $\text{Zn}$  to form  
 a)  $\text{Zn}(\text{OH})_2$  b)  $\text{ZnH}_2$  c)  $\text{Na}_2\text{ZnO}_2$  d)  $\text{ZnO}$
123. Steel is alloy which contains  
 a)  $\text{Fe} + \text{C} + \text{Mn}$  b)  $\text{Fe} + \text{C} + \text{Al}$  c)  $\text{Fe} + \text{Mn}$  d)  $\text{Fe} + \text{Mn} + \text{C}$
124. The major ingredient of Portland cement is  
 a) Silicon of  $\text{Ca}$  b) Aluminium of  $\text{Ca}$  c) Iron (III) oxide d)  $\text{C}_2\text{S}$
125. What are carbide and anode in diaphragm cell to manufacture  $\text{NaOH}$ ?  
 a) Rotten Carbon b) Rotten Carbon c) Carbon anode, Steel cathode d) Carbon cathode, Steel anode
126. Hypo is used in photography to  
 a) Reduce  $\text{AgBr}$  giving metallic  $\text{Ag}$

- b) Convert metallic  $\text{Ag}$  to  $\text{Ag}$  salt
127. Remove undecomposed  $\text{AgBr}$  as a soluble complex  
 a) Remove undecomposed  $\text{AgBr}$  as a soluble complex b) Remove reduced silver c) Distillation d) Crystallization
128. Latest technique for purification, isolation and separation of organic compound is  
 a) Distillation b) Crystallization c) Sublimation d) Chromatography
129. Which of the following compound will give blood red colour with Lassaigne's test?  
 a)  $(\text{NH}_4)_2\text{CO}$  b)  $\text{H}_5\text{N}(\text{CH}_2\text{SO}_3\text{H})$  c)  $\text{CH}_3\text{SO}_3\text{H}$  d)  $\text{CH}_3\text{Cl}$
130. An inorganic salt solution on treatment with  $\text{HCl}$  gives white ppt. Which of the ion is possible?  
 a)  $\text{Ag}^+$  b)  $\text{Cu}^{2+}$  c)  $\text{Zn}^{2+}$  d)  $\text{Cd}^{2+}$
131. The no. of significant figures in 0.0003125 is  
 a) 4 b) 5 c) 6 d) 7
132. If  $\vec{A} \times \vec{B} = \vec{A} \times \vec{B}$ , then angle between A and B will be  
 a)  $30^\circ$  b)  $45^\circ$  c)  $60^\circ$  d)  $90^\circ$
133. If two bodies are projected at  $30^\circ$  and  $60^\circ$  respectively with same speed, then their  
 a) ranges are same b) heights are same c) times of flight are same d) striking velocities are same
134. When two surfaces are coated with a lubricant, then they  
 a) stick to each other b) slide upon each other c) roll upon each other d) are attracted towards each other
135. The masses of two balls are 40 gm and 90 gm respectively. If their kinetic energies are same then the ratio of their momentum will be  
 a) 4/9 b) 9/4 c) 3/2 d) 2/3
136. An electric motor operates at 20 rev/sec. What will be the power delivered by the motor, if it supplies a torque of 75  $\text{Nm}$ ?  
 a) 1500 W b) 4040 W c) 7730 W d) 9420 W
137. At any point below the surface of the earth, acceleration due to gravity varies as (Distance between the centre of earth and the point = r)  
 a)  $r$  b)  $r^2$  c)  $r^3$  d)  $r^4$
138. A train is moving with a constant velocity. If a pendulum is hanging from its roof, then its time period will  
 a) increase b) decrease c) remain the same d) become zero
139. Shearing stress produces a change in  
 a) area b) shape c) length d) volume
140. The shape of liquid meniscus in capillary tube will be concave upwards, if the angle of contact is  
 a) equal to  $0^\circ$  b) equal to  $90^\circ$  c) less than  $90^\circ$  d) greater than  $90^\circ$
141. Absolute zero is the condition at which

NAME \_\_\_\_\_

196. Select the figure which satisfies the same conditions of placement of the dots as in Figure-X.



(X)



(1)



(2)



(3)



(4)

a) 1

b) 2

c) 3

d) 4

197. Choose a figure which would most closely resemble the unfolded form of Figure (Z).



P



X



Y



Z



(1)



(2)



(3)



(4)

a) 1

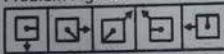
b) 2

c) 3

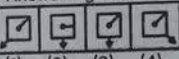
d) 4

198. Select a figure from amongst the Answer Figures which will continue the same series as established by the five Problem Figures.

Problem Figures:



Answer Figures:



(1)

(2)

(3)

(4)

a) 1

b) 2

c) 3

d) 4

199. Find the number of triangles in the given figure.



a) 36

b) 40

c) 44

d) 48

200. Choose the figure which is different from the rest.



(1)

(2)

(3)

(4)

(5)

Result will be published on Sunday

Log on to [www.name.edu.np](http://www.name.edu.np)

or

[www.facebook.com/nameinstitute](http://www.facebook.com/nameinstitute)

Best of Luck

37. d) Transplantation from one to another species is called Xenograft.  
38. b) Barr bodies are inactivated X chromosomes. Since females have 2 X chromosomes, one is inactivated and condensed in the cell and is called Barr bodies.  
39. d) Ozone in the troposphere is a pollutant whereas in the stratosphere it is protective against UV radiation.  
40. d) Previously there were 12 national parks. In 2025, 13th National Park was established in Mizor, It is Chhayanath National Park.

### Botany

41. d) Mycoplasma is the smallest living cell and uniquely lacks a cell wall, making it resistant to penicillin.  
42. c) Azotobacter → Free-living aerobic nitrogen fixer  
Clostridium → Free-living anaerobic nitrogen fixer  
Frankia → Symbiotic nitrogen fixer in some non-leguminous plants  
Deuteromycetes (fungi imperfecti) lack a known sexual stage.  
44. d) The spores of *Marchantia* are produced by meiosis from the sporophyte and are therefore haploid. The elaters are diploid cells in the capsule that aid in spore dispersal; they are part of the sporophyte generation, which is diploid.  
47. b) The prothallus is the haploid gametophyte.  
48. c) Pollen tube directly delivers male gametes to egg, removing dependence on water.  
49. c) Inner wall of ovary → Parietal placentation  
Base of ovary → Basal placentation  
50. b) Margins of carpel → Marginal placentation  
51. c) NPP = CPP - Respiration, so it is always less than CPP.  
52. c) Mosses are often among the first to grow in a burned, cleared area because they can tolerate harsh conditions and colonize barren ground, helping to build new soil.  
53. b) Excess nitrates and phosphates cause algal blooms and oxygen depletion.  
54. c) Sulphur dioxide → Causes acid rain  
55. b) Carbon monoxide → Air pollutant reducing oxygen transport in blood

NAME \_\_\_\_\_

Solutions for MCQs: Bot. Model Exams: Exam Set-4

56. b) Heavy metals → Cause toxicity and biomagnification, not eutrophication.  
57. c) Metacentric: V shape  
58. b) Submetacentric: L shape  
Acrocentric: I shape  
Telocentric: J shape  
59. b) Synapsis occurs during zygotene of prophase I.  
60. c) Spindle fibers are needed to pull sister chromatids to opposite poles during mitosis. If they fail to form, chromosomes cannot separate properly.  
61. b) Amino acids are linked by peptide bonds in proteins.  
62. c) Neither allele is fully dominant, resulting in intermediate expression.  
63. c) RNA polymerase synthesizes RNA from DNA template.  
64. b) AUG codes for methionine and initiates translation.  
65. d) Daughters carrier but phenotypically normal (X<sup>h</sup>X)  
66. b) Sons all hemophilic (X<sup>h</sup>Y)  
67. c) Polyploid plants often show gigantism and increased productivity.  
68. c) Genes on different chromosomes assort independently.  
69. c) Collenchyma has uneven thickening, allowing flexibility.  
70. b) Roots show radial arrangement of xylem and phloem.  
71. c) One glucose produces 2 acetyl-CoA, and each acetyl-CoA in Krebs cycle yields 3 NADH (9 ATP) + 1 FADH<sub>2</sub> (2 ATP) + 1 GTP (1 ATP) = 12 ATP.  
72. c) So, for 2 acetyl-CoA → 12 × 2 = 24 ATP.  
73. b) Gibberellins stimulate enzyme synthesis during germination.  
74. d) Fusion of male gamete with two polar nuclei forms triploid endosperm.  
75. b) Wings/hairs increase surface area and buoyancy.  
76. c) They enhance nutrient availability without harming environment.  
77. b) Golden rice produces β-carotene, a precursor of vitamin A.  
78. b) Golden rice produces β-carotene, a precursor of vitamin A.  
79. c) Golden rice produces β-carotene, a precursor of vitamin A.  
80. c) Golden rice produces β-carotene, a precursor of vitamin A.

# **NAME**

**B**

NAME Institute

In front of Singhdurbar, Putalisadak

Tel: - 01- 5971544

## **MECEE BL**

### **Model Entrance Exam**

### **(Model: III)**

Date: Jestha 16<sup>th</sup> 2083 (Saturday)

Venue: NAME

Duration - 3 hrs

Time - 11:00 AM

#### **INSTRUCTIONS**

There are 200 multiple-choice questions, each having four choices of which only one choice is correct.

Fill ☐ the most appropriate one.

83. In hydrogen spectrum, the series of lines appearing in UV region of electromagnetic spectrum are
- Lyman series
  - Balmer series
  - Paschen series
  - Brackett series
84. Total number of elements in the 4th period of periodic table is
- 18
  - 10
  - 15
  - 32
85. For electron affinity of halogens, which of the following is correct?
- $F > Cl$
  - $F > Br$
  - $F > I$
  - $F > At$
86. Which atomic orbital is always involved in sigma bonding only?
- p
  - d
  - s
  - d
87. For the reaction  $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$  which of the following statement is correct?
- $\Delta H^\circ = \Delta E$
  - $\Delta H^\circ > \Delta E$
  - $\Delta H^\circ < \Delta E$
  - $\Delta H^\circ = 0$
88. The reaction  $2SO_2 + O_2 \rightleftharpoons 2SO_3$  will be favoured by
- high temperature and low pressure
  - low temperature and high pressure
  - high temperature and high pressure
  - low temperature and low pressure
89. pH of  $Ca(OH)_2$  solution is 12. Milli-moles of  $Ca(OH)_2$  present in 100 ml. of solution will be
- 1
  - 0.5
  - 0.05
  - 5
90. Which of the following solution can not act as a buffer?
- $Na_2CO_3 + H_2CO_3$
  - $CH_3COOH + CH_3COONa$
  - $H_2PO_4 + NaH_2PO_4$
  - $HCl + NH_4OH$
91. Compounds A and B have same gases, pressure, volume and temperature of A are all twice that of B, then the ratio of number of molecules of A and B are
- 12
  - 21
  - 14
  - 43
92. Among solids, the highest melting point is exhibited by
- ionic solid
  - molecular solid
  - metallic solid
  - pseudosolid
93. Which of the following does not affect the rate of reaction?
- Amount of reactant
  - Physical state of reactant
  - Size of vessel
  - ΔH of reaction
94. 100 cc of life  $H_2SO_4$  and 200 cc of 0.3 N HCl are mixed together, the normality of the solution is
- 0.2 N
  - 0.4 N
  - 0.8 N
  - 0.6 N
95. The number of electron lost or gained during the change,  $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$  is
- 2
  - 4
  - 6
  - 8
96. For the electrochemical cell,  $Ag | AgCl(s) | KCl(aq) || AgNO_3(aq) | Ag$  The overall cell reaction is
- $Ag^+ + Cl^- \rightarrow AgCl + e^-$
  - $Ag^+ + Cl^- \rightarrow Ag + Cl_2$
  - $AgCl \rightarrow Ag^+ + Cl^-$
  - $Ag + Cl^- \rightarrow AgCl$
97. The isomer of  $H_2C=CH_2$  is

7

8

98. A nucleophilic reagent must necessarily have
- an overall +ve charge
  - an overall -ve charge
  - an unpaired electron
  - a species with complete octet and lone pair of electron
99. The IUPAC name of compound  $CH_3 - CH = CH - COOCH_3$  is
- Ethyl acrylate
  - Ethyl methyl butenoate
  - Ethyl acetoacetate
  - Ethyl 3-methylbut-3-enoate
100. The total number of isomers for compound  $C_4H_8$  is
- 3
  - 4
  - 5
  - 6
101. Which of the following on treatment with hot alkaline  $KMnO_4$  gives benzoic acid?
- Chlorobenzene
  - Nitrobenzene
  - Phenol
  - Ethylbenzene
102. Which of following cannot produce alkane by reacting with sodium in presence of dry ether?
- $CH_3Cl$
  - $CH_3CH_2Cl$
  - $(CH_3)_2CHCl$
  - $(CH_3)_3CCl$
103.  $(CH_3)_3CH \xrightarrow[OH^-]{KMnO_4} A \xrightarrow[OH^-]{O_2} B$ . The compound can be
- Dimethyl sulphate
  - 1,2-dichloroethane
  - 3-chloropropene
  - 1-chloro-2-iodopropane
104. An alcohol on vigorous oxidation is found to give ethanoic acid and propanoic acid. The alcohol may be
- pentan-1-ol
  - pentan-2-ol
  - butan-1-ol
  - butan-2-ol
105. Which of the following is most acidic?
- Phenol
  - Benzyl alcohol
  - m-chlorophenol
  - Chloroacetal
106. Diethyl ether can be decomposed by
- H<sub>2</sub>
  - NaOH
  - Water
  - $KMnO_4$
107. In which of the following reaction, both oxidised and reduced forms of the same compound are obtained?
- Alcohol condensation
  - Canizzaro reaction
  - Reimer-Tiemann reaction
  - Kolbe reaction
108. Addition of  $HBr$  to  $CH_2=CHCOOH$  results in formation of
- β-bromopropionic acid as major product
  - β-bromopropionic acid as major product
  - Both in equal amount
  - $CH_3-CH_2Br$
109. Ethyl alcohol react with acetyl chloride to form
- Ethyl chloride
  - Acetic acid
  - Methyl acetate
  - Ethyl acetate
110. Which is the correct sequence to convert  $R-CH_2OH \rightarrow R-CH_2NH_2$
- $NH_3, H_2O, KCN$
  - $PbO, KCN, H_2/P$
  - $PbO, KCN, H_2/P$
  - $KCN, H_2O, H_2/P$



# **NAME**

**NAME Institute**

In front of Singhdurbar, Putalisadak

Tel: - 01-5971544

## **HINTS AND SOLUTIONS**



# **MECEE BL**

## **Model Entrance Exam**

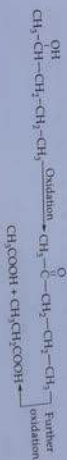
**(Model- III)**

**Date: 2083/02/16**

## Chemistry

81. c)  $nH_2O = \frac{18}{18} = 1$  mole  
No. of  $e^- = 1 \times 6.02 \times 10^{23} \times 10 = 6.02 \times 10^{24}$
82. d) Two elements combine in different ratio to form two or more compound in law of multiple proportion.
83. b) When electron comes to first orbit release UV ray which is Lyman series.
84. c) Straight forward.
85. d) correct order:  $Cl > F > Br > I$
86. c) s and hybrid orbital only form sigma bond.
87. c)  $\Delta H = \Delta E + \Delta nRT$   
 $\Delta nRT = +2$
88. b) Exothermic reaction and less number of moles in product side.
89. b)  $pH = 12$ ,  $pOH = 2$   
 $nOH = 10^{-2} \times 100 \times 10^{-3}$   
 $nCa(OH)_2 = \frac{100}{2} \times 10^{-5} = 0.5$  millimoles
90. c) Buffer must contain weak acid/base and conjugate salt. But HCl is strong.
91. b)  $\frac{P_1 V_1}{P_2 V_2} = \frac{n_1 R T_1}{n_2 R T_2}$   
 $\frac{2P \times 2V}{P'V} = \frac{n_1 \times 2T}{n_2 \times T}$   
 $n_1 n_2 = 2:1$
92. a) Ionic bond is strong.
93. c) Enthalpy change has no effect on rate.
94. b)  $N_{\text{max}} \times V_{\text{max}} = N_1 V_1 + N_2 V_2$   
 $N_{\text{max}} = \frac{100 \times 0.6 + 200 \times 0.3}{300} = 0.4$  N
95. d) Loss or gain of electron = No. of atom  $\times$  change in O.N.  
 $= 3 \times \frac{8}{3} = 8$
96. d) Oxidation half:  $Cl^- \rightarrow \frac{1}{2} Cl_2 + e^-$  reduction half:  $e^- + Ag^+ \rightarrow Ag$
97. b) No. of neutron in Ce =  $76 - 32 = 44$

98. d) Nucleophile must contain lone pair.
99. d) prefix: Ethyl, word root: but, primary suffix: -3-ene.
100. d) alkene - 4 (including cis + trans), cycloalkane - 2.
101. d)
102. d) A 3° haloalkane undergoes elimination.
103. c)  $CH_3-CH-CH_3 \xrightarrow{I} CH_3-CH=CH_2 + HI$
104. b)
105. c) Acidic nature: phenol > alcohol  
Cl is electron withdrawing group increases acidic nature.  
 $CH_3CH_2OCH_2CH_3 + HI \rightarrow CH_3CH_2OH + CH_3CH_2I$
106. a) In Cannizzaro reaction aldehyde without  $\alpha$ -hydrogen undergoes both oxidation & reduction.
107. b)
108. b) Follow anti markovnikov's rule.
109. d)  $C_6H_5OH + CH_3COCl \rightarrow CH_3COOC_6H_5 + HCl$
110. c)  $R_2CH-OH \rightarrow R-CH_2-Br \rightarrow R-CH_2-CN \rightarrow R-CH_2CH_2NH_2$
111. c) Alcohol is acidic in nature.
112. d)
113. a)  $Ca(C_2H_5COO)_2 + Ca(CH_3COO)_2 \rightarrow C_6H_5-C(=O)-CH_3$
114. d) Alkene gives unsaturation test.
115. b)  $ZnO$  is amphoteric.
116. b)  $N_2O_5 + H_2O \rightarrow HNO_3$



117. b)  $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{HOCl}$   
 Nascent oxygen is bleaching agent.
118. c)  $\text{SO}_2 + \text{H}_2\text{S} \rightarrow \text{H}_2\text{O} + \text{S}$
119. b) Al is extracted by electrolytic method.
120. c)
121. b)  $\text{Zn}^{2+} \rightarrow \text{Zn}^{40}$   
 $d^0$  and  $d^{10}$  are colourless.
122. c) German Silver - Alloy of Cu, Zn and Ni.
123. c)  $\text{Zn} + \text{NaOH} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$
124. a) 70-80% of cement consists of trisilicium silicate and Dicalcium silicate.
125. a)
126. b)
127. c)  $\text{AgBr}$  convert into soluble complex.
128. d)
129. b) Organic compound containing both nitrogen and sulphur can form  $\text{Fe}(\text{CNS})_3$ , blood red colour.
130. a)  $\text{Ag}^+$  is group I basic radical which give ppt in  $\text{HCl}(\text{aq})$ .
131. a) No. of significant figure is counted from the first non-zero digit on the left to the last digit on the right, so no. of significant figures in the given number is 4.
132. b)  $|\vec{A} \times \vec{B}| = |\vec{A} \cdot \vec{B}|$ , then angle between  
 $AB \sin \theta = AB \cos \theta$   
 $\tan \theta = 1 = \tan 45^\circ$   
 $\theta = 45^\circ$
133. a) Sum of angles of projection =  $30^\circ + 60^\circ = 90^\circ$   
 So their ranges are same.
134. b) By using lubricant, friction of the surfaces decreases. Therefore they slide upon each other.
135. d)  $P = \sqrt{2mK}$   $K \propto \frac{P^2}{m}$   $\frac{P_1}{P_2} = \sqrt{\frac{m_1}{m_2}}$   $= \sqrt{\frac{40}{90}} = 2.3$
136. d)  $P = 10 = \tau \times 2\pi r = 75 \times 2 \times 3.14 \times 20 = 9420 \text{ W}$
137. a)  $\frac{GM}{R^2} = \frac{R^2}{R^3} = \frac{1}{R}$   
 $g \propto \frac{1}{R}$
138. c)  $T = 2\pi \sqrt{\frac{l}{g}}$   
 $T^2 = 2\pi \sqrt{\frac{l}{g_{\text{actual}}}} = 2\pi \sqrt{\frac{l}{g}}$
139. b)  $g_{\text{actual}} = (g + a) = g$  (since  $a = 0$  ( $v = \text{constant}$ ))  
 Shearing stress produces a change in shape only.
140. c) Less than  $90^\circ$
141. c) Absolute zero temp  $T = 0 \text{ K}$   
 $V_{\text{rms}} = \sqrt{\frac{3RT}{M}} = 0$
142. c) For diatomic gases  $C_p = C_v + R$   
 $C_v = \frac{5R}{2}$   $C_p = \frac{7R}{2}$
143. a)  $\Delta l = \alpha l \Delta T$   
 Since  $\Delta l$  remains same,  $l \propto \frac{1}{\alpha}$   
 $\frac{l_1}{l_2} = \frac{\alpha_2}{\alpha_1}$   
 $l_1 = \frac{\alpha_2}{\alpha_1} \times 10 = 2.5 \text{ cm}$   
 $W = PdV = 0$  if  $dV = 0$  ( $V = \text{const}$ ).  
 Therefore a gas performs no work when it expands isochorically.
144. b)
145. d)  $\eta = \left(1 - \frac{T_2}{T_1}\right) \times 100\% = \left(1 - \frac{273}{373}\right) \times 100\% = 0.268 \times 100\% = 26.8\%$
146. d)  $E = n\Delta T$   
 $E \propto T^4$
148. a)  $f_0 = \frac{v}{\lambda} = f_1 = \frac{v}{\lambda_1}$   $f_0 = f_1$
149. a)  $v = \sqrt{\frac{gH}{M}} = v \propto \sqrt{H}$

150. a) If a glass rod is immersed in a liquid of same refractive index, no refraction takes place. Therefore the glass rod will disappear.
152. c)  $f = \frac{1}{P} = \frac{1}{0.5} \text{ m} = 0.2 \text{ m} = 20 \text{ cm}$
153. d)  $\lambda = \frac{h \cdot m \cdot v}{2\pi} = \frac{6.6 \times 10^{-34} \times 4 \times 10^6}{2} = 50^\circ$
154. a) Parallel beam of light forms plane wavefront.
155. c)  $\theta_p = \tan^{-1}(1.5) = 56.3^\circ$
156. c)  $q = ne$   $q = -20 \times 1.6 \times 10^{-19} = -3.2 \times 10^{-18} \text{ C}$
157. c)  $E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = 9 \times 10^9 \times \frac{5 \times 10^{-18}}{(0.02)^2} = 7 \times 10^4 \text{ N/C}$
158. c)  $C_1 = \frac{C_1 C_2}{C_1 + C_2}$   $C_2 = \frac{3 \times 6}{3 + 6} = \frac{18}{9} = 2 \mu\text{F}$
160. c)  $P = FR = 4^2 \times 12 = 192 \text{ J}$
161. c)  $\tau = \frac{h \cdot 10^{-18}}{1} = \frac{50 \cdot 4.5 \times 10^{-18}}{4.5} = 1.1 \Omega$
162. a) Antimony and bismuth form the pair of metals producing a higher thermo emf in comparison with other pairs of thermocouples.
163. b) In parallel combination,  
 $\frac{1}{L_2} = \frac{R_1}{R_2} = \frac{R_2}{R_1} \cdot L \propto P$   
 Thus 100W bulb will glow more brightly.
164. a)  $B = \mu_0 n I$   
 $= (4\pi \times 10^{-7}) (1000) \times 5 = 2\pi \times 10^{-3} \text{ T}$
165. c)  $I_g = 10\% I = 0.11$   $S = \left(\frac{I_g}{I - I_g}\right) \times G$   
 $= \left(\frac{0.11}{1 - 0.11}\right) \times 99$   
 $= 11 \Omega$
166. a) Diamagnetic materials are repelled when they are placed in a magnetic field.
167. c)  $P = I_{rms} V_{rms} \cos \phi$   
 $0 = I_{rms} V_{rms} \cos \phi$   
 $\cos \phi = 0$   
 $\phi = 90^\circ$
168. a) Transformer core is made by placing soft iron strips one above the other.
169. b)  $\lambda = \frac{h}{mv}$   $\lambda \propto \frac{1}{v}$

170. a)  $K = eV$   
 $= (1.6 \times 10^{-19})(100)$   
 $= 1.6 \times 10^{-17} \text{ J}$
171. c)  $E_{\text{max}} = \phi - \phi_0 = (6 - 2.1) \text{ eV} = 3.9 \text{ eV}$   $eV_s = E_{\text{max}}$   
 $eV_s = 3.9 \text{ eV}$   $V_s = 3.9 \text{ V}$   
 Since  $V_s$  is negative so,  $v_s = -3.9 \text{ V}$
172. a)  $E_n = -\frac{13.6}{n^2} = -\frac{13.6}{(1.5)^2} = -13.6 \text{ eV}$
173. c)  $R \propto A^{1/3} \frac{R_1}{R_2} = \left(\frac{R_1}{R_2}\right)^{1/3}$
174. d) 1 Curie =  $3.7 \times 10^{10} \text{ dps}$
175. a)  $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n$   $\frac{1}{16} = \left(\frac{1}{2}\right)^n$   $\left(\frac{1}{2}\right)^4 = \frac{1}{16}$   $n = 4$   
 $t = n T_{1/2} = 4 \times 100 = 400 \text{ s}$
176. a)  $\lambda_{\text{min}} = \frac{hc}{eV}$   
 $\lambda_{\text{min}} = \frac{12400}{V} \text{ \AA}$   
 $V = 10^4 \text{ volt}$   
 $= 10 \text{ kV}$
177. b) Intrinsic semiconductor
178. a) Electron is lighter than hole.  
 $\mu_e > \mu_h$
179. a)
180. b)  $v = Hr$   $v \propto r$
- MAT
181. d) Position from left =  $(40 + 1) - 14 = 27$
182. b) Direct coding concept.
183. a) First, a candidate submits an application. This is followed by a written examination. After that, an interview is conducted, and once the results are published, if the candidate is successful, they are given an appointment letter. Therefore, the meaningful sequence is 2, 5, 3, 4, 1.
184. b) Husband and Wife exist as a pair (couple), which does not match the relationship in the other options.
185. b) Total no. of handshakes =  $\frac{n(n-1)}{2} = 66$   $\rightarrow n = 12$

NAME \_\_\_\_\_

186. c)  $(0 + 6 + 4 + 2) - 2 = 10$

$(6 + 2 + 10 + 8) - 2 = 24$

$(4 + 14 + 12 + 10) - 2 = 38$

No. of boys =  $\frac{3}{4+3} \times 49 = 21$

187. c) No. of girls =  $\frac{4}{4+3} \times 49 = 28$ ,

When 4 girls leave, Required ratio =  $\frac{24}{21} = \frac{8}{7}$

188. d)

The word is divided into 3 groups of 4 letters each and the letters of each group are written in reverse order to obtain the code.

MATHEMATICAL  $\rightarrow$  MATH/EMAT/ICAL  $\rightarrow$  HTAM/TAME/LACI

189. a) Common coding concept.

190. a) By substituting the letters with mathematical signs as per the question  
 $8 \times 3 + 24 + 12 - 19 = 7$

191. a)

192. a) Total work = LCM(12,36)=36, Eff. of Roshan and Sangita=3:1

Let the number of days as working together be x then  
 $(R+S)x + S \times 4 = Tw$

Total time taken =  $(8 + 4) = 12$  days

193. a) Valid vote =  $\frac{80}{100} \times 7500 = 6000$

Secured vote by second candidate =  $\frac{45}{100} \times 6000 = 45 \times 60 = 2700$

194. c) There is no H in INFRASTRUCTURE. So, CHARTER word cannot be formed.

195. a)  $gf e i i / gf e i i / gf e i i / gf e i i$

196. d)

197. c)

198. c)

The pin rotates 45oCW and 90oCW alternately and moves one space (each space is equal to half-a-side of the square) and two spaces CW alternately. The arrow rotates 90oACW and 45oACW alternately and moves two spaces and one space.

199. d)

200. a) In all other figures, there are two small line segments towards the pin and three small line segments towards the arrow.

Result will be published on Sunday

Best of Luck